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EXERCISE - I (Conceptual Questions) Build Up Your Understanding

Classification

1. The hybrid state of C-atoms which are attached to a single bond with each other in the following structure are : $CH_2=CH—C\equiv CH$

(1) sp^2, sp (2) sp^3, sp
(3) sp^2, sp^2 (4) sp^2, sp^3

CCN001

2. The number of olefinic bonds in the given compound is/are :-

$CH_2=CH—\underset{\underset{O}{\|}}{C}—CH=CH—C\equiv N$

(1) 2 (2) 3 (3) 1 (4) 4

CCN006

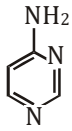
3. The number of acetylinic bonds in the given compound is/are :

$HC\equiv C—\underset{\underset{O}{\|}}{C}—CH=CH—C\equiv N$

(1) 2 (2) 3 (3) 1 (4) 4

CCN007

4. The number of hetero atoms present in the following compound is/are :



(1) 2 (2) 3 (3) 1 (4) 4

CCN010

5. Which of the following compound has sp-hybridised carbon atom :-

(1) CH_3COOH
(2) CH_3COCH_3
(3) CH_3CH_2CN
(4) $CH_2=CH–CH=CH_2$

CCN011

6. In compound $HC\equiv C—CH_2—CH=CH—CH_3$, the $C_2—C_3$ bond is the type of :-

(1) sp – sp^2 (2) sp^3 – sp^3
(3) sp – sp^3 (4) sp^2 – sp^2

CCN012

7. Molecular formula $C_4H_8O_2$ represents :-

(1) An acid only
(2) An ester only
(3) An alcohol only
(4) An acid and an ester also

CCN003

8. The higher homologue of N-methyl methanamine ($CH_3—NH—CH_3$) has the structure :-

(1) $CH_3—\underset{\underset{CH_3}{|}}{N}—CH_3$
(2) $CH_3—CH_2—CH_2—NH_2$
(3) $CH_3—NH—CH_2—CH_3$
(4) $CH_3—\underset{\underset{NH_2}{|}}{CH}—CH_3$

CCN004

9. The third member of the family of alkenynes has the molecular formula :-

(1) C_6H_6 (2) C_5H_6
(3) C_6H_8 (4) C_4H_4

CCN005

10. The number of C-atoms in second member of an ester is/are :

(1) 2 (2) 3 (3) 4 (4) 5

CCN008

11. Which of the following is an example of symmetrical or simple ether :

(1) $CH_3—\underset{\underset{O}{\|}}{C}—CH_3$
(2) $CH_3—O—CH_2—CH_3$
(3) $CH_3—CH_2—O—CH_2—CH_2—CH_3$
(4) $CH_3—\underset{\underset{CH_3}{|}}{CH}—O—\underset{\underset{CH_3}{|}}{CH}—CH_3$

CCN009

12. Which of the following pair of compounds are homologues :-

(1) 1–Propanol & 2–Propanol
(2) Ethanol & Propanal
(3) Acetone & Acetaldehyde
(4) Acetic acid & Butyric acid

CCN013

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13. Which of the following homologous series has incorrect general formula :-

(1) Alkyne C_nH_{2n-2}
(2) Alkanol $C_nH_{2n+2}O$
(3) Alkanal $C_nH_{2n+1}O$
(4) Carboxylic acid $C_nH_{2n}O_2$

CCN014

14. Minimum number of carbon atoms present in a carboxylic acid anhydride.

(1) 2 (2) 1 (3) 4 (4) 3

CCN015

15. Which of the following has general formula C_nH_{2n}

(1) Only Alkyne
(2) Only Alkane
(3) Aromatic hydrocarbon
(4) Alkene & cycloalkane

CCN016

16. Which compound has alkyne group

(1) C_7H_{14} (2) $C_{10}H_{22}$
(3) C_9H_{16} (4) $C_{16}H_{32}$

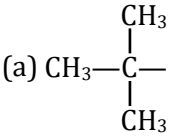
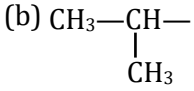
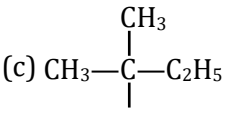
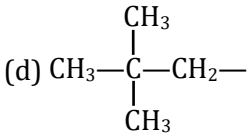
CCN017

17. The third member of the homologous series of aliphatic aldehydes has the structure :-

(1) CH_3CH_2CHO (2) $CH_3(CH_2)_2CHO$
(3) $CH_3COCH_2CH_3$ (4) CH_3COCH_3

CCN002

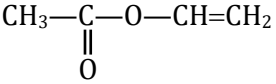
18. Which of the following are tertiary radicals :–

(a) 
(b) 
(c) 
(d) 

(1) a and b (2) b and c
(3) a and c (4) b and d

CCN018

19. Common name of the given compound is :-



(1) Vinyl acetate (2) Acryl acetate
(3) Methyl acrylate (4) Vinyl ethanoate

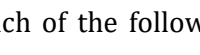
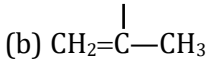
CCN019

20. A primary amine has amino group ($-NH_2$) attached to:-

(1) A primary carbon atom only
(2) A secondary carbon atom only
(3) A tertiary carbon atom only
(4) A primary, secondary or tertiary carbon atom

CCN020

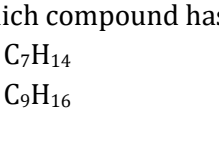
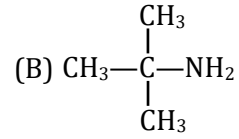
21. Which of the following are secondary radicals:-

(a) 
(b) 
(c) $CH_2=CH-$ (d) $(CH_3)_2CH-$

(1) a, b, c (2) a, d, c
(3) b, c, d (4) a, b, d

CCN021

22. Examine the following structures :–

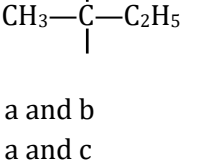
(A)  ; (B) 

Which of the following statement is correct :-

(1) A is tertiary alcohol while B is tertiary amine
(2) A is primary alcohol while B is primary amine
(3) A is tertiary alcohol while B is primary amine
(4) A is primary alcohol while B is tertiary amine

CCN022

23. Which of the following is not a correct match

(1)  ⇒ Neopentyl

(2) ⇒ Neobutyl

(3) $Ph-CH_2-$ ⇒ Benzyl

(4) $CH_2=CH-CH_2-$ ⇒ Allyl

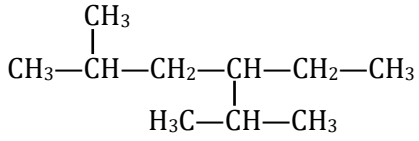
CCN023

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24. The IUPAC name for isobutyl chloride is :-
(1) 2–Methyl-2–chlorobutane
(2) 2–Chloro–2–methylbutane
(3) 1–Chloro–2–methylpropane
(4) 2–Methyl–3–chloropropane

CCN024

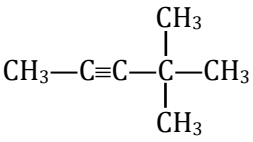
25. The IUPAC name for the given structure is :-



(1) 3–Isopropyl–4–methylhexane
(2) 4–Isopropyl–3–methylhexane
(3) 3–Ethyl–2,5–dimethylhexane
(4) 2–Ethyl–3–isopropylpentane

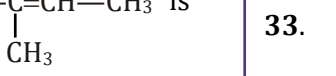
CCN026

26. The IUPAC name of given compound is :-



(1) Methyltertiarybutyl acetylene
(2) t–Butylpropyne
(3) 4,4–Dimethyl–2–pentyne
(4) 1,3,3,3–Tetramethylethyne

CCN029

27. The IUPAC name of  is

(1) 3–Methyl–2–penten–4–yne
(2) 3–Methyl–3–penten–1–yne
(3) 3–Methyl–4–pentyn–1–ene
(4) 3–Methyl pentenyne

CCN033

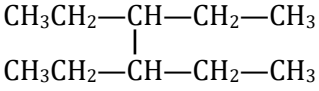
28. IUPAC name of $(CH_3)_2CHCH(CH_3)_2$ is :-
(1) 2,2-Dimethylbutane
(2) 2,3-Dimethylbutane
(3) 2,4-Dimethylbutane
(4) 1-Methylpentane

CCN045

29. IUPAC name of $CH_2=CH—CH_2—Cl$ is :–
(1) Allyl chloride
(2) 1–Chloro–3–propene
(3) 3–Chloropropene
(4) Vinyl chloride

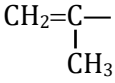
CCN046

30. The IUPAC name of the following compound



(1) 3, 4–Dimethyloctane
(2) 3–sec pentylpentane
(3) 3, 4–Diethylhexane
(4) 3, 4–Dimethylhexane

CCN049

31. The IUPAC name of the following group  is :-

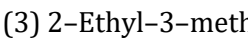
(1) Isopropenyl
(2) 1–Methylethenyl
(3) 2–Methylethylnyl
(4) None of the above

CCN047

32. Which is incorrect IUPAC name :-
(1) 3–Pentyne
(2) 3–Methyl–2–butanone
(3) 2–Ethyl–3–methyl–1–butene
(4) 3–Ethyl-2-methyl pentane

CCN037

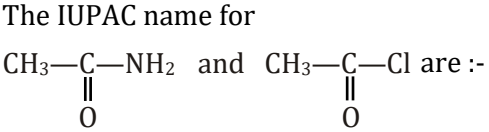
33. The IUPAC name of given compound is :-



(1) 3–Carboxy–2–pentene
(2) 2–Ethylidenebutanoic acid
(3) 2–Ethyl–2–butenoic acid
(4) 3–Ethyl–2–buten–4–oic acid

CCN025

34. The IUPAC name for

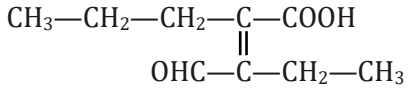


(1) 1–Amino–1–oxo ethane,1–chloro ethanal
(2) 1–Amino ethanal, acetoyl chloride
(3) 1–Oxoethanamine, ethanoyl chloride
(4) Ethanamide, Ethanoyl chloride

CCN027

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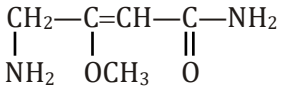
35. The number of carbon atoms in the principle chain of the given compound are :-



(1) 7 (2) 5 (3) 4 (4) 6

CCN028

36. The IUPAC name of the compound is :-



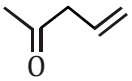
(1) 4–Amino-2–methoxy–1–amino–2–butene
(2) 4–Amino–3–methoxy–2–butenamide
(3) 2–Methoxy–1,4–diamino–2–butenal
(4) 1–Amino–2–methoxy–3–amino propene

CCN030

37. The IUPAC name of $CH_3—CH_2—NH—CH_3$ is :-
(1) Methylethylamine
(2) 1–methylaminoethane
(3) N–methylethanamine
(4) N–ethylmethanamine

CCN031

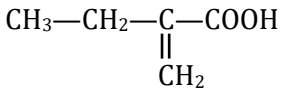
38. The IUPAC name for the compound is :-



(1) 2-Acetyl prop-1-ene (2) Pent-1-en-4-one
(3) Pent-4-en-2-one (4) Formyl propene

CCN036

39. Number of carbon atoms in the principle carbon chain in the given compound are :-



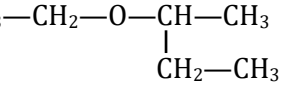
(1) 4 (2) 3 (3) 2 (4) 5

CCN040

40. Wrong IUPAC name is :-
(1) $CH_3CH_2CONH_2$
Propanamide
(2) $CH_3CH_2COOCH_3$
Methyl propanoate

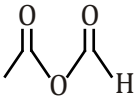


2-Methyl pent-3-ene



2–Ethoxy butane

CCN041

41. The IUPAC name of  is :–

(1) Acetic anhydride
(2) Formyl ethanoate
(3) Butane–2, 4–dione
(4) Ethanoic methanoic anhydride

CCN043

42. Find correct IUPAC name of
$NC—CH_2—CH_2—CH_2—COCl$
(1) 5-cyano pentanoyl chloride
(2) 4-Isocyano butanoyl chloride
(3) 4-cyano butanoyl chloride
(4) 4-cyano butane halo formyl

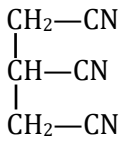
CCN048

43. The IUPAC name of  is :-

(1) Ethoxy formyl chloride
(2) Ethoxy methanoyl chloride
(3) Ethyl chloro methanoate
(4) Ethoxy carbonyl chloride

CCN050

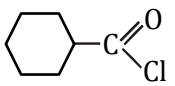
44. The IUPAC name of the given compound is :–



(1) 1,2,3–Tricarbonitrilepropane
(2) Propane–1,1,1–tricarbylamine
(3) Propane–1,2,3–tricarbonitrile
(4) 3–Cyano pentane dicyanide

CCN039

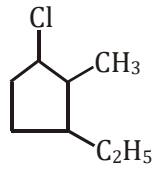
45. The IUPAC name for the compound is :-



(1) Cyclohexanoyl chloride
(2) Cyclohexane carbonyl chloride
(3) 1–Chloro cyclohexanal
(4) Chloro cyclohexyl methanal

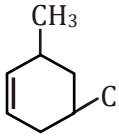
CCN032

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46.  has the IUPAC name :-

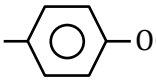
(1) 3-Chloro-1-ethyl-2-methylcyclopentane
(2) 1-Chloro-3-ethyl-2-methylcyclopentane
(3) 4-Chloro-1-ethyl-5-methylcyclopentane
(4) All are correct

CCN034

47. The IUPAC name of  is :-

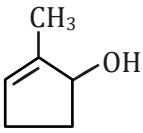
(1) 1–Methyl–5–ethylcyclohex–2–ene
(2) 5–Ethyl–3–methylcyclohex–1–ene
(3) 4–Ethyl–6–methylcyclohex–1–ene
(4) 1–Ethyl–5–methylcyclohex–3–ene

CCN035

48. The IUPAC name of  is :-

(1) 1–Methoxy–4–aminobenzene
(2) Aminophenyl methyl ether
(3) 4–Methoxy aniline
(4) None of the above

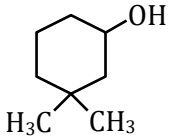
CCN038

49. The IUPAC name of the compound 

is
(1) 2–Methyl cyclopent–1–en–2–ol
(2) 3–Methyl cyclopent–2–en–1–ol
(3) 2–Methyl cyclopent–2–enol
(4) 3–Methyl cyclopent–1–en–2–ol

CCN042

50. The IUPAC name of given compound is :



(1) 3,3–Dimethyl–1–hydroxycyclohexane
(2) 1,1–Dimethyl–3–hydroxycyclohexane
(3) 3,3–Dimethylcyclohexanol
(4) 1,1–Dimethyl–3–cyclohexanol

CCN044

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	3	1	3	3	4	3	3	2	4	4	3	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	3	1	4	4	3	2	3	3	3	2	2	3	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	1	3	4	2	2	3	3	2	3	4	3	3	3	2
Question	46	47	48	49	50										
Answer	2	2	3	3	3										

EXERCISE - II (Previous Year Questions) AIPMT/NEET

AIPMT-2014

1. Which of the following organic compounds has same hybridization as its combustion product CO_2 ?

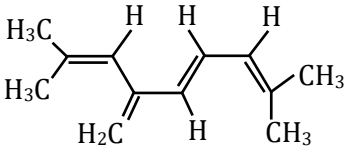
(1) Ethane (2) Ethyne

(3) Ethene (4) Ethanol

CCN059

AIPMT-2015

2. The total number of π-bond electrons in the following structure is :-

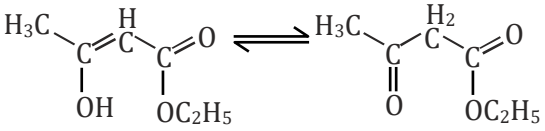


(1) 8 (2) 12

(3) 16 (4) 4

CCN060

3. The enolic form of ethyl acetoacetate as below has:-



Enolic Ketonic

(1) 16 sigma bonds and 1 pi - bond

(2) 9 sigma bonds and 2 pi - bonds

(3) 9 sigma bonds and 1 pi - bond

(4) 18 sigma bonds and 2 pi - bonds

CCN061

NEET-I 2016

4. The pair of electron in the given carbanion, $CH_3C\equiv C^{\ominus}$, is present in which of the following orbitals ?

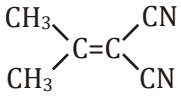
(1) 2p (2) sp^3

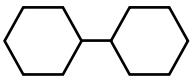
(3) sp^2 (4) sp

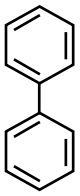
CCN062

NEET-II 2016

5. In which of the following molecules, all atoms are coplanar ?

(1) 

(2) 

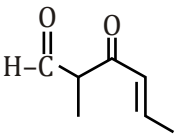
(3) 

(4) 

CCN063

NEET(UG) 2017

6. The IUPAC name of the compound

 is :-

(1) 5-formylhex-2-en-3-one

(2) 5-methyl-4-oxohex-2-en-5-al

(3) 3-keto-2-methylhex-5-enal

(4) 3-keto-2-methylhex-4-enal

CCN064

NEET(UG) 2018

7. Which of the following molecules represents the order of hybridisation sp^2, sp^2, sp, sp from left to right atoms ?

(1) $HC\equiv C—C\equiv CH$

(2) $CH_2=CH—C\equiv CH$

(3) $CH_2=CH—CH=CH_2$

(4) $CH_3—CH=CH—CH_3$

CCN065

NEET(UG) 2019

8. The number of sigma (σ) and pi (π) bonds in pent-2-en-4-yne is :-

(1) 10 σ bonds and 3π bonds

(2) 8 σ bonds and 5π bonds

(3) 11 σ bonds and 2π bonds

(4) 13 σ bonds and no π bond

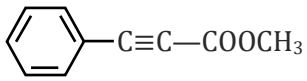
CCN066

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NEET(UG) 2020 (COVID-19)

9. How many (i) sp^2 hybridised carbon atoms and (ii) π bonds are present in the following compound?

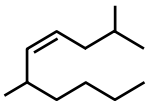


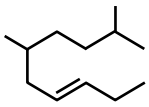
(1) 7, 5 (2) 8, 6 (3) 7, 6 (4) 8, 5

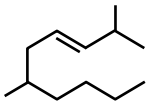
CCN067

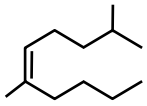
NEET(UG) 2021

10. The correct structure of 2,6-Dimethyl-dec-4-ene is:

(1) 

(2) 

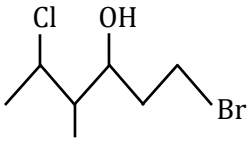
(3) 

(4) 

CCN068

NEET(UG) 2022

11. The correct IUPAC name of the following compound is :



(1) 6-bromo-2-chloro-4-methylhexan-4-ol
(2) 1-bromo-4-methyl-5-chlorohexan-3-ol
(3) 6-bromo-4-methyl-2-chlorohexan-4-ol
(4) 1-bromo-5-chloro-4-methylhexan-3-ol

CCN089

NEET(UG) 2023

12. The number of σ bonds, π bonds and lone pair of electrons in pyridine, respectively are :

(1) 12, 3, 0 (2) 11, 3, 1
(3) 12, 2, 1 (4) 11, 2, 0

CCN090

NEET(UG) 2024

13. Match List I with List II.

List-I (Molecule)		List-II (Number and types of bond/s between two carbon atoms)	
A.	ethane	I.	one σ-bond and two π-bonds
B.	ethene	II.	two π-bonds
C.	carbon molecule, C_2	III.	one σ-bond
D.	ethyne	IV.	one σ-bond and one π-bond

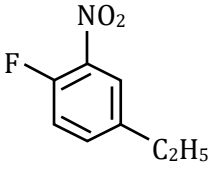
Choose the correct answer from the options given below :

(1) A-I, B-IV, C-II, D-III
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-II, D-I
(4) A-III, B-IV, C-I, D-II

CCN091

NEET (UG) 2024 (Re-examination)

14. The correct IUPAC name of the compound



is:

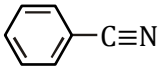
(1) 4-ethyl-1-fluoro-2-nitrobenzene
(2) 4-ethyl-1-fluoro-6-nitrobenzene
(3) 3-ethyl-6-fluoro-1-nitrobenzene
(4) 1-ethyl-4-fluoro-3-nitrobenzene

CCN092

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Answer	2	1	4	4	3	4	2	1	3	1	4	2	3	1

EXERCISE – III (Analytical Questions) Master Your Understanding

1. The number of π bonds and σ bonds in the structure  respectively are :-

(1) 8, 5 (2) 8, 6 (3) 3, 8 (4) 5, 13

CCN069

2. The geometry of the compound is :-

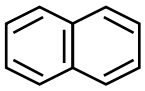
$CH_2=CH-CH=CH_2$

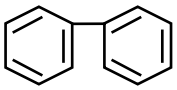
(1) Tetrahedral (2) Linear

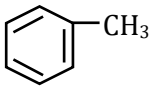
(3) Planar (4) Pyramidal

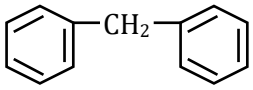
CCN070

3. Which of the following have only 2° H-atom :

(a) 

(b) 

(c) 

(d) 

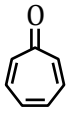
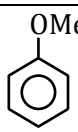
Correct code is :

(1) a and b (2) a, b and d

(3) a, c and d (4) a, b, c and d

CCN071

4. Match the column :-

Column-I (Compounds)		Column-II (Name)	
(a)		(p)	Furan
(b)		(q)	Anisole
(c)		(r)	Pyridine
(d)		(s)	Tropone

(1) a = p, b = s, c = q, d = r

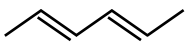
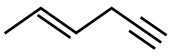
(2) a = s, b = q, c = r, d = p

(3) a = q, b = r, c = p, d = s

(4) a = s, b = p, c = q, d = r

CCN088

5. Match the column :-

Column-I Compound		Column-II Hybridisation relation of carbon	
(a)		(p)	$sp^3 : sp^2 : sp = 1$
(b)		(q)	$\frac{sp}{sp^2}=1$
(c)	$(CH_3)_2CO$	(r)	$\frac{sp^2}{sp^3}=2$
(d)	$H_2C=C=C=CH_2$	(s)	$\frac{sp^3}{sp^2}=2$

(1) a = p, b = r, c = q, d = s

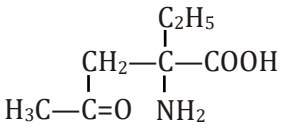
(2) a = r, b = p, c = s, d = q

(3) a = s, b = q, c = r, d = p

(4) a = s, b = p, c = r, d = q

CCN087

6. The IUPAC name of given compound is :-



(1) 2–Oxo–4–amino–4–ethyl pentanoic acid

(2) 2–Amino–4–oxo–2–ethyl pentanoic acid

(3) 3–Amino–3–carboxy–5–hexanone

(4) 2–Amino–2–ethyl–4–oxo pentanoic acid

CCN072

7. The IUPAC name from the incorrect name 4–Amino–3–hydroxy–2–butene is :-

(1) 1–Amino–2–hydroxy–2–butene

(2) 4–Amino–2–buten–3–ol

(3) 1–Amino–2–buten–2–ol

(4) 1–Amino–2–butenol

CCN073

8. The correct IUPAC name of 2–chloro–3–butanol is :-

(1) 3–Chloro–2–hydroxy butane

(2) 3–Chloro–2–butanol

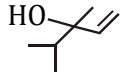
(3) 3–Hydroxy–2–chloro butane

(4) 2–Chloro–3–hydroxy butane

CCN074

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9.  has the IUPAC name :-

(1) 3,4–Dimethyl–1–penten–3–ol
(2) Isopropyl–3–methyl vinyl carbinol
(3) 2,3–Dimethyl–4–penten–3–ol
(4) None of the above

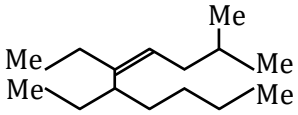
CCN079

10. The IUPAC name of  is :–

(1) 2,3–Dimethylhexane
(2) 2–Ethyl–4–methylpentane
(3) 3–Ethyl –2–methylpentane
(4) 2,4–Dimethylhexane

CCN083

11. The IUPAC name of the structure is :-



(1) 2,4,5–Triethyl–3–nonene
(2) 5,6–Diethyl–2–methyl–4–decene
(3) 2,4,5–Triethyl–3–octene
(4) 3–Ethyl–5–methyl–3–heptene

CCN077

12. Which of the following compounds has wrong IUPAC name ?

(1) $CH_3CH_2CH_2COOCH_2CH_3 \rightarrow$ Ethylbutanoate

(2) $CH_3–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH_2–CHO \rightarrow$ 3–Methylbutanal

(3) $CH_3–\underset{\displaystyle OH}{\underset{|}{CH}}–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH_3 \rightarrow$ 2–Methyl–3–butanol

(4) $CH_3–\underset{\displaystyle CH_3}{\underset{|}{CH}}–\overset{\displaystyle O}{\overset{\|}{C}}–CH_2–CH_3$

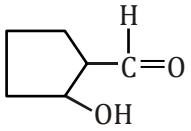
$\rightarrow$ 2–Methyl–3–Pentanone

CCN086

13. The IUPAC name of CH_3CH_2NHCHO is :

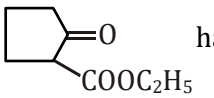
(1) N–formylethanamine
(2) Ethylaminomethanal
(3) N–ethylmethanamide
(4) N-ethylmethanol

CCN076

14. The IUPAC name for  is :-

(1) 2–Hydroxycyclopentanal
(2) 2–Formyl–1–hydroxycyclopentane
(3) 2–Hydroxycyclopentanecarbaldehyde
(4) Cyclopentane–2–ol–1–al

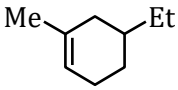
CCN075

15.  has the IUPAC name :

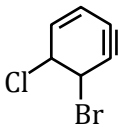
(1) Ethyl–2–ketocyclopentanecarboxylate
(2) 2–Cyclopentanone–1–carbethoxy
(3) 2–Ethylcarbonatecyclopentanone
(4) 1–Keto–2–carbethoxycyclopentanone

CCN078

16. Which of the following has wrong IUPAC name

(1) 

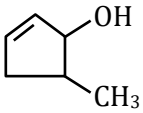
5-Ethyl-1-methylcyclohexene

(2) 

5-Bromo-6-chloro-1-cyclohexen-3-yne

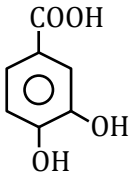
(3) 

1,2-Dimethylcyclopropane

(4) 

2–Methylcyclopent–4–en–1–ol

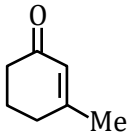
CCN080

17. The IUPAC name of  is :–

(1) 3,4–Dihydroxybenzoic acid
(2) 2,3–Dihydroxybenzoic acid
(3) 4–Carboxy–2–hydroxyphenol
(4) 4–Carboxybenzene–1,2–diol

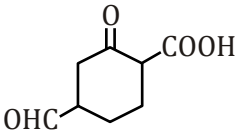
CCN081

18. The IUPAC name of the given compound is :–



(1) 3–Methyl–2–cyclohexenone
(2) 2–Methyl–3–cyclohexenone
(3) 1–Oxo–3–methylcyclohexene
(4) 2–Oxo–6–methylcyclohexene

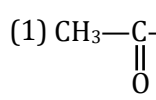
CCN082

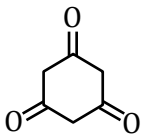
19. IUPAC name of  is :-

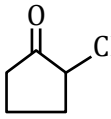
(1) 6–Oxo–4–formylcyclohexanecarboxylic acid
(2) 4–Formyl–2–oxo cyclohexanecarboxylic acid
(3) 4–Formyl–2–oxo cyclohexanoic acid
(4) 1–Carboxy–4–formyl–2–oxo cyclohexane

CCN084

20. Which is correct IUPAC name :-

(1)  1 Bromopropanone

(2)  1,3,5-Cyclohexanetrione

(3)  2-Ketocyclopentanecarbaldehyde

(4) All are correct

CCN085

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	2	4	2	4	3	2	1	4	2	3	3	3	1
Question	16	17	18	19	20										
Answer	4	1	1	2	4										

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Classification and Nomenclature

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (1)

$CH_2=CH-C\equiv CH$

sp^2 sp^2 sp sp

So, Hybridisation of C–atoms which are attached to a single bond with each other is → sp^2, sp

2. Ans. (1)

$CH_2=CH-\underset{\underset{O}{\|}}{C}-CH=CH-C\equiv N$

Olefinic bond → (C=C)

∴ Number of olefinic bond = 2

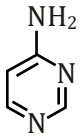
3. Ans. (3)

$HC\equiv C-\underset{\underset{O}{\|}}{C}-CH=CH-C\equiv N$

Acetylinic bond → —C≡C—

So, number of Acetylinic bond = 1

4. Ans. (1)



∴ Atoms other than carbon and hydrogen, when present in cyclic ring are called hetero atoms.

So, number of hetero atoms = 2

5. Ans. (3)

$CH_3-CH_2-C\equiv N$

sp^3 sp^3 sp

6. Ans. (3)

$\overset{1}{H}C\equiv\overset{2}{C}-\overset{3}{C}H_2-\overset{4}{C}H=\overset{5}{C}H-\overset{6}{C}H_3$

sp sp sp^3 sp^2 sp^2 sp^3

So, $C_2—C_3 \Rightarrow sp—sp^3$

7. Ans. (4)

Molecular formula → $C_4H_8O_2$

$CH_3-CH_2-CH_2-\underset{\underset{O}{\|}}{C}-OH$, $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-O-CH_3$

(Carboxylic acid) (Ester)

8. Ans. (3)

$CH_3-NH-CH_3 \xrightarrow{+CH_2} CH_3-NH-CH_2-CH_3$

2° Amine 2° Amine

9. Ans. (3)

Alkenyne and Family

1st Member ⇒ $CH_2=CH-C\equiv CH \rightarrow C_4H_4$

2nd Member ⇒ $CH_3-CH=CH-C\equiv CH \rightarrow C_5H_6$

3rd Member ⇒ $CH_3-CH_2-CH=CH-C\equiv CH \rightarrow C_6H_8$

10. Ans. (2)

$H-\underset{\underset{O}{\|}}{C}-O-CH_3 \xrightarrow{+CH_2} CH_3-\underset{\underset{O}{\|}}{C}-O-CH_3$

1st Member 2nd Member

So, Number of carbon atoms in 2nd member of ester family = 3

11. Ans. (4)

Symmetrical/simple ether ⇒ both alkyl group are same

$CH_3-\underset{\underset{CH_3}{|}}{CH}-O-\underset{\underset{CH_3}{|}}{CH}-CH_3$

Same

12. Ans. (4)

For compounds to be Homologues, Functional group (if present) should be same and the difference of $-CH_2$ or molecular mass =14 should be present.

$CH_3-\underset{\underset{O}{\|}}{C}-OH \xrightarrow{+CH_2} CH_3-CH_2-\underset{\underset{O}{\|}}{C}-OH$

Acetic acid Propionic acid

$\xrightarrow{+CH_2}$

$CH_3-CH_2-CH_2-\underset{\underset{O}{\|}}{C}-OH$

Butyric acid

So, Acetic acid & Butyric acid are pair of Homologues.

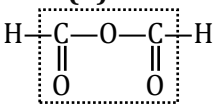
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13. Ans. (3)

$\because$ Alkanal = $C_nH_{2n}O$

14. Ans. (1)



$\therefore$ Minimum number of carbon atoms present in carboxylic acid anhydride = 2

15. Ans. (4)

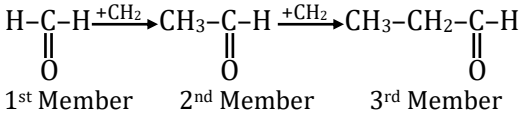
$C_nH_{2n} \rightarrow$ Alkene and cyclic alkane.

16. Ans. (3)

Alkyne $\rightarrow C_nH_{2n-2}$

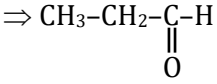
So, $C_9H_{16} \rightarrow$ Alkyne group present.

17. Ans. (1)



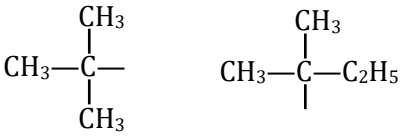
1st Member 2nd Member 3rd Member

So, 3rd member of aliphatic aldehyde homologous series.

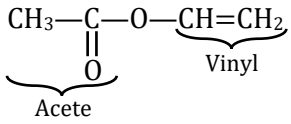
$\Rightarrow$ 

18. Ans. (3)

Tertiary radicals $\rightarrow$



19. Ans. (1)



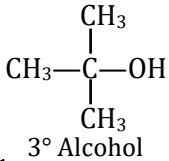
20. Ans. (4)

R—NH_2 where 'R'– can be primary, secondary or (1° amine) tertiary carbon atom.

21. Ans. (4)

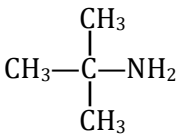
Secondary radical means free valency on 2°C

22. Ans. (3)

(A) 

3° Alcohol

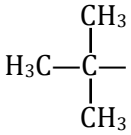
{–OH group attached to that carbon, which is directly attached to 3 other carbon atoms}

(B) 

1° Amine

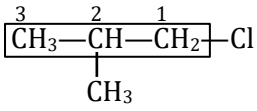
{Nitrogen is attached directly to only one carbon atom}

23. Ans. (2)

 $\Rightarrow$ Tert. butyl

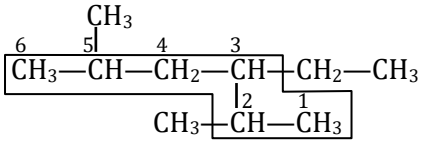
24. Ans. (3)

Isobutyl chloride –



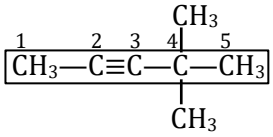
1–Chloro–2–methyl propane.

25. Ans. (3)



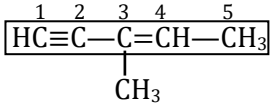
3–Ethyl–2,5–dimethylhexane

26. Ans. (3)



4,4–Dimethyl–2–pentyne

27. Ans. (2)

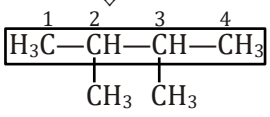


3–Methyl–3–penten–1–yne

28. Ans. (2)

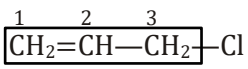
$(CH_3)_2CHCH(CH_3)_2$

$\Downarrow$



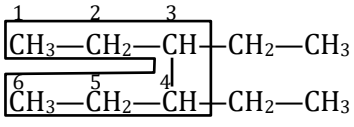
2,3–Dimethylbutane

29. Ans. (3)



3–Chloropropene

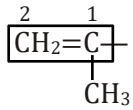
30. Ans. (3)



3, 4– Diethylhexane

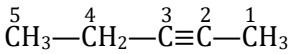
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31. Ans. (2)



1–Methylethenyl

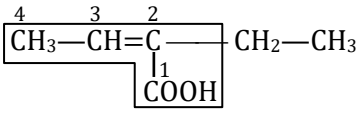
32. Ans. (1)



Incorrect → 3–Pentyne

Correct → 2–Pentyne

33. Ans. (3)

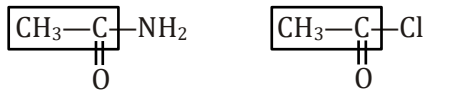


2–Ethyl but –2– enoic acid

or

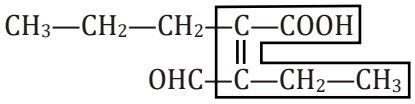
2–Ethyl –2– butenoic acid

34. Ans. (4)



Ethanamide Ethanoyl chloride

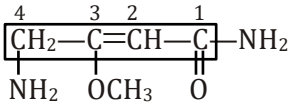
35. Ans. (2)



∴ Total carbon in PCC = 5

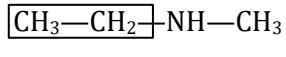
Priority→ Functional group > Multiple Bonds > Number of carbons > Substituent.

36. Ans. (2)



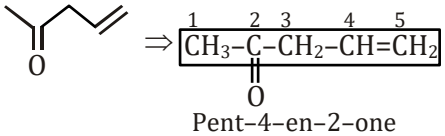
4–Amino–3–methoxy–2–butenamide

37. Ans. (3)



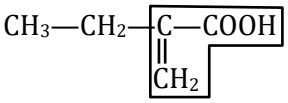
N–methylethanamine

38. Ans. (3)



Pent–4–en–2–one

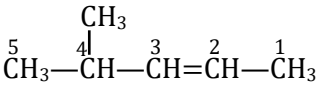
39. Ans. (2)



Priority → F.G. > M.B. > Number of carbon atoms > substituent

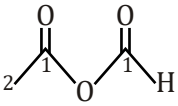
So, Total number of carbon in PCC= 3

40. Ans. (3)



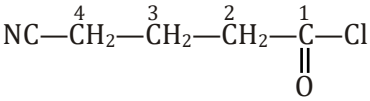
4-Methyl pent-2-ene

41. Ans. (4)



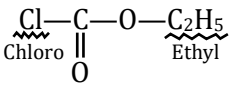
Ethanoic methanoic anhydride

42. Ans. (3)



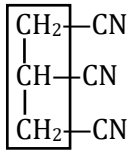
4–Cyano butanoyl chloride

43. Ans. (3)



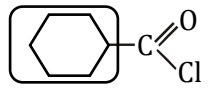
Ethyl Chloro methanoate

44. Ans. (3)



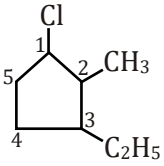
Propane–1,2,3–tricarbonitrile

45. Ans. (2)



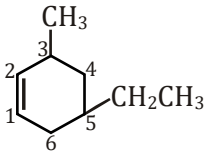
Cyclohexane carbonyl chloride

46. Ans. (2)



1-Chloro-3-ethyl-2-methylcyclopentane

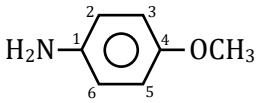
47. Ans. (2)



5–Ethyl–3–methylcyclohex–1–ene

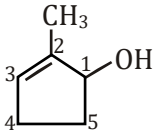
Numbering should start from F.G. > M.B. > Number of substituent.

48. Ans. (3)



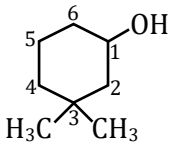
4–Methoxy aniline

49. Ans. (3)



⇒ 2–Methyl cyclopent–2–en–1–ol

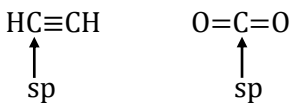
50. Ans. (3)



3,3–Dimethylcyclohexanol

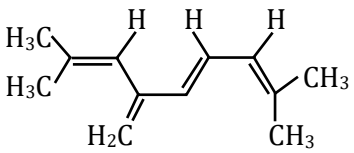
Exercise-II (Previous Year Questions)

1. Ans. (2)



(Ethyne)

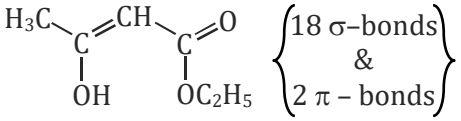
2. Ans. (1)



Total π – bond = 4

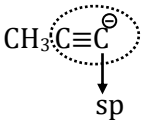
Total π – electrons = 4 × 2 = $8\pi e^-$

3. Ans. (4)



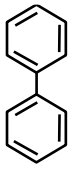
(Enolic form)

4. Ans. (4)



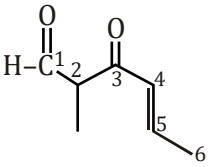
5. Ans. (3)

Coplanar :- All atoms should be in same plane



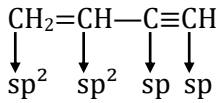
All atoms are sp^2 hybridised.

6. Ans. (4)



3-keto-2-methylhex-4-enal

7. Ans. (2)



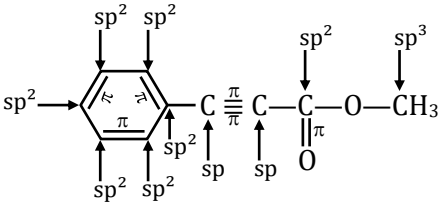
8. Ans. (1)

$CH_3—CH=CH—C\equiv CH$

pent–2–en–4–yne

10 σ–bonds, 3–π–bonds

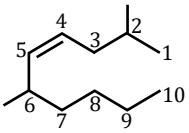
9. Ans. (3)



(i) Total sp^2 hybridised carbon atoms = 7

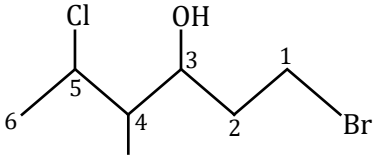
(ii) Total π – bonds = 6

10. Ans. (1)



2, 6 Dimethyldec–4–ene

11. Ans. (4)

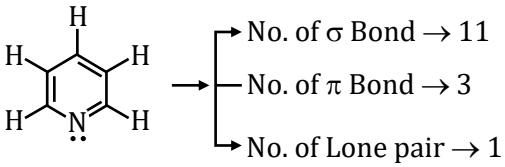


1-Bromo-5-chloro-4-methylhexan-3-ol

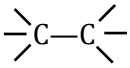
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12. Ans. (2)



13. Ans. (3)

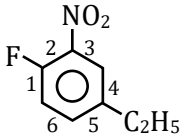
(A) ethane  1σ

(B) ethene  1σ, 1π

(C) C_2 C = C 2π (as per MOT)

(D) ethyne – C ≡ C – 1σ, 2π

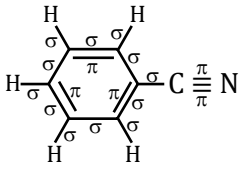
14. Ans. (1)



4-Ethyl-1-Fluoro-2-nitrobenzene

Exercise-III (Analytical Questions)

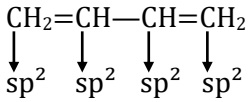
1. Ans. (4)



Number of σ–bonds = 13

Number of π–bonds = 5

2. Ans. (3)

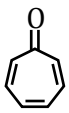


All carbons are sp^2 hybridised so, the compound is planar.

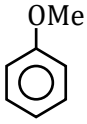
3. Ans. (2)

Hydrogen, attached to 2°C, is known as 2°H.

4. Ans. (4)

(a)  → Tropone

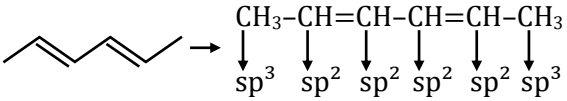
(b)  → Furan

(c)  → Anisole

(d)  → Pyridine

5. Ans. (2)

(a)



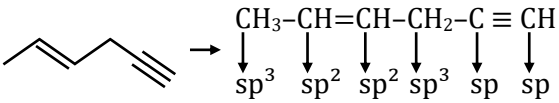
sp^2 'C' = 4

sp^3 'C' = 2

$\frac{sp^2}{sp^3} = \frac{4}{2} = 2$

$\boxed{a \rightarrow r}$

(b)



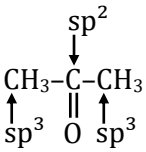
sp^3 'C' = 2

sp^2 'C' = 2

sp 'C' = 2

sp^3 : sp^2 : sp = 1

$\boxed{b \rightarrow p}$

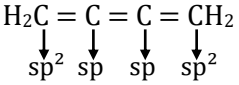
(c) 

sp^3 'C' = 2

sp^2 'C' = 1

So $\frac{sp^3}{sp^2} = \frac{2}{1} = 2$

$\boxed{c \rightarrow s}$

(d) 

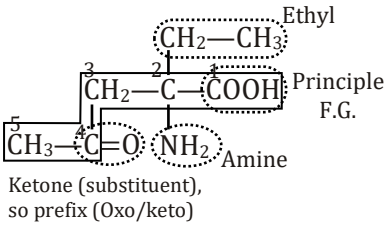
sp 'C' = 2

sp^2 'C' = 2

So $\frac{sp}{sp^2} = \frac{2}{2} = 1$

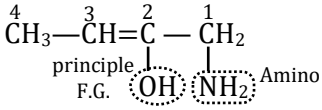
$\boxed{d \rightarrow q}$

6. Ans. (4)



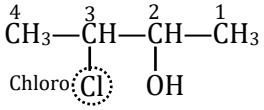
2–Amino–2–ethyl–4–oxo pentanoic acid

7. Ans. (3)



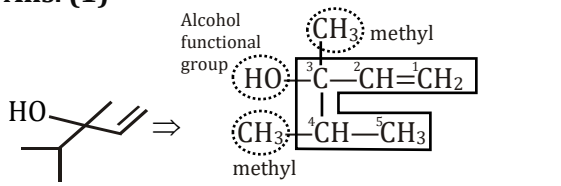
Correct name → 1–Amino–2–buten–2–ol

8. Ans. (2)



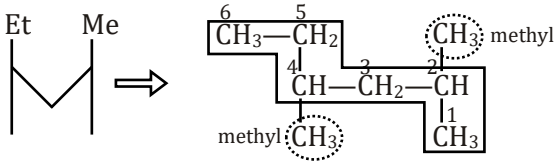
3–Chloro–2–butanol

9. Ans. (1)



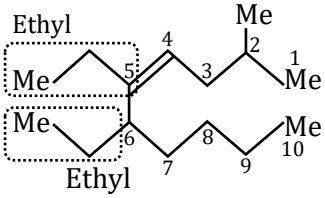
3, 4–Dimethyl–1–penten–3–ol

10. Ans. (4)



2,4–Dimethylhexane

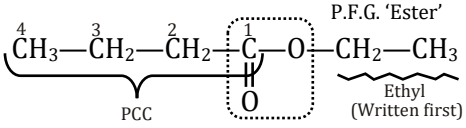
11. Ans. (2)



5, 6–Diethyl–2–methyl–4–decene

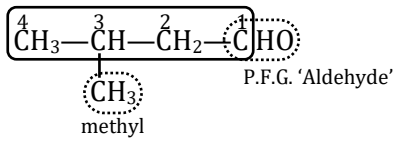
12. Ans. (3)

(1)



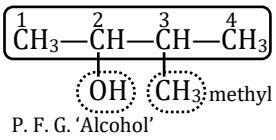
Ethylbutanoate

(2)



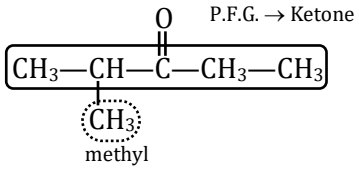
3–Methylbutanal

(3)



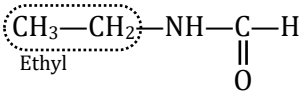
3–Methyl–2–butanol (correct IUPAC name)

(4)



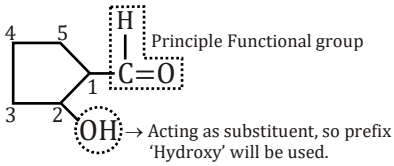
2–Methyl–3–pentanone

13. Ans. (3)



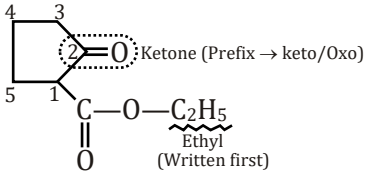
N–ethylmethanamide

14. Ans. (3)



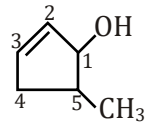
2–Hydroxycyclopentanecarbaldehyde

15. Ans. (1)



Ethyl–2–ketocyclopentanecarboxylate

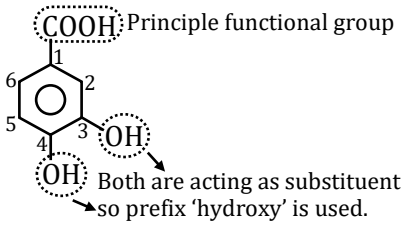
16. Ans. (4)



Priority Rule → Principle functional group > Multiple Bond > Substituent

So, Correct IUPAC name → 5–methyl cyclopent–2–en–1–ol

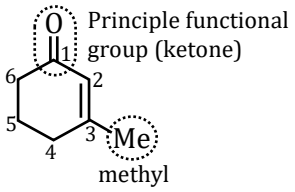
17. Ans. (1)



3,4 – Dihydroxybenzoic acid.

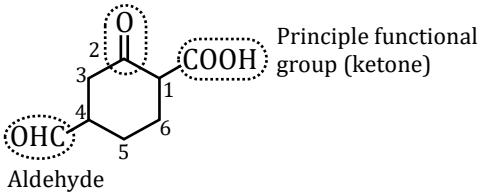
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18. Ans. (1)



3–Methyl–2–cyclohexenone

19. Ans. (2)

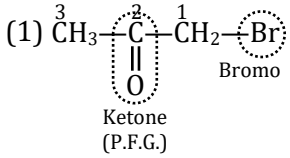


Principle functional group → Since carbon of–COOH group is not in PCC, so suffix 'carboxylic acid'

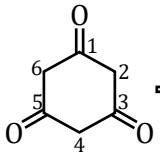
Aldehyde → Acting as substituent, since carbon of aldehyde group is not coming in parent carbon chain, so prefix 'formyl' is used.

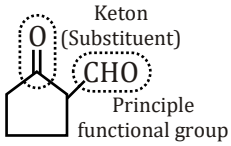
4–Formyl–2–Oxo cyclohexane carboxylic acid.

20. Ans. (4)

(1) 

⇨ 1–Bromopropanone

(2) 
⇨ 1,3,5–Cyclohexanetrione

(3) 

⇒ 2-Ketocylopentanecarbaldehyde

So all are correct

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EXERCISE - I (Conceptual Questions) Build Up Your Understanding

Structural Isomerism

1. $HC\equiv C-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$ & $CH_3-C\equiv C-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$ are

(1) Chain isomer

(2) Homologous

(3) Position isomer

(4) None

CIS001

2. Which similarity is necessary for isomerism–

(1) Molecular formula

(2) Structural formula

(3) Physical properties

(4) Chemical properties

CIS007

3. Which of the following compound is isomeric with propanoic acid :-

(1) $CH_3-\underset{\displaystyle O}{\underset{\|}{C}}-OC_2H_5$

(2) $\underset{\displaystyle OH}{\underset{|}{CH_2}}-\underset{\displaystyle O}{\underset{\|}{CH_2}}-C-H$

(3) $CH_3-CH(OH)-CH_3$

(4) $CH_3O-CH_2-CH_2OH$

CIS002

4. The minimum number of carbon atoms in ketone to show position isomerism :-

(1) 3 (2) 4 (3) 5 (4) 6

CIS005

5. $CH_3—NH—C_2H_5$ and $(CH_3)_3N$ show which type of isomerism :-

(1) Position

(2) Functional

(3) Chain

(4) None

CIS003

6. $\underset{\displaystyle Cl}{\underset{|}{CH_3}}-\underset{\displaystyle H}{\underset{|}{CH}}-CH_2-C=O$ and $CH_3-CH_2-CH_2-\underset{\displaystyle O}{\underset{\|}{C}}-Cl$

are constitute a pair of :-

(1) Position isomers

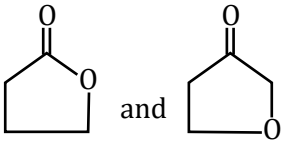
(2) Metamers

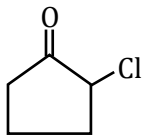
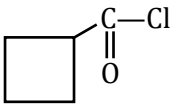
(3) Optical isomers

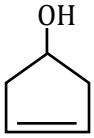
(4) Functional group isomers

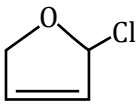
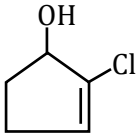
CIS004

7. Which are not functional isomers ?

(1)  and

(2)  and 

(3)  and 

(4)  and 

CIS038

8. Which are metamers :-

(1) $CH_3-O-CH_2CH_2CH_3$, $CH_3-O-CH\begin{<}CH_3\\CH_3\end{>}$

(2) $C_2H_5-O-C_2H_5$, $CH_3CH_2CH_2CH_2OH$

(3) $CH_3-O-C_2H_5$, $CH_3-CH_2-O-CH_3$

(4) $CH_3-\underset{\displaystyle O}{\underset{\|}{C}}-CH_3$, $CH_3-CH_2-\underset{\displaystyle O}{\underset{\|}{C}}-H$

CIS006

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9. Which compound shows metamerism ?

(1) $CH_3—CH_2—O—CH_3$

(2) $CH_3—\underset{\underset{CH_3}{|}}{CH}—OH$

(3) $CH_3—\underset{\underset{O}{||}}{C}—CH_2—CH_3$

(4) $CH_3—CH_2—\underset{\underset{O}{||}}{C}—O—CH_3$

CIS036

10. $CH_3—CH_2—\underset{\underset{O}{||}}{C}—CH_2—CH_3$ and $CH_3—\underset{\underset{O}{||}}{C}—CH_2—CH_2—CH_3$ are

(1) Metamers
(2) Functional isomers
(3) Chain isomers
(4) None

CIS037

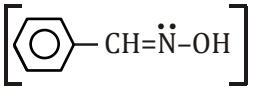
11. Number of structural isomers of C_6H_{14} is -

(1) 3 (2) 4 (3) 5 (4) 6

CIS008

Geometrical Isomerism

12. The isomerism shown by Benzaldoxime

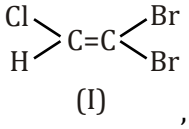


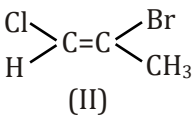
is :-

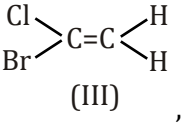
(1) Position (2) Functional
(3) Metamerism (4) Geometrical

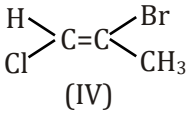
CIS009

13. Which is a pair of geometrical isomers :-


(I)


(II)

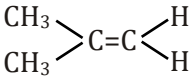

(III)

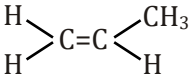

(IV)

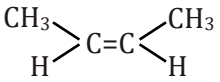
(1) I and II (2) I and III
(3) II and IV (4) III and IV

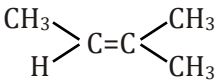
CIS012

14. Which shows 'Geometrical isomerism' :-

(1) 

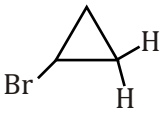
(2) 

(3) 

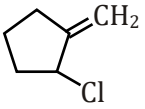
(4) 

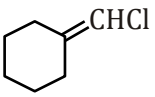
CIS013

15. Geometrical isomerism is shown by :

(1) 

(2) 

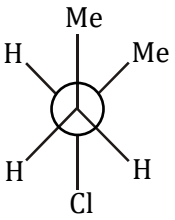
(3) 

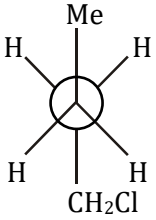
(4) 

CIS014

Conformational Isomerism

16. The pair of structures represents :-





(1) Chain isomer (2) Position isomers
(3) Conformers (4) None

CIS015

17. Which conformation of butane will have the minimum energy :-

(1) Gauche (2) Anti/staggered
(3) Eclipsed (4) None

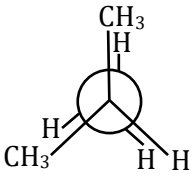
CIS017

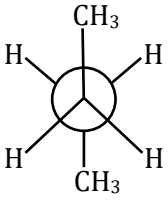
18. Rotational angle require to get maximum stable conformer from minimum stable conformer in n-butane is :

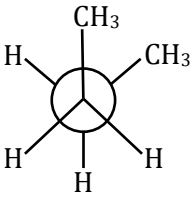
(1) 360° (2) 180° (3) 120° (4) 240°

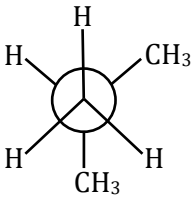
CIS016

19. Max. Stable conformation of isobutane is

(1) 

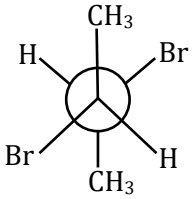
(2) 

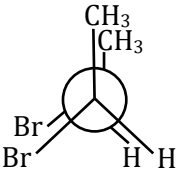
(3) 

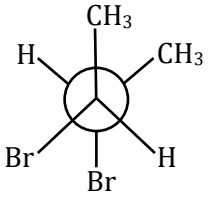
(4) 

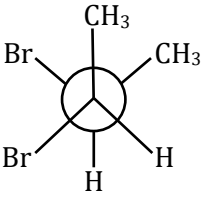
CIS039

20. Following are conformers of each other except ?

(1) 

(2) 

(3) 

(4) 

CIS040

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**1**	**2**	**3**	**2**	**4**	**4**	**1**	**4**	**1**	**3**	**4**	**3**	**3**	**2**
Question	16	17	18	19	20										
Answer	**2**	**2**	**2**	**4**	**3**										

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EXERCISE - II (Previous Year Questions) AIPMT/NEET

Re-AIPMT-2015

1. The number of structural isomers possible from the molecular formula C_3H_9N is :

(1) 2 (2) 3 (3) 4 (4) 5

CIS022

NEET-I 2016

2. The **correct** statement regarding the comparison of staggered and eclipsed conformation of ethane, is :-

(1) The staggered conformation of ethane is less stable than eclipsed conformation, because staggered conformation has torsional strain

(2) The eclipsed conformation of ethane is more stable than staggered conformation, because eclipsed conformation has no torsional strain

(3) The eclipsed conformation of ethane is more stable than staggered conformation even through the eclipsed conformation has torsional strain

(4) The staggered conformation of ethane is more stable than eclipsed conformation, because staggered conformation has no torsional strain.

CIS023

NEET(UG) 2017

3. With respect to the conformers of ethane, which of the following statements is **true** ?

(1) Bond angle changes but bond length remains same

(2) Both bond angle and bond length change

(3) Both bond angles and bond length remains same

(4) Bond angle remains same but bond length changes

CIS024

NEET(UG) 2021

4. Dihedral angle of least stable conformer of ethane is :

(1) 120° (2) 180° (3) 60° (4) 0°

CIS025

5. The compound which shows metamerism is:

(1) C_5H_{12} (2) C_3H_8O

(3) C_3H_6O (4) $C_4H_{10}O$

CIS026

NEET(UG) 2023

6. Which amongst the following compounds will show geometrical isomerism ?

(1) Pent-1-ene

(2) 2,3-Dimethylbut-2-ene

(3) 2-Methylprop-1-ene

(4) 3,4-Dimethylhex-3-ene

CIS041

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6
Answer	**3**	**4**	**3**	**4**	**4**	**4**

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EXERCISE – III (Analytical Questions) Master Your Understanding

1. What is relation between following compounds

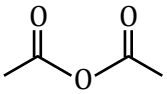
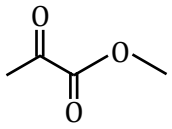
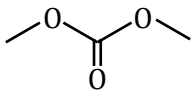
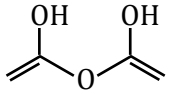
$CH_3-CH_2-\underset{\underset{CN}{|}}{CH}-CH_2-CHO$ &

$CH_3-\underset{\underset{CN}{|}}{CH}-CH_2-CH_2-CHO$

(1) Positional isomers (2) Chain isomers
(3) Optical isomers (4) Metamers

CIS031

2. Following are functional isomers except ?

(1) 
(2) 
(3) 
(4) 

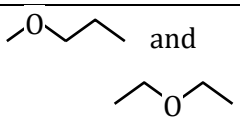
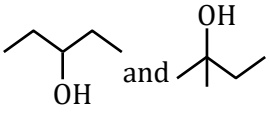
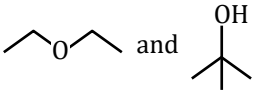
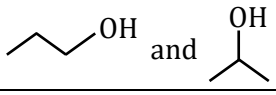
CIS042

3. Which of the following is not a metamer of $C_4H_{10}O$

(1) Diethyl ether
(2) Methyl n-propyl ether
(3) 2–Methoxy propane
(4) Isobutyl alcohol

CIS029

4. Match the column :-

Column-I (Compound)		Column-II (Isomeric relationship)	
(a)		(p)	Positional isomers
(b)		(q)	Chain isomers
(c)		(r)	Functional group isomers
(d)		(s)	Metamers

(1) a = s, b = p, c = q, d = r
(2) a = s, b = q, c = r, d = p
(3) a = p, b = q, c = r, d = s
(4) a = p, b = q, c = s, d = r

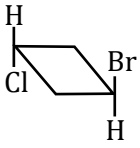
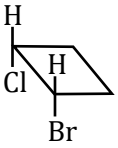
CIS035

5. How many structures of aldehydes are possible for molecular formula $C_5H_{10}O$?

(1) 4 (2) 7 (3) 6 (4) 8

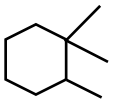
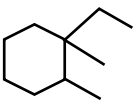
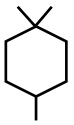
CIS032

6. Incorrect Statement is

(1) Conformations are interconvertable
(2)  and  are geometrical isomerism
(3) Optical isomerism and geometrical isomerism are stereoisomers
(4) All are correct

CIS045

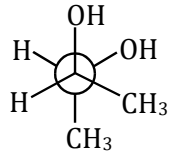
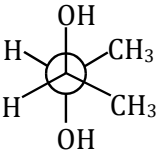
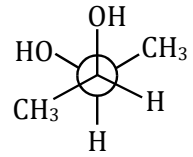
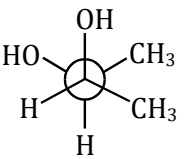
7. Which of these will exhibit geometrical isomerism?

(I) 
(II) 
(III) 

(1) I (2) II
(3) III (4) All of these

CIS030

8. Which one of the following is the most stable conformation of 2, 3–butanediol :–

(1) 
(2) 
(3) 
(4) 

CIS027

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	**2**	**3**	**4**	**2**	**1**	**2**	**2**	**3**

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Important Notes

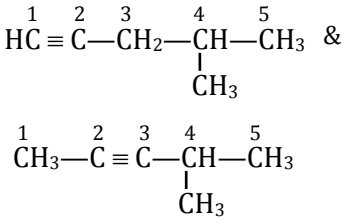
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Isomerism

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (3)



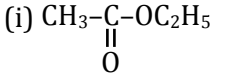
Same M.F. but different position of multiple bond, So position isomers.

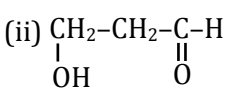
2. Ans. (1)

For isomerism → M.F. should be same but structural formula, Physical & Chemical properties should be different.

3. Ans. (2)

M.F. of CH_3CH_2COOH (Propanoic Acid) ⇒ $C_3H_6O_2$

(i)  M.F.= $C_4H_8O_2$

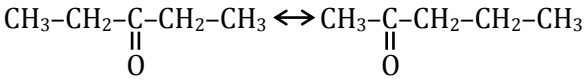
(ii)  M.F.= $C_3H_6O_2$

(iii) $CH_3-CH(OH)-CH_3$ M.F. = C_3H_8O

(iv) $CH_3-O-CH_2-CH_2-OH$ M.F.= $C_3H_8O_2$

4. Ans. (3)

Minimum 5 carbons are required to show P.I. in ketones.



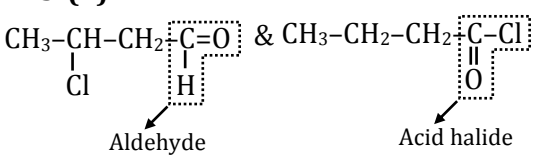
5. Ans. (2)

$CH_3-NH-C_2H_5$ & $(CH_3)_3N$

2° Amine 3° Amine

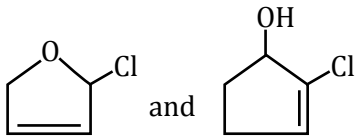
So functional isomers.

6. Ans. (4)



So, functional group isomers.

7. Ans. (4)

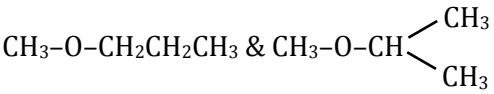


C_4H_5OCl and C_5H_7OCl

Molecular formula different

8. Ans. (1)

Same M.F., but different alkyl group across polyvalent functional group ⇒ Metamers

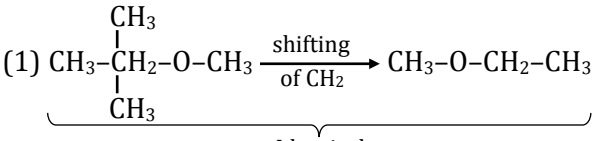


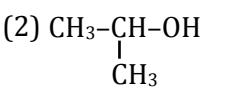
9. Ans. (4)

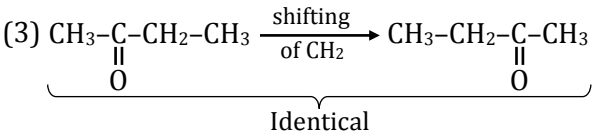
Condition for metamerism

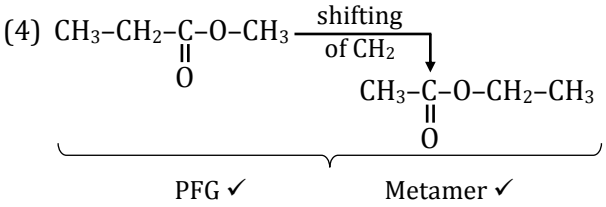
Polyvalent functional group must be present

Shifting of alkyl group should create different molecule

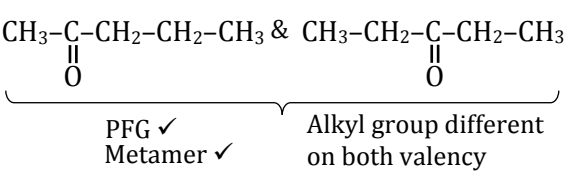
(1) 

(2)  Polyvalent F.G. absent

(3) 

(4) 

10. Ans. (1)



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11. Ans. (3)

Total 5 structural isomers are possible for M.F. → C_6H_{14}

(i) $CH_3–CH_2–CH_2–CH_2–CH_2–CH_3$ (n-hexane)

(ii) $CH_3–CH_2–CH_2–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH_3$ (2-Methyl pentane)

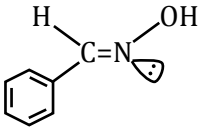
(iii) $CH_3–CH_2–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH_2–CH_3$ (3-Methyl pentane)

(iv) $CH_3–\underset{\displaystyle CH_3}{\underset{|}{CH}}–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH_3$ (2,3-Dimethyl butane)

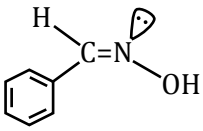
(v) $CH_3–\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}–CH_2–CH_3$ (2,2-Dimethyl butane)

12. Ans. (4)

In Benzaldoxime, the carbon of oxime should be attached with two different groups.

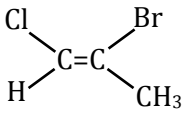


When H and OH are on same side → 'Syn' isomer



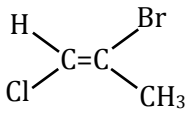
When H and OH are on opposite side→ 'Anti' isomer

13. Ans. (3)



(II)

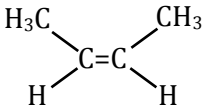
&



(IV)

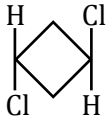
(II) & (IV) are geometrical isomers, due to presence of same M.F. but different relative arrangement across restricted system.

14. Ans. (3)



Shows G.I. due to same M.F. & different relative arrangement across restricted system.

15. Ans. (2)



For cyclic compound, any two carbon in the cycle should be attached with two different groups to show G.I.

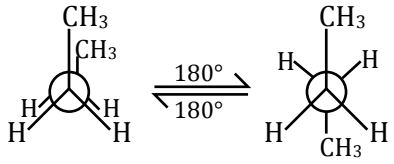
16. Ans. (2)

$H_3\overset{4}{C}–\overset{3}{C}H_2–\underset{\displaystyle Cl}{\overset{2}{\underset{|}{C}}}H–\overset{1}{C}H_3$ & $H_3\overset{4}{C}–\overset{3}{C}H_2–\overset{2}{C}H_2–\underset{\displaystyle Cl}{\overset{1}{\underset{|}{C}}}H_2$

2-chlorobutane 1-chlorobutane

These are position isomers.

17. Ans. (2)

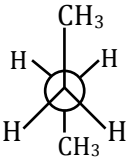


(Fully eclipsed) Minimum Stable

(Anti form) Maximum Stable

18. Ans. (2)

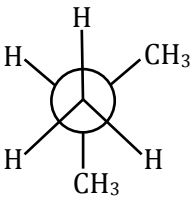
Anti/staggered conformation of butane will have the minimum energy & maximum stability because the methyl groups are as far away from each other as possible.



(Anti/staggered form)

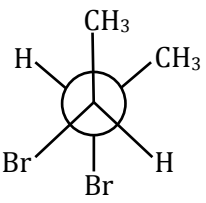
19. Ans. (4)

Isobutane $CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_3$



This structure is most stable conformer of isobutane because bulky group are far away from each other.

20. Ans. (3)



It is not the conformer as this configuration cannot be obtain after rotating other conformers to certain angle.

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Exercise-II (Previous Year Questions)

1. Ans. (3)

M.F. $\Rightarrow C_3H_9N$

(i) $CH_3-CH_2-CH_2-NH_2$ 1° Amine

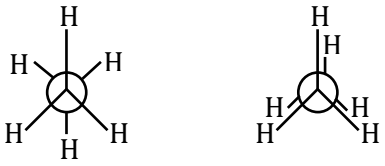
(ii) $CH_3-\underset{\underset{NH_2}{|}}{CH}-CH_3$ 1° Amine

(iii) $H_3C-NH-CH_2-CH_3$ 2° Amine

(iv) $H_3C-\underset{\underset{CH_3}{|}}{N}-CH_3$ 3° Amine

Total (4) structural isomers are possible.

2. Ans. (4)



(Staggered form) (Eclipsed form)

$\therefore$ Staggered form is more stable than eclipsed form, because it has minimum repulsion.

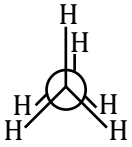
3. Ans. (3)

Conformers are the isomers which are formed by rotation about single bonds (with atleast 3 continuous σ bonds)

So there is no effect on bond angle as well as bond length.

4. Ans. (4)

The dihedral angle of the least stable conformer (Eclipsed) of ethane is 0°



5. Ans. (4)

Compounds having different alkyl groups attached with polyvalent functional Group show metamerism.

(i) $C_5H_{12} \rightarrow$ No polyvalent functional Group, so no metamerism.

(ii) $C_3H_8O \rightarrow CH_3-O-CH_2-CH_3$ (Only one arrangement possible, so no metamerism)

(iii) $C_3H_6O \rightarrow CH_3-\underset{\underset{O}{||}}{C}-CH_3$ (Only one arrangement possible, so no metamerism)

(iv) $C_4H_{10}O \rightarrow CH_3-CH_2-O-CH_2-CH_3$

&

$CH_3-O-CH_2-CH_2-CH_3$

(shows & metamerism)

6. Ans. (4)

$$CH_3-\overset{sp^2}{H_2C}-\overset{sp^2}{\underset{\underset{H_3C}{|}}{C}}=\underset{\underset{CH_3}{|}}{C}-CH_2-CH_3$$

Exercise-III (Analytical Questions)

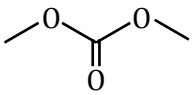
1. Ans. (2)

Isomers, having different no. of carbon atom in the main chain or in side chain, are chain isomers.

$$CH_3-CH_2-\overset{2}{\underset{\underset{^1CN}{|}}{CH}}-\overset{3}{CH_2}-\overset{4}{CHO}$$

$$CH_3-\overset{2}{\underset{\underset{^1CN}{|}}{CH}}-\overset{3}{CH_2}-\overset{4}{CH_2}-\overset{5}{CHO}$$

2. Ans. (3)



It is not a functional group isomer of other because it has different molecular formula.

3. Ans. (4)

$C_4H_{10}O$ can have following metamers except isobutyl alcohol.

(i) $CH_3-CH_2-O-CH_2-CH_3$

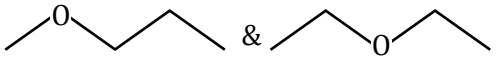
(ii) $CH_3-O-CH_2-CH_2-CH_3$

(iii) $CH_3-O-\underset{\underset{CH_3}{|}}{CH}-CH_3$

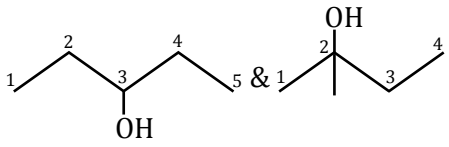
(iv) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-OH$

(Isobutyl alcohol can never be metamer of C_4H_{10}, due to different fn. group)

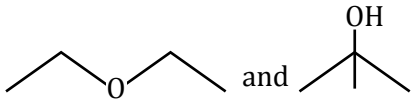
4. Ans. (2)

(a) 

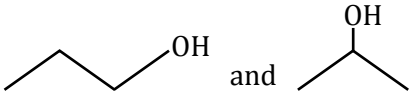
$\rightarrow$ Different alkyl groups across polyvalent fn. group $\Rightarrow$ Metamers (s)

(b) 

→ Same M.F. but diff. no. of 'C' atoms in main chain or in side-chain ⇒ Chain Isomers (q)

(c) 

→ Same M.F., Different fn. group ⇒ fn. group isomers (r)

(d) 

→ Same M.F., but diff. position of fn. group ⇒ positional isomers (p)

a = s, b = q, c = r, d = ps

5. Ans. (1)

These 4 aldehydes are possible with mol. Formula $C_5H_{10}O$

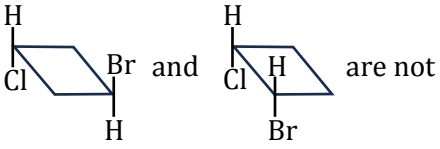
(I) $CH_3-CH_2-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-H$

(II) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-\overset{\overset{O}{\|}}{C}-H$

(III) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-\overset{\overset{O}{\|}}{C}-H$

(IV) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-\overset{\overset{O}{\|}}{C}-H$

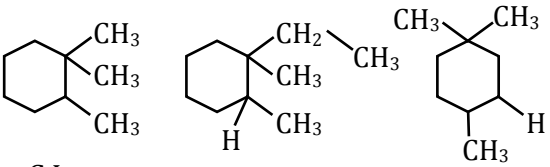
6. Ans. (2)



geometrical isomerism as they have different constitution (structure).

7. Ans. (2)

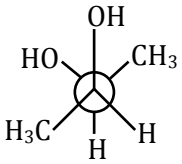
For cyclic compounds, any two carbon in the cycle should be attached with two different groups.



G.I. × G.I. ✓ G.I. ×

8. Ans. (3)

Most stable conformation of 2,3-butanediol is:-

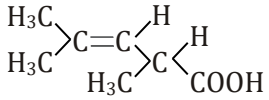


Due to H-bonding & minimum repulsion between two CH_3 groups.

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EXERCISE - I (Conceptual Questions) Build Up Your Understanding

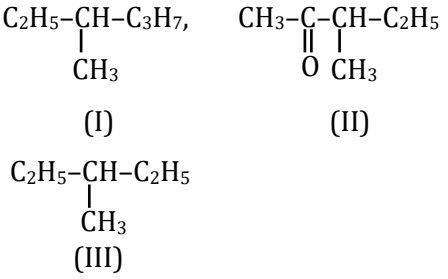
OPTICAL ISOMERISM

1.  Exhibits :-

(1) Tautomerism
(2) Optical isomerism
(3) Geometrical isomerism
(4) Geometrical and optical isomerism

COI001

2. Among the following structure I to III

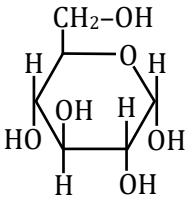


(I) (II) (III)

It is true that :-
(1) All three are chiral compounds
(2) Only I and II are chiral compounds
(3) Only II is chiral compound
(4) Only I and III are chiral compounds

COI004

3. Number of chiral C in

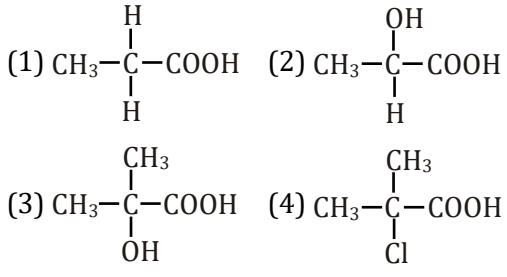


(α-D glucose)

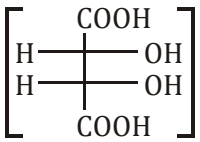
(1) 4 (2) 3
(3) 2 (4) 5

COI019

4. Which compound is optical active –

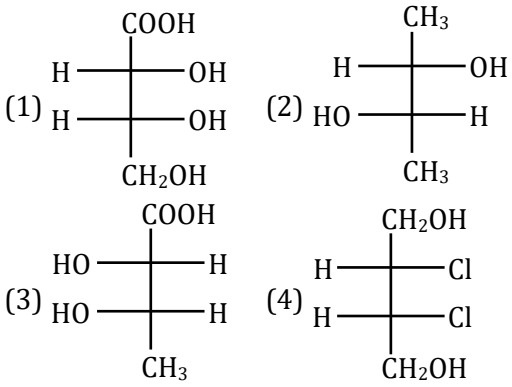


COI006

5. Meso–tartaric acid  is optically inactive due to the presence of :-
(1) Molecular symmetry
(2) Molecular asymmetry
(3) External compensation
(4) Two asymmetric carbon atoms

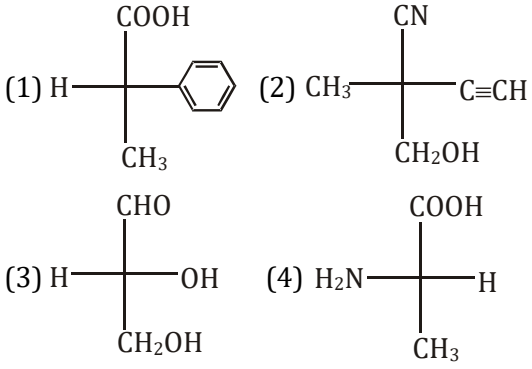
COI002

6. Which one of the following is a meso–compound.

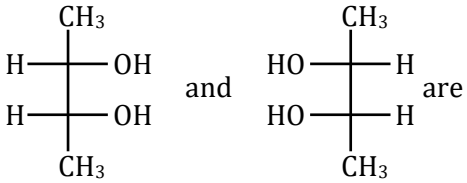


COI005

7. Identify R configuration :



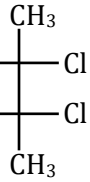
COI003

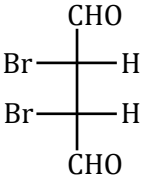
8.  and are
(1) Enantiomers
(2) Diastereomers
(3) Identical
(4) Position isomers

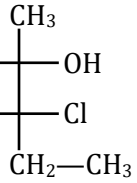
COI022

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9. Of which compound mirror images is non superimposable

(1) 

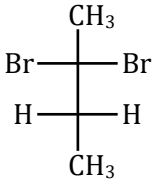
(2) 

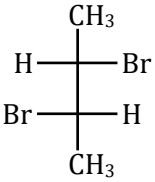
(3) 

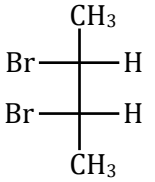
(4) None

COI020

10. Which is Meso 2,3-Dibromobutane ?

(1) 

(2) 

(3) 

(4) All of them

COI021

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	**2**	**2**	**4**	**2**	**1**	**4**	**3**	**3**	**3**	**3**

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EXERCISE - II (Previous Year Questions) AIPMT/NEET

Re-AIPMT-2015

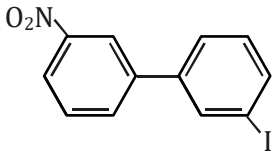
1. Two possible stereo-structures of $CH_3CHOHCOOH$, which are optically active, are called :-

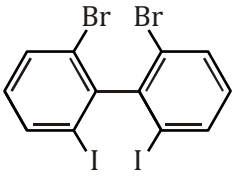
(1) Enantiomers (2) Mesomers
(3) Diastereomers (4) Atropisomers

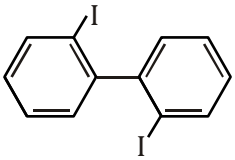
COI011

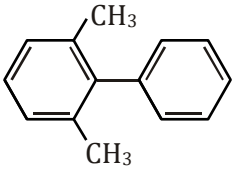
NEET-I 2016

2. Which of the following biphenyls is optically active?

(1) 

(2) 

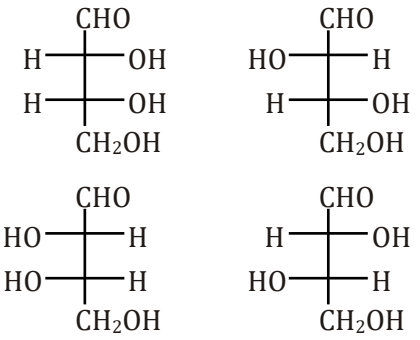
(3) 

(4) 

COI012

NEET-II 2016

3. The **correct** corresponding order names of four aldoses with configuration given below



respectively, is :-

(1) L-erythrose, L-threose, D-erythrose, D-threose
(2) D-erythrose, D-threose, L-erythrose, L-threose
(3) L-erythrose, L-threose, L-erythrose, D-threose
(4) D-threose, D-erythrose, L-threose, L-erythrose

COI013

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3
Answer	**1**	**2**	**2**

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EXERCISE – III (Analytical Questions) Master Your Understanding

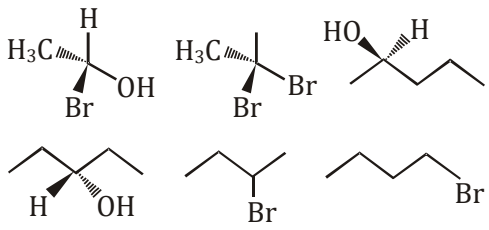
1. Which of the following does not contain any asymmetric carbon but can show enantiomerism:-

(1) Lactic acid (2) 1,3-pentadiene

(3) Tartaric acid (4) 2,3-pentadiene

COI014

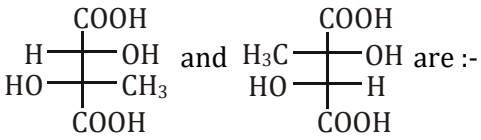
2. How many compounds among the following are chiral ?



(1) 1 (2) 2

(3) 3 (4) 4

COI018

3.  are :-

(1) Enantiomers

(2) Position isomers

(3) Geometrical isomers

(4) Homomers

COI017

4. The total number of configurational isomers of the given compound are :-

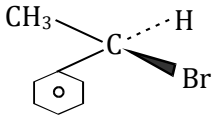
$CH_3–CH=CHCHOHCHOHCH_3$

(1) 2 (2) 4

(3) 6 (4) 8

COI016

5. The complete IUPAC name of the compound:-



(1) (R)-1-Bromo-1-phenyl ethane

(2) (S)-1-Bromo-1-phenyl ethane

(3) (E)-1-Bromo-1-phenyl ethane

(4) (Z)-1-Bromo-1-phenyl ethane

COI015

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5
Answer	**4**	**3**	**4**	**4**	**1**

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Important Notes

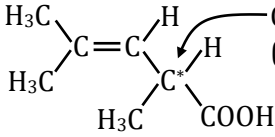
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Optical Isomerism

SOLUTIONS

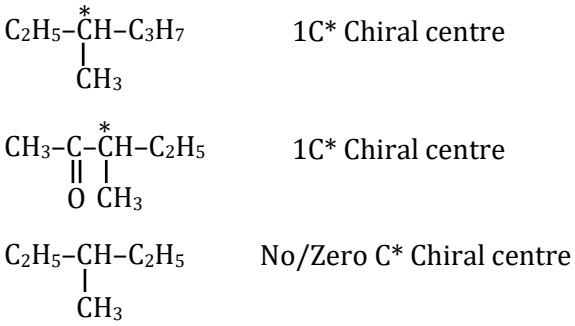
Exercise-I (Conceptual Questions)

1. Ans. (2)

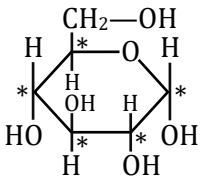


One chiral centre (Asymmetric carbon present) So it will show optical isomerism.

2. Ans. (2)



3. Ans. (4)



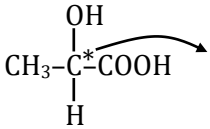
Number of chiral–C ⇒ 5

4. Ans. (2)

For optical active

Molecule should be chiral or unsymmetrical or Non-superimposable mirror image.

(only 2nd option follow)



Only 1 chiral centre optically active.

Option (1) → No Chiral

Option (2) → Chiral (Op. active)

Only 1 chiral carbon

Option (3) → No chiral

Option (4) → No chiral

5. Ans. (1)

Meso compounds

(1) Required min.2 chiral carbon/atom

(2) Symmetry must be present (POS/COS)

(3) Optically inactive due to internal compensation or molecular symmetry

6. Ans. (4)

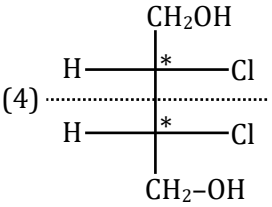
Meso compounds

(1) Required min.2 chiral carbon/atom

(2) Symmetry must be present (POS/COS)

(3) Optically inactive due to internal compensation or molecular symmetry

(4)



2 chiral carbon and symmetry (POS) present

(1) option = 1 ⇒ No POS/COS

(2) option = 2 ⇒ No Symmetry

(3) option = 3 ⇒ No Symmetry

(4) option = 4 ⇒ P.O.S./C.O.S. present

7. Ans. (3)

CIP rule → or sequence rule.

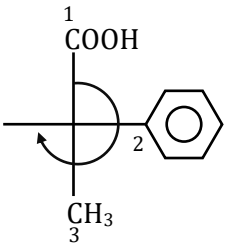
(1) Higher atomic number higher priority

(2) If atomic number is same then priority is decided by atomic mass.

(3) If atomic number or atomic mass both are same then priority is decided by next atom.

According to option

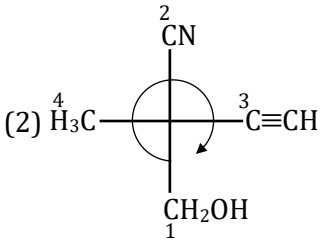
(1)



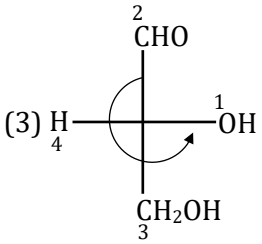
Clock wise but 4th Hz = S

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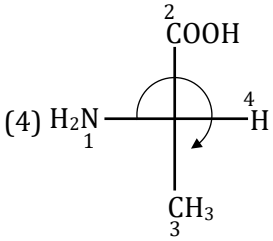
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Clock wise
but 4th Hz = S



Anti-clock wise
but 4th Hz = R



Clock wise
but 4th Hz = S

8. Ans. (3)

Rotation via 180° ∴ similar structure obtain

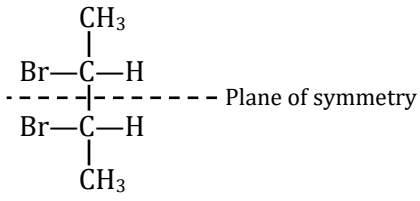
or

Configuration same.

9. Ans. (3)

(1) and (2) have plane of symmetry, so they are optically inactive hence they form super imposable mirror images but (3) option does not have any symmetry element present in it hence it's mirror image is non-super imposable.

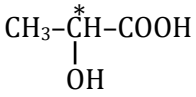
10. Ans. (3)



(Meso) 2, 3-Dibromobutane

Exercise-II (Previous Year Questions)

1. Ans. (1)



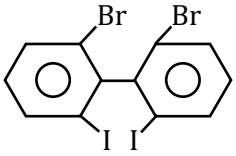
Only 1 Chiral centre/Carbon

R/S optically active enantiomer.

2. Ans. (2)

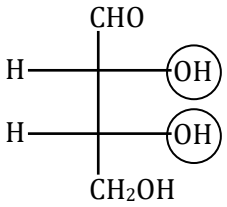
In case of Biphenyl

Ortho-ortho substituted will show optical isomerism (required diff. group each phenyl ring.



(Symmetry POS/COS absent optically active)

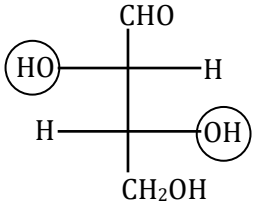
3. Ans. (2)



D-erythrose

↓

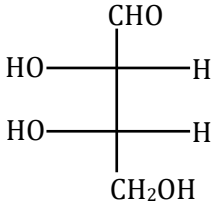
(Same side same group)



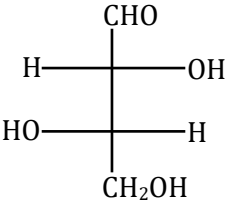
D-threose

↓

(Opp. side same group)



L-erythrose



L-threose

Exercise-III (Analytical Questions)

1. Ans. (4)

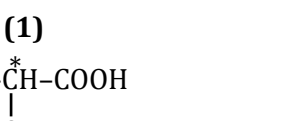
Asymmetric carbon → sp^3 carbon atom attached 4-different group /atom.

Enantiomer → Non-superimposable mirror image of each other

→ Optically active

(1)  → contain asymmetric carbon

(2) $\overset{1}{C}H_2=\overset{2}{C}H-\overset{3}{C}H=\overset{4}{C}H-\overset{5}{C}H_3$ → No Chiral/ Asymmetric carbon Not optically active

(3)  → Chiral carbon present

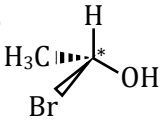
→ Chiral carbon present

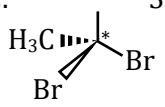
(4) $CH_3–CH=C=CH–CH_3$ → No Chiral/ asymmetric carbon but optically active due to absence of POS/COS

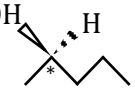
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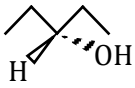
2. Ans. (3)

Chiral Atom-Any sp^3 hybridised atom attached four different group.

1. 
CC=1

2. 
CC=0

3. 
CC=1

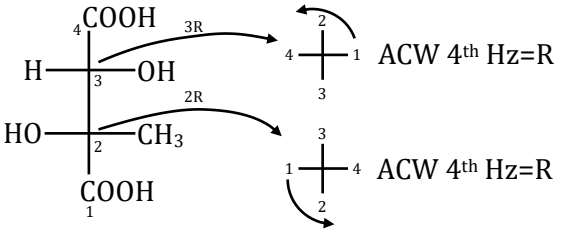
4. 
CC=0

5. 
CC=1

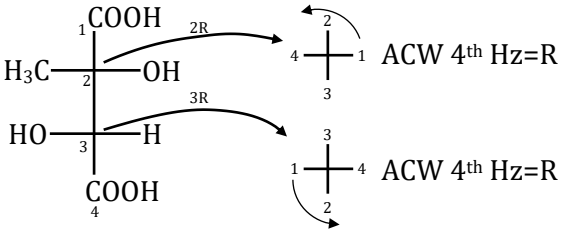
6. 
CC=0

Total CC = 3

3. Ans. (4)



2R, 3R



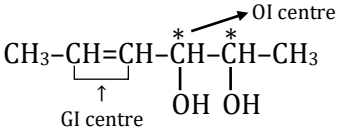
Both are same = 2R 3R

= 2R 3R

= Homomers or identical

4. Ans. (4)

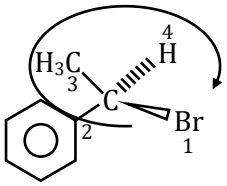
Total No of configuration isomer



n = 3

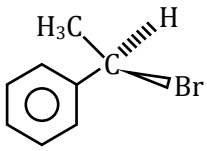
Unsymmetrical compound = $2^n = 2^3 = 8$

5. Ans. (1)



R/S confi. According to CIP Rule

CW 'R'



No ring according to IUPAC

[R]-1—bromo-1-Phenyl/ethane

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EXERCISE - I (Conceptual Questions) Build Up Your Understanding

Alkanes

1. The order of reactivity of alkyl halides in Wurtz reaction is
(1) R–I > R–Br > R–Cl
(2) R–I < R–Br < R–Cl
(3) R–Br > R–I < R–Cl
(4) R–I > R–Cl > R–Br

CHC001

2. Which of the following compound is not suitable to obtain from Wurtz reaction ?
(1) Ethane (2) Butane
(3) Neopentane (4) Hexane

CHC003

3. Which of the following reactions does not involve a C—C bond formation
(1) Hydrolysis of a Grignard reagent
(2) Combination of two alkyl free radicals
(3) Kolbe electrolysis
(4) $RNa + R\text{–}Br \longrightarrow R\text{–}R + NaBr$

CHC007

4. Which of the following reactions of methane is incomplete combustion :-
(1) $2CH_4 + O_2 \xrightarrow{Cu/523K/100atm.} 2CH_3OH$
(2) $CH_4 + O_2 \xrightarrow{Mo_2O_3} HCHO + H_2O$
(3) $CH_4 + O_2 \longrightarrow C(s) + 2H_2O(\ell)$
(4) $CH_4 + 2O_2 \longrightarrow CO_2(g) + 2H_2O(\ell)$

CHC009

5. Pyrolysis of alkanes is a
(1) Nucleophilic addition reaction
(2) Free radical substitution reaction
(3) Electrophilic addition reaction
(4) Free radical elimination reaction

CHC008

6. $CH_3CH_2CH_2Br \xrightarrow[\Delta]{alc.\ KOH} A \xrightarrow[(ii)\ Zn,\ H_2O]{(i)\ O_3} B + C$

In the above reaction A, B and C are given by the set
(1) Propylene, acetone, formaldehyde
(2) Propene, ethanal, methanal
(3) Propyne, acetaldehyde, formaldehyde
(4) Propylene, propionaldehyde, formaldehyde

CHC014

Alkene

7. Ozonolysis of 3–Methyl–1–butene gives a mixture of
(1) Propanal and ethanal
(2) Propanone and ethanal
(3) 2–Methylpropanal and methanal
(4) Butanone and methanal

CHC010

8. Which alkene gives acetone only on ozonolysis
(1) Isobutylene
(2) 2,3–Dimethyl–1–butene
(3) 2,3–Dimethyl–2–butene
(4) 3,3–Dimethyl–1–butene

CHC011

9. Oxidation of isobutylene with acid potassium permanganate gives
(1) Acetone + CO_2
(2) Acetic acid
(3) Acetic acid + CO_2
(4) Acetic acid + acetone

CHC012

10. Which of the following reactions is used for locating the position of double bond in an alkene
(1) Hydroboration
(2) Hydroxylation
(3) Chlorohydroxylation
(4) Ozonolysis

CHC013

11. Which one of the following has the smallest heat of hydrogenation per mole
(1) 1–butene (2) Trans–2–butene
(3) Cis–2–butene (4) Propene

CHC015

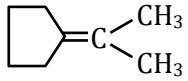
12. An alkene "A" contains three C–C, eigth C–H σ-bonds and one C–C π-bond. "A' on ozonolysis gives two moles of an aldehyde of molar mass 44. Which of the following is the IUPAC name of A.
(1) But-1-ene (2) But-2-ene
(3) Pent-2-ene (4) But-2-yne

CHC016

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13. The molecules having dipole moment are :-
(1) 2,2-dimethylpropane
(2) trans pent-2-ene
(3) cis-hex-3-ene
(4) 2 and 3

CHC017

14. Which of the following alkenes on ozonolysis give a mixture of ketones only?
(a) $CH_3-CH=CH-CH_3$
(b) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH=CH_2$
(c) 
(d) $(CH_3)_2C=C<^{CH_3}_{CH_3}$
(1) a and b (2) b and c
(3) b and d (4) c and d

CHC018

Alkynes

15. Kolbe's electrolysis of sodium maleate at anode gives.
(1) Only acetylene (2) Only CO_2
(3) Acetylene + CO_2 (4) $CO_2 + H_2$

CHC019

16. The pH of solution in Kolbe's electrolysis
(1) Increase with time
(2) Decrease with time
(3) Remains constant
(4) Nothing can be said

CHC020

17. Which of the following compound will not give a precipitate with Tollen's reagent
(1) Ethyne
(2) 1-butyne
(3) 3-methyl -1-butyne
(4) 1-pentene

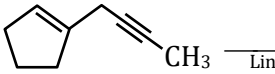
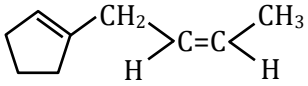
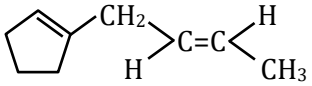
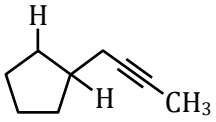
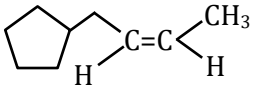
CHC021

18. $B \xleftarrow[H_2O_2/^-OH]{BH_3/THF} CH_3-C\equiv CH \xrightarrow{HgSO_4/H_2SO_4} A$,
A and B are
(1) CH_3CH_2CHO, CH_3COCH_3
(2) CH_3COCH_3, CH_3CH_2CHO
(3) CH_3COCH_3 both
(4) CH_3COCH_3, $CH_3CH_2CH_2OH$

CHC022

19. $MeCH_2C\equiv CH \xrightarrow{NH_3/NaNH_2} A \xrightarrow{Et\ Br} B$,
A and B are
(1) $MeCH_2C\equiv CNa$, $MeCH_2C\equiv C—Et$
(2) $MeCH_2CH=CH_2$, $MeCH_2—CHEt—CH_3$
(3) $MeCH_2CH=CHNH_2$,$MeCH_2CH=CH-NHBr$
(4) $MeCH_2C\equiv C—NH_2$, $MeC\equiv C—NH—Br$

CHC023

20.  product will be :
(1) 
(2) 
(3) 
(4) 

CHC024

21. To distinguish between propene and propyne, the reagent would be -
(1) Bromine
(2) Alkaline $KMnO_4$
(3) Ammonical silver nitrate
(4) Ozone

CHC025

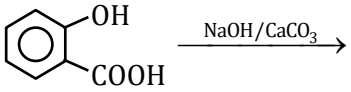
22. When treated with ammonical cuprous chloride, which one among the following forms red precipitate ?
(1) C_2H_6 (2) C_2H_4
(3) C_2H_2 (4) C_6H_6

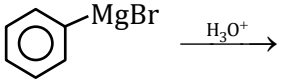
CHC026

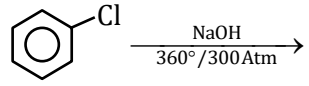
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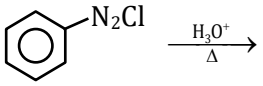
Aromatic Hydrocarbons

23. In which of the following reactions benzene is obtained

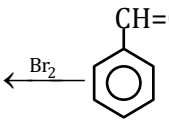
(1) 

(2) 

(3) 

(4) 

CHC027

24. A 

B ; A and B respectively are

(1) o-bromo styrene, benzoic acid
(2) p-bromostryrene, benzaldehyde
(3) m-bromostyrene, benzaldehyde
(4) Styrene dibromide, benzoic acid.

CHC028

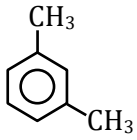
EXERCISE-I (Conceptual Questions) ANSWER KEY

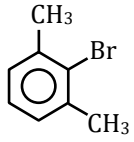
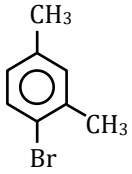
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**1**	**3**	**1**	**3**	**4**	**2**	**3**	**3**	**1**	**4**	**2**	**2**	**4**	**4**	**3**
Question	16	17	18	19	20	21	22	23	24						
Answer	**1**	**4**	**2**	**1**	**1**	**3**	**3**	**2**	**4**						

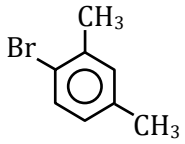
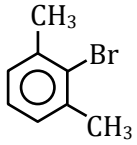
EXERCISE - II (Previous Year Questions) AIPMT/NEET

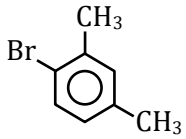
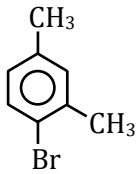
AIPMT 2014

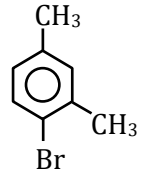
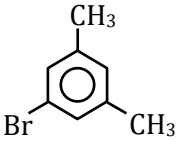
1. What products are formed when the following compound is treated with Br_2 in the presence of $FeBr_3$?



(1)  and 

(2)  and 

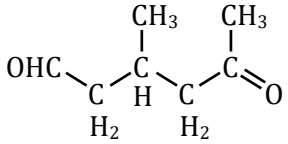
(3)  and 

(4)  and 

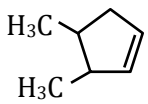
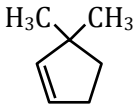
CHC046

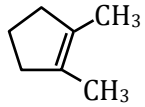
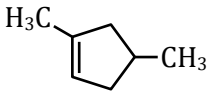
AIPMT 2015

2. A single compound of the structure :-



is obtainable from ozonolysis of which of the following cyclic compounds ?

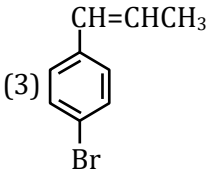
(1)  (2) 

(3)  (4) 

CHC047

3. The reaction of $C_6H_5CH = CHCH_3$ with HBr produces:-

(1) $C_6H_5CH_2\underset{\underset{Br}{|}}{C}HCH_3$ (2) $C_6H_5CH_2CH_2CH_2Br$

(3)  (4) $C_6H_5\underset{\underset{Br}{|}}{C}HCH_2CH_3$

CHC048

Re-AIPMT 2015

4. 2,3–Dimethyl–2–butene can be prepared by heating which of the following compounds with a strong acid ?

(1) $(CH_3)_2C{=}CH{-}CH_2{-}CH_3$

(2) $(CH_3)_2CH{-}CH_2{-}CH{=}CH_2$

(3) $(CH_3)_2CH{-}\underset{\underset{CH_3}{|}}{C}H{-}CH{=}CH_2$

(4) $(CH_3)_3C{-}CH{=}CH_2$

CHC049

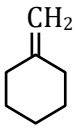
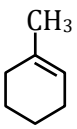
5. The oxidation of benzene by V_2O_5 in the presence of air produces :

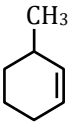
(1) Benzoic acid (2) Benzaldehyde

(3) Benzoic anhydride (4) Maleic anhydride

CHC050

6. In the reaction with HCl, an alkene reacts in accordance with the Markovnikov's rule, to give a product 1-chloro-1 methylcyclohexane. The possible alkene is :-

(1)  (2) 

(3) (1) and (2) (4) 

CHC051

NEET-I 2016

7. In the reaction

$$H{-}C{\equiv}CH \xrightarrow[(2)CH_3CH_2Br]{(1)NaNH_2\,/\,liq.NH_3} X$$

$$X \xrightarrow[(2)CH_3CH_2Br]{(1)NaNH_2\,/\,liq.NH_3} Y$$

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X and Y are :
(1) X = 1-Butyne ; Y = 3-Hexyne
(2) X = 2-Butyne ; Y = 3-Hexyne
(3) X = 2-Butyne ; Y = 2-Hexyne
(4) X = 1-Butyne ; Y = 2-Hexyne

CHC052

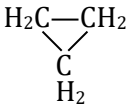
8. Consider the nitration of benzene using mixed conc. H_2SO_4 and HNO_3. If a large amount of $KHSO_4$ is added to the mixture, the rate of nitration will be:-
(1) Faster (2) Slower
(3) Unchanged (4) Doubled

CHC053

NEET-II 2016

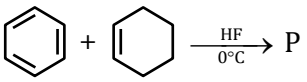
9. Which of the following compounds shall not produced propene by reaction with HBr followed by elimination or direct only by elimination reaction ?

(1) $H_2C=C=O$ (2) $H_3C-\overset{H_2}{\overset{|}{C}}-CH_2Br$

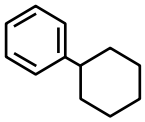
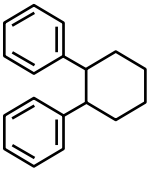
(3)  (4) $H_3C-\overset{H_2}{\overset{|}{C}}-CH_2OH$

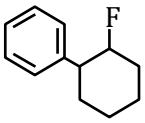
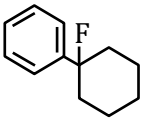
CHC054

10. In the given reaction



the product P is :-

(1)  (2) 

(3)  (4) 

CHC055

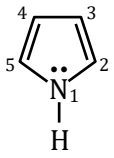
11. The compound that will react most readily with gaseous bromine has the formula
(1) C_4H_{10} (2) C_2H_4 (3) C_3H_6 (4) C_2H_2

CHC056

12. Which of the following can be used as the halide component for Friedel-Crafts reaction ?
(1) Chloroethene (2) Isopropyl chloride
(3) Chlorobenzene (4) Bromobenzene

CHC057

13. In pyrrole



The electron density is maximum on :-
(1) 2 and 4 (2) 2 and 5
(3) 2 and 3 (4) 3 and 4

CHC058

NEET(UG) 2017

14. Predict the correct intermediate and product in the following reaction :

$H_3C-C\equiv CH \xrightarrow[HgSO_4]{H_2O,\ H_2SO_4}$ Intermediate (A) $\longrightarrow$ product (B)

(1) A : $H_3C-\overset{OH}{\overset{|}{C}}=CH_2$ B : $H_3C-\overset{SO_4}{\overset{|}{C}}=CH_2$

(2) A : $H_3C-\overset{O}{\overset{\|}{C}}-CH_3$ B : $H_3C-C\equiv CH$

(3) A : $H_3C-\overset{OH}{\overset{|}{C}}=CH_2$ B : $H_3C-\overset{O}{\overset{\|}{C}}-CH_3$

(4) A : $H_3C-\overset{SO_4}{\overset{|}{C}}=CH_2$ B : $H_3C-\overset{O}{\overset{\|}{C}}-CH_3$

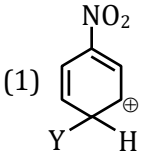
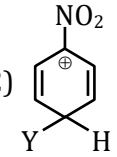
CHC059

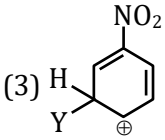
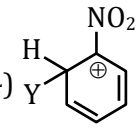
NEET(UG) 2018

15. Hydrocarbon (A) reacts with bromine by substitution to form an alkyl bromide which by Wurtz reaction is converted to gaseous hydrocarbon containing less than four carbon atoms. (A) is
(1) $CH\equiv CH$ (2) $CH_2=CH_2$
(3) CH_3-CH_3 (4) CH_4

CHC060

16. Which of the following carbocations is expected to be most stable ?

(1)  (2) 

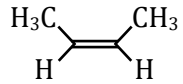
(3)  (4) 

CHC061

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NEET(UG) 2019

17. The most suitable reagent for the following conversion is :-

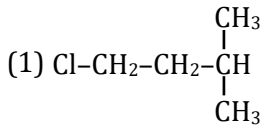
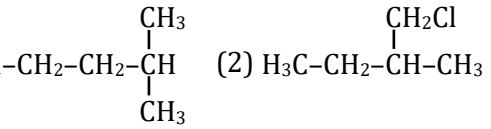
$H_3C-C\equiv C-CH_3 \longrightarrow$ 

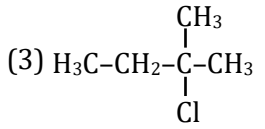
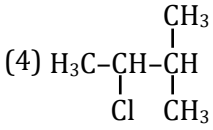
cis-2-butene

(1) Na/liquid NH_3 (2) H_2, Pd/C, quinoline
(3) Zn/HCl (4) Hg^{2+}/H^+, H_2O

CHC062

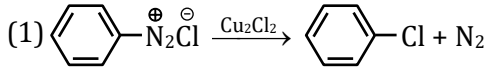
18. An alkene "A" on reaction with O_3 and $Zn-H_2O$ gives propanone and ethanal in equimolar ratio. Addition of HCl to alkene "A" gives "B" as the major product. The structure of product "B" is :-

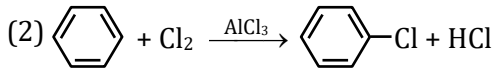
(1)  (2) 

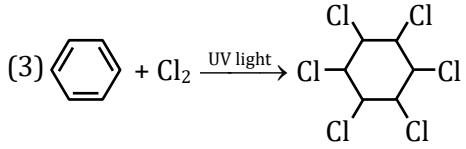
(3)  (4) 

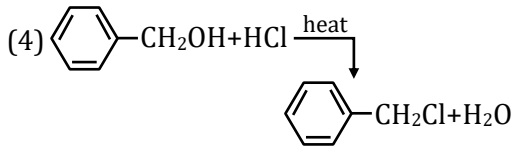
CHC063

19. Among the following, the reaction that proceeds through an electrophilic substitution is :

(1) 

(2) 

(3) 

(4) 

CHC064

NEET(UG) 2019 (ODISHA)

20. The alkane that gives only one mono-chloro product on chlorination with Cl_2 in presence of diffused sunlight is :-

(1) 2,2-dimethylbutane
(2) Neopentane
(3) n-pentane
(4) Isopentane

CHC065

21. In the following reaction,

$$H_3C-C\equiv CH \xrightarrow[873\,K]{\text{red hot iron tube}} A,$$

the number of sigma(σ) bonds present in the product A is :-

(1) 21 (2) 9 (3) 24 (4) 18

CHC066

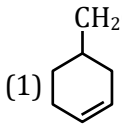
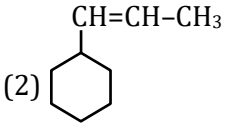
NEET(UG) 2020

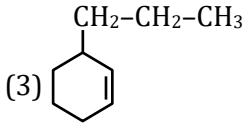
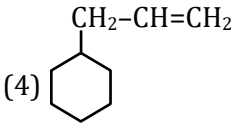
22. Which of the following alkane cannot be made in good yield by Wurtz reaction ?

(1) n-Butane (2) n-Hexane
(3) 2,3-Dimethylbutane (4) n-Heptane

CHC067

23. An alkene on ozonolysis gives methanal as one of the product. Its structure is :

(1)  (2) 

(3)  (4) 

CHC069

NEET(UG) 2020 (COVID-19)

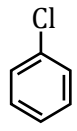
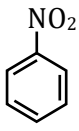
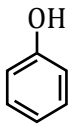
24. $CH_3CH_2CH = CH_2 \xrightarrow[H_2O_2,OH^-]{B_2H_6} Z$

What is Z ?

(1) $CH_3CH_2CH_2CH_2OH$
(2) $CH_3CH_2\underset{\displaystyle OH}{\underset{|}{C}}HCH_3$
(3) $CH_3CH_2CH_2CHO$
(4) $CH_3CH_2CH_2CH_3$

CHC070

25. Which of the following compound is most reactive in electrophilic aromatic substitution?

(1)  (2)  (3)  (4) 

CHC071

26. Which of the following is a free radical substitution reaction ?

(1) Benzene with $Br_2/AlCl_3$
(2) Acetylene with HBr
(3) Methane with Br_2/hv
(4) Propene with $HBr/(C_6H_5COO)_2$

CHC072

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NEET(UG) 2021

27. The major product formed in dehydrohalogenation reaction of 2-Bromo pentane is Pent-2-ene. This product formation is based on ?

(1) Saytzeff's Rule (2) Hund's Rule
(3) Hoffmann Rule (4) Huckel's Rule

CHC073

28. $CH_3CH_2COO^-Na^+ \xrightarrow[\text{Heat}]{NaOH, +?} CH_3CH_3 + Na_2CO_3$.

Consider the above reaction and identify the missing reagent/chemical.

(1) B_2H_6 (2) Red Phosphorus
(3) CaO (4) DIBAL-H

CHC074

29. The major product of the following chemical reaction is:

$$\begin{matrix}CH_3\\CH_3\end{matrix}\!\!>\!CH-CH=CH_2+HBr \xrightarrow{(C_6H_5CO)_2O_2} ?$$

(1) $\begin{matrix}CH_3\\CH_3\end{matrix}\!\!>\!CH-CH_2-CH_2-Br$

(2) $\begin{matrix}CH_3\\CH_3\end{matrix}\!\!>\!CH-CH_2-CH_2-O-COC_6H_5$

(3) $\begin{matrix}CH_3\\CH_3\end{matrix}\!\!>\!CH-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_3$

(4) $\begin{matrix}CH_3\\CH_3\end{matrix}\!\!>\!CBr-CH_2-CH_3$

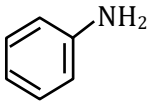
CHC075

NEET(UG) 2022

30. Which of the following is suitable to synthesize chlorobenzene ?

(1) Phenol, $NaNO_2$, HCl, CuCl

(2)  , HCl

(3)  , HCl, Heating

(4) Benzene, Cl_2, anhydrous $FeCl_3$

CHC156

31. Compound X on reaction with O_3 followed by Zn/H_2O gives formaldehyde and 2-methyl propanal as products. The compound X is :

(1) 2-Methylbut-l-ene (2) 2-Methylbut-2-ene
(3) Pent-2-ene (4) 3-Methylbut-l-ene

CHC157

NEET(UG) 2023

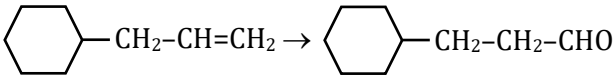
32. Weight (g) of two moles of the organic compound, which is obtained by heating sodium ethanoate with sodium hydroxide in presence of calcium oxide is :

(1) 32 (2) 30 (3) 18 (4) 16

CHC158

NEET(UG) 2024

33. Identify the correct reagents that would bring about the following transformation



(1) (i) H_2O/H^+ (ii) CrO_3

(2) (i) BH_3 (ii) H_2O_2 / $\overset{1}{O}H$ (iii) PCC

(3) (i) BH_3 (ii) $H_2O_2/\overset{1}{O}H$ (iii) Alk. $KMnO_4$ (iv) $H_3O^{\oplus}$

(4) (i) H_2O/H^+ (ii) PCC

CHC159

34. A compound with a molecular formula of C_6H_{14} has two tertiary carbons. Its IUPAC name is :

(1) n-hexane
(2) 2-methylpentane
(3) 2, 3-dimethylbutane
(4) 2, 2-dimethylbutane

CHC160

35. Given below are two statements :

Statement-I : The boiling point of three isomeric pentanes follows the order
n-pentane > isopentane > neopentane

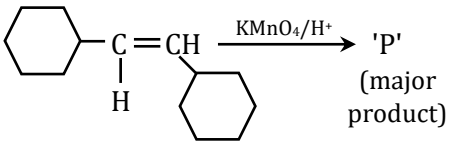
Statement-II : When branching increases, the molecule attains a shape of sphere. This results in smaller surface area for contact, due to which the intermolecular forces between the spherical molecules are weak, thereby lowering the boiling point.

In the light of the above statements, choose the *most appropriate* answer from the options given below :

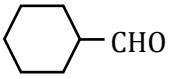
(1) Both statement-I and Statement-II are correct
(2) Both Statement-I and Statement-II are incorrect
(3) Statement-I is correct but Statement-II is incorrect
(4) Statement-I is incorrect but Statement-II is correct

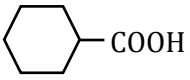
CHC161

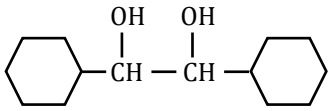
36. For the given reaction

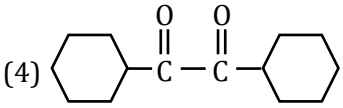


'P' is

(1) 

(2) 

(3) 

(4) 

CHC162

NEET(UG) 2024 (Re-examination)

37. Baeyer's reagent is

(1) acidic potassium permanganate solution

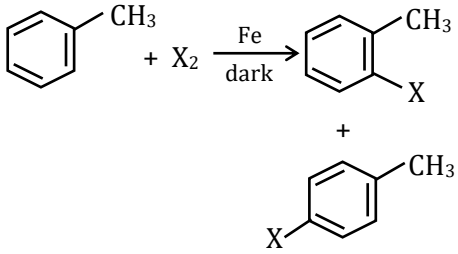
(2) acidic potassium dichromate solution

(3) cold, dilute, aqueous solution of potassium permanganate

(4) hot, concentrated solution of potassium permanganate

CHC163

38. The following reaction method

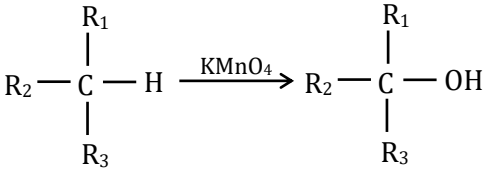


is **not** suitable for the preparation of the corresponding haloarene products, due to high reactivity of halogen, when X is :

(1) F (2) I (3) Cl (4) Br

CHC164

39. The alkane that can be oxidized to the corresponding alcohol by $KMnO_4$ as per the equation



is, when :

(1) $R_1 = H; R_2 = H; R_3 = H$

(2) $R_1 = CH_3; R_2 = CH_3; R_3 = CH_3$

(3) $R_1 = CH_3; R_2 = H; R_3 = H$

(4) $R_1 = CH_3; R_2 = CH_3; R_3 = H$

CHC165

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**4**	**4**	**4**	**4**	**3**	**1**	**2**	**1**	**1**	**3**	**2**	**2**	**3**	**4**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**3**	**2**	**3**	**2**	**2**	**1**	**4**	**4**	**1**	**4**	**3**	**1**	**3**	**1**	**4**
Question	31	32	33	34	35	36	37	38	39						
Answer	**4**	**1**	**2**	**3**	**1**	**2**	**3**	**1**	**2**						

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EXERCISE – III (Analytical Questions) Master Your Understanding

1. The chain propagating step is fastest in the reaction of an alkane with
(1) Fluorine free radical
(2) Chlorine free radical
(3) Iodine free radical
(4) Bromine free radical

CHC097

2. What is the chief product obtained when n-butane is treated with bromine in the presence of light at 130°C ?
(1) $CH_3–CH_2–CH_2–CH_2–Br$
(2) $CH_3–CH_2–\underset{\displaystyle CH_3}{\underset{|}{CH}}–Br$
(3) $CH_3–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH_2–Br$
(4) $CH_3–\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}–CH_2–Br$

CHC100

3. Which gives maximum yield of C_2H_5Cl
(1) $C_2H_6 + Cl_2 \xrightarrow[\text{excess}]{hv,\ light} C_2H_5Cl + HCl$
(2) $C_2H_6 + Cl_2 \xrightarrow[\text{excess}]{\substack{hv\\light}} C_2H_5Cl + HCl$
(3) $C_2H_6 + Cl_2 \xrightarrow[light]{hv} C_2H_5Cl$
(4) $C_2H_6 + Cl_2 \xrightarrow{Dark} C_2H_5Cl + HCl^-$

CHC122

4. Match the column-I with column-II :-

Column-I (Reaction)		Column-II (Product)	
(a)	$2CH_3CH_2Br \xrightarrow[dry\ ether]{2Na}$	(p)	
(b)	$CH_3COO^{\ominus}Na^{\oplus} \xrightarrow[\Delta]{CaO+NaOH}$	(q)	$CH_3CH_2CH_2CH_3$
(c)	$CH_3(CH_2)_4CH_3 \xrightarrow[773K,\ 10-20atm]{V_2O_5}$	(r)	$2CH_3OH$
(d)	$2CH_4 + O_2 \xrightarrow[100atm]{Cu/523K}$	(s)	CH_4

(1) a – q, b – p, c – r, d – s
(2) a – q, b – s, c – p, d – r
(3) a – q, b – p, c – s, d – r
(4) a – p, b – q, c – r, d – s

CHC133

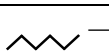
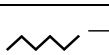
5. Match the column-I with column-II :-

Column-I (Reaction)		Column-II (Product)	
(a)	$CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3}$	(p)	$(CH_3)_3C – OH$
(b)	$2CH_3CH_3 + 3O_2 \xrightarrow[\Delta]{(CH_3COO)_2Mn}$	(q)	$HCHO + H_2O$
(c)	$(CH_3)_3CH \xrightarrow[Oxidation]{KMnO_4}$	(r)	$CO + 3H_2$
(d)	$CH_4 + H_2O \xrightarrow[\Delta]{Ni}$	(s)	$2CH_3COOH + 2H_2O$

(1) a – q, b – p, c – r, d – s
(2) a – r, b – s, c – p, d – r
(3) a – q, b – s, c – p, d – r
(4) a – p, b – q, c – r, d – s

CHC134

6. Match the column-I with column-II :-

Column-I (Reaction)		Column-II (Type of reaction)	
(a)	$HC \equiv CH \xrightarrow[Pd]{H_2}$	(p)	Pyrolysis
(b)		(q)	Reduction
(c)		(r)	Free radical substitution
(d)	$CH_3CH_3 \xrightarrow[hv]{Cl_2}$	(s)	Isomerisation

(1) a – q, b – s, c – p, d – r
(2) a – p, b – q, c – s, d – r
(3) a – q, b – p, c – s, d – r
(4) a – p, b – q, c – r, d – s

CHC135

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7. Which of the following statements is/are incorrect :-

(A) In alkenes first three members are gases, the next 14 are liquid & the higher ones are solids.

(B) Except ethene all other alkenes are colourless & odourless.

(C) Straight chain alkenes have higher boiling point than isomeric branched chain compounds.

(D) Generally alkenes are soluble in water & insoluble in non polar solvents.

(1) A, B & C only (2) Only D

(3) B & D only (4) B, C & D only

CHC154

8. Which of the following is maximum reactive for NBS

(1) $PhCH_3$ (2) Ph CH_2—CH_3

(3) $PhCH_2$—CH=CH_2 (4) Ph–$\underset{\displaystyle CH_3}{\underset{|}{CH}}$–CH=$CH_2$

CHC077

9. Chlorination of toluene in the presence of light and heat followed by treatment with aqueous NaOH gives

(1) o-cresol

(2) p-cresol

(3) 2, 4-dihydroxy toluene

(4) Benzyl alcohol

CHC121

10. Only two isomeric monochloro derivatives are possible for (exclude stereo isomers)

(1) n-butane

(2) 2, 4-dimethyl pentane

(3) Benzene

(4) 2-methyl butane

CHC099

11. Isobutylene $\xrightarrow[H_2O_2]{HBr}$ "product". The product is

(1) Isobutyl bromide

(2) Tert. butyl bromide

(3) Tert. butyl alcohol

(4) isobutyl alcohol

CHC095

12. Match the column-I with column-II :-

Column-I		Column-II	
(a)	$CH_3-CH=CH_2 \xrightarrow[H_2O]{H^{\oplus}} CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$	(p)	Hydro-genation
(b)	$CH_3-CH=CH_2 \xrightarrow[R_2O_2]{HBr} CH_3-CH_2-CH_2-Br$	(q)	Hydration
(c)	$CH_3-CH=CH_2 \xrightarrow[CCl_4]{Br_2} CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_2-Br$	(r)	Halogenation
(d)	$CH_2=CH_2 \xrightarrow[Pd]{H_2} CH_3-CH_3$	(s)	Hydro-halogenation

(1) a – s, b – r, c – q, d – p

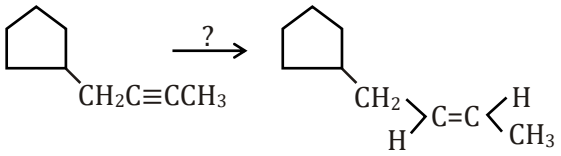
(2) a – q, b – s, c – r, d – p

(3) a – r, b – s, c – p, d – q

(4) a – p, b – q, c – r, d – s

CHC139

13. Reagent used to carry out following alkyne to alkene is

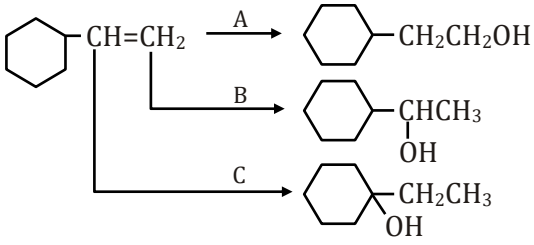


(1) Pd-C/H_2 (2) Na/NH_3

(3) Pd/H_2 (4) Ni/H_2

CHC079

14.



Schemes; A, B and C are :-

(1) Simple hydration

(2) Hydroboration, Hg^{+2} / H_2O and $NaBH_4$, hydration

(3) Hydration, hydroboration, Hg^{+2} / H_2O and $NaBH_4$,

(4) Hg^{+2} / H_2O and $NaBH_4$, hydration, hydroboration

CHC080

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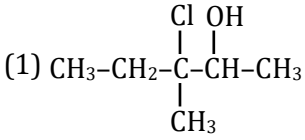
15. Match the column-I with column-II :-

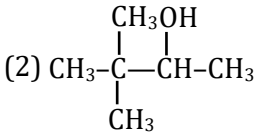
Column-I		Column-II	
(a)	Lindlar's catalyst	(p)	Na/liq. NH_3
(b)	Birch catalyst	(q)	$H_2/Pd\text{-}CaCO_3$
(c)	Baeyer's Reagent	(r)	$HBr/(PhCO)_2O_2$
(d)	Kharasch effect	(s)	Dil. $KMnO_4/273K$

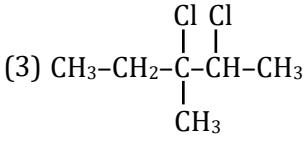
(1) a – q, b – p, c – r, d – s
(2) a – p, b – q, c – s, d – r
(3) a – q, b – p, c – s, d – r
(4) a – p, b – q, c – r, d – s

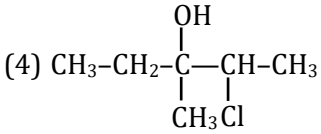
CHC132

16. The predominant product formed when 3–methyl–2–pentene reacts with HOCl is

(1) 

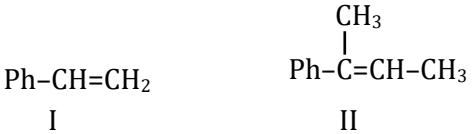
(2) 

(3) 

(4) 

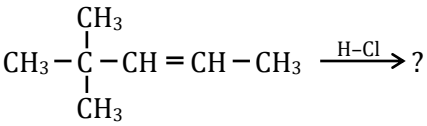
CHC090

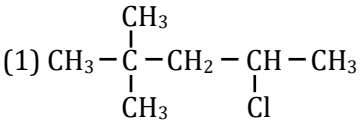
17. Correct reactivity order for EAR of following compounds is

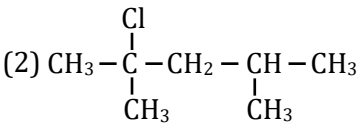


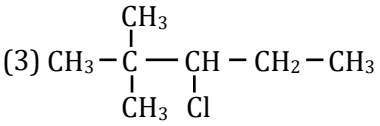
(1) IV > I > II > III
(2) III > II > I > IV
(3) II > III > I > IV
(4) II > III > IV > I

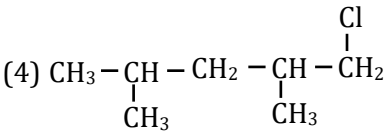
CHC082

18. 

(1) 

(2) 

(3) 

(4) 

CHC081

19. Which of the following alkene is most reactive for hydration
(1) Ethene (2) Propene
(3) 1–butane (4) 2–methyl propene

CHC086

20. Which of the following alkenes on hydration gives a tertiary alcohol
(1) 2–Butene (2) Isobutylene
(3) Ethene (4) α–Butylene

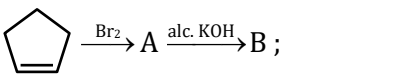
CHC089

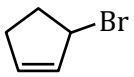
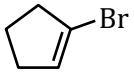
21. Propene on addition with HI, gives
(1) $CH_3—CHI—CH_3$ (2) $CH_3—CH_2—CH_2I$
(3) $CH_3—CHI—CH_2I$ (4) None of the above

CHC091

22. 3–Phenyl propene on reaction with HBr gives (as a major product)
(1) $C_6H_5CH_2CH(Br)CH_3$
(2) $C_6H_5CH(Br)CH_2CH_3$
(3) $C_6H_5CH_2CH_2CH_2Br$
(4) $C_6H_5CH(Br)CH=CH_2$

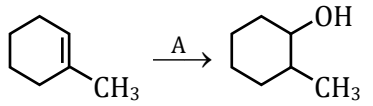
CHC093

23. 

What is the structure of B
(1)  (2) 
(3)  (4) 

CHC076

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24.  ; Reagent 'A' is

(1) $BH_3, H_2O_2/\overset{\ominus}{O}H$ (2) $H_2O/H^{\oplus}$

(3) $Hg^{+2}, H_2O/NaBH_4$ (4) Cl_2/aq. NaOH

CHC088

25. $CH_3-CH=CH_2 \xrightarrow{dil/H_2SO_4} A$

$CH_3-CH=CH_2 \xrightarrow{B_2H_6} \xrightarrow[^{-}OH]{H_2O_2} B$

Wrong statement about the product is

(1) A and B have the same functional group

(2) A and B are position isomers.

(3) A and B show chain isomerism

(4) Mixed ether is the isomer of both A and B

CHC085

26. $CH_3-\underset{CH_3}{\underset{|}{CH}}-CH=CH_2 \xrightarrow[-]{\overset{\oplus}{D}/H_2O} A$;

A (major product) is

(1) $CH_3-\underset{CH_3}{\underset{|}{CH}}-CHD-CH_2-OH$

(2) $CH_3-\underset{CH_3}{\underset{|}{CH}}-\underset{OH}{\underset{|}{CH}}-CH_2-D$

(3) $CH_3-\underset{CH_3}{\underset{|}{C}}(OH)\,CHDCH_3$

(4) $CH_3-\underset{CH_3}{\underset{|}{C}}(OH)\,CH_2-CH_2-D$

CHC113

27. The intermediate during the addition of HCl to propene in the presence of peroxide is

(1) $CH_3\overset{\bullet}{C}H\,CH_2Cl$ (2) $CH_3\overset{\oplus}{C}H-CH_3$

(3) $CH_3-CH_2-\overset{\oplus}{C}H_2$ (4) $CH_3\overset{\bullet}{C}HCH_3$

CHC083

28. Reaction of HBr with propene in the presence of peroxide gives

(1) 3–bromo propane (2) Allyl bromide

(3) n–propyl bromide (4) Isopropyl bromide

CHC094

29. In given statements which is/are correct :-

(A) Cis & Trans isomers have same structure but have different configuration (arrangement of atoms or groups)

(B) Cis-2-butene has more boiling point than trans-2-butene.

(C) Trans-2-butene has more dipole moment than cis-2-butene.

(D) Geometrical isomerism is also shown by alkenes of the types XY C = C XZ and XY C = C ZW.

(1) A, B, C & D (2) Only A, B

(3) Only A, B & C (4) Only A, B & D

CHC153

30. $Ph-\underset{CH_3}{\underset{|}{CH}}=\underset{CH_3}{\underset{|}{C}}-CH_3 + HBr \xrightarrow{Peroxide} (X)$

'X' is

(1) $Ph-\underset{Br}{\underset{|}{CH}}-\underset{CH_3}{\underset{|}{CH}}-CH_3$

(2) $Ph-CH_2-\underset{CH_3}{\underset{|}{\underset{Br}{\underset{|}{C}}}}-CH_3$

(3) $Ph-CH_2-\underset{CH_3}{\underset{|}{CH}}-CH_2-Br$

(4) $Ph-CH=\underset{CH_3}{\underset{|}{C}}-CH_2-Br$

CHC112

31. Match the column :-

	Column-I (Reaction)		Column-II (Product)
(a)	$HC\equiv CH + H_2O \rightarrow$	(p)	Benzene
(b)	$HC\equiv CH + H_2O \xrightarrow{Hg^{+2}/H^{\oplus}}$	(q)	Ammonia
(c)	$HC\equiv CH + NaNH_2 \rightarrow$	(r)	No reaction
(d)	$HC\equiv CH \xrightarrow[Fe\,tube]{Red\ hot}$	(s)	Acetaldehyde

(1) a – q, b – p, c – r, d – s

(2) a – r, b – s, c – q, d – p

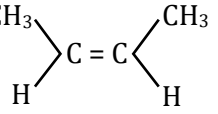
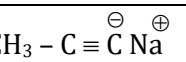
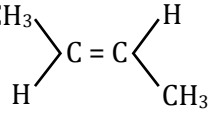
(3) a – q, b – p, c – s, d – r

(4) a – p, b – q, c – r, d – s

CHC128

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32. Match the column-I with column-II :-

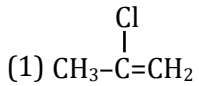
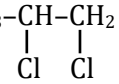
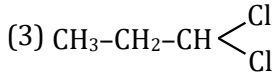
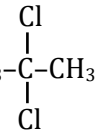
Column-I (Reaction)		Column-II (Product)	
(a)	$CH_3-C\equiv C-CH_3 \xrightarrow[Pd/C]{H_2}$	(p)	
(b)	$CH_3-C\equiv C-CH_3 \xrightarrow{Na/liq.NH_3}$	(q)	
(c)	$CH_3-C\equiv C-CH_3 \xrightarrow[Quinoline]{H_2/Pd-C}$	(r)	$CH_3CH_2CH_2CH_3$
(d)	$CH_3-C\equiv CH \xrightarrow{Na/liq.NH_3}$	(s)	

(1) a – s, b – r, c – q, d – p
(2) a – p, b – s, c – r, d – q
(3) a – r, b – s, c – p, d – q
(4) a – p, b – q, c – r, d – s

CHC136

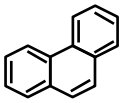
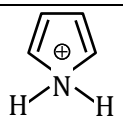
33. What is the main product of this reaction?

$CH_3-C\equiv CH \xrightarrow[Excess]{HCl(g)} ?$

(1) 
(2) 
(3) 
(4) 

CHC092

34. Match the column :-

Column-I		Column-II	
(a)	Aromatic	(p)	
(b)	Anti aromatic	(q)	
(c)	Non-aromatic	(r)	
		(s)	

(1) a – q, b – p, c – r,s
(2) a – p, b – q, c – r,s
(3) a – r, b – q, c – p,s
(4) a – p,s, b – q, c – r

CHC130

35. Match the column-I with column-II :-

Column-I (Compound)		Column-II (Boiling point)	
(a)	2-Methyl propane	(p)	309.1 K
(b)	Pentane	(q)	282.5 K
(c)	2-Methyl butane	(r)	261 K
(d)	2,2-Dimethylpropane	(s)	300.9 K

(1) a – r, b – p, c – s, d – q
(2) a – q, b – r, c – s, d – p
(3) a – r, b – s, c – q, d – p
(4) a – s, b – r, c – q, d – p

CHC137

36. Correct order of reactivity of following compound with an electrophile :-

(I) $p-CH_3-C_6H_4-CH_3$ (II) $C_6H_5-CH_3$
(III) $p-CH_3-C_6H_4-NO_2$ (IV) $p-O_2N-C_6H_4-NO_2$
(1) I > II > III > IV (2) II > I > IV > III
(3) III > II > I > IV (4) IV > III > II > I

CHC103

37. Toluene is more reactive than benzene towards electrophilic reagents due to :-

(1) Inductive effect only
(2) Hyperconjugative effect only
(3) Both inductive as well as hyperconjugative effects
(4) Strong mesomeric effect

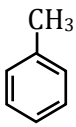
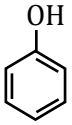
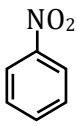
CHC104

38. Nitration of benzene is

(1) Nucleophilic substitution
(2) Nucleophilic addition
(3) Electrophilic substitution
(4) Electrophillic addition

CHC105

39. Consider the following compounds :

(I)  (II)  (III)  (IV) 

Correct order of their reactivity in electrophilic substitution reactions would be:-

(1) I > II > III > IV (2) IV > III > II > I
(3) III > II > I > IV (4) III > IV > I > II

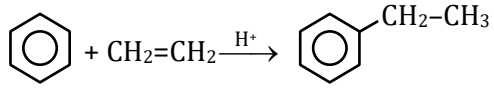
CHC106

40. The active species in the nitration of benzene is

(1) $\overset{+}{N}O_2$ (2) HNO_3

(3) $N\bar{O}_3$ (4) $N\bar{O}_2$

CHC107

41. 

Incorrect statement about this reaction

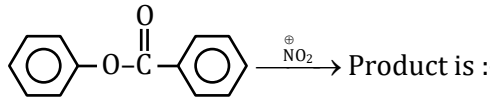
(1) Benzene is substrate

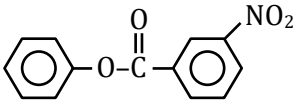
(2) Ethene is reagent

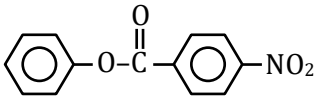
(3) Reaction is EAR with respect to ethene

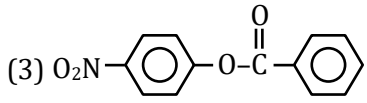
(4) Reaction is NSR for benzene

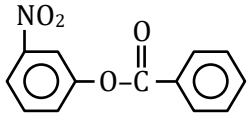
CHC110

42.  Product is :-

(1) 

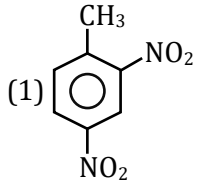
(2) 

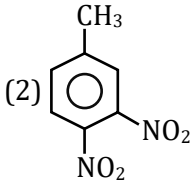
(3) 

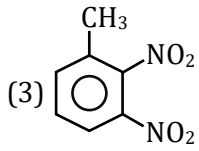
(4) 

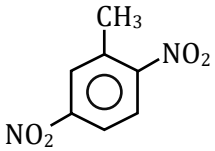
CHC116

43. p- nitro toluene on further nitration gives

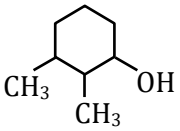
(1) 

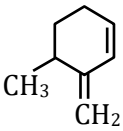
(2) 

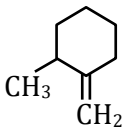
(3) 

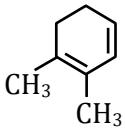
(4) 

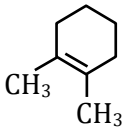
CHC117

44.  X (major) 'A' is

(1) 

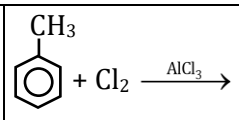
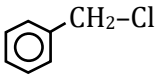
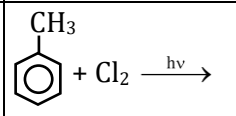
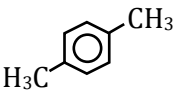
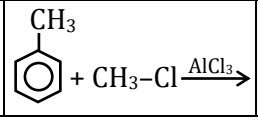
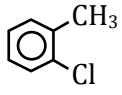
(2) 

(3) 

(4) 

CHC125

45. Match the column :-

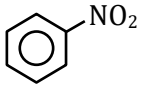
Column-I (Reaction)		Column-II (Product)	
(a)		(p)	
(b)		(q)	
(c)		(r)	

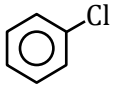
(1) a – q, b – p, c – r (2) a – r, b – p, c – q

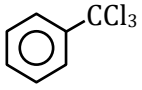
(3) a – q, b – r, c – p (4) a – p, b – q, c – r

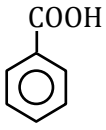
CHC131

46. In which of the following compound the electrophile attack on o– and p– positions :

(1) 

(2) 

(3) 

(4) 

CHC109

47. Which of the following reactions show that there are three C=C bond in benzene

(a) $C_6H_6 + Br_2 \xrightarrow{FeBr_3}$

(b) $C_6H_6 + HNO_3 \xrightarrow{H_2SO_4}$

(c) $C_6H_6 + 3O_3 \longrightarrow$

(d) $C_6H_6 + 3H_2 \xrightarrow{Ni}$

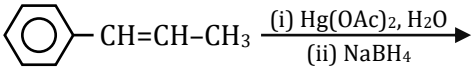
(1) a, c (2) b, d (3) b, c, d (4) c, d

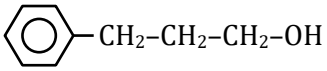
CHC078

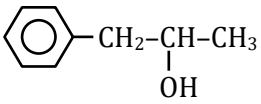
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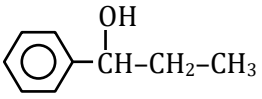
AIM : FREE EDUCATION TO ALL APUL

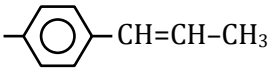
48. The major product of the following reaction is



(1) 

(2) 

(3) 

(4) 

CHC087

49. Halogenation of alkanes gives a mixture of monohalo products. The ease of substitution follows the order

(1) Tertiary H > primary H > secondary H

(2) Primary H < secondary H < tertiary H

(3) Primary H > secondary H > tertiary H

(4) Secondary H > tertiary H < primary H

CHC119

50. Which of the following reaction gives t-butyl benzene:

(1) C_6H_6 + t–butyl chloride, $\xrightarrow{AlCl_3}$

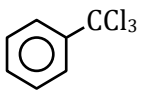
(2) $C_6H_6 + (CH_3)_2C{=}CH_2 \xrightarrow{HF}$

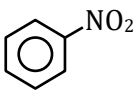
(3) C_6H_6 + t–butyl alcohol $\xrightarrow{H_2SO_4}$

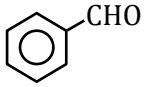
(4) All of them

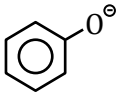
CHC118

51. Electrophile $\overset{\oplus}{N}O_2$ attacks the following :

(I) 

(II) 

(III) 

(IV) 

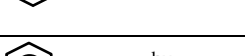
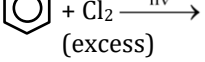
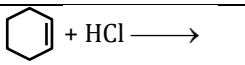
in which cases $\overset{\oplus}{N}O_2$ will attack at meta position

(1) II and IV (2) I, II and III

(3) III and IV (4) I only

CHC123

52. Match the column :-

Column-I (Reaction)		Column-II (Type of reaction)	
(a)		(p)	Free radical addition
(b)	 (excess)	(q)	Free radical substitution
(c)	$CH_4 + Br_2 \xrightarrow{h\nu}$	(r)	Electrophilic addition
(d)		(s)	Electrophilic substitution

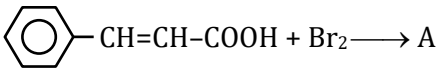
(1) a – p, b – q, c – s, d – r

(2) a – q, b – p, c – s, d – r

(3) a – p, b – q, c – r, d – s

(4) a – s, b – p, c – q, d – r

CHC129

53. 

the number of chiral carbons in 'A' are

(1) 1 (2) 2 (3) 3 (4) 4

CHC084

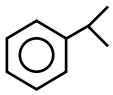
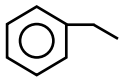
54. The strongest deactivating effect on aromatic ring is

(1) $-CH_2Cl$ (2) $-OCH_3$ (3) $-CH_3$ (4) $-CCl_3$

CHC101

55. Which of the following is maximum reactive towards E.S.R. :-

(1)  (2) 

(3)  (4) 

CHC102

56. The function of anhydrous $AlCl_3$ in the Friedel craft's reaction

(1) To absorb water

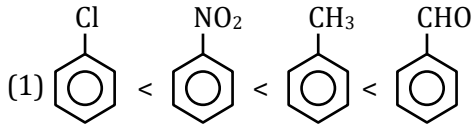
(2) To absorb HCl

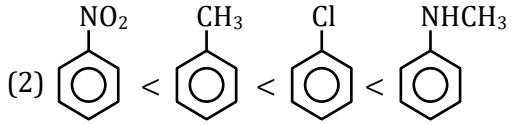
(3) To produce electrophile

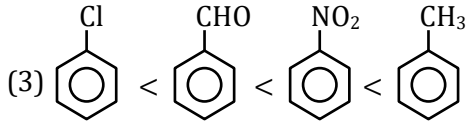
(4) To produce Nucleophile

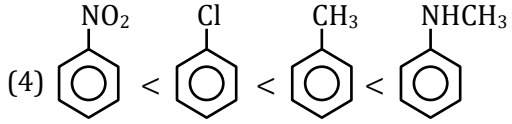
CHC108

57. Identify correct order of reactivity for electrophilic substitution reaction of the following compounds :–

(1) 

(2) 

(3) 

(4) 

CHC115

58. Which is not correct about nitration ?

(1) $\overset{\oplus}{NO_2}$ is an attacking electrophile

(2) In general, nitration is monosubstitution process

(3) In nitrating mix, HNO_3 acts as acid, & H_2SO_4 acts as base

(4) In nitrating mix, HNO_3 acts as base, & H_2SO_4 acts as an acid

CHC126

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**1**	**2**	**2**	**2**	**3**	**1**	**2**	**4**	**4**	**1**	**1**	**2**	**2**	**2**	**3**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**4**	**2**	**1**	**4**	**2**	**1**	**2**	**4**	**1**	**3**	**4**	**2**	**3**	**4**	**2**
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	**2**	**3**	**4**	**3**	**1**	**1**	**3**	**3**	**3**	**1**	**4**	**3**	**1**	**4**	**2**
Question	46	47	48	49	50	51	52	53	54	55	56	57	58		
Answer	**2**	**4**	**3**	**2**	**4**	**2**	**4**	**2**	**4**	**2**	**3**	**4**	**3**		

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Hydrocarbons

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (1)

Reactivity order of alkyl halide in Wurtz reaction

R—I > R—Br > R—Cl > R—F

2. Ans. (3)

Wurtz reaction is not suitable for preparation of Unsymmetrical alkane.

Ethan → CH_3-CH_3 Symmetrical

Butane → $CH_3CH_2-CH_2CH_3$ Symmetrical

Neo pentane → $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\underset{|}{\overset{|}{C}}}}-CH_3$

Hexane→$CH_3CH_2CH_2-CH_2CH_2CH_3$

Symmetrical

3. Ans. (1)

C—C Bond formation

(I) $R-MgX \xrightarrow{H_3O^+} R-H + Mg\begin{matrix}X\\OH\end{matrix}$ (Grignard reagent)

(II) $\dot{R} + \dot{R} \longrightarrow R-R$

(III) Corey house reaction

$R-X \xrightarrow[(iii)\ R'-X]{(i)\ Li\ (ii)\ CuX} R-R'$

(IV) $R\,Na + R-Br \longrightarrow R-R$

4. Ans. (3)

$CH_4 + O_2 \xrightarrow{Cu/523K/\ 200\ atm} CH_3OH$ (Alcohol)

$\xrightarrow{Mo_2O_3}$ H–CHO (Aldehyde)

Both are catalytic oxidation

$CH_4 + O_2 \longrightarrow C + H_2O$ (In complete oxidation)

↓

Carbon Black formation

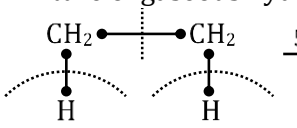
$CH_4 + O_2 \longrightarrow CO_2 + H_2O$ (complete oxidation)

5. Ans. (4)

Pyrolysis of alkanes

Free Radical elimination reaction

Liquid Hydrocarbon can be convert into a mixture of gaseous hydrocarbon.



$CH_2 = CH_2 + H_2$

+

CH_4

6. Ans. (2)

$CH_3CH_2CH_2Br \xrightarrow[\Delta]{alc.\ KOH} CH_3-CH=CH_2$ (Elimination)

(A)

$\xrightarrow[(ii)\ Zn,\ H_2O]{(i)\ O_3} CH_3-\underset{\displaystyle O}{\overset{\displaystyle H}{\underset{||}{\overset{|}{C}}}} + H-\underset{\displaystyle O}{\overset{\displaystyle H}{\underset{||}{\overset{|}{C}}}}$

(B) (C)

A → Propene; B → Ethanal; C → Methanal

7. Ans. (3)

3-Methyl-1-butene

$CH_2=CH-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\underset{|}{\overset{|}{CH}}}}$

$\downarrow O_3/H_2O/Zn$

$H-\underset{\displaystyle O}{\overset{\displaystyle H}{\underset{||}{\overset{|}{C}}}} + O=\overset{1}{\underset{\displaystyle H}{\overset{\displaystyle H}{\underset{|}{\overset{|}{C}}}}}-\overset{2}{\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\underset{|}{\overset{|}{CH}}}}}-\overset{3}{CH_3}$

Methanol 2-Methyl propanol

8. Ans. (3)

$CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\underset{|}{\overset{|}{C}}}}=\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\underset{|}{\overset{|}{C}}}}-CH_3 \xrightarrow{O_3/H_2O/Zn} 2CH_3-\underset{\displaystyle O}{\overset{\displaystyle CH_3}{\underset{||}{\overset{|}{C}}}}-CH_3$

2,3-Dimethyl-2-butene Acetone

9. Ans. (1)

$CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_2}{\underset{|}{\overset{||}{C}}}} \xrightarrow{KMnO_4} CH_3-\underset{\displaystyle O}{\overset{\displaystyle CH_3}{\underset{||}{\overset{|}{C}}}} + H-\underset{\displaystyle O}{\overset{\displaystyle H}{\underset{||}{\overset{|}{C}}}}$

$\longrightarrow H-\underset{\displaystyle O}{\overset{\displaystyle OH}{\underset{||}{\overset{|}{C}}}} + H_2O + CO_2$

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10. Ans. (4)

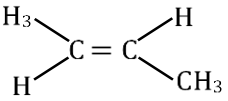
Ozonolysis $\longrightarrow$

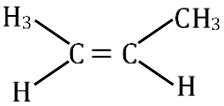
11. Ans. (2)

H.O.H. $\propto$ No. of π Bond

$\propto \frac{1}{\text{stability}}$ (If No. of π Bond same)

(i) $CH_2=CH-CH_2-CH_3$

(ii) 

(iii) 

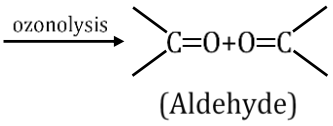
(iv) $CH_2=CH-CH_3$

12. Ans. (2)

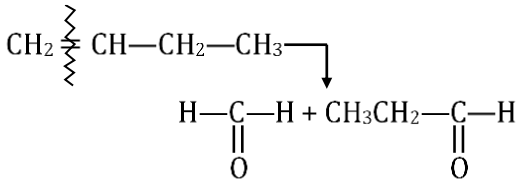
3 C—C

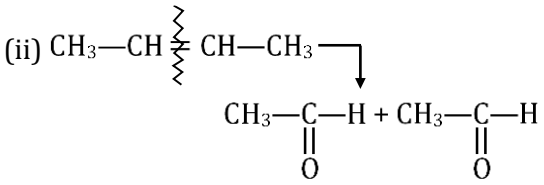
8 C—H σ Bond

1 C—C π Bond



(Aldehyde)

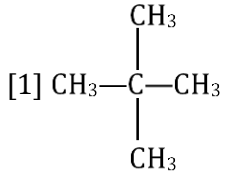
(i) 

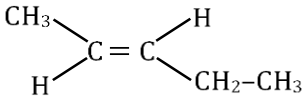
(ii) 

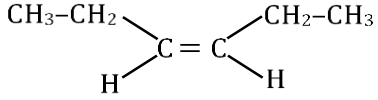
(iii) $CH_3-CH=CH-CH_2-CH_3$

(iv) $CH_3-C\equiv C-CH_3$

13. Ans. (4)

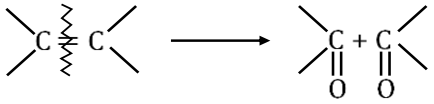
[1]  $\mu = 0$

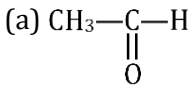
[2]  $\mu \neq 0$

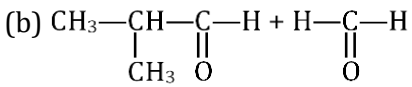
[3]  $\mu \neq 0$

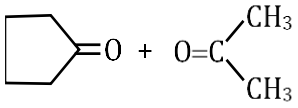
14. Ans. (4)

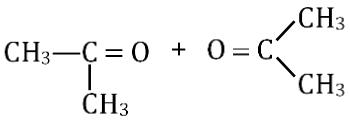
Ozonolysis



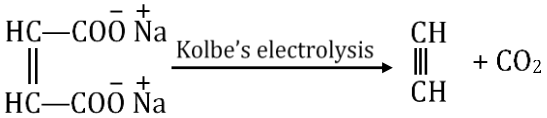
(a) 

(b) 

(c) 

(d) 

15. Ans. (3)



at Anode

16. Ans. (1)

Kolbe's electrolysis

at Anode $\rightarrow$ Hydrocarbon + CO_2

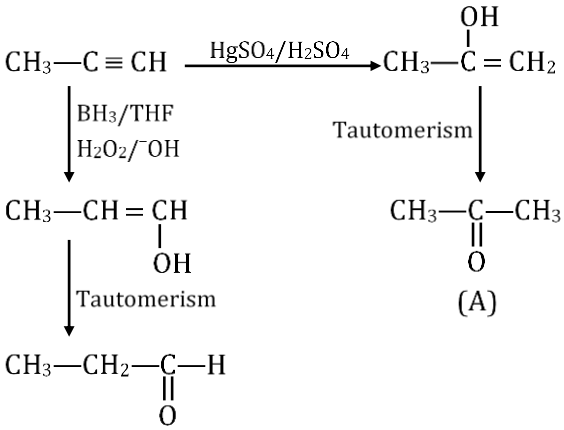
at Cathode $\rightarrow$ H_2 + NaOH

$\therefore$ pH of solution increase with time

17. Ans. (4)

Terminal alkynes react with tollen's reagent and give a P.P.T.

18. Ans. (2)



(A)

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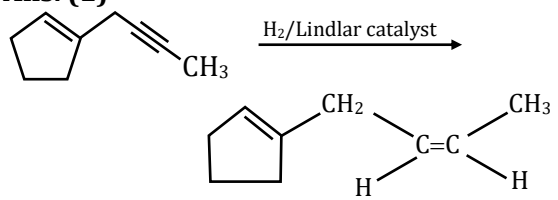
19. Ans. (1)

$Me-CH_2-C\equiv CH \xrightarrow{NH_3/NaNH_2} Me-CH_2-C\equiv \bar{C}\,\overset{+}{Na}$ (A)

$\xrightarrow{CH_3CH_2Br}$

$Me-CH_2-C\equiv C-CH_2CH_3$ (B)

20. Ans. (1)



Cis-alkene

21. Ans. (3)

Propene	Propyne
$CH_2=CH-CH_3$	$HC\equiv C-CH_3$

(1) Bromine water, (2) Alkaline $KMnO_4$ — Test of Unsaturation ($\therefore$ both gives the test)

(3) Ammonical silver nitrate → Propyne ✓

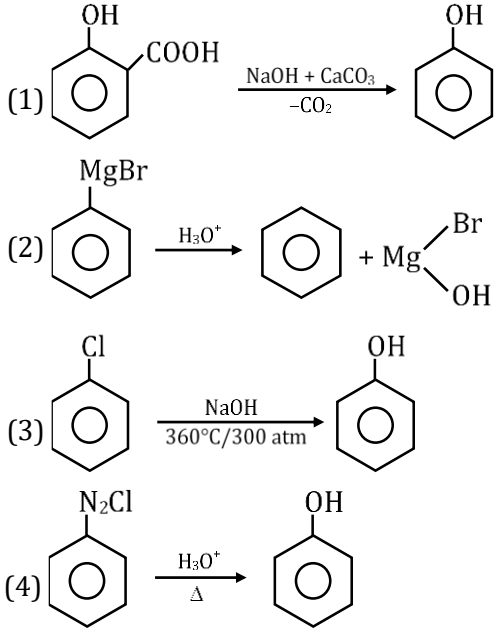
(Terminal alkyne gives this test)→Propene ×

22. Ans. (3)

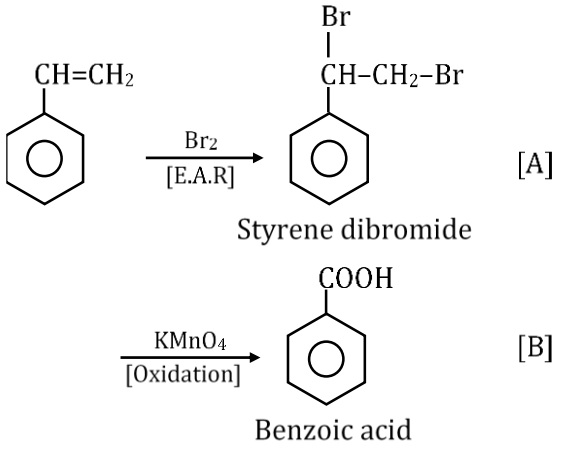
Terminal alkyne $\xrightarrow[Cu_2Cl_2]{\text{Ammonical}}$ $HC\equiv \overset{\ominus}{C}\,\overset{\oplus}{Cu}$ (Red P.P.T.)

$HC\equiv CH$

23. Ans. (2)

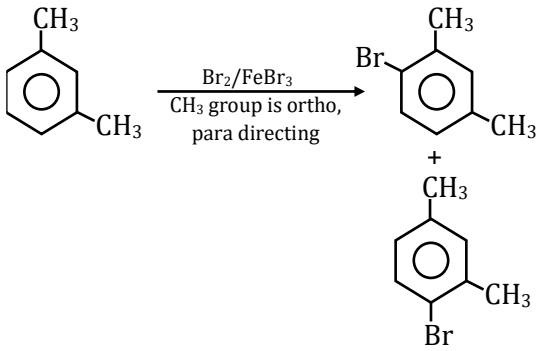


24. Ans. (4)

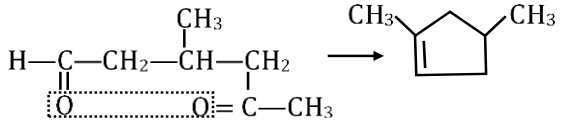


Exercise-II (Previous Year Questions)

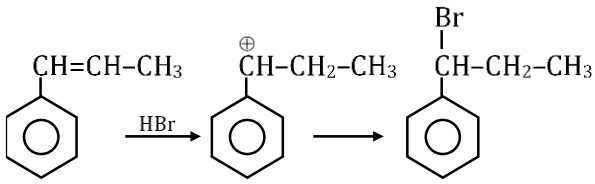
1. Ans. (3)



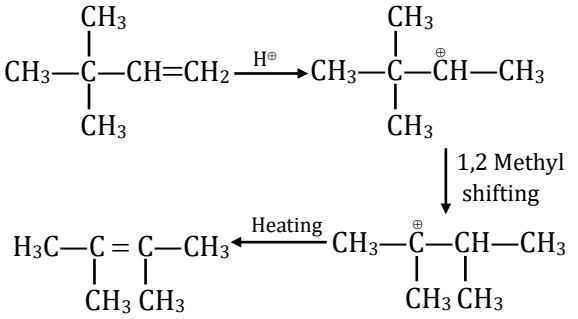
2. Ans. (4)



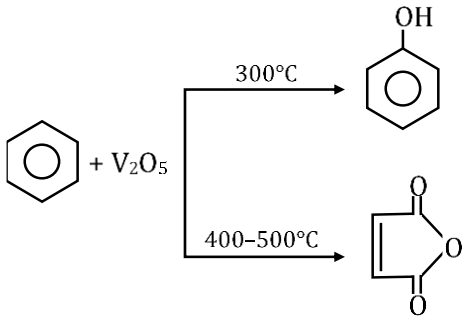
3. Ans. (4)



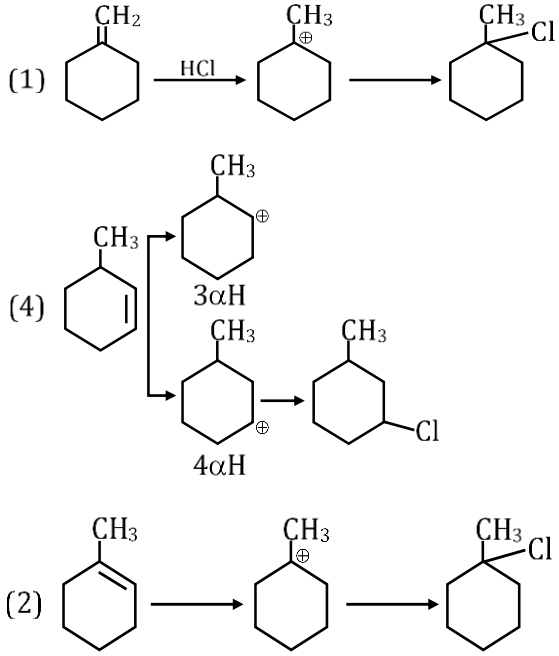
4. Ans. (4)



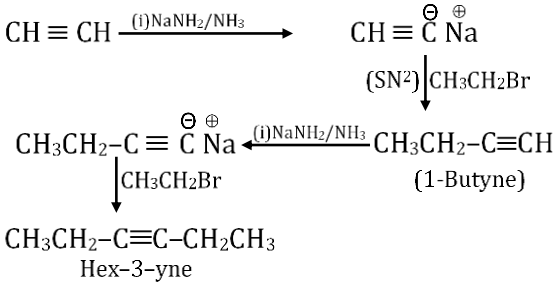
5. Ans. (4)



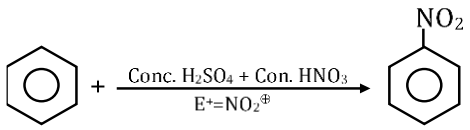
6. Ans. (3)



7. Ans. (1)

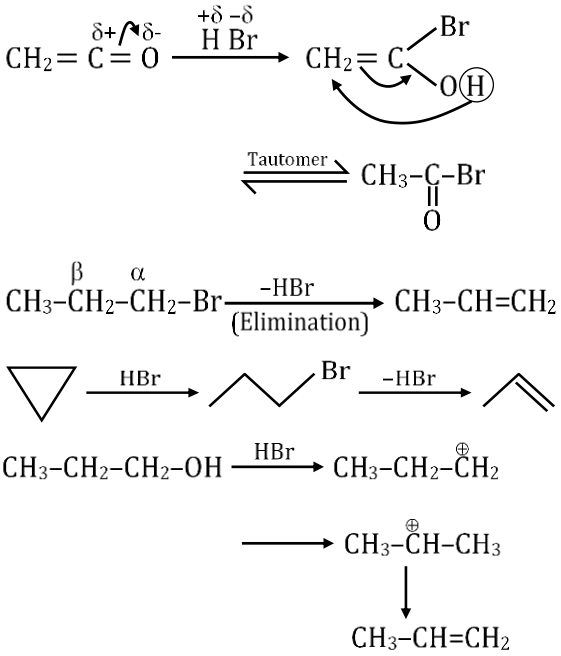


8. Ans. (2)

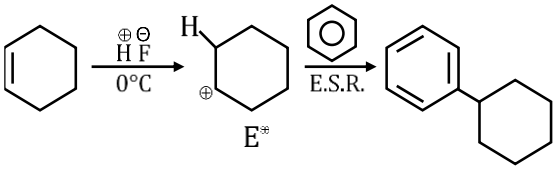


If $KHSO_4$ added, more K^+ HSO_4^- ion form, conc. of $E^{\oplus}$ decrease. So rate of ESR=decrease.

9. Ans. (1)



10. Ans. (1)



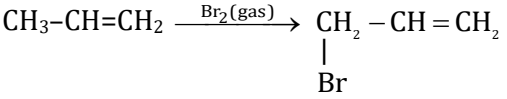
11. Ans. (3)

(1) $CH_3CH_2CH_2CH_3$ (2) $CH_2=CH_2$

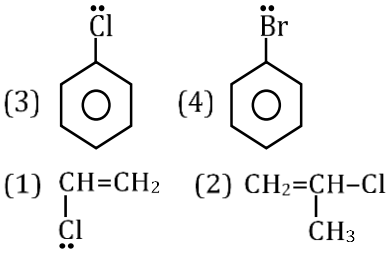
(3) $H_3C–HC=CH_2$ (4) $HC\equiv CH$

Bromine gas react with alkene to give allylic substituted product by free radical mechanism.

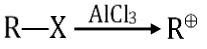
Reaction of free Radical depends upon → Stability of free radical



12. Ans. (2)



Friedel-Crafts reaction

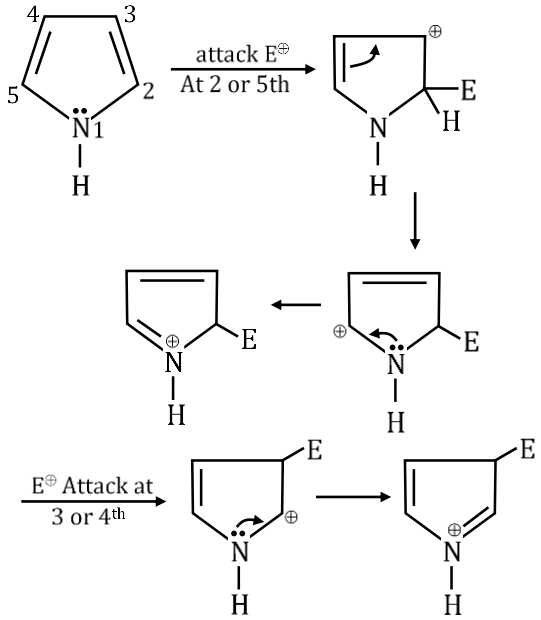


Option-1,3,4 are Reso. stable.

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13. Ans. (2)



More resonating structure

When $E^{\oplus}$ attack on 2nd or 5th position

So Maxi electron density on 2nd and 5th position.

14. Ans. (3)

$$CH_3-C\equiv CH \xrightarrow[HgSO_4]{H_2O,\ H_2SO_4} CH_3-\underset{\underset{O(H)}{}}{C}=CH_2$$

Tautomerism

$$CH_3-\underset{\underset{O}{\|}}{C}-CH_3$$

15. Ans. (4)

(1) $CH\equiv CH \xrightarrow{Br_2} \underset{Br}{\overset{Br}{CH=CH}}$

(2) $CH_2=CH_2 \xrightarrow{HBr} CH_3-CH_2Br \xrightarrow{Na/dry\ ether} CH_3CH_2CH_2CH_3$

(3) $CH_3-CH_3 \xrightarrow{HBr} CH_3CH_2Br \xrightarrow{Na/dry\ ether} CH_3CH_2CH_2CH_3$

(4) $CH_4 \xrightarrow{Br_2} CH_3-Br \xrightarrow{Na/dry\ ether} CH_3-CH_3$

[Carbon less than 4]

16. Ans. (3)

NO_2 is meta directing

$\therefore$ $E^{\oplus}$ Y attack on meta position

17. Ans. (2)

$$CH_3-C\equiv C-CH_3 \longrightarrow \underset{H}{\overset{CH_3}{}}C=C\underset{H}{\overset{CH_3}{}}$$

Cis-product

$H_2/Pd/C$ quinoline → Lindlar catalyst.

18. Ans. (3)

$$\underset{CH_3}{\overset{CH_3}{}}C=CH-CH_3 \xrightarrow{O_3/Zn/H_2O} \underset{CH_3}{\overset{CH_3}{}}C=\boxed{O+O}=CH-CH_3$$

[A] Propanone Ethanal

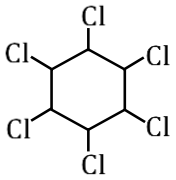
$$[A] \xrightarrow{HCl} CH_3-\underset{CH_3}{\overset{Cl}{\overset{|}{C}}}-CH_2-CH_3$$

19. Ans. (2)

Electrophilic Substitution reaction

[1] Sandmeyer reaction

[2] Friedal-Craft's reaction

[3]  Gammexane Or Lindane or BHC — FRAR reaction

[4] NSR reaction

20. Ans. (2)

$$CH_3-\underset{CH_3}{\overset{CH_3}{\overset{|}{C}}}-CH_2-CH_3 \xrightarrow{-H} \text{3 Radical}$$

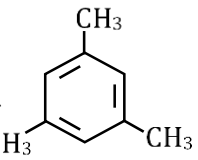
$$CH_3-\underset{CH_3}{\overset{CH_3}{\overset{|}{C}}}-CH_3 \longrightarrow \text{1 Radical}$$

$CH_3CH_2CH_2CH_2CH_3 \longrightarrow$ 3 Radical

$CH_3-\underset{CH_3}{CH}-CH_2-CH_3 \longrightarrow$ 4 Radical

21. Ans. (1)

$$CH_3-C\equiv CH \xrightarrow[873K]{\text{Red not iron tube}} A$$



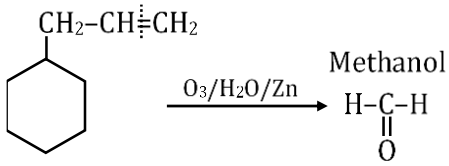
σ Bond = 2^1

π Bond = 3

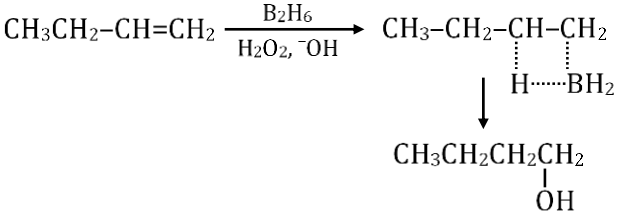
22. Ans. (4)

Wurtz reaction is not suitable for preparation of Unsymmetrical alkane.

23. Ans. (4)



24. Ans. (1)



25. Ans. (4)

ESR ∝ + M ∝ + H ∝ + I

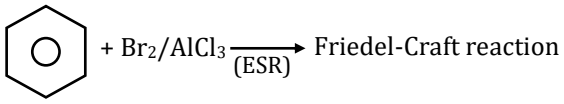
(1) –I effect of Cl (2) –M effect of NO_2

(3) Zero (4) +M effect of –OH

Most reactive option (4)

26. Ans. (3)

(1)

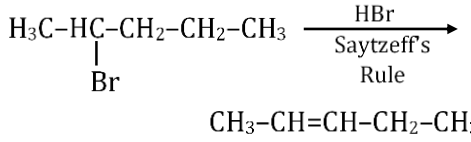


(2) $CH_2=CH_2 \xrightarrow[(EAR)]{HBr}$

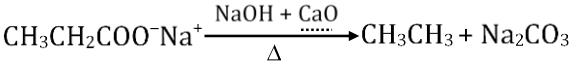
(3) $CH_4 \xrightarrow{Br_2/h\nu}$ FRSR

(4) $CH_3-CH=CH_2 \xrightarrow[(C_6H_5COO)_2]{HBr}$ Peroxide effect (FRAR)

27. Ans. (1)

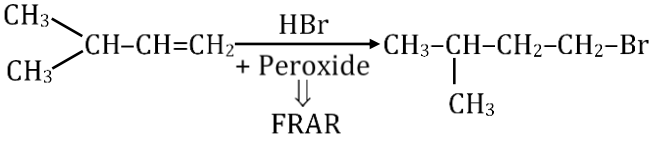


28. Ans. (3)



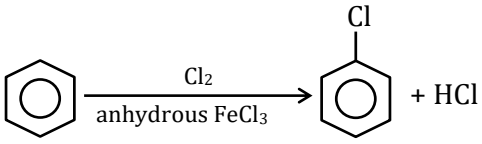
(Soda lime decarboxylation)

29. Ans. (1)

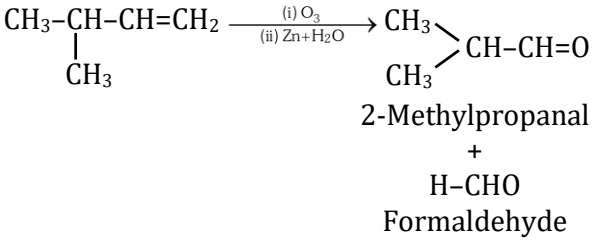


[1st attack by $\dot{B}r$]

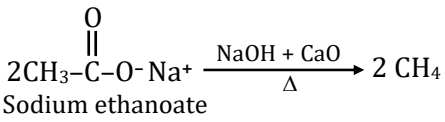
30. Ans. (4)



31. Ans. (4)

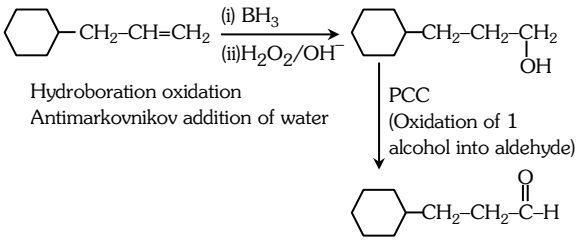


32. Ans. (1)



Weight = 2 × 16 = 32 g

33. Ans. (2)



34. Ans. (3)

(1) $CH_3-CH_2-CH_2-CH_2-CH_2-CH_3$

No tertiary carbon

(2) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-CH_2-CH_3$

One tertiary carbon

(3) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$

Two tertiary carbon

(4) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-CH_2-CH_3$

No tertiary carbon

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35. Ans. (1)

For alkanes

Boiling point $\propto$ molecular mass

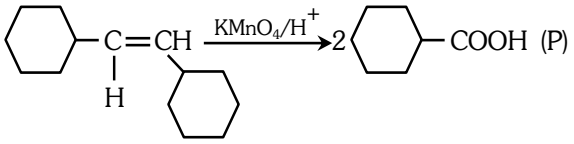
$\propto \frac{1}{\text{Branching}}$ (if molecular mass is same)

Boiling point

$CH_3-CH_2-CH_2-CH_2-CH_3$ > $CH_3-CH(CH_3)-CH_2-CH_3$ > $CH_3-C(CH_3)_2-CH_3$

n-pentane, isopentane, neopentane

36. Ans. (2)



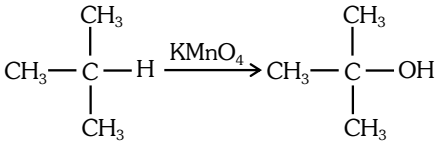
37. Ans. (3)

Bayer's Reagent is cold dilute alkaline aqueous $KMnO_4$.

38. Ans. (1)

F_2 is highly reactive so it is not suitable for the preparation of corresponding haloarene products.

39. Ans. (2)



$R_1 = CH_3$, $R_2 = CH_3$, $R_3 = CH_3$

Exercise-III (Analytical Questions)

1. Ans. (1)

Alkane → FRSR

Reactivity of Reaction $\propto$ stability of Radical

Reactivity of Halogen = $F_2 > Cl_2 > Br_2 > I_2$

2. Ans. (2)

$CH_3CH_2CH_2CH_3 \xrightarrow[\text{(FRSR)}]{Br_2/\text{light at } 130^\circ} CH_3-\underset{Br}{CH}-CH_2-CH_3$

3. Ans. (2)

$\underset{\Downarrow}{C_2H_6} + Cl_2 \xrightarrow{h\nu} \underset{\Downarrow}{C_2H_5Cl} + HCl$

Excess — Maximum yield

4. Ans. (2)

(a) $2CH_3CH_2Br \xrightarrow[\text{(Wurtz reaction)}]{2Na/\text{dry ether}} CH_3-CH_2-CH_2-CH_3$

(b) $CH_3COO^{\ominus}Na^{\oplus} \xrightarrow[\text{Soda lime decarboxylation}]{CaO + NaOH,\ \Delta} CH_4$ $(-CO_2)$

(c) $CH_3(CH_2)_4CH_3 \xrightarrow[773K,\ 10\text{-}20\ atom]{V_2O_5}$ 

(d) $2CH_4 + O_2 \xrightarrow[\text{100 atm catalytic oxidation}]{Cu/523K} CH_3-OH$

5. Ans. (3)

(a) $CH_4 + O_2 \xrightarrow[\text{(catalytic oxidation)}]{Mo_2O_3,\ \Delta} HCHO$ (Aldehyde)

(b) $CH_3CH_3O_2 \xrightarrow[\Delta]{(CH_3COO)_2Mn} CH_3COOH$ (Acid)

(c) $(CH_3)_3CH \xrightarrow{KMnO_4} (CH_3)_3CH-OH$ (Alcohol)

Only 3°C—H

(d) $CH_4 + H_2O \xrightarrow{Ni,\ \Delta} CO + 3H_2$

6. Ans. (1)

(a) $CH{\equiv}CH \xrightarrow{H_2,\ Pd}$ (Reduction)

(b)  $\xrightarrow{\text{Anhy. } AlCl_3 + HCl}$ (Isomerisation)

(c)  $\xrightarrow[\text{(High temp.)}]{773\ K}$ Pyrolysis

(d) $CH_3CH_3 \xrightarrow{Cl_2,\ h\nu}$ F.R.S.R.

7. Ans. (2)

Incorrect statement

(A) Alkene → First 3 member gas

Next 14 are liquid

Higher are solid True

(B) True

(C) Straight chain alkenes > Branched chain compound

$\boxed{BP \propto M.W.}$ $\boxed{BP \propto \frac{1}{\text{Branching}}}$

(D) False

8. Ans. (4)

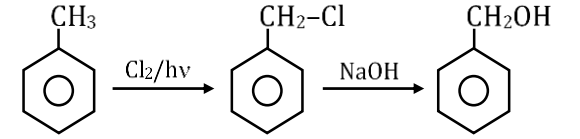
Reactivity towards NBS $\propto$ Stability of free radical.

(Free radical substitution Reaction)

Remove Benzylic and Allylic Hydrogen then check stability of free radical

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9. Ans. (4)



(Toluene) (Benzyl alcohol)

10. Ans. (1)

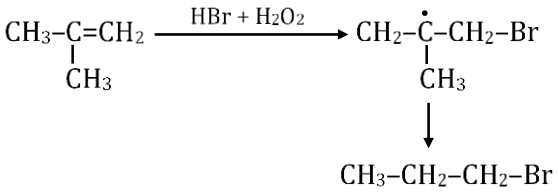
(1) $CH_3-CH_2-CH_2-CH_3$ → 2 Radical possible

(2) $CH_3-\underset{CH_3}{CH}-CH_2-\underset{CH_3}{CH}-CH_3$ → 3 Radical possible

(3)  ⟶ 1 Radical possible

(4) $CH_3-\underset{CH_3}{CH}-CH_2-CH_3$ ⟶ 4 Radical possible

11. Ans. (1)



Isobutylbromide

12. Ans. (2)

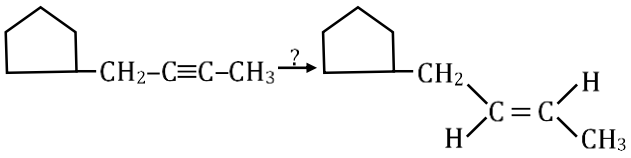
(a) Hydration (q)

(b) Peroxide effect ∴ FRAR
(s) Hydrohalogenation

(c) EAR ∴ Halogenation (r)

(d) Reduction ∴ Hydrogenation (p)

13. Ans. (2)

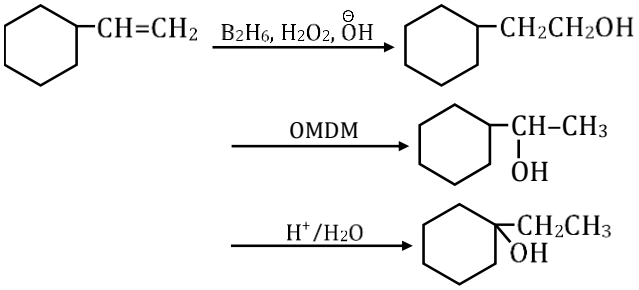


Trans-Alkene

∴ Birch reduction

Reagent = Na/NH_3

14. Ans. (2)



15. Ans. (3)

(a) Lindlar's catalyst ⟶ $H_2/Pd\text{-}CaCO_3$

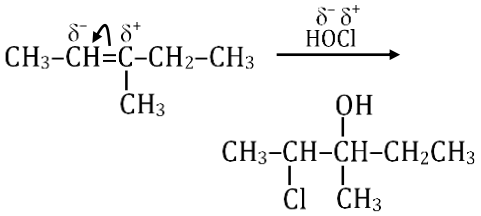
(b) Birch catalyst ⟶ $Na/liq.\ NH_3$

(c) Baeyer's reagent ⟶ $Dil/\ KMnO_4/273k$

(d) Kharash effect ⟶ $HBr + (PhCO)_2O_2$ (Peroxide)

16. Ans. (4)

3–Methyl-2-Pentene



17. Ans. (2)

Reactivity towards EAR ∝ Stability of $C^{\oplus}$

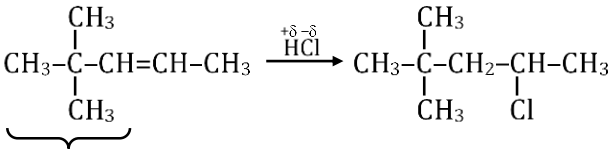
(1) $Ph-\overset{\oplus}{C}H-CH_3$ Reso. + 3αH

(2) $Ph-\overset{\oplus}{C}(CH_3)-CH_2-CH_3$ Reso. + 5αH

(3) $Ph-\overset{\oplus}{C}(Ph)-CH_2-CH_3$ More reso.

(4) $\overset{\oplus}{C}H_2-CH_2-NO_2$ 3 > 2 > 1 > 4

18. Ans. (1)



Bulky group (Hofmann product

19. Ans. (4)

Hydration means H^+/H_2O or Addition of water from carbocation and check reactivity.

(1) $CH_3-\overset{\oplus}{C}H_2$

(2) $CH_3-\overset{\oplus}{C}H-CH_3$

(3) $CH_3-\overset{\oplus}{C}H-CH_2CH_3$

(4) $CH_3-\overset{\oplus}{C}(CH_3)-CH_3$

Maximum αH ∴ Maximum stable

20. Ans. (2)

$H_3C-CH=CH-CH_3 \xrightarrow{H^+/H_2O} CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CH_3$

$CH_3-\underset{\underset{CH_3}{|}}{C}=CH_2 \xrightarrow{H^+/H_2O} CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{OH}{|}}{C}}-CH_3$

$CH_2=CH_2 \xrightarrow{H^+/H_2O} CH_3-CH_2-OH$

$CH_2=CH-CH_2CH_3 \xrightarrow{H^+/H_2O} CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CH_3$

d-(Butylene)

21. Ans. (1)

$CH_3-CH=CH_2 \xrightarrow[(EAR)]{\overset{\delta^+}{H}\ \overset{\delta^-}{I}} CH_3-\overset{\oplus}{C}H-CH_3 \xrightarrow{\overset{\delta^-}{I}} CH_3-\underset{\underset{I}{|}}{CH}-CH_3$

(Propene)

22. Ans. (2)

$\overset{\delta^-}{H_2C}=\overset{\delta^+}{CH}-CH_2-Ph \xrightarrow[(EAR)]{\overset{\delta^+}{H}\overset{\delta^-}{Br}} CH_3-\overset{\oplus}{C}H-CH_2-Ph$

(3-Phenyl Propene)

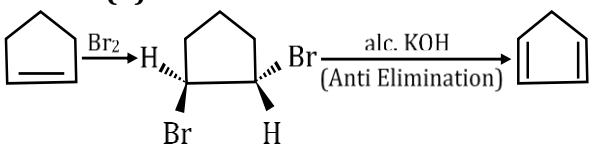
Rearrangment

$CH_3-CH_2-\overset{\oplus}{C}H-Ph$

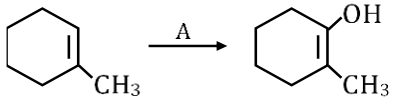
$Br^{\ominus}$

$CH_3-CH_2-\underset{\underset{Br}{|}}{CH}-Ph$

23. Ans. (4)



24. Ans. (1)



25. Ans. (3)

$CH_3-CH=CH_2 \xrightarrow{dil.\ H_2SO_4} CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$

$CH_3-CH=CH_2 \xrightarrow{B_2H_6/H_2O_2/\overset{\ominus}{O}H} CH_3-CH_2-\underset{\underset{OH}{|}}{CH_2}$

A and B → Same f. g. ✓

→ Position isomer ✓

→ Chain isomer ×

26. Ans. (4)

$CH_3-\underset{\underset{CH_3}{|}}{CH}-CH=CH_2 \xrightarrow{D^{\oplus}/H_2O} CH_3-\underset{\underset{CH_3}{|}}{CH}-\overset{\oplus}{C}H-\underset{\underset{D}{|}}{CH_2}$

Rearrangement

$CH_3-\underset{\underset{CH_3}{|}}{\overset{\oplus}{C}}-CH_2-CH_2-D$

$CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{OH}{|}}{C}}-CH_2-CH_2-D \xleftarrow{H_2O}$

27. Ans. (2)

$CH_3-CH=CH_2 \xrightarrow[\text{No effect of peroxide}]{HCl+H_2O_2} CH_3-\overset{\oplus}{C}H-CH_3$

$\therefore$ EAR Mech.

28. Ans. (3)

$CH_2=CH-CH_3 \xrightarrow[\text{Peroxide effect/ Kharash effect}]{HBr + Peroxide} CH_2-\underset{\underset{Br}{|}}{\overset{\bullet}{C}}H-CH_3$

(Propene)

FRAR

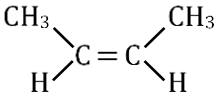
$CH_2-CH_2-CH_3$ (Br)

n-Propyl bromide

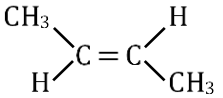
29. Ans. (4)

(A) Cis-trans → Geometrical isomer
Same Molecular formula
Same structural formula
Different configuration or arrangement of atom

(B)



Cis-2-butene

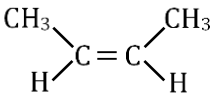


trans-2-butene

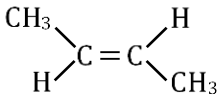
⇓ μ≠0, B.P. more

⇓ μ=0, B.P. less

(C)

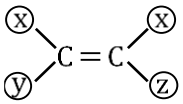


μ≠0

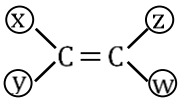


μ=0

(D)



G.I ✓



G.I ✓

A ✓ B ✓ C × D ✓

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30. Ans. (2)

$$Ph-CH=\underset{\underset{CH_3}{|}}{C}-CH_3 + HBr \xrightarrow[\text{Kharash rule}\\ \text{Anti M.K.}\\ \text{I}^{st}\text{ attack by }\dot{Br}]{\text{Peroxide}}$$

$$Ph-\dot{\underset{H}{C}}-\underset{\underset{CH_3}{|}}{\overset{Br}{\overset{|}{C}}}-CH_3 \longrightarrow Ph-CH_2-\underset{\underset{CH_3}{|}}{\overset{Br}{\overset{|}{C}}}-CH_3$$

31. Ans. (2)

(i) $CH \equiv CH + H_2O \xrightarrow{\text{(More acidic)}}$ No reaction

Acid Base Reaction

(ii)

$$CH\equiv CH + H_2O + Hg^{2+} \longrightarrow \underset{\underset{OH}{|}}{CH_2=CH} \longrightarrow \underset{\underset{O}{||}}{CH_3-C-H}$$

Tautomer (Acetaldehyde)

(iii) $CH\equiv CH + NaNH_2 \longrightarrow CH\equiv C^{\ominus}\overset{\oplus}{Na} + NH_3$

(iv) $CH\equiv CH \xrightarrow[\text{Tube}]{\text{Red Hot}}$ Benzene

32. Ans. (3)

(a) $CH_3-C\equiv C-CH_3 \xrightarrow[\text{(Reduction)}]{H_2, Pd/C} CH_3CH_2CH_2CH_3$

(b) $CH_3-C\equiv C-CH_3 \xrightarrow[\text{(Birch reduction)}]{Na/liq.\ NH_3}$ Trans-alkene

(c) $CH_3-C\equiv C-CH_3 \xrightarrow[\text{Quinoline}]{H_2/Pd\text{-}C}$ Cis-alkene

(Partial reduction)

(d) $CH_3-C\equiv CH \xrightarrow{Na/liq.\ NH_3} CH_3-C\equiv \overset{\ominus\oplus}{C}Na$

(Terminal alkyne)

33. Ans. (4)

$$CH_3-C\equiv CH \xrightarrow[\text{Excess}]{HCl(g)} \underset{\underset{Cl}{|}}{CH_3-C=CH_2} \xrightarrow{HCl} \underset{\underset{Cl}{|}}{\overset{Cl}{\overset{|}{CH_3-C-CH_3}}}$$

(M.K. rule)

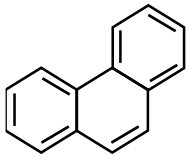
34. Ans. (3)

(p) 

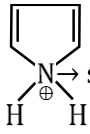
Non-aromatic

(q) 

Anti-aromatic

(r) 

Aromatic

(s) 

$\rightarrow sp^3$ ∴ Non aromatic

a → r ; b → q ; c → p, s

35. Ans. (1)

(a) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_3$

2-Methyl propane

(b) $CH_3CH_2CH_2CH_2CH_3$

Pentane

(c) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3$

2-Methyl butane

(d) $CH_3-\underset{\underset{CH_3}{|}}{\overset{CH_3}{\overset{|}{CH}}}-CH_3$

2,2-Dimethyl propane

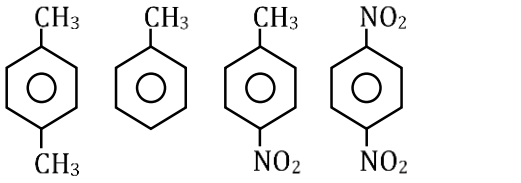
B.P. ∝ M.W.

$$\propto \frac{1}{\text{Branching}}$$

$$\underline{b} > \underline{c} > \underline{d} > \underline{a}$$

309.1 300.9 282.5 261

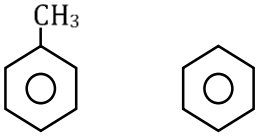
36. Ans. (1)



+H −M −M of two group

1 > 2 > 3 > 4

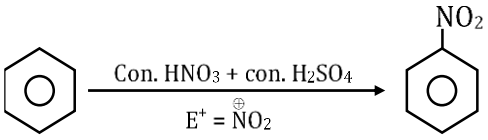
37. Ans. (3)



Toluene Benzene

+H effect of CH_3 group

38. Ans. (3)



E.S.R

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39. Ans. (3)

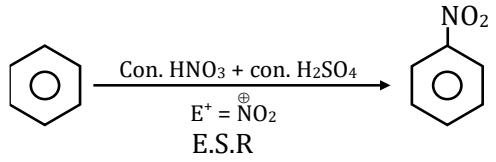
Reactivity towards ESR $\propto$ + M $\propto$ +H $\propto$ +I

(I) No effect (II) +H effect of CH_3

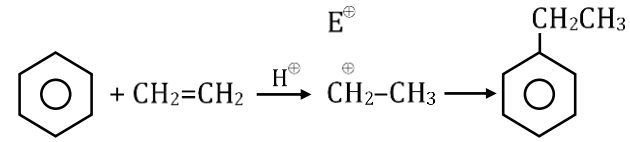
(III) +M effect OH (IV) –M of NO_2 group

3 > 2 > 1 > 4

40. Ans. (1)

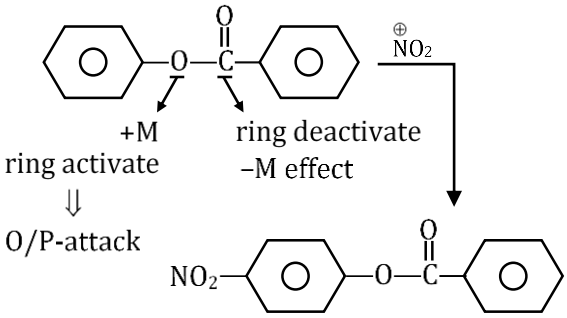


41. Ans. (4)

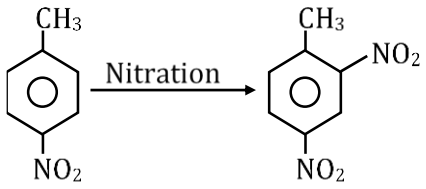


∴ attack on Benzene (ESR mech)

42. Ans. (3)

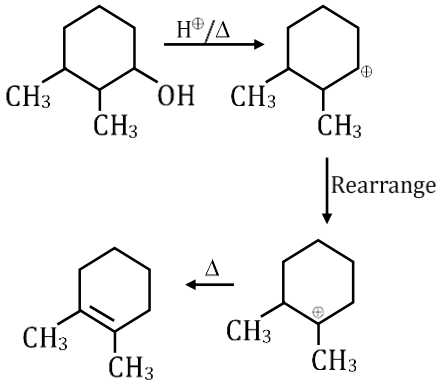


43. Ans. (1)

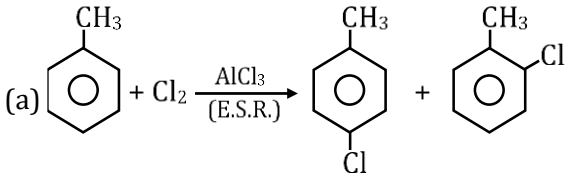


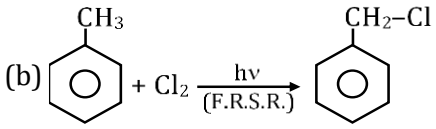
(p-nitro toluene)
ortho attack with
respect to-CH_3 group

44. Ans. (4)

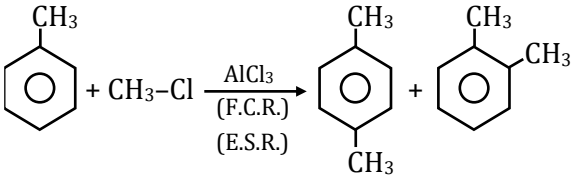


45. Ans. (2)

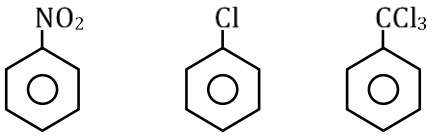




(c)



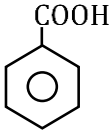
46. Ans. (2)



–M effect
∴ Metal directing

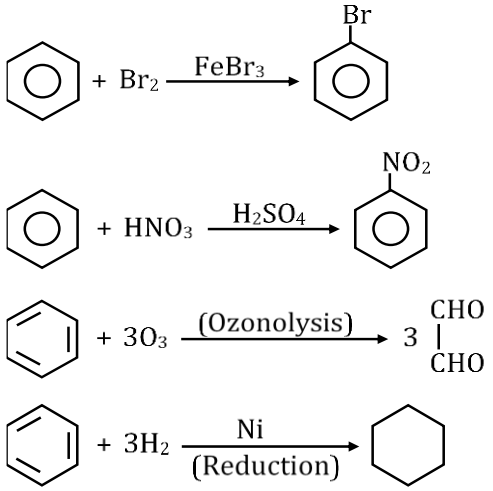
+M
O/P-directing

–H
Meta directing

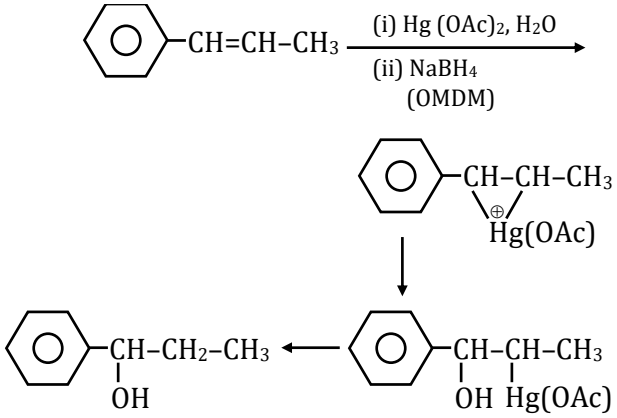


–M effect
∴ Meta directing

47. Ans. (4)



48. Ans. (3)



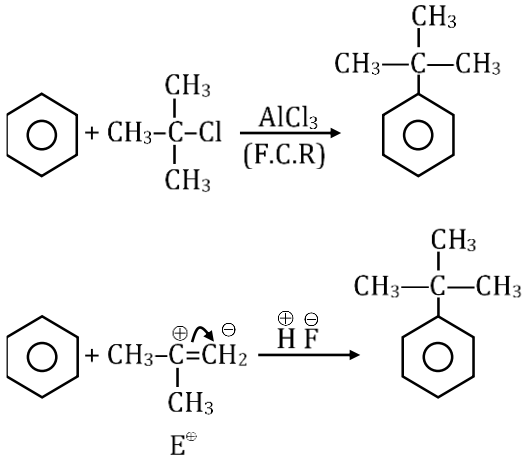
HBO Reaction

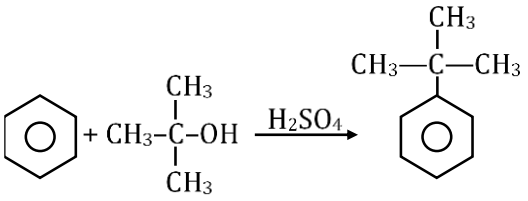
49. Ans. (2)

Halogenation of Alkane → FRSR

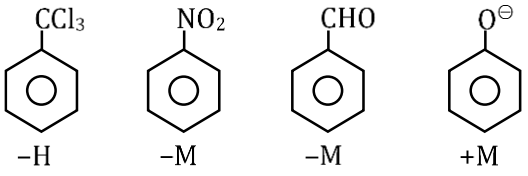
1°H < 2°H < 3°H for monohalo Product

50. Ans. (4)

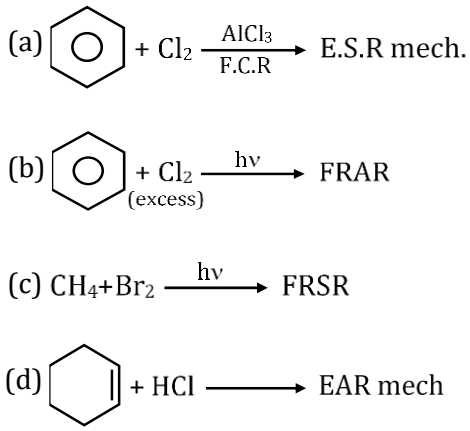




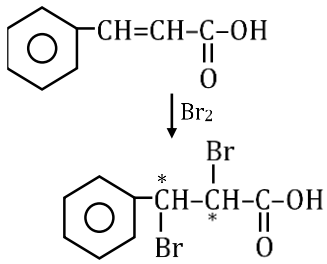
51. Ans. (2)



52. Ans. (4)

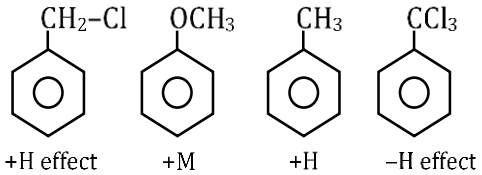


53. Ans. (2)



No. of chiral carbon = 2

54. Ans. (4)



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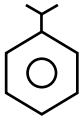
55. Ans. (2)

Reactivity towards E.S.R. $\propto$ e^- density in ring

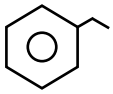
$\propto \frac{+M}{-M}$

(1) 

(2) 

(3) 

1αH

(4) 

2αH

Option (2) is Maximum reactive

56. Ans. (3)

Friedel craft reaction

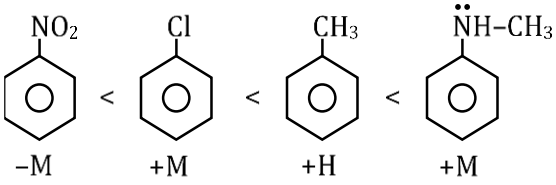
$R—Cl + AlCl_3 \longrightarrow \overset{\oplus}{R}$

$R—\underset{\underset{O}{\|}}{C}—Cl + AlCl_3 \longrightarrow R—\overset{\oplus}{\underset{\underset{O}{\|}}{C}}$

Produce Electrophile

57. Ans. (4)

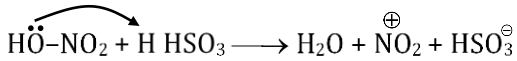
Reactivity towards E.S.R. $\propto \frac{+M}{-M} \propto \frac{+H}{-H} \propto \frac{+I}{-I}$



–I (Dominate)

58. Ans. (3)

$HNO_3 + H_2SO_4$



Attacking $E^{\oplus} = \overset{\oplus}{N}O_2$

$HNO_3 \rightarrow$ Base

$H_2SO_4 \rightarrow$ Acid

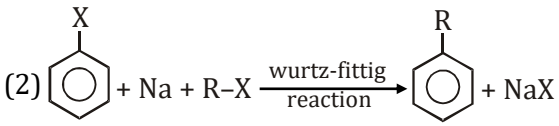
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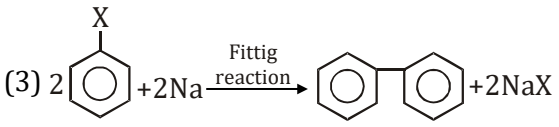
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EXERCISE-I (Conceptual Questions) Build Up Your Understanding

1. Which of the following is correct

(1) $2RX + 2Na \xrightarrow{\text{wurtz reaction}} R–R + 2NaX$

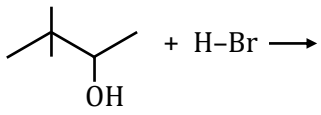
(2) 

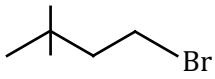
(3) 

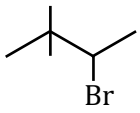
(4) All of these

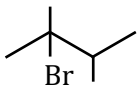
CHH024

2. Which halide is not obtained from following reaction ?



(1) 

(2) 

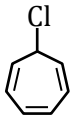
(3) 

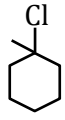
(4) All

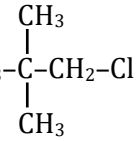
CHH128

3. Arrange the following compounds in decreasing order of reactivity in S_N1 reaction:-

(a) $Ph–CH_2–Cl$

(b) 

(c) 

(d) 

(1) a > c > b > d

(2) c > d > b > a

(3) a > b > c > d

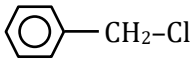
(4) b > a > c > d

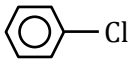
CHH007

4. Which of the following undergoes nucleophilic substitution by S_N1 mechanism at fastest rate :

(1) $CH_3–CH_2–Cl$

(2) 

(3) 

(4) 

CHH015

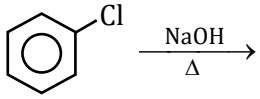
5. The oxidation of $CHCl_3$ by air, gives-

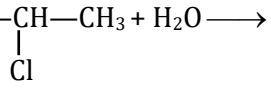
(1) HCOOH (2) CH_3OH

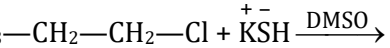
(3) $COCl_2$ (4) CCl_4

CHH008

6. Which is not two step reaction ?

(1) 

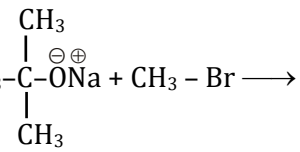
(2) 

(3) 

(4) All of the above

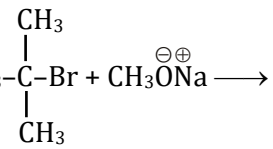
CHH135

7. Which reaction gives elimination as a major product

(1) 

(2) $CH_3–CH_2–Br + NaCN \xrightarrow{DMSO}$

(3) $CH_3–CH_2–Br + NaI \xrightarrow{Dryacetone}$

(4) 

CHH011

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8. When a haloalkane with β-hydrogen is heated with alcoholic solution of KOH the product and the type of mechanism is-

(1) Alcohol, S_N1

(2) Alkene, α-elimination

(3) Alcohol, S_N2

(4) Alkene, β-elimination

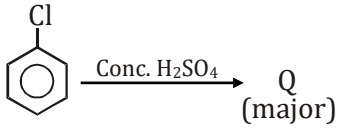
CHH019

9. Hydrolysis of optically active 2-bromobutane gives-

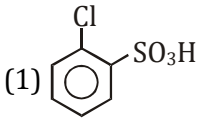
(1) (d)-butan-2-ol (2) (ℓ)-butan-2-ol

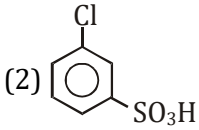
(3) (dℓ)-butan-2-ol (4) 2- butene

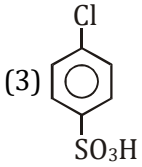
CHH018

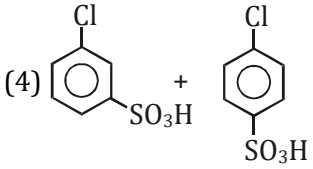
10. 

(major)

(1) 

(2) 

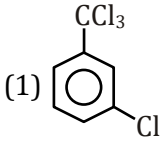
(3) 

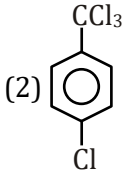
(4) 

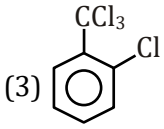
CHH022

11. $C_6H_5CCl_3 \xrightarrow[Fe]{Cl_2} X$

In the above reaction X is

(1) 

(2) 

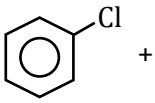
(3) 

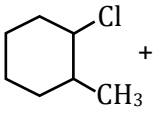
(4) None of these

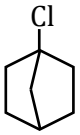
CHH016

12. Which gives S_N2 product as a major product ?

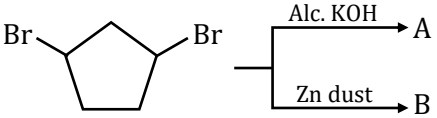
(1) $CH_3—CH_2—CH_2—Cl + NaI \xrightarrow[acetone]{dry}$

(2)  $+ H_2O \longrightarrow$

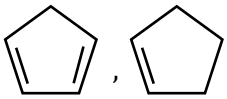
(3)  $+ H_2O \longrightarrow$

(4)  $+$ aq. KOH $\longrightarrow$

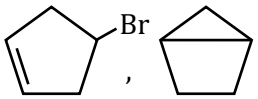
CHH133

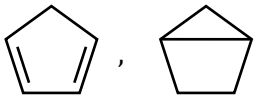
13. 

Product A and B respectively are

(1) 

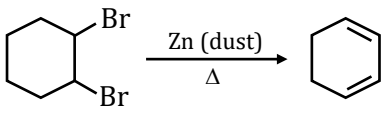
(2)  both A and B

(3) 

(4) 

CHH132

14. Which reaction product is wrong (major) product

(1) 

(2) $CH_3CH_2CH_2Br \xrightarrow[dry\ Acetone]{NaI} CH_3CH_2CH_2I$

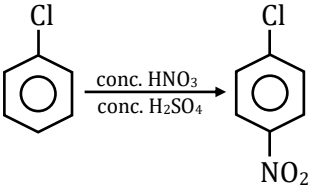
(3) $CH_3\underset{Br}{\underset{|}{C}}H\underset{Br}{\underset{|}{C}}HCH_3 \xrightarrow[\Delta]{Zn(dust)} CH_3CH=CHCH_3$

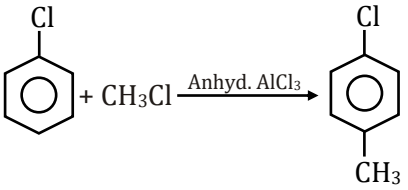
(4) $CH_3CH_2CHCl_2 \xrightarrow[(ii)\ H^+]{(i)\ NaNH_2(excess)} CH_3C\equiv CH$

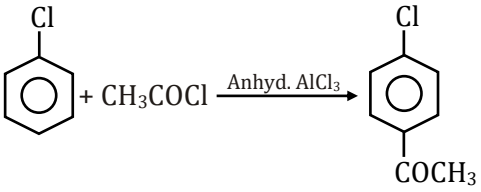
CHH014

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15. The correct reaction is -

(1) 

(2) 

(3) 

(4) All of these

CHH023

16. $CH_3—CH_2—CH=CH_2 + H—Br \xrightarrow[\Delta]{R_2O_2} A \xrightarrow[\Delta]{aq.\ KOH} B$?

(1) $CH_3—CH_2—\underset{\displaystyle OH}{\underset{|}{CH}}—CH_3$

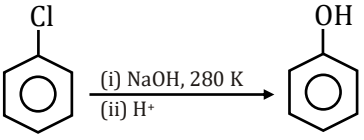
(2) $CH_3—CH_2—CH_2—CH_2—OH$

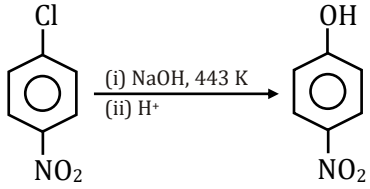
(3) $CH_3—CH_2—CH=CH_2$

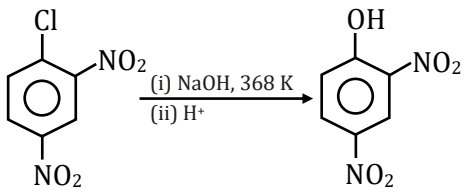
(4) $CH_3—\underset{\displaystyle CH_3}{\underset{|}{C}}=CH_2$

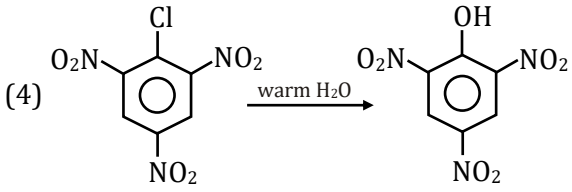
CHH131

17. The incorrect reaction is -

(1) 

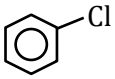
(2) 

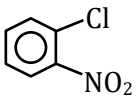
(3) 

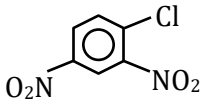
(4) 

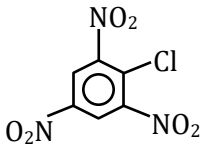
CHH021

18. Which of the following undergoes hydrolysis most easily

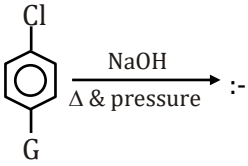
(1) 

(2) 

(3) 

(4) 

CHH009

19. 

Rate of reaction is maximum if G is :-

(1) $-OCH_3$ (2) $-CH_3$ (3) $-NO_2$ (4) $-H$

CHH017

20. $CH_3—CH_2—CH_2—Cl + CH_3MgBr \longrightarrow$?

(1) $CH_3—CH_2—CH_2—CH_3$

(2) $CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_3$

(3) $CH_3—CH_2—CH_2—CH_2—CH_2—CH_3$

(4) CH_4

CHH134

21. In RMgX, C–Mg bond is-

(1) Non polar covalent

(2) Polar covalent

(3) Ionic

(4) Coordinate

CHH020

22. When alkyl magnesium halide reacts with R'-NH_2, the product is

(1) R—R (2) R— H

(3) R_2NH (4) R—X

CHH012

23. Which of the following is used as insecticide

(1) D.D.T. (2) Chloretone

(3) $CHCl_3$ (4) All of them

CHH010

24. Incorrect match is-

(1) Iodoform - Antiseptic

(2) Pyrene - Fire extinguisher

(3) Freon 12 - aerosol propellants

(4) DDT - Fat insoluble

CHH025

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**1**	**4**	**3**	**3**	**3**	**4**	**4**	**3**	**3**	**1**	**1**	**4**	**1**	**4**
Question	16	17	18	19	20	21	22	23	24						
Answer	**3**	**1**	**4**	**3**	**1**	**2**	**2**	**1**	**4**						

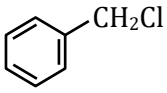
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EXERCISE-II (Previous Year Questions) AIPMT/NEET

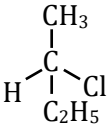
AIPMT 2014

1. Which of the following compounds will undergo racemisation when hydrolysed

(1) 

(2) $CH_3CH_2CH_2Cl$

(3) $H_3C-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2Cl$

(4) 

CHH037

2. Identify Z in the sequence of reactions:

$CH_3CH_2CH=CH_2 \xrightarrow{HBr/H_2O_2} Y \xrightarrow{C_2H_5ONa} Z$

(1) $CH_3-(CH_2)_3-O-CH_2CH_3$

(2) $(CH_3)_2CH_2-O-CH_2CH_3$

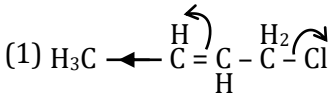
(3) $CH_3(CH_2)_4-O-CH_3$

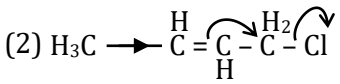
(4) $CH_3CH_2-CH(CH_3)-O-CH_2CH_3$

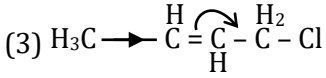
CHH038

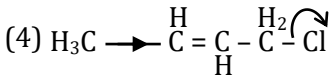
AIPMT 2015

3. Which of the following is the most **correct** electron displacement for a nucleophilic reaction to take place?

(1) 

(2) 

(3) 

(4) 

CHH039

4. In an S_N1 reaction on chiral centres, there is :

(1) 100% retention

(2) 100% inversion

(3) 100% racemization

(4) Inversion more than retention leading to partial racemization

CHH040

5. Which of the following reaction(s) can be used for the preparation of alkyl halides ?

(I) $CH_3CH_2OH + HCl \xrightarrow{anhy.ZnCl_2}$

(II) $CH_3CH_2OH + HCl \longrightarrow$

(III) $(CH_3)_3COH + HCl \longrightarrow$

(IV) $(CH_3)_2CHOH + HCl \xrightarrow{anhy.ZnCl_2}$

(1) (IV) only

(2) (III) and (IV) only

(3) (I), (III) and (IV) only

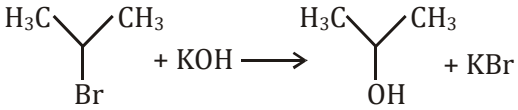
(4) (I) and (II) only

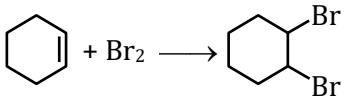
CHH041

NEET-II 2016

6. For the following reactions :-

(a) $CH_3CH_2CH_2Br + KOH \rightarrow CH_3CH=CH_2 + KBr + H_2O$

(b) 

(c) 

Which of the following statement is **correct** ?

(1) (a) and (b) are elimination reaction and (c) is addition reaction

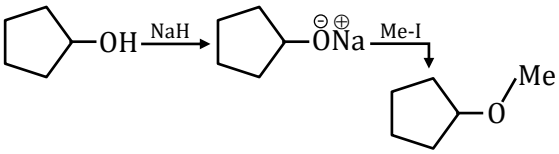
(2) (a) is elimination, (b) is substitution and (c) is addition reaction

(3) (a) is elimination, (b) and (c) are substitution reactions

(4) (a) is substitution, (b) and (c) are addition reaction

CHH042

7. The reaction



Can be classified as :-

(1) Williamson ether synthesis reaction

(2) Alcohol formation reaction

(3) Dehydration reaction

(4) Williamson alcohol synthesis reaction

CHH043

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8. Consider the reaction

$CH_3CH_2CH_2Br + NaCN \rightarrow CH_3CH_2CH_2CN + NaBr$

This reaction will be the fastest in

(1) N, N'-dimethylformamide (DMF)

(2) Water

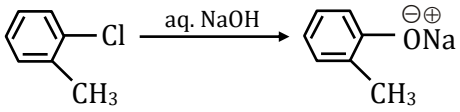
(3) Ethanol

(4) Methanol

CHH044

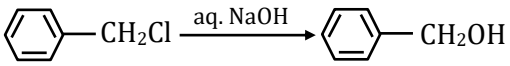
NEET(UG) 2019 (ODISHA)

9. The hydrolysis reaction that takes place at the slowest rate, among the following is :-

(1) 

(2) $H_3C-CH_2-Cl \xrightarrow{\text{aq. NaOH}} H_3C-CH_2-OH$

(3) $H_2C=CH-CH_2Cl \xrightarrow[\text{NaOH}]{\text{aq.}} H_2C=CH-CH_2OH$

(4) 

CHH045

NEET(UG) 2020

10. Elimination reaction of 2-Bromopentane to form pent-2-ene is:

(a) β-Elimination reaction

(b) Follow Zaitsev rule

(c) Dehydrohalogenation reaction

(d) Dehydration reaction

(1) (a), (b), (d) (2) (a), (b), (c)

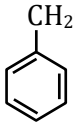
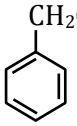
(3) (a), (c), (d) (4) (b), (c), (d)

CHH046

NEET(UG) 2020 (Covid-19)

11. Which of the following will **NOT** undergo S_N1 reaction with $\bar{O}H$?

(1) $CH_2 = CH - CH_2Cl$ (2) $(CH_3)_3CCl$

(3)  (4) 

CHH047

NEET(UG) 2021

12. The correct sequence of bond enthalpy of 'C-X' bond is

(1) $CH_3-F < CH_3-Cl < CH_3-Br < CH_3-I$

(2) $CH_3-F > CH_3-Cl > CH_3-Br > CH_3-I$

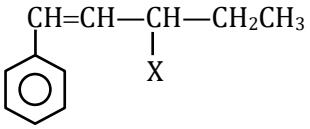
(3) $CH_3-F < CH_3-Cl > CH_3-Br > CH_3-I$

(4) $CH_3-Cl > CH_3-F > CH_3-Br > CH_3-I$

CHH048

NEET(UG) 2023

13. The given compound



is an example of ____.

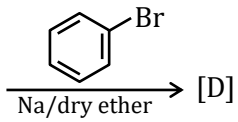
(1) aryl halide (2) allylic halide

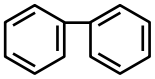
(3) vinylic halide (4) benzylic halide

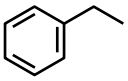
CHH136

14. Identify the final product [D] obtained in the following sequence of reactions.

$CH_3CHO \xrightarrow[(ii)\ H_3O^+]{(i)\ LiAlH_4} [A] \xrightarrow[\Delta]{H_2SO_4} [B] \xrightarrow{HBr} [C]$



(1)  (2) C_4H_{10}

(3) $HC \equiv C^-Na^+$ (4) 

CHH137

15. Which amongst the following reactions of alkyl halides produces isonitrile as a major product ?

(A) $R - X + HCN \rightarrow$

(B) $R - X + AgCN \rightarrow$

(C) $R - X + KCN \rightarrow$

(D) $R - X + NaCN \xrightarrow[C_2H_5OH]{H_2O}$

Choose the **most appropriate** answer from the options given below :

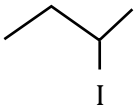
(1) (D) only (2) (C) and (D) only

(3) (B) only (4) (A) and (B) only

CHH138

16. The correct order for the rate of α, β-dehydrohalogenation for the following compounds is ___.

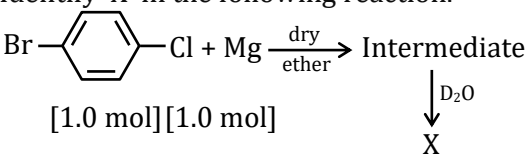
(i)  (ii) 

(iii) 

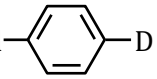
(1) (i) < (ii) < (iii)
(2) (ii) < (i) < (iii)
(3) (iii) < (ii) < (i)
(4) (ii) < (iii) < (i)

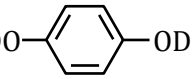
CHH139

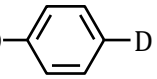
17. Identify 'X' in the following reaction.

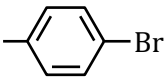


[1.0 mol] [1.0 mol]

(1) 

(2) 

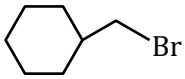
(3) 

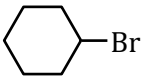
(4) 

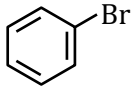
CHH140

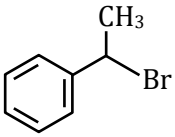
NEET(UG) 2024

18. The compound that will undergo S_N1 reaction with the fastest rate is :

(1) 

(2) 

(3) 

(4) 

CHH141

19. The products A and B obtained in the following reactions, respectively, are

$3ROH + PCl_3 \rightarrow 3RCl + A$

$ROH + PCl_5 \rightarrow RCl + HCl + B$

(1) $POCl_3$ and H_3PO_3 (2) $POCl_3$ and H_3PO_4
(3) H_3PO_4 and $POCl_3$ (4) H_3PO_3 and $POCl_3$

CHH142

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**1**	**2**	**4**	**3**	**2**	**1**	**1**	**1**	**2**	**3**	**2**	**2**	**4**	**3**
Question	16	17	18	19											
Answer	**4**	**1**	**4**	**4**											

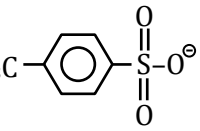
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EXERCISE-III (Analytical Questions) Master Your Understanding

1. Consider the following statements-
(A) In allylic halides halogen atom is bonded to an sp^3 carbon adjacent to C=C
(B) In benzylic halides halogen atom is bonded to sp^3 carbon attached to an aromatic ring
(C) In vinylic halides halogen atom is bonded to a sp^2 carbon of a C=C
(D) In aryl halides halogen atom is directly bonded to a sp^2 carbon of an aromatic ring
Which statement(s) is/are correct
(1) Only A, B, C (2) Only A, C
(3) Only A, C, D (4) All

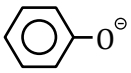
CHH057

2. Which of the following has maximum nucleophilicity :-

(1) 

(2) $CH_3-O^{\ominus}$

(3) $CF_3-\underset{\underset{O}{\|}}{C}-O^{\ominus}$

(4) 

CHH100

3. Which halide ion is the best nucleophile in DMF -
(1) $F^{\ominus}$ (2) $Cl^{\ominus}$
(3) $Br^{\ominus}$ (4) $I^{\ominus}$

CHH062

4. The product in the following reaction is
$Ph—Cl + Fe / Br_2 \longrightarrow$ Product
(1) o–bromo chloro benzene
(2) p–bromo chloro benzene
(3) Both the above
(4) 2,4,6–tribromo chloro benzene

CHH052

5. Isobutyl magnesium bromide with dry ether and absolute alcohol gives
(1) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2OH$ and CH_3CH_2MgBr
(2) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_2-CH_3$ and Mg (OH) Br
(3) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_3$, $CH_2=CH_2$ and Mg(OH)Br
(4) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_3$ and CH_3CH_2OMgBr

CHH053

6. Identify 'Z' in the following reaction series,
$CH_3.CH_2CH_2Br \xrightarrow{aq.NaOH}$ (X) $\xrightarrow[Heat]{Al_2O_3}$ (Y) $\xrightarrow{Cl_2/H_2O}$ (Z)
(1) Mixture of $CH_3-\underset{\underset{Cl}{|}}{CH}-\underset{\underset{Cl}{|}}{CH_2}$ and $CH_3-\underset{\underset{OH}{|}}{CH}-\underset{\underset{OH}{|}}{CH_2}$
(2) $CH_3-\underset{\underset{OH}{|}}{CH}-\underset{\underset{Cl}{|}}{CH_2}$
(3) $CH_3-\underset{\underset{Cl}{|}}{CH}-\underset{\underset{OH}{|}}{CH_2}$
(4) $CH_3-\underset{\underset{Cl}{|}}{CH}-\underset{\underset{Cl}{|}}{CH_2}$

CHH054

7. Which alkane gives 1-chloro-2,2-dimethyl propane on monochlorination-
(1) Isopentane
(2) Isobutane
(3) Neopentane
(4) 2,3-dimethyl butane

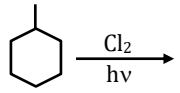
CHH058

8. Correct order of rate of photo bromination for following compounds-
(I) CH_3-CH_3 (II) CD_3-CD_3
(III) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-H$
(1) II < I < III (2) I < II < III
(3) III < I < II (4) II < III < I

CHH060

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9.  Monochlorination product

Number of possible monochloro derivatives excluding stereo isomers is/are -

(1) 4 (2) 5 (3) 3 (4) 6

CHH061

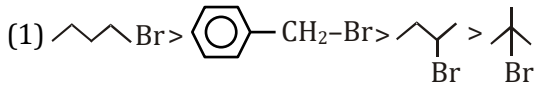
10. In S_N1 the first step involves the formation of

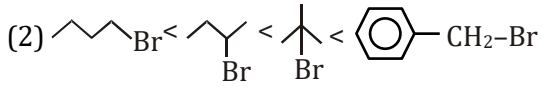
(1) Free radical (2) Carbanion

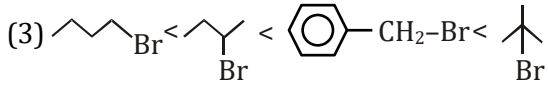
(3) Carbocation (4) Final product

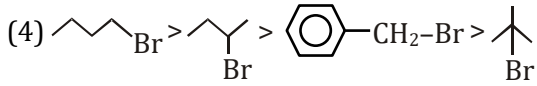
CHH072

11. The correct reactivity order for S_N1 reactions is-

(1) 

(2) 

(3) 

(4) 

CHH068

12. S_N1 reaction on an optically active substrate having only one chiral centre which is also reaction centre, gives :-

(1) Retention in configuration

(2) Inversion in configuration

(3) Partially racemised product

(4) Complete racemised product

CHH083

13. Compound is most reactive towards NaOH.

(1) CH_3Cl (2) $CH_2=CHCl$

(3) C_6H_5Cl (4) $C_6H_5CH_2Cl$

CHH081

14. Most stable carbocation formed from $(CH_3)_3C-Br$, $(C_6H_5)_3CBr$, $(C_6H_5)_2CHBr$ and $C_6H_5CH_2Br$ would be

(1) $C_6H_5\overset{\oplus}{C}H_2$ (2) $(CH_3)_3\overset{\oplus}{C}$

(3) $(C_6H_5)_3\overset{\oplus}{C}$ (4) $(C_6H_5)_2\overset{\oplus}{C}H$

CHH082

15. The hydrolysis of alkyl halides by aqueous NaOH is best termed as

(1) Electrophilic substitution reaction

(2) Electrophilic addition reaction

(3) Nucleophilic addition reaction

(4) Nucleophilic substitution reaction

CHH084

16. Which of the following S_N2 reaction is the slowest-

(1) $CH_3-CH_2-CH_2-Br \xrightarrow{\overset{\ominus}{O}H} CH_3-CH_2-CH_2-OH + Br^-$

(2) $CH_3-CH_2-CH_2-Cl \xrightarrow{\overset{\ominus}{O}H} CH_3-CH_2-CH_2-OH + Cl^-$

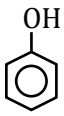
(3) $CH_3-CH_2-CH_2-F \xrightarrow{\overset{\ominus}{O}H} CH_3-CH_2-CH_2-OH + F^-$

(4) $CH_3-\underset{F}{\underset{|}{CH}}-CH_3 \xrightarrow{\overset{\ominus}{O}H} CH_3-\underset{OH}{\underset{|}{CH}}-CH_3 + F^-$

CHH063

17. Rate of S_N2 reaction will be maximum in which of the following solvent -

(1) H_2O (2) CH_3OH

(3)  (4) DMSO

CHH064

18. Which of the following is most reactive for S_N2 reactions.

(1) $(C_6H_5)_2C(CH_3)Br$ (2) $C_6H_5CH(CH_3)Br$

(3) $(C_6H_5)_3CBr$ (4) $C_6H_5CH_2Br$

CHH069

19. The reaction, $CH_3Br + OH^- \longrightarrow CH_3OH + Br^-$ obeys the mechanism

(1) S_N1 (2) S_N2 (3) E_1 (4) E_2

CHH077

20. Butanenitrile may be prepared by heating

(1) Propyl alcohol with KCN

(2) Butyl alcohol with KCN

(3) Butyl chloride with KCN

(4) Propyl chloride with KCN

CHH078

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21. The given reaction is an example of

$C_2H_5Br + KCN(aq.) \longrightarrow C_2H_5CN + KBr$

(1) Elimination
(2) Nucleophilic substitution
(3) Electrophilic substitution
(4) Redox change

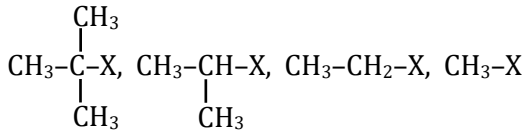
CHH079

22. The least reactive chlorine is present in

(1) Methyl chloride (2) Allyl chloride
(3) Ethyl chloride (4) Vinyl chloride

CHH086

23. Arrange the following compounds in the increasing order of their S_N2 reactivity?



(a) (b) (c) (d)

(1) (a) < (b) < (c) < (d)
(2) (a) < (c) < (d) < (b)
(3) (d) < (c) < (b) < (a)
(4) (b) < (d) < (c) < (a)

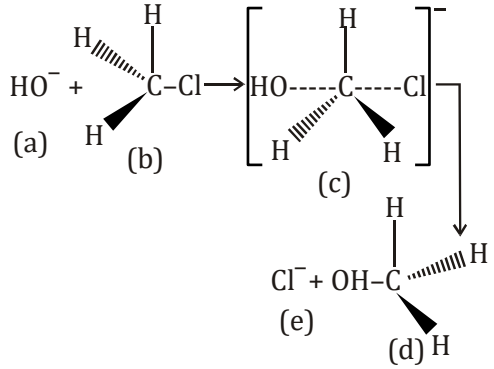
CHH087

24. The Williamson synthesis involves :-

(1) A nucleophilic addition
(2) An electrophilic substitution
(3) S_N2 displacement
(4) S_N1 displacement

CHH090

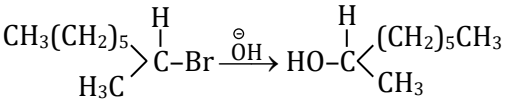
25. Which of the following statement is correct for S_N^2 mechanism



(1) In (c) carbon atom is sp^3 hybridized
(2) In (c) carbon atom is sp^2 hybridised
(3) (a) and (e) are electrophiles
(4) (c) is more stable then (d)

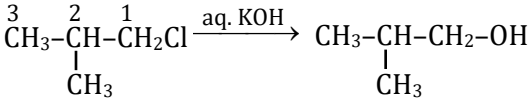
CHH102

26. Following reaction is



(1) E_1 (2) S_N1 (3) E_2 (4) S_N^2

CHH103

27. 

In the above reaction the attack of a nucleophile would be from which side

(1) On the front side of Cl of C_1 carbon
(2) On C_2 carbon
(3) On the rear side of Cl of C_1 carbon
(4) On C_3 carbon

CHH104

28. Which of the following ether can not be prepared by Williamson's method :-

(1) Ditert.butyl ether
(2) Methyltert.butyl ether
(3) Anisole
(4) Both (1) and (2)

CHH105

29. With conc. HBr ethyl phenyl ether yields :–

(1) Phenol and ethyl bromide.
(2) Bromobenzene and ethanol
(3) Phenol and ethane
(4) Bromobenzene and ethane

CHH092

30. Reaction of sodium ethoxide and ethyl iodide will give :-

(1) Ether (2) Ethyl alcohol
(3) Acetaldehyde (4) Acetic acid

CHH085

31. The reaction of ethyl iodide with sodium ethoxide is

(1) An electrophilic substitution reaction
(2) A nucleophilic addition reaction
(3) A nucleophilic substitution reaction
(4) A free radical substitution reaction

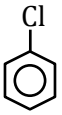
CHH089

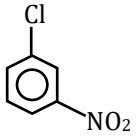
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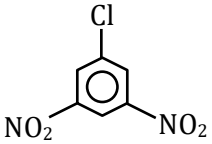
32. In the Williamson synthesis of ethers given by the general equation –
R–X + R'ONa ⟶ R–O–R' the yield from R–X follows the sequence :-
(1) $CH_3X > 1° > 2° > 3°$
(2) $CH_3X < 1° < 2° < 3°$
(3) $CH_3X < 1° < 2° > 3°$
(4) $CH_3X > 1° < 2° < 3°$

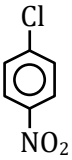
CHH091

33. Arrange the following compound in increasing order of reactivity towards aromatic nucleophilic substitution reaction.

(a) 

(b) 

(c) 

(d) 

(1) c > d > b > a (2) c > b > a > d
(3) d > c > b > a (4) a > d > b > c

CHH055

34. To form alkane isonitrile, alkyl halide is reacted with:
(1) KCN (2) AgCN
(3) NaCN (4) NH_4CN

CHH073

35. Alkyl fluorides are synthesised by
(1) Finkelstein reaction
(2) Swart reaction
(3) Kolbe reaction
(4) Wurtz reaction

CHH074

36. The products of reaction of alcoholic silver nitrite with ethyl bromide are
(1) Ethane (2) Ethene
(3) Ethyl alcohol (4) Nitro ethane

CHH075

37. The unimolecular elimination involves formation of
(1) A free radical (2) A carbanion
(3) A carbocation (4) A biradical

CHH094

38. Major product of the reaction :-
$(CH_3)_3C–Cl + C_2H_5ONa$ ⟶ would be :-
(1) $(CH_3)_2C–OC_2H_5$
(2) $(CH_3)_3C–C_2H_5$
(3) $(CH_3)_2C=CH_2$
(4) $CH_3–CH=CH–C_2H_5$

CHH106

39. In following two columns :

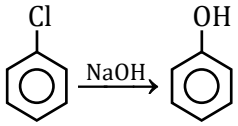
Column-I

(a) $CH_3–CH_2–Br + HS^- \rightarrow CH_3–CH_2–SH + Br^-$

(b) $(CH_3)_2C=CH_2 + HOCl \rightarrow (CH_3)_2\overset{OH}{\overset{|}{C}}–CH_2–Cl$

(c) $CH_3–CH_2–Br + {}^-OH \rightarrow CH_2=CH_2+H_2O + Br^-$

(d) $(CH_3)_3C–CH_2–OH + H–Br \rightarrow (CH_3)_2CBrCH_2–CH_3 + H_2O$

(e) 

Column-II
(1) Nucleophilic substitution
(2) Elimination
(3) Electrophilic substitution
(4) Electrophilic addition
(5) Rearrangement reaction
(1) a, d, e → 1, b → 4, c → 2
(2) a, d, e → 1, b → 4, c → 3
(3) a, e → 1, b → 4, c → 2, d → 5
(4) a, c → 1, b → 4, e → 3, d → 5

CHH107

40. Match the column :-

Column-I (Conversion)		**Column-II** (Reagent)	
(a)	1–Butene→ 2–Butanol	(p)	BH_3/THF, CH_3COOH
(b)	1–Butene → 1–Butanol	(q)	HBr, aq KOH
(c)	Propene → Propane	(r)	Alc.KOH, HBr/R_2O_2
(d)	2–Bromopropane → 1–Bromo propane	(s)	HBr/H_2O_2, aq KOH

(1) a – q, b – p, c – s, d – r
(2) a – p, b – s, c – q, d – r
(3) a – q, b – r, c – s, d – p
(4) a – q, b – s, c – p, d – r

CHH108

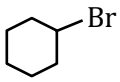
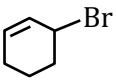
41. The reactivity of alkyl halides in E_2 elimination reactions follows the order
(1) R–I < R–Br < R–Cl < R–F
(2) R–F < R–Cl < R–Br < R–I
(3) R–I > R–Cl > R–Br < R–F
(4) R–I < R–Br < R–F < R–Cl

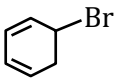
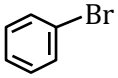
CHH093

42. Which of the following alkyl halides gives a mixture of alkenes on dehydrohalogenation
(1) n–Propyl halide (2) Isopropyl halide
(3) s–Butyl bromide (4) t–Butyl bromide

CHH095

43. Arrange the following in order of ease of dehydrohalogenation:

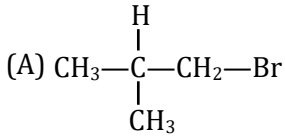
(i)  (ii) 

(iii)  (iv) 

(1) (iii) > (iv) > (ii) > (i)
(2) (iii) > (ii) > (i) > (iv)
(3) (ii) > (iii) > (i) > (iv)
(4) (i) > (ii) > (iii) > (iv)

CHH051

44. Arrange the following alkyl halides in decreasing order of the rate of elimination reaction with alcoholic KOH.

(A) 

(B) $CH_3—CH_2—Br$
(C) $CH_3–CH_2–CH_2–Br$
(1) A > B > C (2) C > B > A
(3) B > C > A (4) A > C > B

CHH098

45. Alkaline hydrolysis of 2,2-Dichloropropane gives
(1) Acetone (2) 2,2-Propanediol
(3) Propane (4) Propan-2-ol

CHH101

46. Which of the following has highest dipole moment:
(1) CH_3Cl (2) CH_3F (3) CH_3Br (4) CH_3I

CHH049

47. Chloroform when treated with benzene in presence of anhydrous $AlCl_3$, the product formed is
(1) Chlorobenzene
(2) Toulene
(3) Mixture of ortho and para chlorotoluene
(4) Triphenyl methane

CHH056

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**2**	**1**	**3**	**4**	**2**	**3**	**1**	**2**	**3**	**2**	**3**	**4**	**3**	**4**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**4**	**4**	**4**	**2**	**4**	**2**	**4**	**1**	**3**	**2**	**4**	**3**	**1**	**1**	**1**
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	**3**	**1**	**3**	**2**	**2**	**4**	**3**	**3**	**1**	**4**	**2**	**3**	**2**	**4**	**1**
Question	46	47													
Answer	**1**	**4**													

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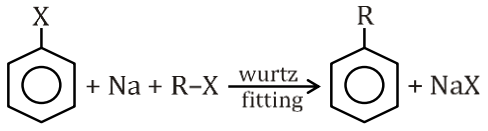
Haloalkanes and Haloarenes

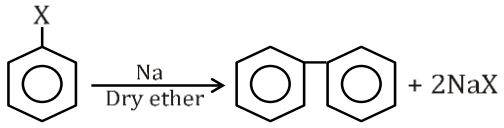
SOLUTIONS

Exercise-I (Conceptual Questions)

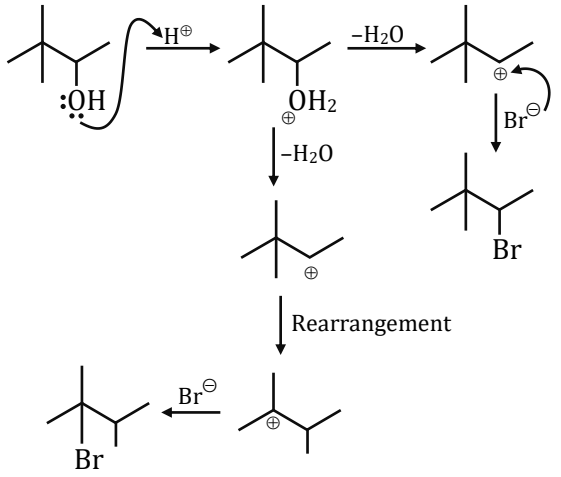
1. Ans. (4)

$$R{-}X \xrightarrow[\text{Dry ether}]{Na} R{-}R + 2NaX$$





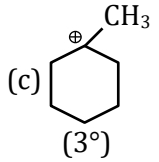
2. Ans. (1)



3. Ans. (4)

(a) $Ph{-}\overset{\oplus}{C}H_2$ Resonance

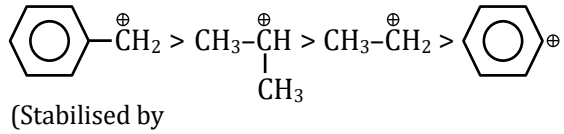
(b)  Aromatic

(c)  (3°)

(d) $CH_3{-}\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}{-}\overset{\oplus}{C}H_2$ (1°)

b > a > c > d

4. Ans. (3)



(Stabilised by Resonance)

5. Ans. (3)

Chloroform is slowly oxidised by air is the presence of light to an extremely poisonous gas, carbonyl chloride also known as phosgene.

6. Ans. (3)

$$CH_3{-}CH_2{-}CH_2{-}Cl + \overset{\oplus}{K}\overset{\ominus}{SH} \xrightarrow{DMSO} CH_3{-}CH_2{-}CH_2{-}SH + KCl$$

This reaction follow S_N2 mechanism, involve one step reaction.

7. Ans. (4)

$$CH_3{-}\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}{-}\overset{\ominus}{O}\overset{\oplus}{Na} + CH_3{-}Br \longrightarrow CH_3{-}\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}{-}O{-}CH_3$$

1° Alkyl halide (S_N2)

$$CH_3{-}CH_2{-}Br + NaCN \xrightarrow{DMSO} CH_3{-}CH_2{-}CN \quad (S_N2)$$

$$CH_3{-}CH_2{-}Br + NaI \xrightarrow{\text{Dry acetone}} CH_3{-}CH_2{-}I \quad (S_N2)$$

$$CH_3{-}\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}{-}Br + CH_3{-}O{-}Na \longrightarrow CH_3{-}\underset{CH_3}{\overset{CH_2}{\underset{|}{\overset{\|}{C}}}}$$

3° Alkyl halide (E_2) Elimination

8. Ans. (4)

$$R{-}\overset{\beta}{C}H_2{-}\overset{\alpha}{C}H_2{-}X \xrightarrow{\text{Alc. KOH}} R{-}CH{=}CH_2$$

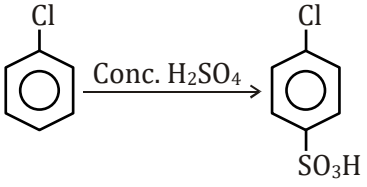
(Alkene)
[β-elimination]

9. Ans. (3)

$$CH_3{-}\overset{*}{C}H(Br){-}CH_2{-}CH_3 \xrightarrow{H_2O} CH_3{-}\overset{*}{C}H(OH){-}CH_2{-}CH_3$$

(2-Bromobutane) (Butan-2-ol)

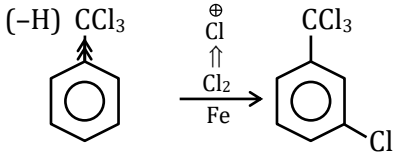
10. Ans. (3)



$\{E^{\oplus} \Rightarrow SO_3\}$
+M of Cl ∴ ortho/para directing
–I of Cl ∴ deactivating in nature

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11. Ans. (1)

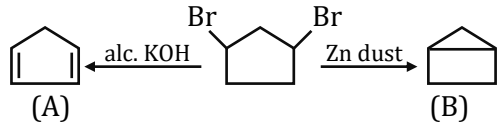


12. Ans. (1)

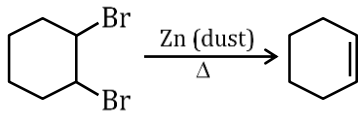
$CH_3—CH_2—CH_2—Cl + NaI \xrightarrow[\text{acetone}]{\text{dry}} CH_3—CH_2—CH_2—I$

Finkelstein reaction is a type of S_N2 reaction that involves the exchange of halide ion.

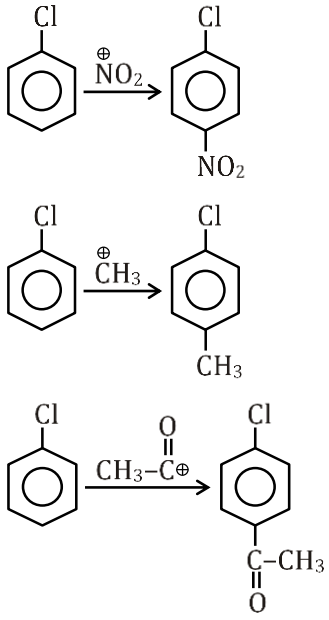
13. Ans. (4)



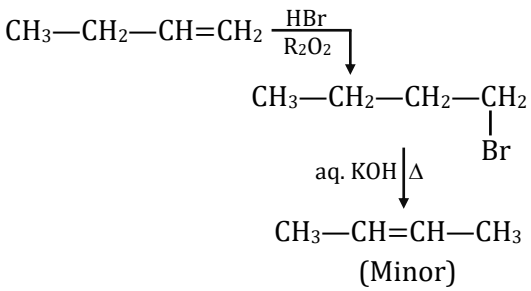
14. Ans. (1)



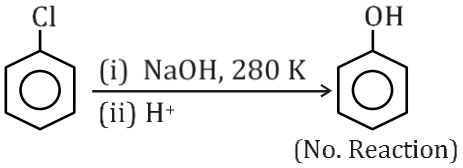
15. Ans. (4)



16. Ans. (3)

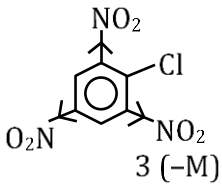


17. Ans. (1)



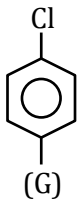
$\therefore$ High temperature required

18. Ans. (4)



$SnAr \propto \frac{-M}{+M} \propto \frac{-H}{+H} \propto \frac{-I}{+I}$

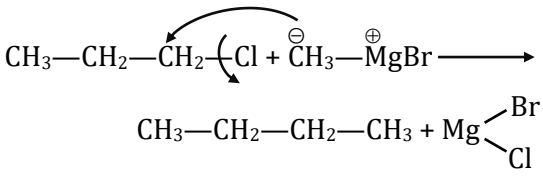
19. Ans. (3)



$SnAr \propto \frac{-M/-H/-I}{+M/+H/+I}$

Here ; G must be electron withdrawing group i.e. $-NO_2$

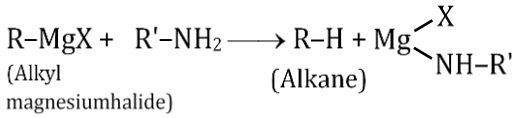
20. Ans. (1)



21. Ans. (2)

C–Mg bond is polar covalent.

22. Ans. (2)



It is a reaction of active-H.

23. Ans. (1)

DDT is used as Insecticide

24. Ans. (4)

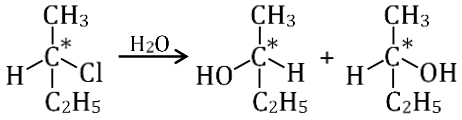
D.D.T. $\Rightarrow$ fat soluble in nature

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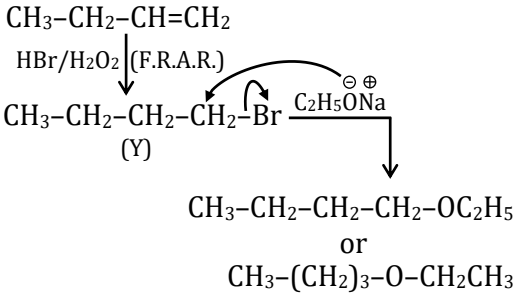
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Exercise-II (Previous Year Questions)

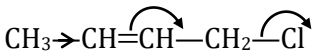
1. Ans. (4)



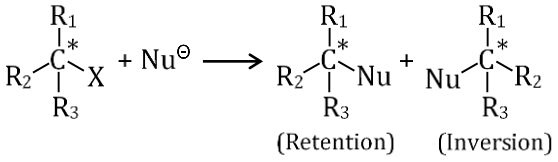
2. Ans. (1)



3. Ans. (2)

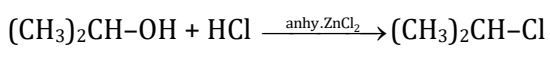


4. Ans. (4)

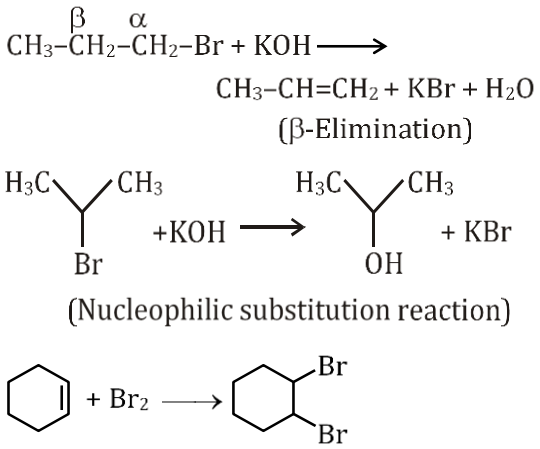


Both inversion and retention take place. But, inversion is more than retention leading to partial racemization.

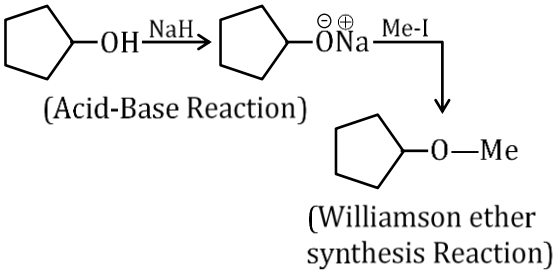
5. Ans. (3)



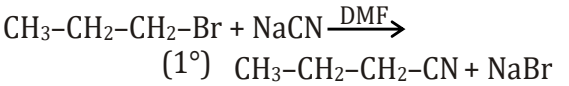
6. Ans. (2)



7. Ans. (1)

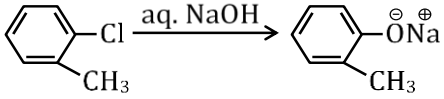


8. Ans. (1)



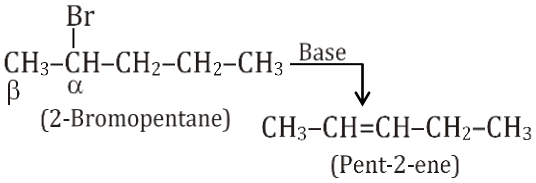
Path is S_N2. Hence, this reaction will be fastest in D.M.F.

9. Ans. (1)



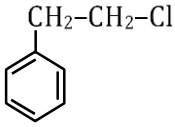
Because aryl halides are less reactive than alkyl halides.

10. Ans. (2)



It is β-elimination.
It follows Zaitsev (saytzeff's) rule.
It is Dehydrohalogenation reaction.

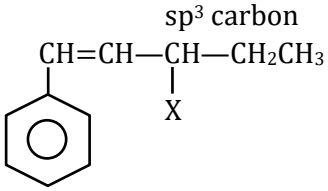
11. Ans. (3)

Amongst the given options  is 1° alkyl halide where S_N1 does not take place.

12. Ans. (2)

$CH_3–F > CH_3–Cl > CH_3–Br > CH_3–I$
[C–X bond enthalpy]
The size of halogen atom increases from F to I i.e. F < Cl < Br < I hence; bond length from C–F to C–I increases. That's why bond enthalpy from CH_3–F to CH_3–I decreases.

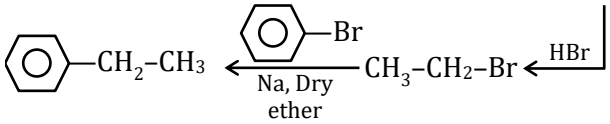
13. Ans. (2)



Allylic halide → Halogen attached to sp^3 carbon which is next to C=C

14. Ans. (4)

$CH_3-CH=O \xrightarrow[H_3O^+]{LiAlH_4} CH_3-CH_2-OH \xrightarrow[\Delta]{H^+} CH_2=CH_2$



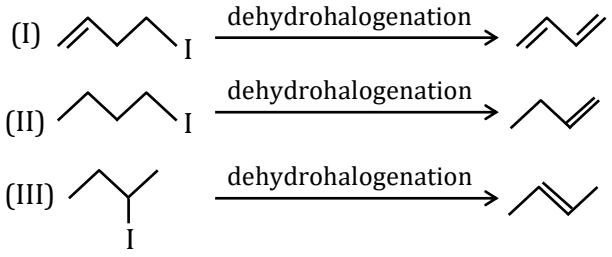
(Wurtz – Fitting reaction)

15. Ans. (3)

$R-X + Ag-C\equiv N \rightarrow R-NC$

Isonitrile

16. Ans. (4)



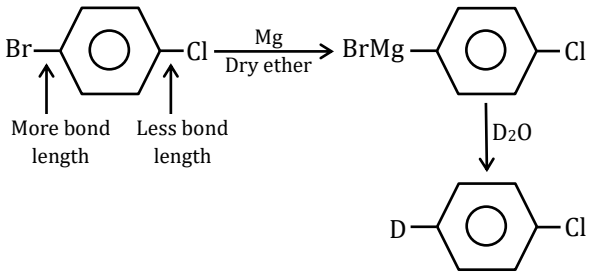
Rate of Dehydrohalogenation :

II < III < I

↓ ↓ ↓

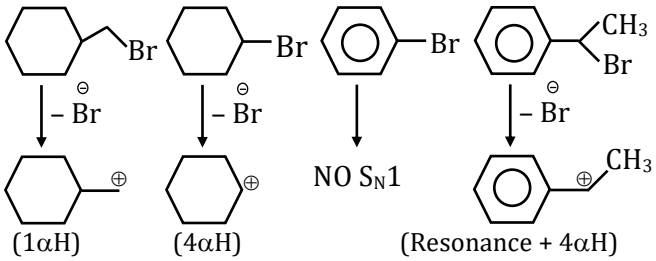
2αH 6αH Resonance

17. Ans. (1)



18. Ans. (4)

Reactivity towards S_N1 ∝ stability of carbocation



19. Ans. (4)

$3R-OH + PCl_3 \rightarrow 3RCl + H_3PO_3$

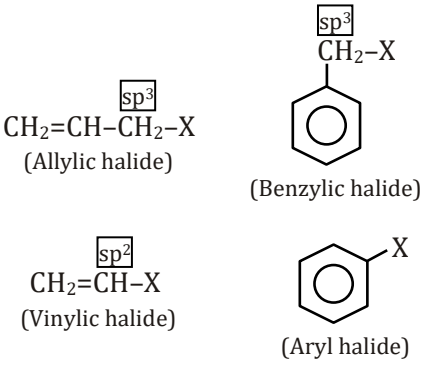
$R-OH + PCl_5 \rightarrow R-Cl + HCl + POCl_3$

$A \rightarrow H_3PO_3$

$B \rightarrow POCl_3$

Exercise-III (Analytical Questions)

1. Ans. (4)



2. Ans. (2)

Nucleophilicity ⇒ $CH_3-O^{\ominus}$

If e^- donor is same atom then

$$\left\{N.S. \propto \frac{1}{\text{stablity of anion}}\right\}$$

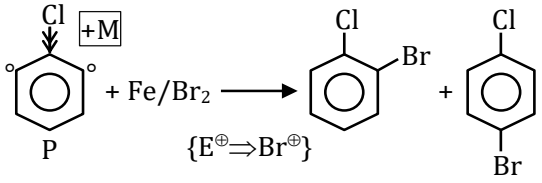
3. Ans. (1)

$F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$

Polar aprotic solvent DMF is given

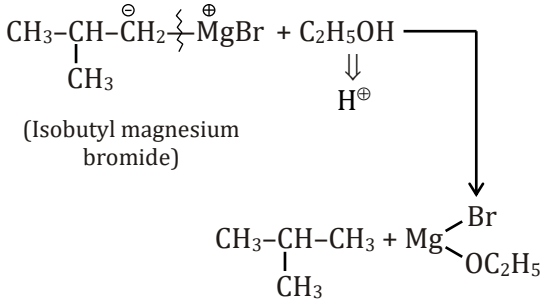
$$\left[\text{Nucleophilicity} \propto \frac{1}{\text{size}}\right]$$

4. Ans. (3)

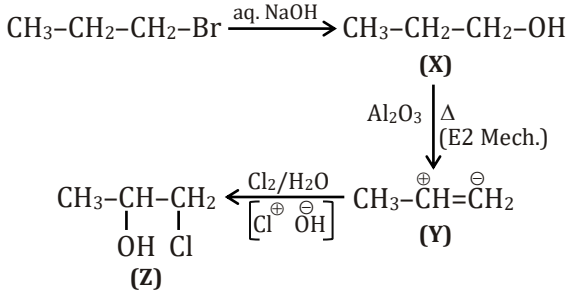


5. Ans. (4)

Reaction of active –H



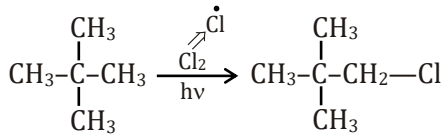
6. Ans. (2)



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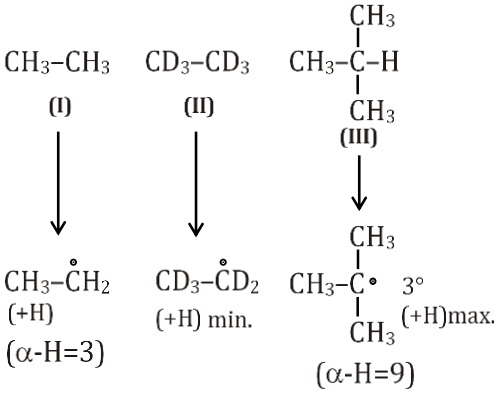
7. Ans. (3)



(Neopentane) (1-Chloro-2,2-dimethyl propane)

8. Ans. (1)

{Rate of photo bromination (i.e. towards FRSR) ∝ stability of formed free radical}

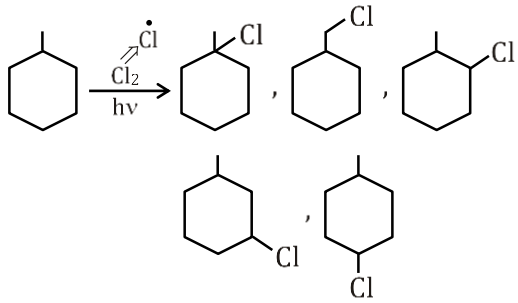


(+H) (α-H=3) (+H) min. 3° (+H)max. (α-H=9)

III > (I, II)

III > I > II

9. Ans. (2)



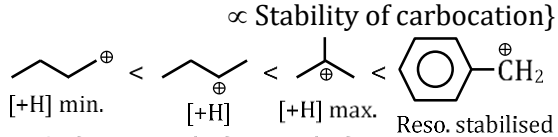
(5 Possible product)

10. Ans. (3)

In S_N1, the first step involves the formation of carbocation.

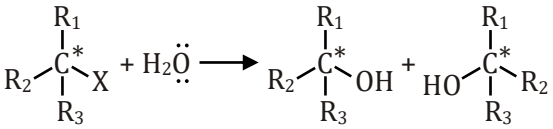
11. Ans. (2)

{Reactivity towards S_N1 reaction ∝ Stability of carbocation}



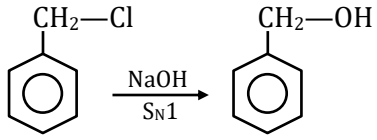
[+H] min. (1°) [+H] (2°) [+H] max. (3°) Reso. stabilised

12. Ans. (3)



(Partially Racemised Product)

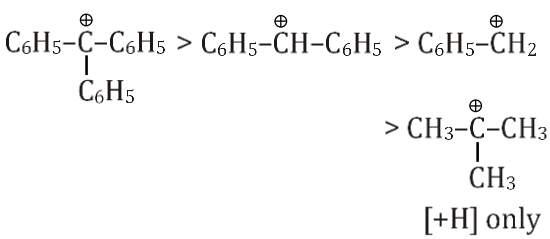
13. Ans. (4)



(Benzyl Chloride)

14. Ans. (3)

$$\left[\text{Stability of carbocation} \propto \frac{+M}{-M} \propto \frac{+H}{-H} \propto \frac{+I}{-I}\right]$$



[+H] only

15. Ans. (4)



(Alkyl halide)

16. Ans. (4)

F is weaker base

For 2° alky halide, steric hindrance is more than 1° alkyl halide

∴ slowest S_N2 reaction.

17. Ans. (4)

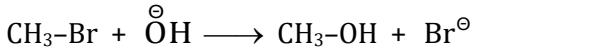
S_N2 reaction requires polar aprotic solvent (PAS) i.e. DMSO

18. Ans. (4)

$$\left\{\text{Rectivity towards } S_{N^2} \propto \frac{1}{\text{Steric Hindrance}}\right\}$$

$C_6H_5CH_2–Br$ is 1° alkyl halide.

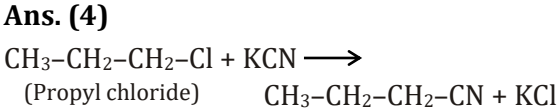
19. Ans. (2)



1° Alkyl Halide

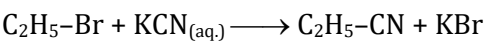
It follows S_N2 mechanism.

20. Ans. (4)

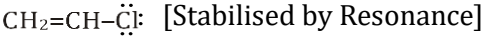


(Propyl chloride) (Butanenitrile)

21. Ans. (2)



22. Ans. (4)



[Stabilised by Resonance]

(Vinyl chloride)

23. Ans. (1)

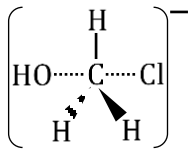
$$CH_3{-}X > CH_3{-}CH_2{-}X > CH_3{-}\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{CH}}{-}X > CH_3{-}\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{C}}{-}X$$

24. Ans. (3)

$$\underset{(1°)}{R{-}X} + R{-}\overset{\ominus}{O}\overset{\oplus}{Na} \longrightarrow R{-}O{-}R \quad [S_N2]$$

(Williamson synthesis)

25. Ans. (2)



Transition state

It is sp^2 hybrid.

26. Ans. (4)

$$CH_3(CH_2)_5\underset{\displaystyle H_3C}{\overset{\displaystyle H}{C}}{-}Br \xrightarrow{\overset{\ominus}{O}H} HO{-}\underset{\displaystyle CH_3}{\overset{\displaystyle H}{C}}{-}(CH_2)_5CH_3$$

(2°)

$\overset{\ominus}{O}H$ is strong nucleophile and it is 2° alkyl halide hence inversion taking place, so it follows S_N2 path.

27. Ans. (3)

$$CH_3{-}\underset{\displaystyle CH_3}{CH}{-}CH_2{-}Cl \xrightarrow[[S_N2]]{\text{aq. KOH}} CH_3{-}\underset{\displaystyle CH_3}{CH}{-}CH_2{-}OH$$

(1°)

On the rear side of Cl of C_1 carbon.

28. Ans. (1)

$$CH_3{-}\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{C}}{-}O{-}\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{C}}{-}CH_3$$

(Di tert. butyl ether)

bulky group

$\therefore$ more steric hindrance

29. Ans. (1)

$$C_2H_5\{O{-}Ph \xrightarrow[[S_N2]]{\overset{\oplus}{H}\overset{\ominus}{Br}} Ph{-}OH + C_2H_5{-}Br$$

30. Ans. (1)

$$\underset{\text{(Sodium ethoxide)}}{C_2H_5{-}ONa} + C_2H_5{-}I \longrightarrow \underset{\text{(Ether)}}{C_2H_5{-}O{-}C_2H_5} + NaI$$

31. Ans. (3)

$$\underset{\text{(ethyl iodide)}}{C_2H_5{-}I} + \underset{\text{(Sodium ethoxide)}}{C_2H_5ONa} \longrightarrow C_2H_5{-}O{-}C_2H_5 + NaI$$

It is a Nucleophile substitution Reaction.

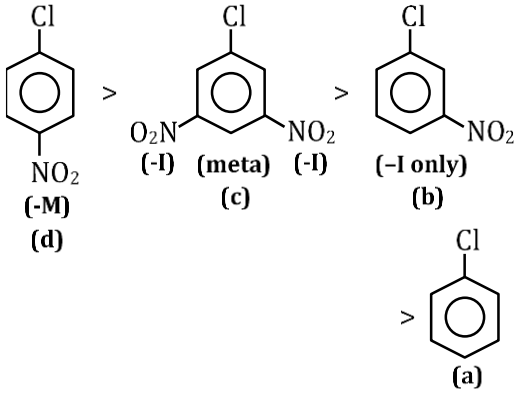
32. Ans. (1)

$R{-}X + R'{-}ONa \longrightarrow R{-}O{-}R'$

$\{R{-}X \Rightarrow CH_3{-}X > 1° > 2° > 3°\}$

33. Ans. (3)

$$\left\{\text{Reactivity towards SnAr} \propto \frac{-M}{+M} \propto \frac{-H}{+H} \propto \frac{-I}{+I}\right\}$$



34. Ans. (2)

$R{-}X + AgCN \longrightarrow R{-}NC + AgX$

35. Ans. (2)

$R{-}Cl + Hg_2F_2 \longrightarrow R{-}F + Hg_2Cl_2$

[Swartz reaction]

36. Ans. (4)

$C_2H_5{-}Br + AgNO_2 \longrightarrow C_2H_5{-}NO_2 + AgBr\downarrow$

37. Ans. (3)

E_1 elimination involves formation of carbocation.

$$CH_3{-}CH_2{-}OH \xrightarrow[\Delta]{\overset{\oplus}{H}} CH_3{-}\overset{\oplus}{C}H_2 \longrightarrow CH_2{=}CH_2$$

38. Ans. (3)

Though it is a 3° alkyl halide hence S_N2 not possible.

$$\underset{\text{(3°)}}{CH_3{-}\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{C}}{-}Cl} + C_2H_5ONa \longrightarrow CH_3{-}\underset{\displaystyle CH_3}{\overset{\displaystyle CH_2}{C}} \quad \text{or} \quad (CH_3)_2C{=}CH_2$$

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39. Ans. (1)

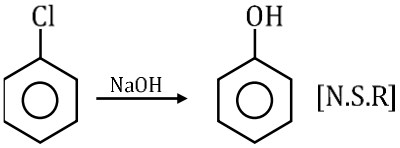
$CH_3-CH_2-Br + HS^{\ominus} \longrightarrow CH_3-CH_2-SH + Br^{\ominus}$ [N.S.R.]

$(CH_3)_2-C=CH_2 + HOCl \longrightarrow$

$(CH_3)_2-\underset{OH}{\underset{|}{C}}-CH_2-Cl$ [E.A.R]

$CH_3-CH_2-Br + \overset{\ominus}{O}H \rightarrow CH_2=CH_2 + Br^{\ominus} + H_2O$ [Elimination]

$(CH_3)_3CCH_2-OH + HBr \longrightarrow (CH_3)_3CBrCH_2CH_3 + H_2O$ [N.S.R.]



[N.S.R]

40. Ans. (4)

$CH_2=CH-CH_2-CH_3 \xrightarrow[\text{Aq. KOH}]{HBr} CH_3-\underset{OH}{\underset{|}{CH}}-CH_2-CH_3$

$CH_2=CH-CH_2-CH_3 \xrightarrow[\text{Aq. KOH}]{HBr/H_2O_2}$

$CH_2(OH)-CH_2-CH_2-CH_3$

$CH_2=CH-CH_3 \xrightarrow[\text{(ii) } CH_3COOH]{\text{(i) } BH_3/THF} CH_3-CH_2-CH_3$

$CH_3-\underset{Br}{\underset{|}{CH}}-CH_3 \xrightarrow[HBr/R_2O_2]{\text{Alc. KOH}} \underset{Br}{\underset{|}{CH_2}}-CH_2-CH_3$

41. Ans. (2)

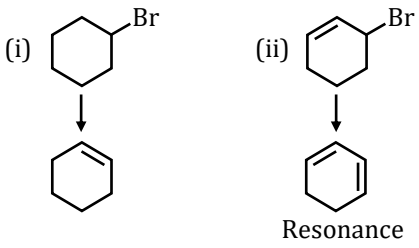
Reactivity order $\Rightarrow$ R–I > R–Br > R–Cl > R–F

42. Ans. (3)

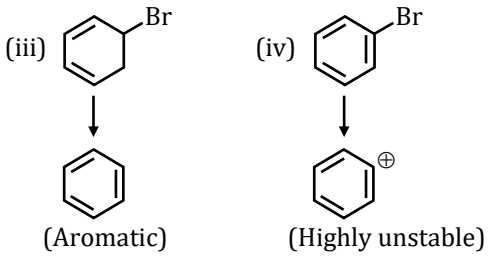
$CH_3-\underset{\beta_1}{\underset{|}{\overset{Br}{\overset{|}{CH}}}}-\underset{\alpha}{CH_2}-\underset{\beta_2}{CH_3} \xrightarrow[(-HBr)]{\Delta} CH_2=CH-CH_2-CH_3 + CH_3-CH=CH-CH_3$

(Sec. butyl bromide)

43. Ans. (2)



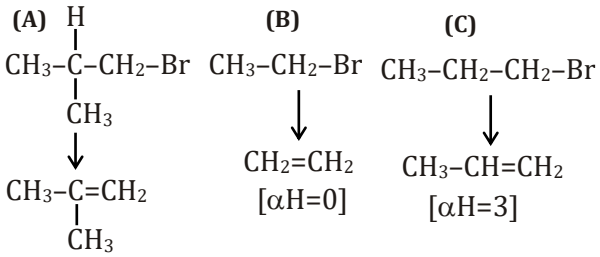
Resonance



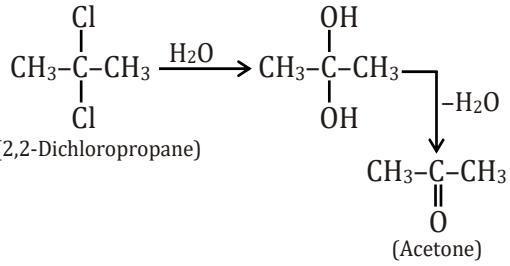
(Aromatic) (Highly unstable)

(iii) > (ii) > (i) > (iv)

44. Ans. (4)

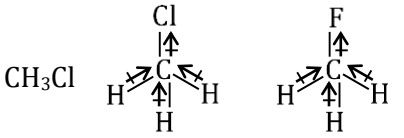


45. Ans. (1)



(2,2-Dichloropropane) (Acetone)

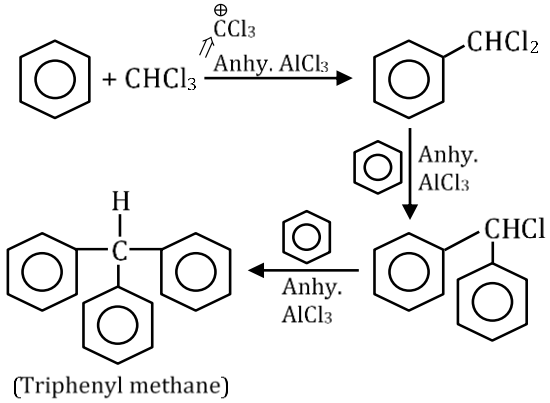
46. Ans. (1)



(μ=1.94D) (μ=1.82D)

The charge separation is larger in CH_3Cl as compared to CH_3-F.

47. Ans. (4)



(Triphenyl methane)

EXERCISE-I (Conceptual Questions) Build Up Your Understanding

1. Select the incorrect option for the following reaction:

$CH_3-CH=CH_2 + H_2O \xrightarrow{H^+} CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$

(1) This is an example of NAR of alkene
(2) In the first step, protonation of alkene takes place
(3) In the second step, Nucleophilic attack of water takes place on carbocation
(4) In the last step deprotonation takes place to form an alcohol

CAE006

2. For the reduction of aldehydes and ketones into alcohol the reagent which can be used is/are :
(1) H_2 in presence of Ni, Pt or Pd
(2) $NaBH_4$
(3) $LiAlH_4$
(4) All

CAE007

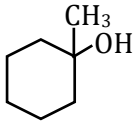
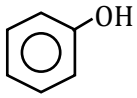
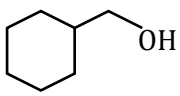
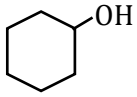
3. The compound A, B and C in the reaction sequence

$CH_3CH_2OH \xrightarrow{PBr_3} A \xrightarrow[\Delta]{alc.KOH} B \xrightarrow{Br_2} C$

are given by the set
(1) C_2H_5Br, CH_3CH_2OH, CH_3CHBr_2
(2) C_2H_5Br, $CH\equiv CH$, $CH_2=CHBr$
(3) C_2H_5Br, $CH_2=CH_2$, CH_2Br-CH_2Br
(4) C_2H_5Br, CH_3CH_2OH, $BrCH_2-CH_2Br$

CAE001

4. Which is most reactive with H-X ?

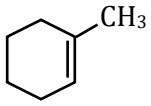
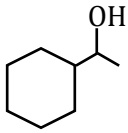
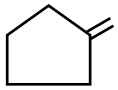
(1) 
(2) 
(3) 
(4) 

CAE084

5. Alkyl halides can be obtained by all methods excepts
(1) $CH_3-CH_2-OH + HCl \xrightarrow{ZnCl_2}$
(2) $CH_2=CH-CH_3 + HBr \longrightarrow$
(3) $C_2H_5OH + NaCl \longrightarrow$
(4) $CH_3-CH_2-OH + NaBr \xrightarrow{H_2SO_4}$

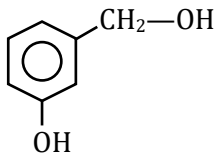
CAE124

6. In which reaction 3° Halide is not obtained as a major product ?

(1) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-OH + H-Cl \longrightarrow$
(2)  $+ H-Br \longrightarrow$
(3)  $+ H-Br \longrightarrow$
(4)  $+ H-Cl \longrightarrow$

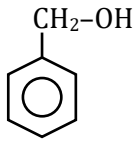
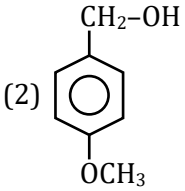
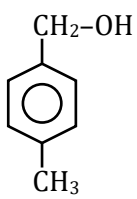
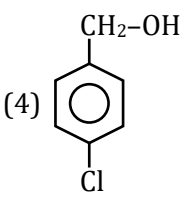
CAE125

7. Which reaction gives only dihalide

(1) $CH_3-CH=CH_2 + Br_2 \longrightarrow$
(2) $CH_3-\underset{\underset{CH_3}{|}}{C}=CH_2 + Cl_2 \xrightarrow{H_2O}$
(3) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3 + PCl_3 \longrightarrow$
(4)  $+ H-I \longrightarrow$ excess

CAE126

8. Which is maximum reactive towards H-X ?

(1) 
(2) 
(3) 
(4) 

CAE127

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9. p, s and t-alcohols can be distinguished by :-
(1) Reimer-Tieman reaction
(2) Tollen's reagent
(3) Lucas test
(4) Lassaigne's test

CAE004

10. Which of the following does not turn orange colour of chromic acid to green
(1) 1° alcohol (2) 2° alcohol
(3) 3° alcohol (4) Allyl alcohol

CAE003

11. Consider the following reaction :
$C_2H_5OH + H_2SO_4 \rightarrow$ Product
Among the following, which one cannot be formed as a product under any conditions ?
(1) $C_2H_5OSO_3H$
(2) $H_2C{=}CH_2$
(3) $HC{\equiv}CH$
(4) $CH_3{-}CH_2{-}O{-}CH_2{-}CH_3$

CAE005

12. Which of the following compounds shows intramolecular hydrogen bonding :-
(1) p–Nitrophenol (2) Ethanol
(3) o–Nitrophenol (4) Methanamine

CAE013

13. Phenol on treatment with methyl chloride in the presence of anhydrous $AlCl_3$ gives chiefly:-
(1) o–cresol (2) m–cresol
(3) Anisole (4) p–cresol

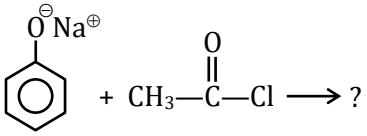
CAE017

14. Phenol on heating with $NaNO_2$ and a few drops of conc. H_2SO_4 mainly gives :-
(1) p–nitrophenol
(2) p–nitrosophenol
(3) o–nitrophenol
(4) m–nitrosophenol

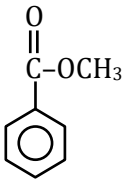
CAE018

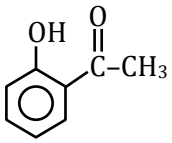
15. Phenol and benzoic acid are distinguished by:-
(1) Lucas reagent
(2) Victor Meyer test
(3) Caustic soda
(4) Sodium bicarbonate

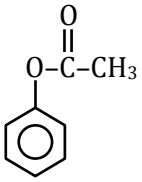
CAE019

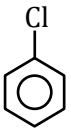
16. 

Sodium phenate

(1) 

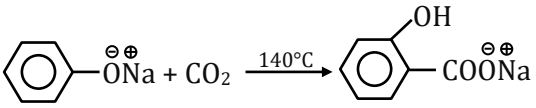
(2) 

(3) 

(4) 

CAE014

17. The reaction



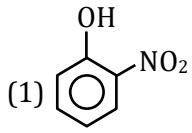
is called :-
(1) Schotten Bauman reaction
(2) Kolbe Schmidt reaction
(3) Reimer–Tiemann reaction
(4) Lederer–Manasse reaction

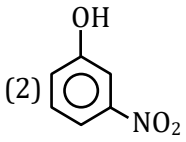
CAE015

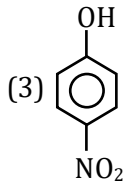
18. Phenol can be distinguished from ethanol by reactions with the following except :-
(1) Iodine and alkali
(2) Ferric chloride
(3) Acetyl chloride
(4) Bromine water

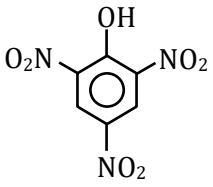
CAE016

19. Nitration of phenol with conc. H_2SO_4 followed by nitric acid gives :-

(1) 

(2) 

(3) 

(4) 

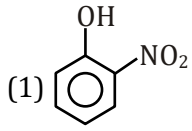
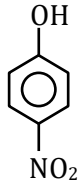
CAE011

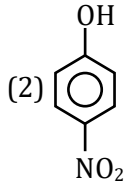
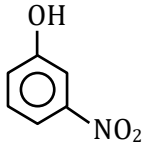
20. Deoxygenation of phenol can be achieved by distillation with :-

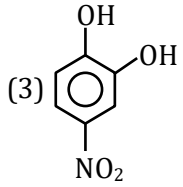
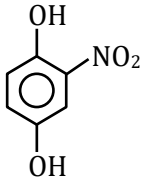
(1) Raney nickel
(2) Lithium aluminium hydride
(3) Sodium borohydride
(4) Zinc dust

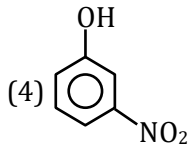
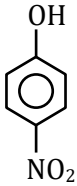
CAE012

21. Phenol on treatment with dil HNO_3 at low temp (298 K) gives two products P and Q. P is steam volatile but Q is not. P and Q are respectively.

(1)  and 

(2)  and 

(3)  and 

(4)  and 

CAE020

22. The preparation of ethers from alcohols by using sulphuric acid is called :-

(1) Williamson's ether Synthesis
(2) Williamson's continuous etherification process
(3) Ziesel's method
(4) Zerewitinoff method

CAE021

23. In the reaction sequence

$A \xrightarrow{HBr} B \xrightarrow{C_2H_5ONa}$ Ethoxyethane,

A and B are :-

(1) C_2H_6, C_2H_5Br
(2) CH_4, CH_3Br
(3) $CH_2=CH_2$, C_2H_5Br
(4) $CH\equiv CH$, $CH_2=CHBr$

CAE023

24. $CH_3–CH_2–OH + Ph–OH \xrightarrow[140°C]{H^{\oplus}}$ which is not obtained?

(1) $CH_3–CH_2–O–CH_2–CH_3$
(2) $CH_3–CH_2–O–Ph$
(3) Both (1) and (2)
(4) Ph–O–Ph

CAE024

25. Which of the following does not react with aq. NaOH :-

(1) Phenol (2) Benzoic acid
(3) CH_3COOH (4) $CH_3–O–C_6H_5$

CAE026

26. A student tried two reactions for preparing tert-butyl ethyl ether :

(I) $C_2H_5ONa + CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-Cl \longrightarrow$

(II) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-ONa + CH_3-CH_2-Cl \longrightarrow$

Which reaction will give better yield of tert-butyl ethyl ether ?

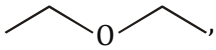
(1) Only I (2) Only II
(3) Both I & II (4) Neither I nor II

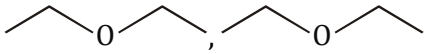
CAE027

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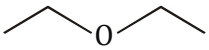
27. $CH_3CH_2OH \xrightarrow[443K]{H_2SO_4} A$, $CH_3CH_2OH \xrightarrow[413K]{H_2SO_4} B$ A and B are

(respectively)

(1)  , $CH_2 = CH_2$

(2) 

(3) $CH_2 = CH_2$, $CH_2 = CH_2$

(4) $CH_2 = CH_2$, 

CAE009

28. $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3 \xrightarrow{PBr_3} X \xrightarrow{Mg/ether} Y \xrightarrow{H_2O/H^+}$

The final product is :-

(1) $CH_3-\underset{\underset{CH_3}{|}}{CH}-OH$

(2) $CH_3-CH_2-CH_2-OH$

(3) $CH_3-\underset{\underset{CH_3}{|}}{CH}-O-\underset{\underset{CH_3}{|}}{CH}-CH_3$

(4) $CH_3-CH_2-CH_3$

CAE022

29. Oxonium ion of ether has the structure :-

(1) $C_2H_5-O-\underset{\underset{CH_3}{|}}{CH}-\overset{\oplus}{\underset{\underset{H}{|}}{O}}-H$

(2) $CH_3-CH_2-\overset{\oplus}{\underset{\underset{H}{|}}{O}}-CH_2-CH_3$

(3) $(C_2H_5)_2O$

(4) $CH_3-CH_2-O-CH_2-CH_2-CH_2-\overset{\oplus}{\underset{\underset{H}{|}}{O}}-O-H$

CAE025

30. Which of the following is insoluble in water ?

(1) Ethanol (2) Ethoxyethane

(3) Phenol (4) Pentane

CAE010

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**1**	**4**	**3**	**1**	**3**	**1**	**1**	**2**	**3**	**3**	**3**	**3**	**4**	**2**	**4**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**3**	**2**	**3**	**4**	**4**	**1**	**2**	**3**	**4**	**4**	**2**	**4**	**4**	**2**	**4**

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EXERCISE-II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Among the following sets of reactants which one produces anisole?

(1) CH_3CHO ; RMgX

(2) C_6H_5OH ; NaOH ; CH_3I

(3) C_6H_5OH ; neutral $FeCl_3$

(4) $C_6H_5 - CH_3$; CH_3COCl; $AlCl_3$

CAE037

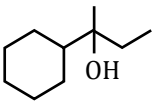
RE-AIPMT 2015

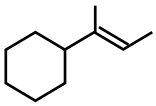
2. Reaction of phenol with chloroform in presence of dilute sodium hydroxide finally introduces which one of the following functional group ?

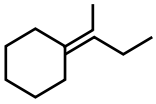
(1) $-CHCl_2$ (2) $-CHO$

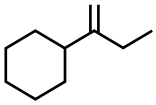
(3) $-CH_2Cl$ (4) $-COOH$

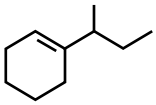
CAE038

3. Which of the following is not the product of dehydration of  ?

(1) 

(2) 

(3) 

(4) 

CAE039

NEET(UG) 2017

4. The heating of phenyl–methyl ethers with HI produces

(1) Iodobenzene (2) Phenol

(3) Benzene (4) Ethyl chlorides

CAE040

NEET(UG) 2018

5. The compound A on treatment with Na gives B, and with PCl_5 gives C. B and C react together to give diethyl ether. A, B and C are in the order

(1) C_2H_5OH, C_2H_6, C_2H_5Cl

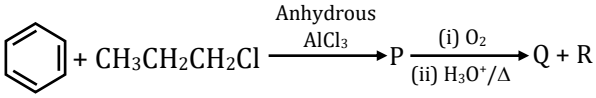
(2) C_2H_5OH, C_2H_5Cl, C_2H_5ONa

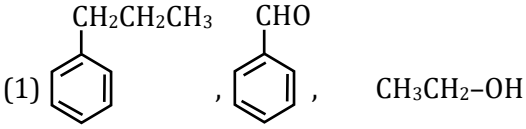
(3) C_2H_5Cl, C_2H_6, C_2H_5OH

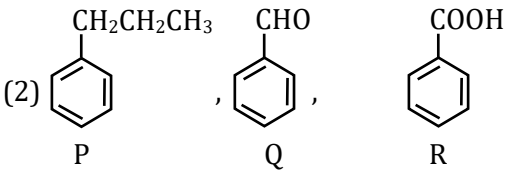
(4) C_2H_5OH, C_2H_5ONa, C_2H_5Cl

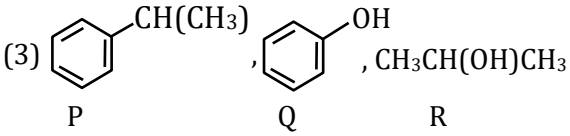
CAE041

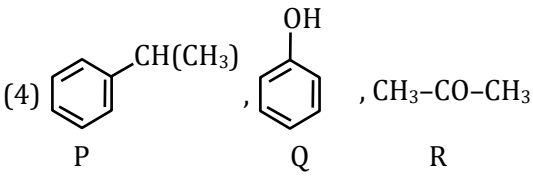
6. Identify the major products P, Q and R in the following sequence of reaction :



(1) 

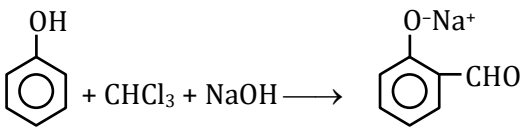
(2) 

(3) 

(4) 

CAE042

7. In the reaction



the electrophile involved is

(1) Dichloromethyl cation ($\overset{\oplus}{C}HCl_2$)

(2) Formyl cation ($\overset{\oplus}{C}HO$)

(3) Dichloromethyl anion ($\overset{\ominus}{C}HCl_2$)

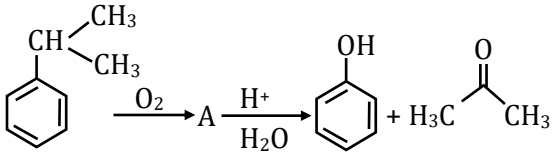
(4) Dichlorocarbene ($:CCl_2$)

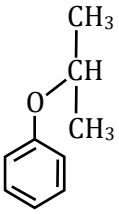
CAE043

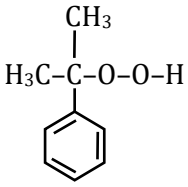
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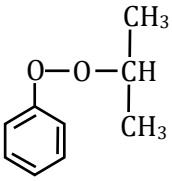
NEET(UG) 2019

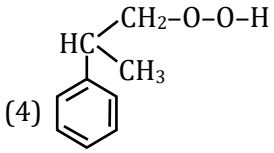
8. The structure of intermediate A in the following reaction is :-



(1) 

(2) 

(3) 

(4) 

CAE044

NEET(UG) 2019 (ODISHA)

9. The major products C and D formed in the following reactions respectively are :-

$H_3C–CH_2–CH_2–O–C(CH_3)_3 \xrightarrow[\Delta]{\text{excess HI}} C + D$

(1) $H_3C–CH_2–CH_2–I$ and $I–C(CH_3)_3$

(2) $H_3C–CH_2–CH_2–OH$ and $I–C(CH_3)_3$

(3) $H_3C–CH_2–CH_2–I$ and $HO–C(CH_3)_3$

(4) $H_3C–CH_2–CH_2–OH$ and $HO–C(CH_3)_3$

CAE045

NEET(UG) 2020

10. Reaction between acetone and methylmagnesium chloride followed by hydrolysis will give :

(1) Isobutyl alcohol

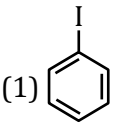
(2) Isopropyl alcohol

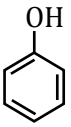
(3) Sec-butyl alcohol

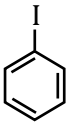
(4) Tert-butyl alcohol

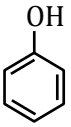
CAE046

11. Anisole on cleavage with HI gives:

(1)  + C_2H_5OH

(2)  + CH_3I

(3)  + CH_3OH

(4)  + C_2H_5I

CAE047

NEET(UG) 2021

12. What is the IUPAC name of the organic compound formed in the following chemical reaction ?

Acetone $\xrightarrow[(ii)\ H_2O,\ H^+]{(i)\ C_2H_5MgBr,\ \text{dry Ether}}$ Product

(1) 2-methyl propan-2-ol

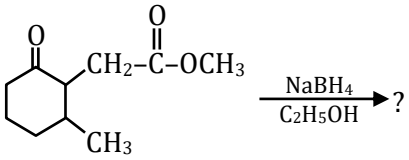
(2) pentan-2-ol

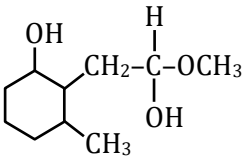
(3) pentan-3-ol

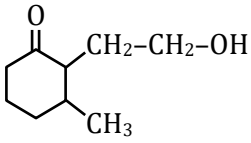
(4) 2-methyl butan-2-ol

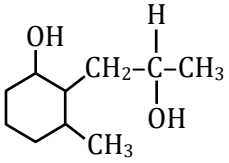
CAE048

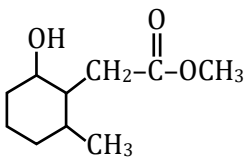
13. The product formed in the following chemical reaction is



(1) 

(2) 

(3) 

(4) 

CAE049

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NEET(UG) 2022

14. Given below are two statements :

Statement-I : In Lucas test, primary, secondary and tertiary alcohols are distinguished on the basis of their reactivity with cone. HCl + $ZnCl_2$, known as Lucas Reagent.

Statement-II : Primary alcohols are most reactive and immediately produce turbidity at room temperature on reaction with Lucas Reagent.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct

(4) Both Statement-I and Statement-II are correct

CAE085

NEET(UG) 2023

15. Consider the following reaction and identify the product (P).

$CH_3-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{OH}{|}}{CH}-CH_3 \xrightarrow{HBr}$ Product (P)

3-Methylbutan-2-ol

(1) $CH_3CH=CH-CH_3$

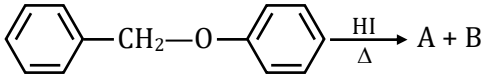
(2) $CH_3-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{Br}{|}}{CH}-CH_3$

(3) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_2Br$

(4) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{Br}{|}}{C}}-CH_2-CH_3$

CAE086

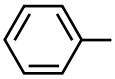
16. Consider the following reaction

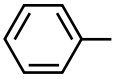


Identify products A and B :-

(1) A =  and B = 

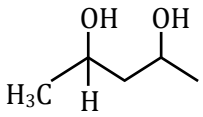
(2) A =  and B = 

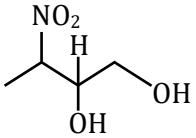
(3) A =  and B = 

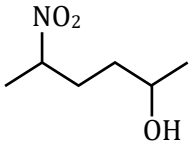
(4) A =  and B = 

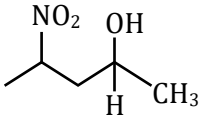
CAE087

17. Which amongst the following will be most readily dehydrated under acidic conditions ?

(1) 

(2) 

(3) 

(4) 

CAE088

18. $R-COOH \xrightarrow[(ii)\ H_2O/HCl]{(i)\ "X"} R-CH_2OH$

$R-CH=CH_2 \xrightarrow[(ii)\ H_2O,NaOH,H_2O_2]{(i)\ "X"} R-CH_2-CH_2-OH$

Identify 'X' in above reactions

(1) B_2H_6 (2) $LiAlH_4$

(3) $NaBH_4$ (4) H_2/Pd

CAE089

19. Reagents which can be used to convert alcohols to carboxylic acids, are

(A) $CrO_3 - H_2SO_4$ (B) $K_2Cr_2O_7 + H_2SO_4$

(C) $KMnO_4 + KOH/H_3O^+$ (D) Cu, 573 K

(E) CrO_3, $(CH_3CO)_2O$

Choose the **most appropriate** answer from the options given below :

(1) (B), (C) and (D) only

(2) (B), (D) and (E) only

(3) (A), (B) and (C) only

(4) (A), (B) and (E) only

CAE090

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NEET(UG) 2024

20. Which one of the following alcohols reacts instantaneously with Lucas reagent ?

(1) $CH_3—CH_2—CH_2—CH_2OH$

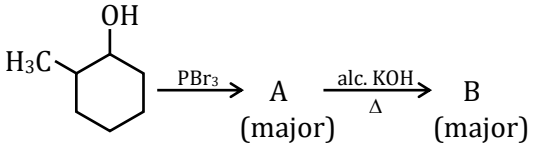
(2) $CH_3—CH_2—\underset{\displaystyle CH_3}{\underset{|}{CH}}—OH$

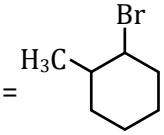
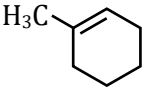
(3) $CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_2OH$

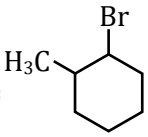
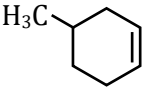
(4) $CH_3—\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}—OH$

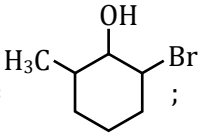
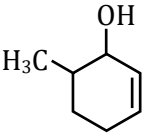
CAE091

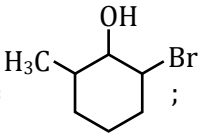
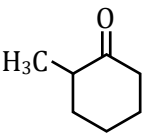
21. Major products A and B formed in the following reaction sequence, are



(1) A =  ; B = 

(2) A =  ; B = 

(3) A =  ; B = 

(4) A =  ; B = 

CAE092

NEET(UG) 2024 (Re-examination)

22. The major product C in the below mentioned reaction is :

$CH_3CH_2CH_2Br \xrightarrow[\Delta]{\text{alc.KOH}} A \xrightarrow{HBr} B \xrightarrow[\Delta]{\text{aq.KOH}} C$

(1) Propan-1-ol (2) Propan-2-ol

(3) Propane (4) Propyne

CAE094

23. Given below are two statements :

Statement-I : Propene on treatment with diborane gives an addition product with the formula $((CH_3)_2–CH)_3B$

Statement-II : Oxidation $((CH_3)_2–CH)_3B$ with hydrogen peroxide in presence of NaOH gives propan-2-ol.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I is correct but Statement-II is incorrect

(2) Statement-I is incorrect but Statement-II is correct

(3) Both Statement-I and Statement-II are correct

(4) Both Statement-I and Statement-II are incorrect

CAE093

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**2**	**4**	**2**	**4**	**4**	**4**	**2**	**1**	**4**	**2**	**4**	**4**	**2**	**4**
Question	16	17	18	19	20	21	22	23							
Answer	**2**	**1**	**1**	**3**	**4**	**1**	**2**	**2**							

EXERCISE-III (Analytical Questions) Master Your Understanding

1. Compound A and C in the following reaction are

$CH_3CHO \xrightarrow[(ii)\ H_2O]{(i)\ CH_3MgBr} (A) \xrightarrow[\Delta]{H_2SO_4} (B) \xrightarrow{\text{Hydroboration oxidation}} (C)$

(1) Identical (2) Functional isomer
(3) Positional isomer (4) Optical isomer

CAE057

2. Which of the following gives 2°-alcohol ?

(A) $CH_3CH=CH_2 \xrightarrow[(ii)\ H_2O_2/\overset{\ominus}{O}H]{(i)\ BH_3/THF}$

(B) $CH_3CH=CH_2 \xrightarrow[(ii)\ NaBH_4/\overset{\ominus}{O}H]{(i)\ (CH_3COO)_2Hg/H_2O}$

(C) $CH_3CH=CH_2 \xrightarrow{Conc.H_2SO_4}$

(D) $CH_3-CH=CH_2 \xrightarrow{dil\ H_2SO_4}$

(1) A, B and C are correct
(2) A and B are correct
(3) A and C are correct
(4) B and D are correct

CAE076

3. Which of the following reactions does not give aldehyde as major product ?

(1) $RCH_2OH \xrightarrow[N]{CrO_3.\ HCl}$

(2) $RCH_2OH \xrightarrow{anhy.\ CrO_3}$

(3) $RCH_2OH \xrightarrow{Cu/573K}$

(4) $RCH_2OH \xrightarrow{KMnO_4,\ H^+}$

CAE060

4. Methanol can be distinguished from ethanol by the following except
(1) Reaction with iodine and alkali
(2) Reaction with salicylic acid and H_2SO_4
(3) Reaction with Lucas reagent
(4) Boiling point

CAE050

5. Ethanol on heating with acetic acid in the presence of a few drops of sulphuric acid gives the smell of
(1) Oil of wintergreen
(2) Oil of mustard
(3) An ester
(4) Oil of bitter almonds

CAE051

6. The compounds A, B and C in the reaction sequence are given by the set :-

$CH_3CH_2OH \xrightarrow[\Delta]{HBr} A \xrightarrow{Mg} B \xrightarrow[(ii)H_3O^{\oplus}]{(i)CH_3CHO} C$

(1) CH_3CH_2Br, CH_3CH_2MgBr, $(CH_3)_3C—OH$
(2) CH_3CH_2Br, $(CH_3CH_2)_2Mg$, $(CH_3)_2CHCH_2OH$
(3) CH_3CH_2Br, CH_3CH_2MgBr, $CH_3CH(OH)CH_2CH_3$
(4) CH_3CHBr_2, $CH_3CH(MgBr)_2$, $CH_3CH(OH)CH_3$

CAE052

7. Which of the following product will be obtained when neopentyl alcohol is treated with conc. HCl in presence of $ZnCl_2$.
(1) t– butyl chloride
(2) Isobutylene
(3) t– pentyl chloride
(4) Neo pentyl chloride

CAE128

8. Which alcohol produces turbidity with Lucas reagent most slowly
(1) 2–Butanol
(2) t–Butyl alcohol
(3) Isobutyl alcohol
(4) Diphenyl methanol

CAE129

9. $CH_3-CH_2-OH \xrightarrow[H_2SO_4]{NaBr} (A) \xrightarrow[\Delta]{Alc.KOH} (B) \xrightarrow{HBr} (C)$

A, B and C are respectively -
(1) $CH_3CH_2HSO_4$, $CH_2=CH_2$, $BrCH_2-CH_2-Br$
(2) CH_3-CH_2-Br, $CH_2=CH_2$, CH_3-CH_2-Br
(3) $CH_3-CH_2-OCH_2-CH_3$, $CH_2=CH_2$, $CH_3-\underset{\displaystyle Br}{\underset{|}{CH_2}}$
(4) $CH_2=CH_2$, CH_3-CH_2-OH, CH_3-CH_2-Br

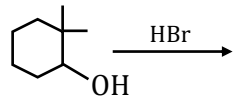
CAE130

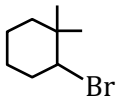
10. Best method to convert alcohol into alkyl chloride is -
(1) $ROH + SOCl_2 \longrightarrow R-Cl + SO_2 + HCl$
(2) $R-OH + PCl_3 \longrightarrow R-Cl$
(3) $R-OH + PCl_5 \longrightarrow R-Cl$
(4) $R-OH + HCl \longrightarrow R-Cl + H_2O$

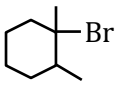
CAE131

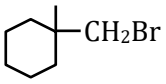
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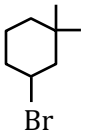
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11.  Major product ?

(1) 

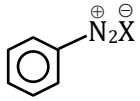
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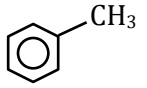
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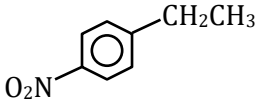
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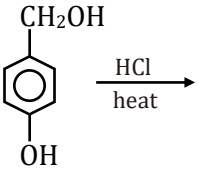
CAE132

12. Identify incorrect match in the following -

(1)  ESR

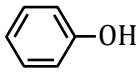
(2)  ESR

(3)  FRSR

(4)  NSR

CAE133

13. The correct reactivity order towards H–X will be

(I) 

(II) $CH_2=CH-CH_2OH$

(III) CH_3-CH_2-OH

(IV) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$

(1) II > I > III > IV (2) IV > III > II > I

(3) II > IV > I > III (4) II > IV > III > I

CAE134

14. For the reaction,

$C_2H_5OH + HX \longrightarrow C_2H_5X$, the order of reactivity is

(1) HI > HCl > HBr (2) HI > HBr > HCl

(3) HCl > HBr > HI (4) HBr > HI > HCl

CAE135

15. Arrange the following alkanols A, B and C in order of their reactivity towards acid catalysed dehydration:-

(A) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_2-OH$

(B) $CH_3-\overset{\overset{CH_3}{|}}{\overset{OH}{|}}{C}-CH_2-CH_3$

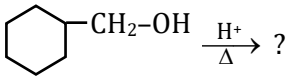
(C) $CH_3-\overset{\overset{CH_3}{|}}{CH}-\overset{\overset{OH}{|}}{CH}-CH_3$

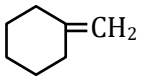
(1) A > B > C (2) B > A > C

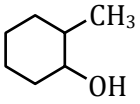
(3) B > C > A (4) C > B > A

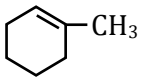
CAE136

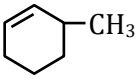
16. The major product in the following reaction is



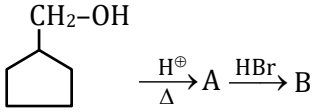
(1) 

(2) 

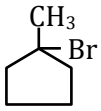
(3) 

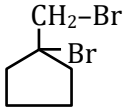
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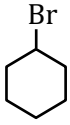
CAE137

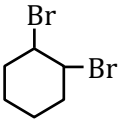
17. 

What is the structure of B :-

(1) 

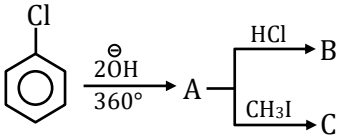
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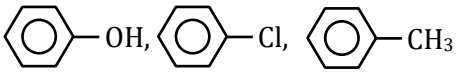
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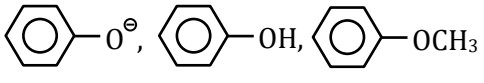
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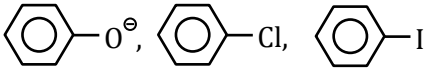
CAE138

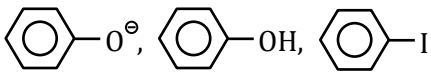
18. The structures of the compounds/ions A, B and C in the reaction sequence are given by the set :-



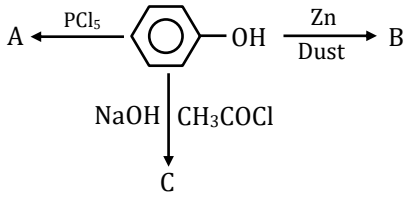
(1) 

(2) 

(3) 

(4) 

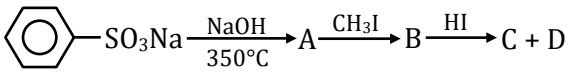
CAE054

19. 

The compounds A, B and C in the above reaction sequence are :-
(1) Chlorobenzene, benzene, methyl benzoate
(2) Chlorobenzene, benzene, phenyl acetate
(3) Benzyl chloride, benzene, phenyl acetate
(4) Benzyl chloride, benzene, phenylacetyl chloride

CAE055

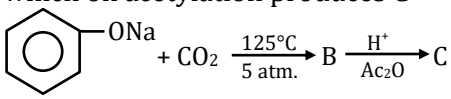
20. In the reaction sequence –



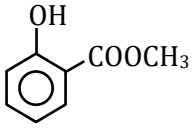
A, B, C and D are given by the set :-
(1) Sodium phenate, anisole, C_6H_5I, CH_3OH
(2) Sodium phenate, phenetole, C_2H_5I, C_6H_5OH
(3) Sodium phenate, anisole, C_6H_5OH, CH_3I
(4) Sodium phenate, phenetole, C_6H_5I, C_2H_5OH

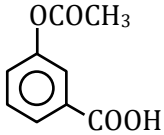
CAE056

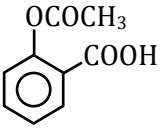
21. Sodium phenoxide when heated with CO_2 under pressure at 125°C yields a product which on acetylation produces C

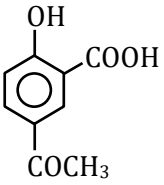


The major product C would be :

(1) 

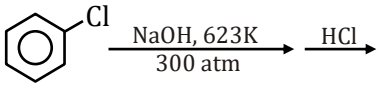
(2) 

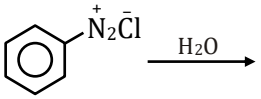
(3) 

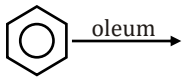
(4) 

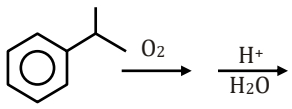
CAE058

22. Which of the following reaction sequence does not give phenol ?

(1) 

(2) 

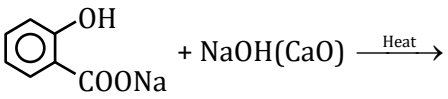
(3) 

(4) 

CAE059

23. Which of the following statement is/are correct ?
(A) Phenol is less acidic than ethanol
(B) Phenol is less acidic than acetic acid
(C) Phenol is more acidic than acetic acid
(D) Acetic acid is more acidic than ethanol
(1) A, B and C are correct
(2) A, B are correct
(3) B and D are correct
(4) A and C are correct

CAE078

24. Which of the following reactions will not lead to a phenol :-
(1) $C_6H_5SO_3Na + NaOH \xrightarrow[ii)H^+]{i)\ Fuse/\Delta}$
(2) $C_6H_5N_2Cl + H_2O \xrightarrow{Boil}$
(3) $C_6H_5ONa + RX \xrightarrow[Heat]{NaOH}$
(4) 

CAE053

25. The final product in the reaction is
$Ph—OH + CCl_4 \xrightarrow[(2)\ H^{\oplus}]{(1)\ KOH,\Delta}$ product
(1) Salicylaldehyde (2) Salicylic acid
(3) Methyl salicylate (4) Benzyl chloride

CAE139

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	4	3	3	3	3	3	2	1	2	1	4	2	3
Question	16	17	18	19	20	21	22	23	24	25					
Answer	3	3	2	2	3	3	3	3	3	2					

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Alcohols, Phenols and Ethers

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (1)

$CH_3-CH=CH_2 + H_2O \xrightarrow{H^+} CH_3-\underset{OH}{CH}-CH_3$

This is an example of EAR.

2. Ans. (4)

$R-\underset{\underset{O}{\|}}{C}-H/R \xrightarrow[\text{or } NaBH_4 \text{ or } H_2/Ni/Pd/Pt]{LiAlH_4 \text{ or}} R-CH_2-OH/R-\underset{OH}{CH}-R$

(Aldehyde /Ketone) (Alcohol)

3. Ans. (3)

$CH_3-CH_2-OH \xrightarrow{PBr_3} CH_3-CH_2-Br \xrightarrow[\Delta]{Alc.KOH}$

(A)

$CH_2-CH_2 \xrightarrow{Br_2} \underset{Br}{CH_2}-\underset{Br}{CH_2}$

(B) (C)

4. Ans. (1)

Reactivity of HX $\propto$ Degree of alcohol

$3° > 2° > 1°$

5. Ans. (3)

$CH_3-CH_2-OH + HCl \xrightarrow{ZnCl_2} CH_3-CH_2-Cl$

$CH_2=CH-CH_3 + HBr \longrightarrow CH_3-\underset{Br}{CH}-CH_3$

$C_2H_5OH + NaCl \longrightarrow$ No Reaction

$CH_3-CH_2-OH + NaBr \xrightarrow{H_2SO_4} CH_3-CH_2-Br$

6. Ans. (1)

$CH_3-\underset{CH_3}{CH}-CH_2-OH + HCl \xrightarrow{S_N2}$ (1°Alcohol)

$CH_3-\underset{CH_3}{CH}-CH_2-Cl + H_2O$

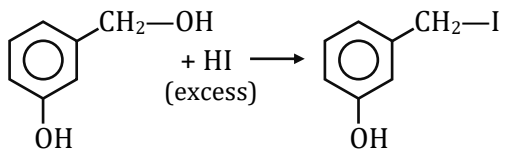
Primary alcohol on reacting with HCl follow S_N2 mechanism, Hence No carbocation form.

7. Ans. (1)

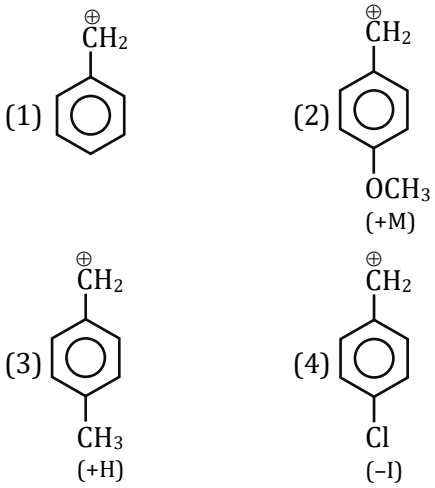
$CH_3-CH=CH_2 + Br_2 \rightarrow CH_3-\underset{Br}{CH}-\underset{Br}{CH_2}$

$CH_3-\overset{\oplus}{C}=\overset{\ominus}{C}H_2 + Cl_2 \xrightarrow[Cl^{\oplus}\ OH^{\ominus}]{H_2O} CH_3-\underset{CH_3}{\overset{OH}{C}}-CH_2-Cl$

$CH_3-\underset{OH}{CH}-CH_2 + PCl_3 \longrightarrow CH_3-\underset{Cl}{CH}-CH_3$



8. Ans. (2)



In option (2) carbocation is stabilise by +M effect.

9. Ans. (3)

1°, 2°, 3° Alcohols can be distinguished by Lucas test.

10. Ans. (3)

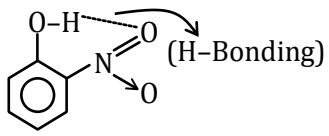
3° Alcohol doesn't turn orange colour to green.

11. Ans. (3)

$C_2H_5-OH + H_2SO_4 \rightarrow CH_2=CH_2 + \underset{+}{C_2H_5OSO_3H}$

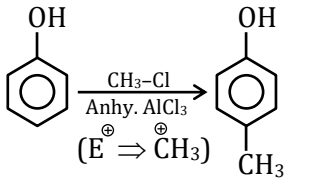
Et–O–Et

12. Ans. (3)



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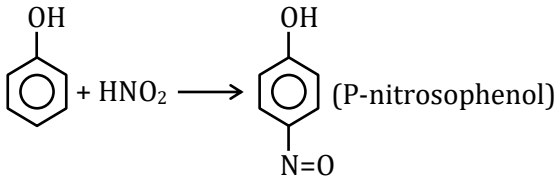
13. Ans. (4)



(P-cresol)

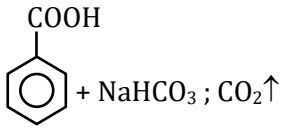
14. Ans. (2)

$NaNO_2 + H_2SO_4 \rightarrow HNO_2$



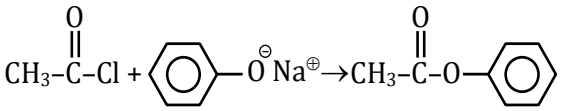
(P-nitrosophenol)

15. Ans. (4)

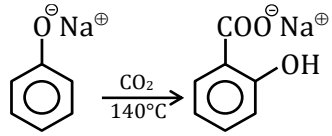


While phenol doesn't react.

16. Ans. (3)

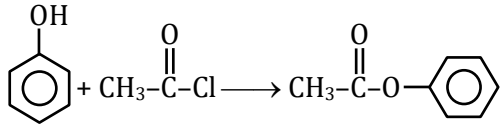


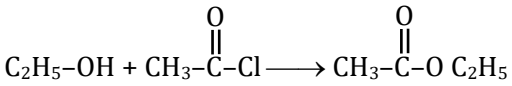
17. Ans. (2)



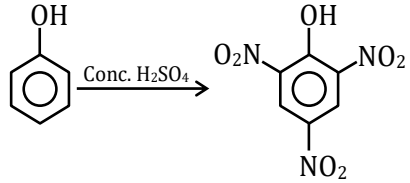
[Kolbe's Schmidt section]

18. Ans. (3)



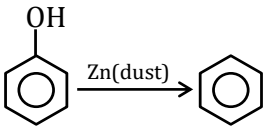


19. Ans. (4)

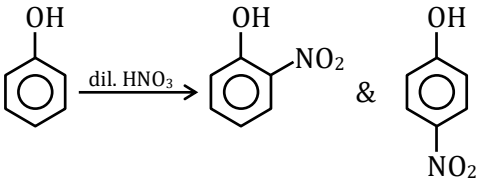


(Picric acid)

20. Ans. (4)



21. Ans. (1)

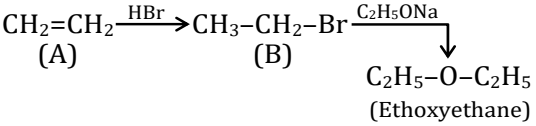


22. Ans. (2)

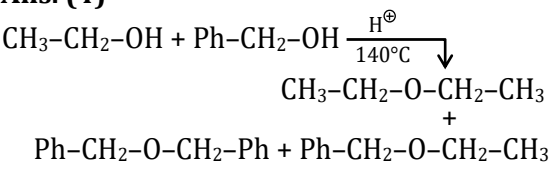


(Williamson's continuous etherification process)

23. Ans. (3)



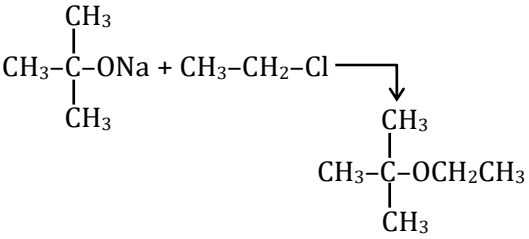
24. Ans. (4)



25. Ans. (4)

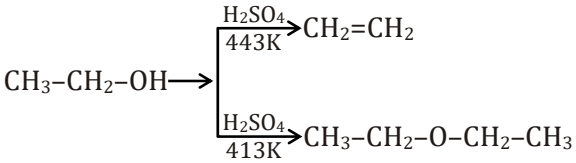
$CH_3–O–C_6H_5$ doesn't react with aq. NaOH.

26. Ans. (2)

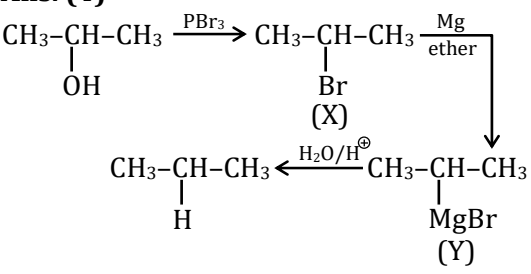


(tert-butyl ethyl ether)

27. Ans. (4)



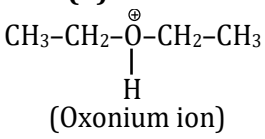
28. Ans. (4)



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29. Ans. (2)



(Oxonium ion)

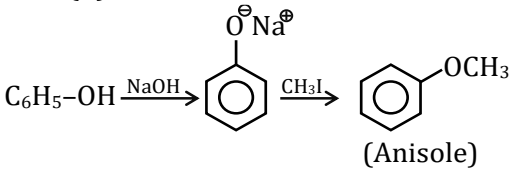
30. Ans. (4)

Like dissolves like.

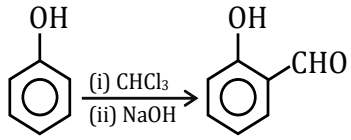
Hence, Pentane in insoluble in water.

Exercise-II (Previous Year Questions)

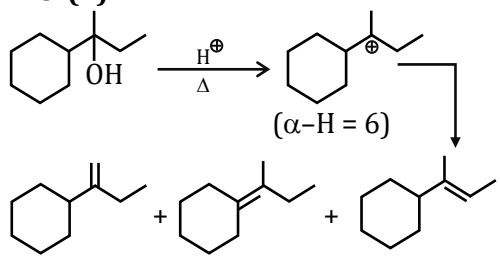
1. Ans. (2)



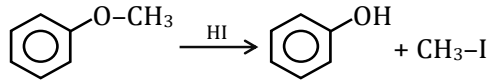
2. Ans. (2)



3. Ans. (4)

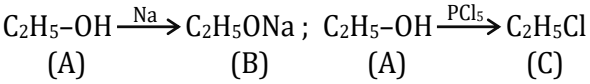


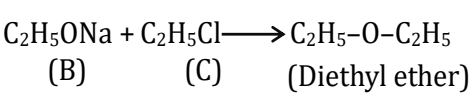
4. Ans. (2)



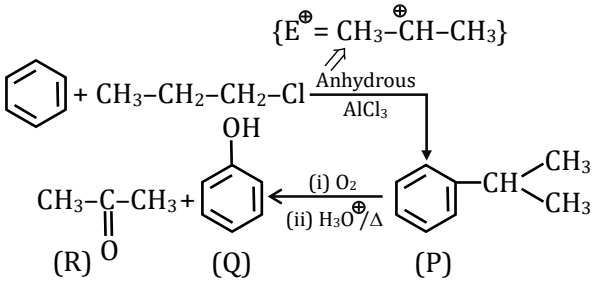
(Phenyl methyl ether)

5. Ans. (4)

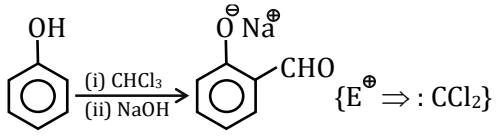




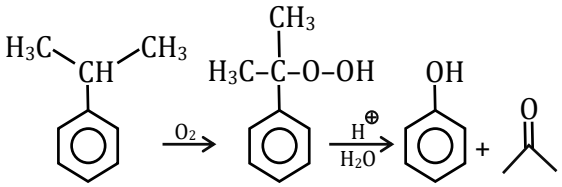
6. Ans. (4)



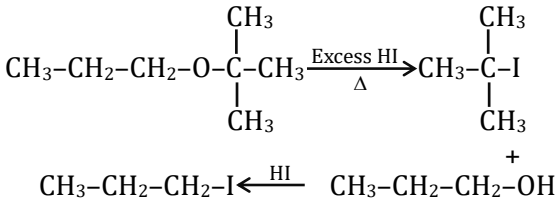
7. Ans. (4)



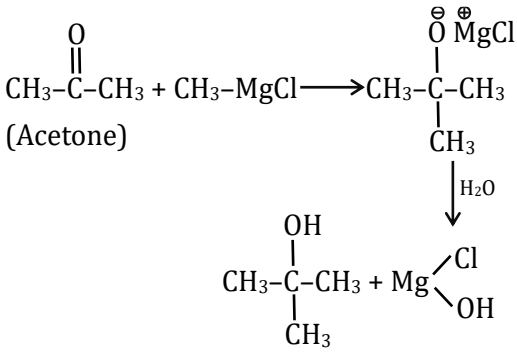
8. Ans. (2)



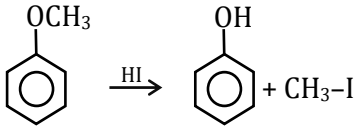
9. Ans. (1)



10. Ans. (4)

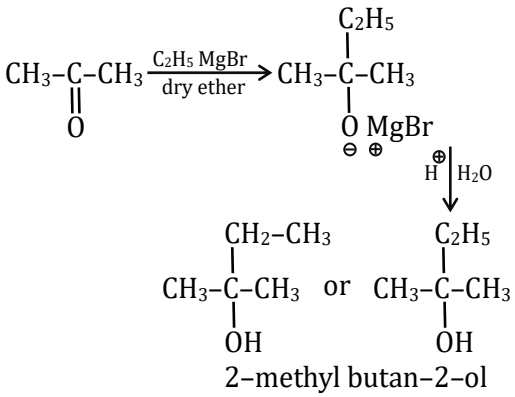


11. Ans. (2)



(Anisole)

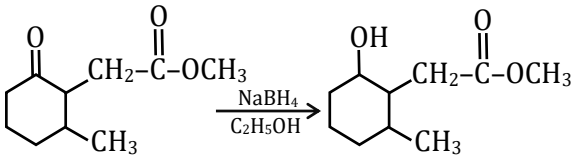
12. Ans. (4)



2–methyl butan–2–ol

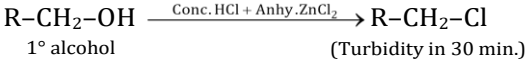
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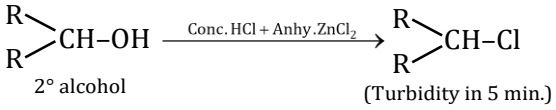
13. Ans. (4)

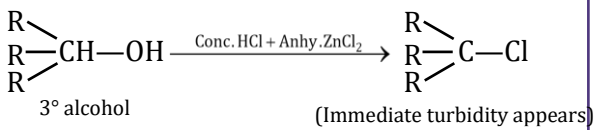


14. Ans. (2)

1°, 2°, 3° Alcohol are distinguished by Lucas test on the basis of the time taken for turbidity to appear



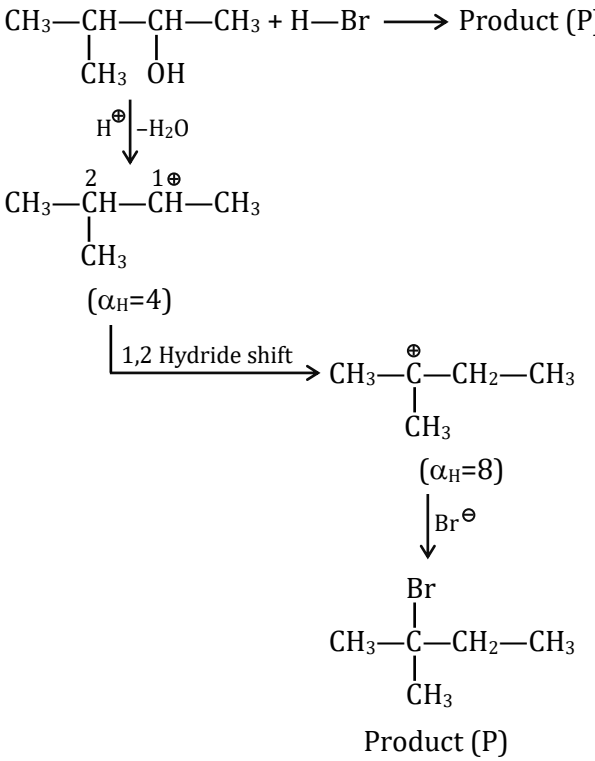




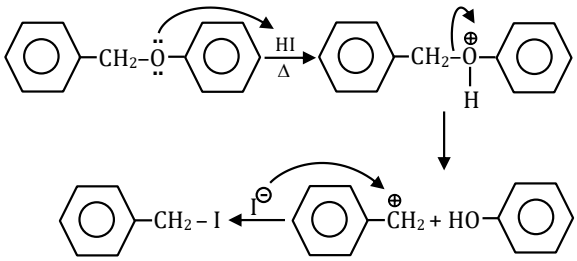
Reactivity of alcohol towards Lucas reagent

$\Rightarrow$ 3° > 2° > 1° Alcohol

15. Ans. (4)

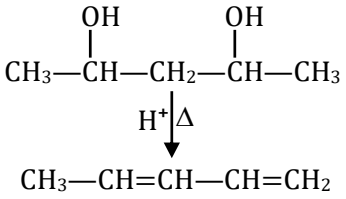


16. Ans. (2)

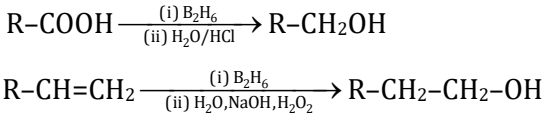


17. Ans. (1)

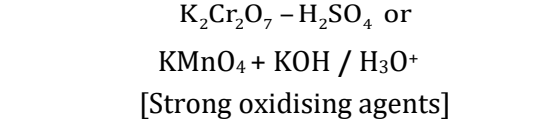
Due to presence of conjugation in product.



18. Ans. (1)

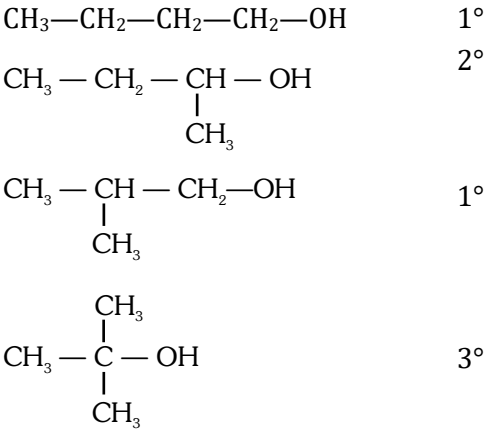


19. Ans. (3)

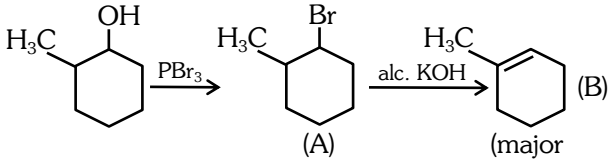


20. Ans. (4)

Reactivity order of alcohols towards Lucas' reagent 3° alcohol > 2° alcohol > 1° alcohol



21. Ans. (1)



22. Ans. (2)

$CH_3CH_2CH_2-Br \xrightarrow[\Delta]{\text{alc KOH}} \underset{(A)}{CH_3-CH=CH_2} \xrightarrow{HBr} CH_3-\underset{Br}{\underset{|}{CH}}-CH_3 \xrightarrow[\Delta]{\text{aq.KOH}} CH_3-\underset{OH}{\underset{|}{CH}}-CH_3$

(B) (C)

23. Ans. (2)

$CH_3-CH=CH_2 \xrightarrow{B_2H_6} (CH_3-CH_2-CH_2-)_3B$

$\left(\begin{matrix}CH_3\\CH_3\end{matrix}\!\!>\!CH-\right)_3 B \xrightarrow{H_2O_2/NaOH} \begin{matrix}CH_3\\CH_3\end{matrix}\!\!>\!CH-OH$

Propan-2-ol

Exercise-III (Analytical Questions)

1. Ans. (3)

$CH_3-\overset{O}{\overset{\|}{C}}-H \xrightarrow{CH_3MgBr} CH_3-\underset{CH_3}{\underset{|}{\overset{\ominus}{O}\overset{\oplus}{M}gBr}}-H \xrightarrow{H_2O} \underset{(A)}{CH_3-\underset{CH_3}{\underset{|}{C}}-H}$ (OH)

$\xrightarrow[\Delta]{H_2SO_4}$

$CH_3-\underset{H}{\underset{|}{CH}}-\underset{OH}{\underset{|}{CH_2}} \xleftarrow{H.B.O.} \underset{(B)}{CH_3-CH=CH_2}$

(C)

A is Propan–2–ol

B is Propan–1–ol Both are P.I.

2. Ans. (4)

$CH_3-CH=CH_2 \xrightarrow[(ii)\,H_2O_2/\overset{\ominus}{O}H]{(i)\,BH_3/THF} CH_3-\underset{H}{\underset{|}{CH}}-\underset{OH}{\underset{|}{CH_2}}$

(1° Alcohol)

$CH_3-CH=CH_2 \xrightarrow[(ii)\,NaBH_4/\overset{\ominus}{O}H]{(i)\,Hg(OAC)_2/H_2O} CH_3-\underset{OH}{\underset{|}{CH}}-CH_2-H$ (2° Alcohol)

$CH_3-CH=CH_2 \xrightarrow{Conc.\,H_2SO_4} CH_3-\underset{OSO_3H}{\underset{|}{CH}}-CH_2-H$

$CH_3-CH=CH_2 \xrightarrow{dil.\,H_2SO_4} CH_3-\underset{OH}{\underset{|}{CH}}-CH_3$

(2° Alcohol)

3. Ans. (4)

$R-CH_2-OH \xrightarrow[H^{\oplus}]{KMnO_4} \underset{\text{(Carboxylic acid)}}{R-COOH}$

4. Ans. (3)

$CH_3-OH \longrightarrow \times$

$CH_3-CH_2-OH \longrightarrow \times$

(Turbidity only on heating after long time i.e. 30 min.)

5. Ans. (3)

$C_2H_5-OH + CH_3-\overset{O}{\overset{\|}{C}}-OH \xrightarrow[-H_2O]{H^{\oplus}} CH_3-\overset{O}{\overset{\|}{C}}-OC_2H_5$

(ethanol) (Acetic acid) (Ester)

6. Ans. (3)

$CH_3-CH_2-OH \xrightarrow[\Delta]{HBr} \underset{(A)}{CH_3-CH_2-Br} \xrightarrow{Mg} \underset{(B)}{CH_3-CH_2-MgBr} \xrightarrow{CH_3CHO}$

$CH_3-\underset{OH}{\underset{|}{CH}}-CH_2-CH_3$ or $CH_3-\underset{OH}{\underset{|}{\overset{CH_2CH_3}{\overset{|}{C}}}}-H \xleftarrow{H_3O^{\oplus}} CH_3-\underset{\overset{\ominus}{O}\overset{\oplus}{M}gBr}{\underset{|}{\overset{CH_2CH_3}{\overset{|}{C}}}}-H$

7. Ans. (3)

$CH_3-\underset{CH_3}{\underset{|}{\overset{CH_3}{\overset{|}{C}}}}-CH_2-\ddot{O}H \xrightarrow{H^{\oplus}} CH_3-\underset{CH_3}{\underset{|}{\overset{CH_3}{\overset{|}{C}}}}-\overset{\oplus}{C}H_2 \xrightarrow{\sim\overset{\ominus}{C}H_3}$

$CH_3-\underset{CH_3}{\underset{|}{\overset{Cl}{\overset{|}{C}}}}-CH_2-CH_3 \longleftarrow CH_3-\underset{CH_3}{\underset{|}{\overset{\oplus}{C}}}-CH_2-CH_3$

8. Ans. (3)

$CH_3-\underset{CH_3}{\underset{|}{CH}}-CH_2-OH$

(1° Alcohol)

{Reactivity towards Lucas Reagent

$\Rightarrow 3°>2°>1°$}

9. Ans. (2)

$CH_3-CH_2-OH \xrightarrow[H_2SO_4]{NaBr} \underset{\textbf{(A)}}{CH_3-CH_2-Br}$

$\xrightarrow[\Delta]{\text{Alc. KOH}}$

$CH_3-\underset{Br}{\underset{|}{CH_2}} \xleftarrow{HBr} \underset{\textbf{(B)}}{CH_2=CH_2}$

(C)

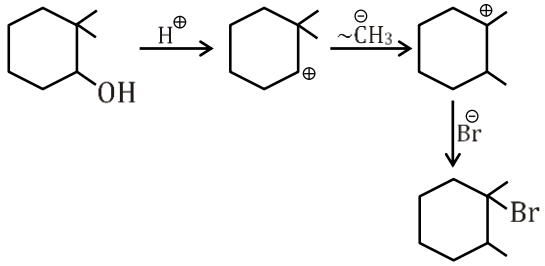
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10. Ans. (1)

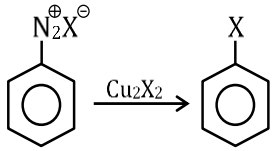
$R-OH + SOCl_2 \longrightarrow R-Cl + SO_2\uparrow + HCl\uparrow$

(Alcohol) (Alkyl chloride)

11. Ans. (2)

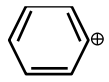


12. Ans. (1)



It is N.S.R.

13. Ans. (4)

(I) 

(II) $CH_2=CH-\overset{\oplus}{C}H_2$ Reso.

(III) $CH_3-\overset{\oplus}{C}H_2$ [3α-H]

(IV) $CH_3-\overset{\oplus}{C}H-CH_3$ [6α-H]

II > IV > III > I

14. Ans. (2)

$C_2H_5-OH + HX \longrightarrow C_2H_5-X$

The order of reactivity ⇒ HI > HBr > HCl

15. Ans. (3)

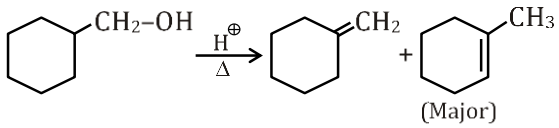
(A) $CH_3-\underset{\underset{(2\alpha\text{-H})}{|}}{\overset{CH_3}{CH}}-CH_2-\overset{\oplus}{C}H_2$

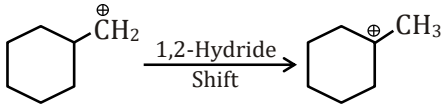
(B) $CH_3-\overset{\oplus}{\underset{\underset{CH_3}{|}}{C}}-CH_2-CH_3$ (8α-H)

(C) $CH_3-\underset{\underset{(4\alpha\text{-H})}{|}}{\overset{CH_3}{CH}}-\overset{\oplus}{C}H-CH_3$

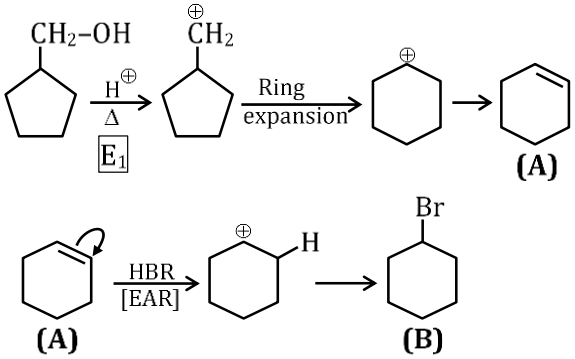
B > C > A

16. Ans. (3)

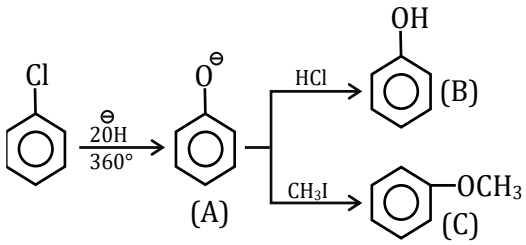


Mech. 

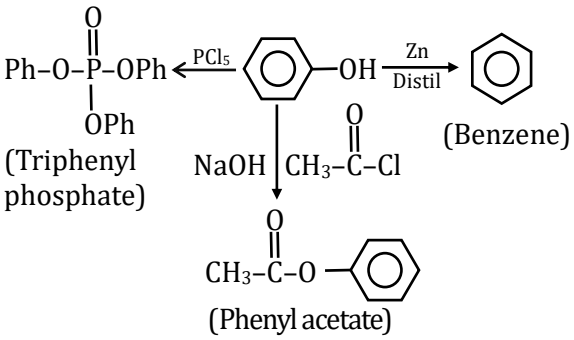
17. Ans. (3)



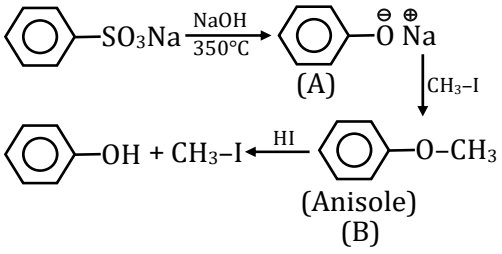
18. Ans. (2)



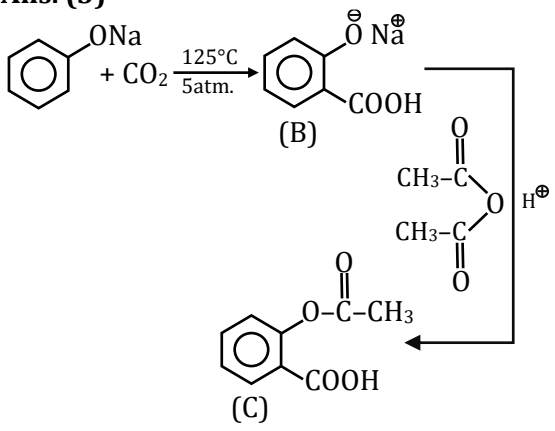
19. Ans. (2)



20. Ans. (3)

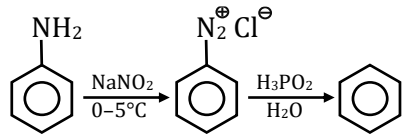


21. Ans. (3)

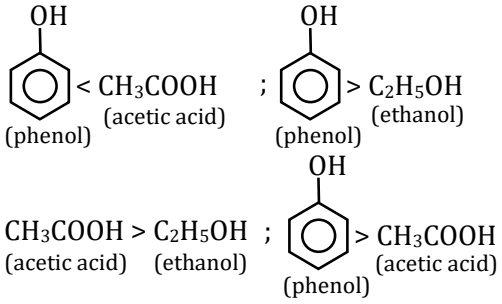


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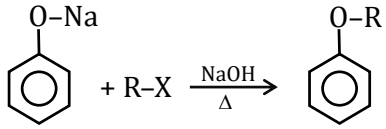
22. Ans. (3)



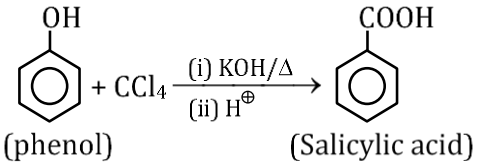
23. Ans. (3)



24. Ans. (3)



25. Ans. (2)



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EXERCISE-I (Conceptual Questions) Build Up Your Understanding

Amines

1. Among the following which one is not formed in Hoffmann degradation

(1) RNCO (2) $R-\ddot{N}H_2$

(3) $RCO\ddot{N}HBr$ (4) RNC

CAM001

2. $CH_3CH_2CONH_2 \xrightarrow[Br_2]{NaOH} A$,

Aqueous solution of A

(1) Turns blue litmus to red

(2) Turns red litmus to blue

(3) Does not affect the litmus

(4) Decolourise the litmus

CAM002

3. Ethanamine can be obtained if the following compound is heated with [KOH + Br_2]

(1) Ethanamide (2) Methanamide

(3) Propionamide (4) All the above

CAM003

4. Tertiary amine is obtained in the reaction :-

(1) Aniline $\xrightarrow{CH_3I}$ $\xrightarrow{CH_3I}$

(2) Aniline $\xrightarrow{CH_3I}$

(3) Nitrobenzene $\xrightarrow{Sn/HCl}$

(4) None of the above

CAM005

5. $C_2H_5NH_2$ cannot be prepared by the reduction of

(1) $C_2H_5NO_2$ (2) $CH_3CH{=}NOH$

(3) C_2H_5NC (4) CH_3CN

CAM006

6. Gabriel reaction for the synthesis of amines, involves the use of

(1) 1° amide (2) 2° amide

(3) Imides (4) Aliphatic amide

CAM007

7. Gabriel phthalimide reaction is used in the synthesis of

(1) Primary aromatic amines

(2) Secondary amines

(3) Primary aliphatic amines

(4) Tertiary amines

CAM008

8. The reaction: [C_2H_5Br + NH_3] is in fact an example of

(1) Ammonolysis only

(2) Nucleophilic substitution only

(3) Ammonolysis as well as nucleophilic substitution

(4) None

CAM009

9. $CH_3CONH_2 \xrightarrow[I]{P_2O_5} A \xrightarrow[II]{Na/EtOH} B$

Product 'B' is

(1) $CH_3—CH_2—OH$

(2) $CH_3—NH_2$

(3) $CH_3—CH_2—NH_2$

(4) $CH_3—CHO$

CAM004

10. The product obtained by the alkaline hydrolysis of C_2H_5—NCO when treated with t–butyl magnesiumbromide, the compound obtained will be

(1) t–butylamine (2) n–butylamine

(3) Isobutane (4) n–butane

CAM013

11. Aniline can be obtained by :-

(1) Benzoyl chloride and ammonia

(2) Reduction of benzamide

(3) Phenol and ammonia in presence of $ZnCl_2$

(4) Benzoic anhydride and ammonia

CAM029

12. Melting points are normally the highest for

(1) Tertiary amides (2) Secondary amides

(3) Primary amides (4) Amines

CAM010

13. Solubility of ethylamine in water is due to

(1) Low molecular weight

(2) Ethyl group is present in ethyl amine

(3) Formation of H–bonding with water

(4) Being a derivative of ammonia

CAM011

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14. Hydrogen attached to nitrogen is released in the reaction

(1) $RCONH_2 + NaNH_2$ (2) $RNH_2 + Na$

(3) Both the above (4) None of the above

CAM017

15. This compound does not respond to carbylamine reaction :-

(1) $CH_3-\underset{\displaystyle CH_3}{\overset{}{\underset{|}{CH}}}-NH_2$ (2) $C_2H_5-NH-C_2H_5$

(3) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\underset{|}{\overset{|}{C}}}}-NH_2$ (4) $CH_3-\underset{\displaystyle NH_2}{\underset{|}{CH}}-CH_2-CH_3$

CAM020

16. Chloroform and ethanolic KOH is used as a reagent in the following reaction :-

(a) Hoffmann carbylamine reaction

(b) Hoffmann degradation reaction

(c) Reimer–Tieman reaction

(d) Kolbe's reaction

Code is :-

(1) Only for a (2) Only for a and b

(3) Only for b and d (4) Only for a and c

CAM033

17. Reaction of RCN with sodium and alcohol leads to the formation of :-

(1) $RCONH_2$ (2) $RCOO^-NH_4^+$

(3) RCH_2NH_2 (4) $R(CH_2)_3NH_2$

CAM041

18. Which of the following when heated with KOH and primary amine gives carbylamine test

(1) $CHCl_3$ (2) CH_2Cl_2

(3) CH_3OH (4) CCl_4

CAM046

19. Chloroform reacts with aniline and aqueous KOH gives :-

(1) Ph–N ≡ C (Phenyl isocyanide)

(2) Benzene

(3) Phenyl cyanide

(4) None of these

CAM047

20. Acidic nature of amino group is shown by the reaction :-

(1) $R-NH_2 + NaCl \rightarrow NaNH_2$

(2) $2RNH_2 + 2Na \rightarrow 2RNH.Na + H_2$

(3) $R.CH_2NH_2 + HNO_2 \rightarrow R.CH_2OH + N_2 + H_2O$

(4) $R.NH_2 + HCl \rightarrow \overset{\oplus}{R}\overset{}{NH_3}\ \overset{\ominus}{Cl}$

CAM015

21. Which one of the following amine compound gives alcohol with HNO_2 :-

(1) N, N–Dimethylaniline

(2) Benzylamine

(3) N–methylaniline

(4) Aniline

CAM022

22. Which of the following compound liberates CO_2 when treated with $NaHCO_3$

(1) $CH_3COCH_2NH_2$ (2) CH_3NH_2

(3) $(CH_3)_4\overset{\oplus}{N}OH$ (4) $CH_3\overset{\oplus}{N}H_3Cl$

CAM012

23. If primary amines are treated with ketones the product is

(1) Urea (2) Guanidine

(3) Amide (4) Schiff's base

CAM018

24. Blue litmus can be turned to red by the compound

(1) ROH (2) RNH_2

(3) $R\overset{\oplus}{N}H_3OH$ (4) $R\overset{\oplus}{N}H_3Cl$

CAM021

25. Reaction $C_6H_5NH_2 + HAuCl_4 \longrightarrow [C_6H_5\overset{\oplus}{N}H_3]Au\overset{\ominus}{Cl}_4$ shows ... behaviour of aniline:-

(1) Acidic (2) Neutral

(3) Basic (4) Amphoteric

CAM024

26. Which of the following does not reduce Tollen's reagent :-

(1) CH_3CHO (2) HCOOH

(3) $CH_3-\underset{\displaystyle O}{\overset{}{\underset{\|}{C}}}-CH_2-OH$ (4) $C_6H_5NH_2$

CAM028

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27. Aromatic nitriles (ArCN) are not prepared by reaction

(1) ArX + KCN (2) $ArN_2^+ + CuCN$

(3) $ArCONH_2 + P_2O_5$ (4) $ArCONH_2 + POCl_3$

CAM035

28. $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-NC \xrightarrow{\text{reduction}} ?$

(1) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-NH_2$

(2) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-NH-CH_3$

(3) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-NH-CH_2CH_3$

(4) None

CAM040

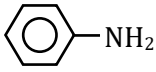
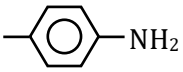
29. Which of the following is used as a solvent in the Friedel–Crafts reaction :-

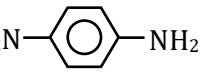
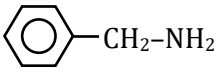
(1) Toluene (2) Nitrobenzene

(3) Benzene (4) Aniline

CAM044

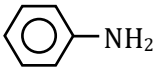
30. Which compound does not show diazo coupling reaction :-

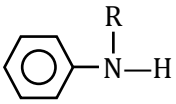
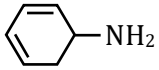
(1)  (2) 

(3)  (4) 

CAM026

31. Which of the following amines give N–nitroso derivative with $NaNO_2$ and HCl :-

(1) $C_2H_5NH_2$ (2) 

(3)  (4) 

CAM027

32. $C_6H_5NH_2 \xrightarrow[0-5°C]{NaNO_2/HCl}$ A, Which is the incorrect structure of the product 'A' :-

(1) $[C_6H_5-\overset{\oplus}{N}=N]\overset{\ominus}{Cl}$ (2) $[C_6H_5\overset{\oplus}{N}_2]\overset{\ominus}{Cl}$

(3) $[C_6H_5-\overset{\oplus}{N}\equiv N]\overset{\ominus}{Cl}$ (4) $[C_6H_5-N\equiv\overset{\oplus}{N}]\overset{\ominus}{Cl}$

CAM032

33. Aniline in a set of reactions yielded end product D

 $\xrightarrow[0-5°C]{NaNO_2+HCl}$ A $\xrightarrow{CuCN}$ B $\xrightarrow[Ni]{H_2}$ C $\xrightarrow{HNO_2}$ D

The structure of the product D would be

(1) $C_6H_5CH_2OH$ (2) $C_6H_5CH_2NH_2$

(3) C_6H_5NHOH (4) $C_6H_5NHCH_2CH_3$

CAM036

34. $\phi-X \xrightarrow{NaNO_2/HCl} C_6H_5N_2Cl \xrightarrow[\Delta]{Water} \phi-Y$,

In the above sequence X and Y are :-

(1) o–, p– and m–directing

(2) o–, p– and o–, p–directing

(3) m and m directing

(4) m and o, p directing

CAM037

35. $C_6H_5NO_2 \xrightarrow{Sn/HCl} A \xrightarrow[0°C]{NaNO_2/HCl} B$;

In the above sequence Benzene from B, is suitably obtained by using :-

(1) Ethanol (2) H_3PO_2

(3) Both the above (4) Methanol

CAM042

36. Which reagent is used to get iodo benzene from benzene diazonium acid sulphate $[C_6H_5N_2HSO_4]$:

(1) CuBr, Δ (2) Cu powder + HI

(3) KI, Δ (4) None

CAM043

37. In the Sandmeyer's reaction, –N=N–X group of diazonium salt is replaced by :-

(1) Halide group (2) Nitro group

(3) —OH group (4) $—NHNH_2$ group

CAM045

38. $C_6H_5NH_2 \xrightarrow{Br_2/CCl_4}$? The product is :-
(1) Only o– bromoaniline
(2) 2, 4, 6–tribromoaniline
(3) o–and p–bromoaniline
(4) Only p–bromoaniline

CAM023

39. Aniline on treatment with bromine water yields white precipitate of :-
(1) o–Bromoaniline
(2) p–Bromonailine
(3) 2, 4, 6–Tribromoaniline
(4) m–Bromoaniline

CAM025

40. Aniline on direct nitration produces :-
(1) o–Nitroaniline (2) m–Nitroaniline
(3) p–Nitroaniline (4) All

CAM030

41. Nitration of acetanilide followed by hydrolysis gives
(1) o–Nitroaniline only
(2) p–Nitroaniline only
(3) o– & p–Nitroaniline
(4) o–Nitroanilinium ion

CAM031

42. Acetanilide when treated with bromine in acetic acid mainly gives :-
(1) o–Bromoacetanilide
(2) N–Bromoacetanilide
(3) p–Bromoacetanilide
(4) m–Bromoacetanilide

CAM034

43. Which of the following compound gives an explosive on decarboxylation :-
(1) 2,4, 6–Trinitrobenzoicacid
(2) 2, 4–Dinitrobenzoicacid
(3) o–Aminobenzoicacid
(4) o–Hydroxybenzoicacid

CAM038

EXERCISE-I (Conceptual Questions) ANSWER KEY

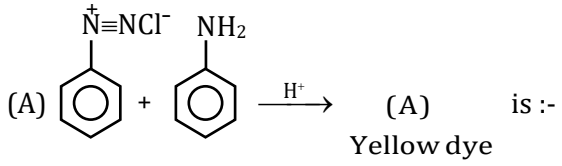
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**2**	**3**	**1**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**2**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**4**	**3**	**1**	**1**	**2**	**2**	**4**	**4**	**4**	**3**	**4**	**1**	**2**	**2**	**4**
Question	31	32	33	34	35	36	37	38	39	40	41	42	43		
Answer	**3**	**4**	**1**	**2**	**3**	**3**	**1**	**2**	**3**	**4**	**3**	**3**	**1**		

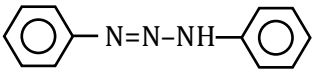
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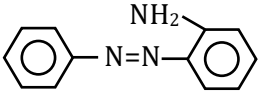
EXERCISE-II (Previous Year Questions) AIPMT/NEET

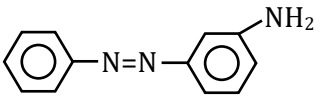
AIPMT 2014

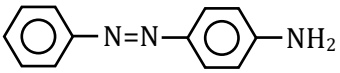
1. In the following reaction, the product



(1) 

(2) 

(3) 

(4) 

CAM063

2. Which of the following will be most stable diazonium salt $RN_2^+X^-$?

(1) $CH_3N_2^+X^-$ (2) $C_6H_5N_2^+X^-$

(3) $CH_3CH_2N_2^+X^-$ (4) $C_6H_5CH_2N_2^+X^-$

CAM064

AIPMT 2015

3. The electrolytic reduction of nitrobenzene in strongly acidic medium produces :-

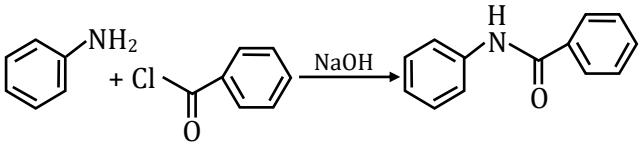
(1) Azoxybenzene (2) Azobenzene

(3) Aniline (4) p-Aminophenol

CAM065

RE-AIPMT 2015

4. The following reaction



is known by the name :

(1) Acetylation reaction

(2) Schotten-Baumann reaction

(3) Friedel–Craft's reaction

(4) Perkin's reaction

CAM066

5. Method by which Aniline cannot be prepared is:-

(1) Reduction of nitrobenzene with H_2/Pd in ethanol

(2) Potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution

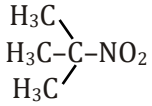
(3) Hydrolysis of phenyl isocyanide with acidic solution

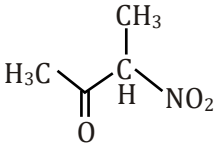
(4) Degradation of benzamide with bromine in alkaline solution

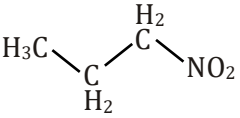
CAM067

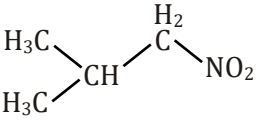
NEET-II 2016

6. Which one of the following nitro-compounds does not react with nitrous acid ?

(1) 

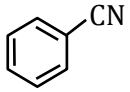
(2) 

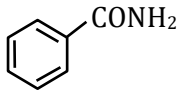
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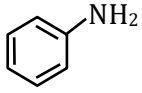
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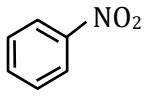
CAM068

7. A given nitrogen-containing aromatic compound (A) reacts with Sn/HCl, followed by HNO_2 to give an unstable compound (B). (B), on treatment with phenol, forms a beautiful coloured compound (C) with the molecular formula $C_{12}H_{10}N_2O$. The structure of compound (A) is :-

(1) 

(2) 

(3) 

(4) 

CAM069

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NEET(UG) 2017

8. Which of the following reactions is appropriate for converting acetamide to methanamine ?

(1) Hoffmarnn hypobromamide reaction

(2) Stephens reaction

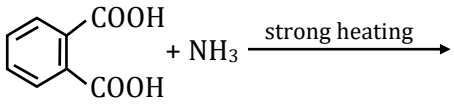
(3) Gabriels phthalimide synthesis

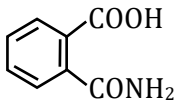
(4) Carbylamine reaction

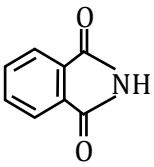
CAM070

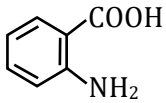
NEET(UG) 2019

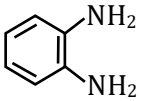
9. The major product of the following reaction is :



(1) 

(2) 

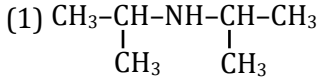
(3) 

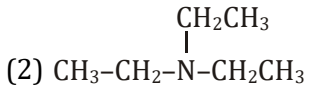
(4) 

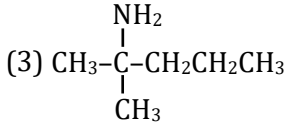
CAM071

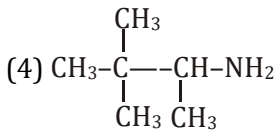
NEET(UG) 2019 (ODISHA)

10. The amine that reacts with Hinsberg's reagent to give an alkali insoluble product is:-

(1) 

(2) 

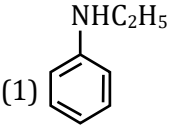
(3) 

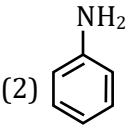
(4) 

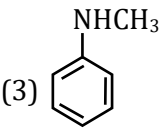
CAM072

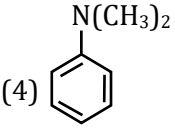
NEET(UG) 2020

11. Which of the following amine will give the carbylamine test?

(1) 

(2) 

(3) 

(4) 

CAM073

NEET(UG) 2020 (COVID-19)

12. Reaction of propanamide with ethanolic sodium hydroxide and bromine will give

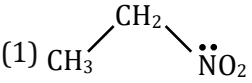
(1) Ethylamine (2) Methylamine

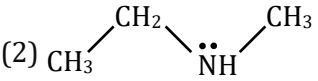
(3) Propylamine (4) Aniline

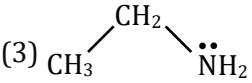
CAM074

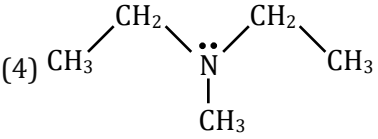
NEET(UG) 2021

13. Identify the compound that will react with Hinsberg's reagent to give a solid which dissolves in alkali :

(1) 

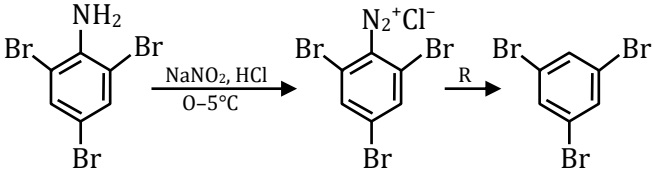
(2) 

(3) 

(4) 

CAM075

14. The reagent 'R' in the given sequence of chemical reaction is :



(1) H_2O (2) CH_3CH_2OH

(3) HI (4) CuCN/KCN

CAM076

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NEET(UG) 2022

15. Given below are two statements :

Statement-I : Primary aliphatic amines react with HNO_2 to give unstable diazonium salts.

Statement-II : Primary aromatic amines react with HNO_2 to form diazonium salts which are stable even above 300 K.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both Statement-I and Statement-II are incorrect.

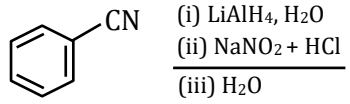
(2) Statement-I is correct but Statement-II is incorrect.

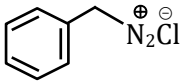
(3) Statement-I is incorrect but Statement-II is correct.

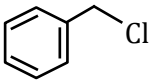
(4) Both Statement-I and Statement-II are correct.

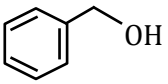
CAM131

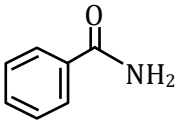
16. The product formed from the following reaction sequence is



(1) 

(2) 

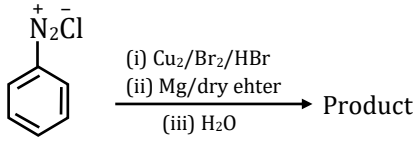
(3) 

(4) 

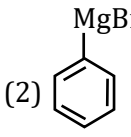
CAM132

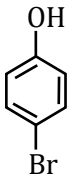
NEET(UG) 2023

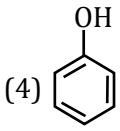
17. Identify the product in the following reaction:



(1) 

(2) 

(3) 

(4) 

CAM133

18. Which of the following reactions will NOT give primary amine as the product?

(1) $CH_3CN \xrightarrow[(ii)H_3O^{\oplus}]{(i)LiAlH_4}$ Product

(2) $CH_3NC \xrightarrow[(ii)H_3O^{\oplus}]{(i)LiAlH_4}$ Product

(3) $CH_3CONH_2 \xrightarrow[(ii)H_3O^{\oplus}]{(i)LiAlH_4}$ Product

(4) $CH_3CONH_2 \xrightarrow{Br_2/KOH}$ Product

CAM134

19. Choose the **correct** sequence of reagents in the conversion of 4-nitrotoluene to 2-bromotoluene.

(1) $NaNO_2/HCl$; Sn/HCl; Br_2; H_2O/H_3PO_2

(2) Sn/HCl; $NaNO_2/HCl$; Br_2; H_2O/H_3PO_2

(3) Br_2; Sn/HCl; $NaNO_2/HCl$; H_2O/H_3PO_2

(4) Sn/HCl; Br_2; $NaNO_2/HCl$; H_2O/H_3PO_2

CAM135

NEET(UG) 2024

20. Given below are two statements :

Statement-I : Aniline does not undergo Friedel- Crafts alkylation reaction

Statement-II : Aniline cannot be prepared through Gabriel synthesis.

In the light of the above statements, choose the *correct* answer from the options given below :

(1) Both Statement-I and Statement-II are true.

(2) Both Statement-I and Statement-II are false.

(3) Statement-I is correct but Statement-II is false.

(4) Statement-I is incorrect but Statement-II is true.

CAM136

21. Identify the major product C formed in the following reaction sequence:

$CH_3 - CH_2 - CH_2 - I \xrightarrow{NaCN} A \xrightarrow[\text{Partial hydrolysis}]{OH^-} B \xrightarrow[Br_2]{NaOH} \underset{\text{(Major)}}{C}$

(1) propylamine

(2) butylamine

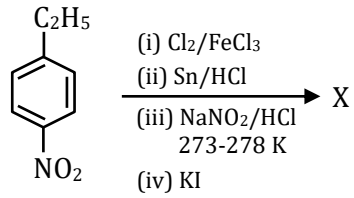
(3) butanamide

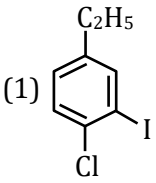
(4) α – bromobutanoic acid

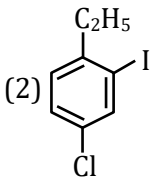
CAM137

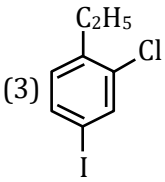
NEET(UG) 2024 (Re-examination)

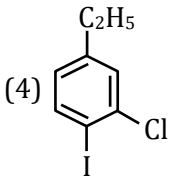
22. The major product X formed in the following reaction sequence is :



(1) 

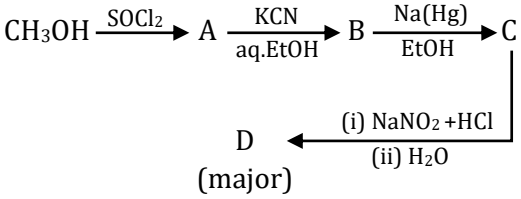
(2) 

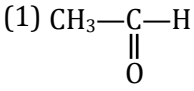
(3) 

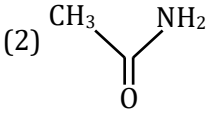
(4) 

CAM138

23. The major product D formed in the following reaction sequence is:



(1) 

(2) 

(3) CH_3CH_2OH

(4) CH_3CH_2Cl

CAM139

24. The compound that does not undergo Friedel-Crafts alkylation reaction but gives a positive carbylamine test is :

(1) aniline (2) pyridine

(3) N-methylaniline (4) triethylamine

CAM140

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**2**	**4**	**2**	**2**	**1**	**4**	**1**	**2**	**1**	**2**	**1**	**3**	**2**	**2**
Question	16	17	18	19	20	21	22	23	24						
Answer	**3**	**1**	**2**	**3**	**1**	**1**	**3**	**3**	**1**						

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EXERCISE-III (Analytical Questions) Master Your Understanding

1. $A \xrightarrow{NH_3} B \xrightarrow{\Delta} C \xrightarrow{Br_2+KOH} CH_3CH_2NH_2$

A is :-

(1) CH_3COOH (2) $CH_3CH_2CH_2COOH$

(3) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COOH$ (4) CH_3CH_2COOH

CAM077

2. $C_2H_5Br \xrightarrow{AgCN} A \xrightarrow{H_3O^+} HCOOH + B;$

$B \xrightarrow[KOH]{CHCl_3} A \xrightarrow{Reduction} C$

A, B, C respectively in the above sequence are

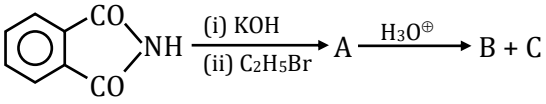
(1) Ethane amine, methane nitrile and diethyl amine

(2) Carbyl amino ethane, ethane amine and primary amine

(3) Ethyl isocyanide, ethyl amine and methyl isocyanate

(4) Carbylamino ethane, ethanamine and ethyl methyl amine

CAM080

3. 

B and C in the above sequence are

(1) Benzoic acid + aniline

(2) Phthalic acid + ethylamine

(3) Phthalic acid + aniline

(4) Benzoic acid + ethylamine

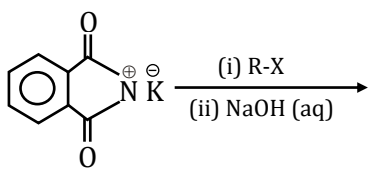
CAM081

4. Which of following will not gives only primary amine

(1) $C_2H_5X \xrightarrow{NH_3(Alc)}$

(2) $CH_3-CN \xrightarrow[C_2H_5OH]{Na(Hg)}$

(3) $CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-NH_2 \xrightarrow[(ii)\ H_2O]{(i)\ LiAlH_4}$

(4) 

CAM092

5. Which one of the following tests can be used to identify primary amino group in a given organic compound

(1) Iodoform test

(2) Victor Meyer's test

(3) Carbylamine reaction

(4) Libermann's reaction

CAM086

6. What is decreasing order of H-bonding in water for alkyl substituted ammonium cation?

(1) $R\overset{\oplus}{N}H_3 > R_2\overset{\oplus}{N}H_2 > R_3\overset{\oplus}{N}H$

(2) $R_3\overset{\oplus}{N}H > R_2\overset{\oplus}{N}H_2 > R\overset{\oplus}{N}H_3$

(3) $R_2\overset{\oplus}{N}H_2 > R_3\overset{\oplus}{N}H > R\overset{\oplus}{N}H_3$

(4) $R_2\overset{\oplus}{N}H_2 > R\overset{\oplus}{N}H_3 > R_3\overset{\oplus}{N}H$

CAM095

7. $A \xrightarrow{H_2NOH} B \xrightarrow{reduction} C \xrightarrow{NOCl} CH_3CH_2Cl,$

A and C in the above sequence respectively are:-

(1) Methanal, Methylamine

(2) Acetone, ethanamine

(3) Ethanal, dimethylamine

(4) Acetaldehyde, ethylamine

CAM083

8. Basic nature of amine in aqueous medium can be explained by :-

(1) Inductive effect (2) Solvation effect

(3) Steric hinderance (4) All of these

CAM093

9. The correct set of the products obtained in the following reactions

$RCN \xrightarrow{reduction} (A),\ RCN \xrightarrow[(ii)H_2O]{(i)\ CH_3MgBr} (B),$

$RNC \xrightarrow{hydrolysis} (C),\ RNH_2 \xrightarrow{HNO_2} (D)$

The answer is

	A	B	C	D
(1)	2°Amine	Methyl ketone	1°Amine	Alcohol
(2)	1°Amine	Methyl ketone	1°Amine	Alcohol
(3)	2°Amine	Methyl ketone	2°Amine	Acid
(4)	1°Amine	Methyl ketone	2° Amine	Aldehyde

CAM078

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10. Which of following reaction is not acylation ?

(1) $C_2H_5NH_2 + CH_3-\overset{\overset{O}{\|}}{C}-Cl \longrightarrow$

(2) $C_2H_5NH_2 + CH_3-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-CH_3 \longrightarrow$

(3) $C_2H_5NH_2 + CH_3\overset{\overset{O}{\|}}{C}-OCH_3 \longrightarrow$

(4) $C_2H_5NH_2 + CH_3-\overset{\overset{O}{\|}}{C}-OH \longrightarrow$

CAM096

11. Which of following will not give acylation reaction?

(1) $H_5C_2 - NH_2$ (2) $H_5C_2 - NH - C_2H_5$

(3) $C_2H_5-\underset{\underset{C_2H_5}{|}}{\overset{\overset{C_2H_5}{|}}{N}}$ (4) $H_5C_6 - NH_2$

CAM097

12. Which of following is not correct match?

(1) $H_5C_2NH_2 + CH_3-\overset{\overset{O}{\|}}{C}-Cl \longrightarrow$

$H_5C_2-NH-\overset{\overset{O}{\|}}{C}-CH_3+HCl$

(Acetylation)

(2) $H_5C_2NH_2 + H_5C_6-\overset{\overset{O}{\|}}{C}-Cl \longrightarrow$

$H_5C_2-NH-\overset{\overset{O}{\|}}{C}-C_6H_5+HCl$

(Benzoylation)

(3) $(H_5C_2)_2 NH + CHCl_3 + 3KOH \longrightarrow H_5C_2-NC + 3KCl + 3H_2O$

(Carbyl amine reaction)

(4) $H_5C_2 - NH_2 + CH_3COOH \longrightarrow H_5C_2N^{\oplus}H_3\ CH_3COO^{-}$

(Salt formation or Neutralisation)

CAM098

13. Which of following can be use to distinguish primary, secondary and tertiary amine?

(1) Carbyl amine reaction

(2) Reaction with Nitrous acid

(3) Reaction with Hinsberg's reagent followed by NaOH

(4) Both (2) and (3)

CAM099

14. Which of following is insoluble in alkali?

(1) $H_5C_6-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-\underset{\underset{C_2H_5}{|}}{N}-C_2H_5$

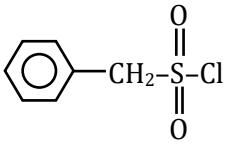
(2) $H_5C_6-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-\underset{\underset{H}{|}}{N}-CH_3$

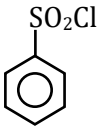
(3) $H_5C_6-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-\underset{\underset{H}{|}}{N}-C_2H_5$

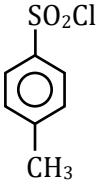
(4) All of these

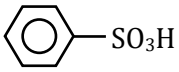
CAM100

15. These days benzene sulphonyl chloride replaced by which compound in Hinsberg test :-

(1) 

(2) 

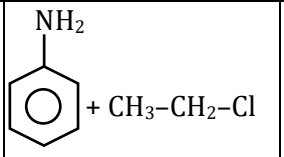
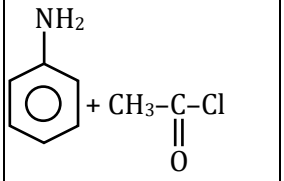
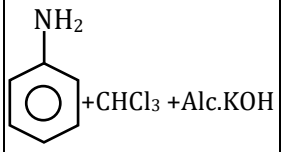
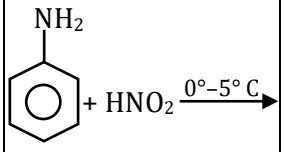
(3) 

(4) 

CAM101

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16. Match the column

Column-I (Reaction)		Column-II (Process)	
(a)		(p)	Acetylation
(b)		(q)	Alkylation
(c)		(r)	Diazotization
(d)		(s)	Carbylamine reaction
		(t)	Halogenation

Correct match is

(1) (a) → (q), (b) → (p), (c) → (s), (d) → (r)

(2) (a) → (t), (b) → (p), (c) → (s), (d) → (r)

(3) (a) → (q), (b) → (p), (c) → (t), (d) → (s)

(4) (a) → (t), (b) → (q), (c) → (s), (d) → (r)

CAM117

17. The end–product in the reaction sequence would be :

Ethyl amine $\xrightarrow{HNO_2}$ A $\xrightarrow{PCl_5}$ B $\xrightarrow{NH_3}$ C

(1) Ethyl cyanide (2) Ethyl amine

(3) Methyl amine (4) Acetamide

CAM082

18. In the reaction sequence A, B and C are :-

$A \xrightarrow[HCl]{Sn} B \xrightarrow[0°C]{NaNO_2/HCl} C \xrightarrow{H_2O/\Delta} C_6H_5OH$

(1) Benzene, nitrobenzene, aniline

(2) Nitrobenzene, aniline and azo–compound

(3) Nitrobenzene, benzene, aniline

(4) Benzene, amino compound, aniline

CAM085

19. $CH_3Cl \xrightarrow{KCN} \xrightarrow{+4H} A \xrightarrow{NaNO_2/HCl}$ Ethanol+?

Apart from ethanol as the main product, the other products would be

(1) $H_2C=CH_2$

(2) $CH_3CH_2–ONO$

(3) $CH_3CH_2–Cl$

(4) All the above

CAM087

20. Identify B, X and R respectively in the following sequence of reactions

$C_2H_5MgBr \xrightarrow{ClCN} A \xrightarrow{H_3O^+} \boxed{B}$,

$CH_3COCH_3 \xrightarrow[NaOH]{I_2} \boxed{X} \xrightarrow{Ag} CH\equiv CH$

$C_6H_5NH_2 \xrightarrow[0–5°C]{NaNO_2/HCl} P \xrightarrow{CuCN} Q \xrightarrow[EtOH]{Na} \boxed{R}$

(1) C_2H_5COOH, CHI_3, $C_6H_5CH_2NH_2$

(2) C_2H_5COOH, CH_3I, C_6H_5COOH

(3) $C_2H_5CH_2NH_2$, CH_3I, C_6H_5COOH

(4) C_2H_5COOH, C_2H_5I, $C_6H_5CONH_2$

CAM088

21. Total structural isomeric amines possible for molecular formula $C_4H_{11}N$

(1) 7 (2) 8

(3) 9 (4) 6

CAM090

22. Which of following is correct regarding basic nature:-

(1) $R – NH_2 > R_2NH > R_3N$ (Gaseous phase)

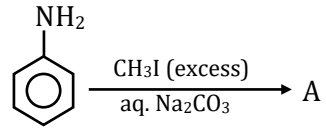
(2) $CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N$ (Aqueous solutions)

(3) $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5–NH_2 > NH_3$ (Aqueous solutions)

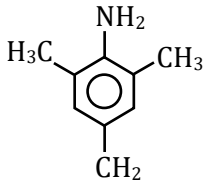
(4) $(C_2H_5)_2NH > PhNH_2 > Ph – NH – CH_3$ (Aqueous solution)

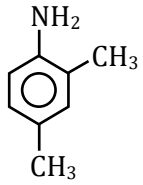
CAM094

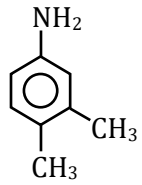
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23. 

What will be "A"

(1) 

(2) 

(3) 

(4) $Ph-\overset{\oplus}{N}(CH_3)_3\overset{\ominus}{I}$

CAM102

24. Total number of isomers of molecular formula C_3H_9N which will liberate N_2 gas on treatment with nitrous acid?

(1) 2 (2) 3 (3) 4 (4) 1

CAM103

25. Match the column

	Column-I (Compound)		Column-II (Pkb)
(i)	N,N-Dimethyl aniline	(a)	9.38
(ii)	Benzenamine	(b)	8.92
(iii)	N-Methyl methanamine	(c)	3.38
(iv)	Methanamine	(d)	3.27

Correct match is -

(1) (i) → b, (ii) → a, (iii) → c, (iv) → d

(2) (i) → c, (ii) → d, (iii) → a, (iv) → b

(3) (i) → b, (ii) → a, (iii) → d, (iv) → c

(4) (i) → d, (ii) → a, (iii) → c, (iv) → b

CAM113

26. Some statements are given about following order of basic strength in aqueous medium.

$(CH_3)_2NH > CH_3-NH_2 > (CH_3)_3N > NH_3$

(a) More substituted ammonium ion gets stablised due to +I effect of alkyl group.

(b) The greater the size of ammonium ion, more will be solvation, so more stable ion so more basicity.

(c) More the alkyl groups, more steric hindrance to H-bonding in ammonium ion.

(d) Basic strength in aqueous medium is explained by more than one type of effects.

Correct statements is/are

(1) a, b, c, d (2) a, b, d

(3) a, c, d (4) a and d only

CAM115

27. Which of following is water insoluble and stable at room temperature.

(1) $C_6H_5N_2^{\oplus}Cl^{\ominus}$ (2) $C_6H_5N_2^{\oplus}BF_4^{\ominus}$

(3) $C_6H_5N_2^{\oplus}HSO_4^{\ominus}$ (4) All

CAM104

28. Which of following reaction do not displace N_2 from benzenediazonium salt?

(1) $ArN_2^{\oplus}Cl^- \xrightarrow{Cu_2Cl_2/HCl}$

(2) $ArN_2^{\oplus}Cl^- \xrightarrow{Cu/HCl}$

(3) $ArN_2^{\oplus}Cl^- + KI \longrightarrow$

(4) $ArN_2^{\oplus}Cl^- + HBF_4 \xrightarrow{0-5°C}$

CAM105

29. Which of following group can not be introduced in the benzene ring by sandmeyer reaction?

(1) –Cl (2) –Br (3) –CN (4) –I

CAM106

30. Diazo coupling reaction can be use to distinguish

(1) Aniline and Benzyl amine

(2) Ethylamine and N-methyl aniline

(3) Aniline and Phenol

(4) All of these

CAM109

31. $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-OH$ is obtained from following method

(1) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-Cl + H_2O$

(2) $CH_3CHO, CH_3-CH_2MgBr, H_2O$

(3) $H-CHO, CH_3-\underset{\underset{CH_3}{|}}{CH}MgBr, H_2O$

(4) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-NH_2 + HNO_2$

CAM112

32. Using Fe/HCl, which one of the following reaction is possible [Here $\phi = C_6H_5$]

(1) $\phi-NO_2 \rightarrow \phi-NH-NH-\phi$

(2) $\phi-NO_2 \rightarrow \phi-NH_2$

(3) $\phi-NO_2 \rightarrow \phi-NHOH$

(4) $\phi-NO_2 \rightarrow \phi-N=N-\phi$

CAM089

33. What is correct sequence of reagent to form 2,4,6-tibromofluorobenzene from Aniline?

(1) $NaNO_2/HCl$ (0-5°C), HBF_4/Δ, Br_2/H_2O

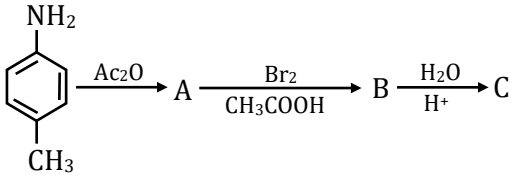
(2) Br_2/H_2O, $NaNO_2/HCl$ (0-5°C), HBF_4/Δ

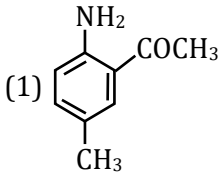
(3) $NaNO_2/HCl$ (25°C), HBF_4/Δ, Br_2/H_2O

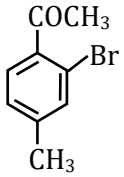
(4) Br_2/H_2O, $NaNO_2/HCl$ (0-5°C), HBF_4 (Room temperature)

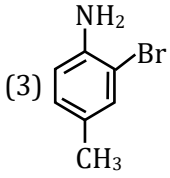
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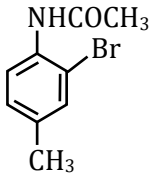
34. The final product C, obtained in this reaction would be



(1) 

(2) 

(3) 

(4) 

CAM079

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**4**	**2**	**1**	**3**	**1**	**4**	**4**	**2**	**4**	**3**	**3**	**4**	**1**	**3**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**1**	**2**	**2**	**4**	**1**	**2**	**3**	**4**	**1**	**3**	**3**	**2**	**4**	**4**	**4**
Question	31	32	33	34											
Answer	**3**	**2**	**2**	**3**											

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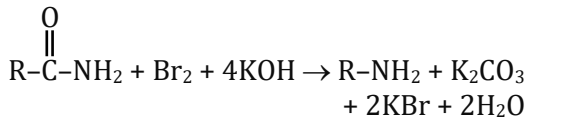
Amines

SOLUTIONS

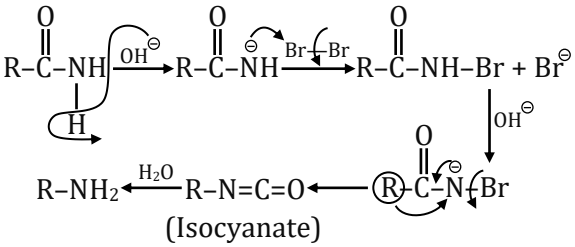
Exercise-I (Conceptual Questions)

1. Ans. (4)

Hoffmann bromamide degradation reaction

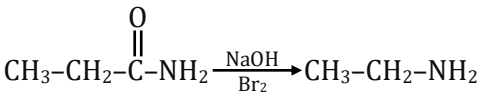


Mechanism:-



Therefore, R–NC is not formed in the Hoffmann Bromamide degradation reaction.

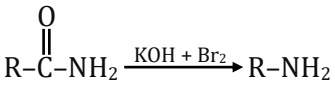
2. Ans. (2)



Since product obtained is amine which is basic in nature, it turns red litmus to blue.

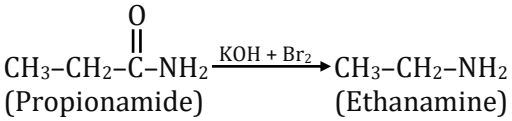
3. Ans. (3)

In Hoffmann bromamide degradation reaction.



The total number of carbons in product is reduced by one carbon.

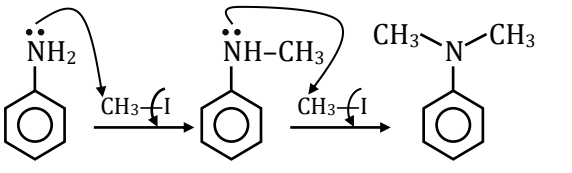
∴ To obtain Ethanamine as product,



(Propionamide) (Ethanamine)

4. Ans. (1)

Tertiary amine :- when nitrogen is directly attached with three carbons, it is called tertiary amine

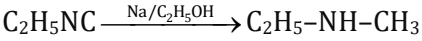


(Aniline) (Tertiary Amine)

5. Ans. (3)

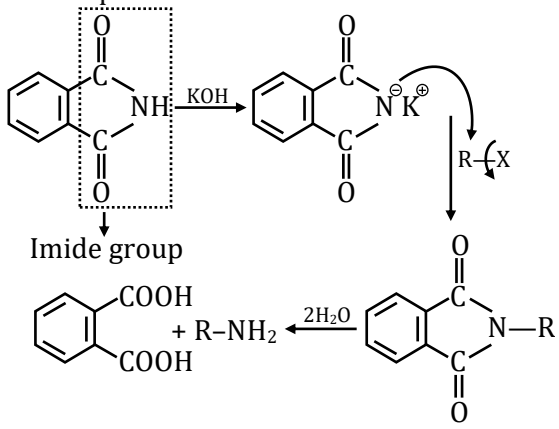
Reduction of Alkyl Isocyanides gives secondary amines

For Example,



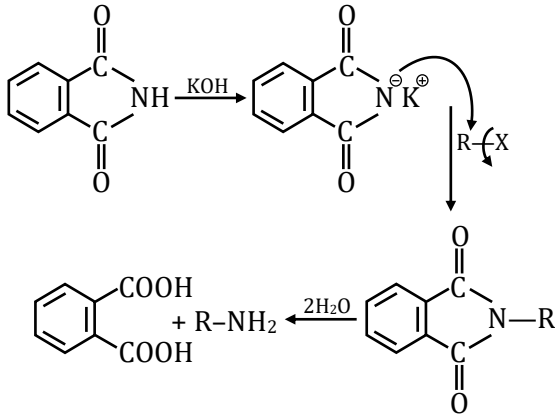
6. Ans. (3)

Gabriel phthalimide reaction

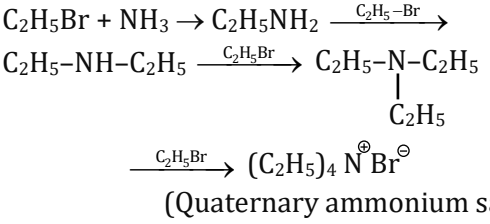


7. Ans. (3)

Gabriel phthalimide reaction is used in the formation of pure Aliphatic primary amines.



8. Ans. (3)



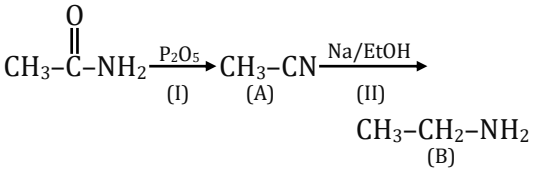
(Quaternary ammonium salt)

The above reaction is called ammonolysis and also it involves attack of ammonia as a Nucleophile which substitutes –Br, Therefore, it is also an example of nucleophilic substitution.

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9. Ans. (3)

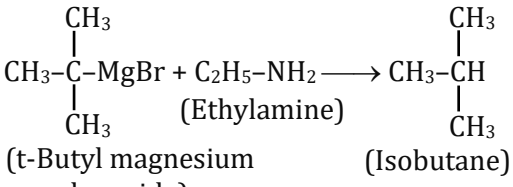


Therefore, the given reaction II is Mendius reaction which involves the reduction of the organic compound containing the cyanide group in the presence of an alkali metal and alcohol which results in the formation of amines.

10. Ans. (3)

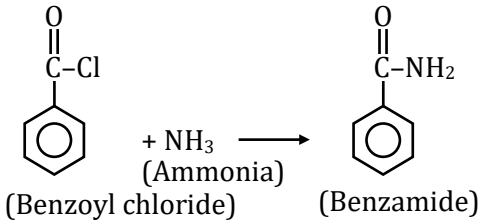
$C_2H_5–NCO \xrightarrow[\text{Hydrolysis}]{\text{Alkaline}} C_2H_5–NH_2$ (Ethylamine)

When ethylamine is treated with t-butyl magnesium bromide, the compound obtained will be Isobutane.

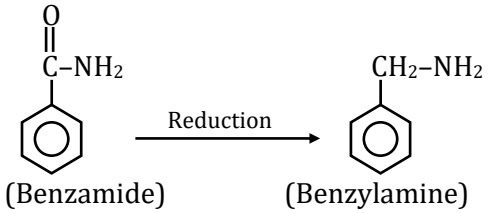


11. Ans. (3)

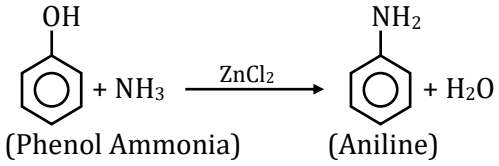
Option 1 : Benzoyl chloride and ammonia



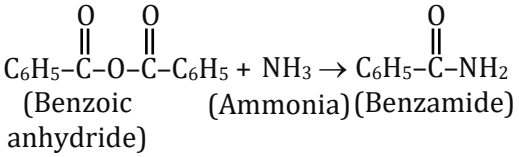
Option 2 : Reduction of benzamide



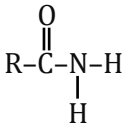
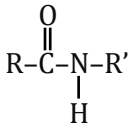
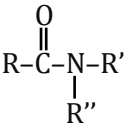
Option 3 : Phenol and ammonia in presence of $ZnCl_2$



Option 4 : Benzoic Anhydride and ammonia



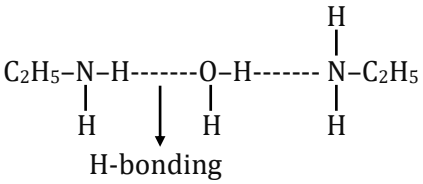
12. Ans. (3)

Primary amide	Secondary amide	Tertiary amide
		
Two H can participate in H-bonding with another molecule	Only one H can participate in H-bonding	No H-bonding

Therefore, primary amides have maximum boiling point

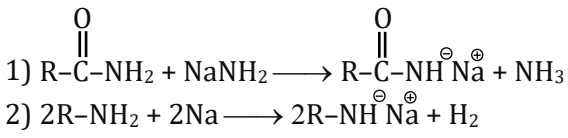
13. Ans. (3)

Ethylamine in water shows H-bonding which involves formation of H-bond with water.



Therefore, solubility of ethylamine is due to formation of H–bonding with water.

14. Ans. (3)



Therefore, the hydrogen attached to nitrogen is released in both of the above reaction.

15. Ans. (2)

$C_2H_5–NH–C_2H_5$ is a secondary amine, therefore, it will not give carbylamine test.

16. Ans. (4)

Chloroform and ethanolic KOH can be used as a reagent in the following reactions.

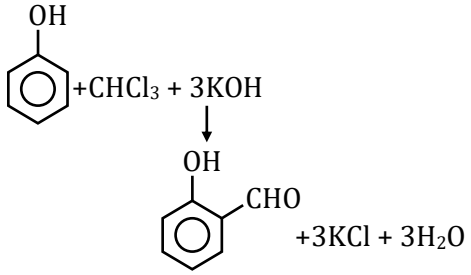
(a) Hoffmann carbylamine reaction

$C_6H_5NH_2 + CHCl_3 + 3KOH \longrightarrow C_6H_5NC + 3KCl + 3H_2O$

(b) Hoffmann degradation reaction

$$R-\overset{\overset{O}{\|}}{C}-NH_2 + Br_2 + 4KOH \longrightarrow R-NH_2 + K_2CO_3 + 2KBr + 2H_2O$$

(c) Reimer-Tiemann reaction



(d) Hoffmann mustard oil reaction

$$C_6H_5NH_2 + CS_2 \xrightarrow[\Delta]{HgCl_2} C_6H_5-N=C=S$$

Therefore, option 4 only a & c is correct.

17. Ans. (3)

$$R-C\equiv N \xrightarrow[\text{Alcohol}]{Na} R-CH_2-NH_2$$

(Mendius Reaction)

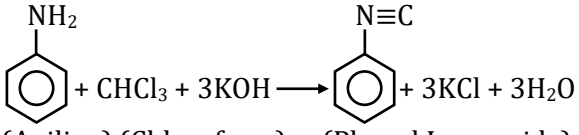
18. Ans. (1)

Carbylamine Test

$$R-NH_2 + \underbrace{CHCl_3} + 3KOH \rightarrow R-NC + 3KCl + 3H_2O$$

(Primary amine) (Alkyl Isocyanide)

19. Ans. (1)



(Aniline) (Chloroform) (Phenyl Isocyanide)

20. Ans. (2)

Acidic nature of amines is shown by its acidic hydrogen.

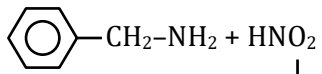
For example :-

$$2R-NH_2 + 2Na \longrightarrow 2R-NH^{\ominus}Na^{\oplus} + H_2$$

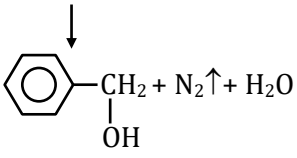
21. Ans. (2)

$$R-NH_2 + HNO_2 \longrightarrow R-OH + N_2\uparrow + H_2O$$

(1° amine)

$\therefore$  $-CH_2-NH_2 + HNO_2$

(benzylamine)



(Alcohol)

22. Ans. (4)

$CH_3\overset{\oplus}{N}H_3\overset{\ominus}{Cl}$ with react with $NaHCO_3$ to given H_2CO_3 which further liberates CO_2 and H_2O

$$CH_3\overset{\oplus}{N}H_3\overset{\ominus}{Cl} + NaHCO_3 \longrightarrow CH_3NH_2 + NaCl + \underset{\downarrow\quad\downarrow}{\overset{H_2CO_3}{}} \; H_2O \quad CO_2$$

Note:- Amines and Alcohols are less acidic than H_2CO_3, therefore they will not react with $NaHCO_3$.

23. Ans. (4)

When primary amines are treated with ketone, schiff's base is obtained.

$$R-NH_2 + R'-\underset{\underset{O}{\|}}{C}-R'' \longrightarrow R'-\underset{\underset{R''}{|}}{C}=N-R$$

(Primary amine) (Ketone) (Schiff's base)

24. Ans. (4)

Since $R\overset{\oplus}{N}H_3\overset{\ominus}{Cl}$ is a acidic salt, it turns blue litmus into red.

Note:- R–OH, $R-NH_2$ & $R-\overset{\oplus}{N}H_3\overset{\ominus}{O}H$ are basic in nature.

25. Ans. (3)

In the given reaction

$$C_6H_5NH_2 + HAuCl_4 \longrightarrow [C_6H_5\overset{\oplus}{N}H_3]Au\overset{\ominus}{Cl}_4$$

Here, $-NH_2$ group of Aniline donates $e^{\ominus}$ Pair to $HAuCl_4$, Therefore It acts as Lewis base. So, Aniline shows basic nature.

26. Ans. (4)

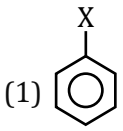
Tollen's reagent $\rightarrow$ Ammonical silver nitrate solution

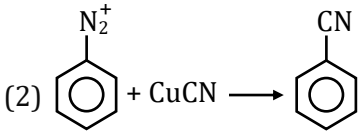
Aldehydes, Alpha Hydroxy Ketones and Formic acid are able to give Tollen's test.

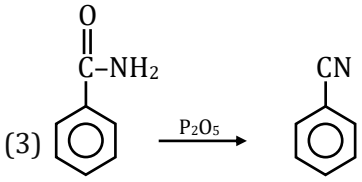
$C_6H_5NH_2$ will not reduce Tollen's reagent.

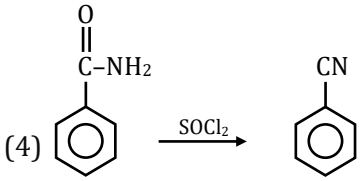
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27. Ans. (1)

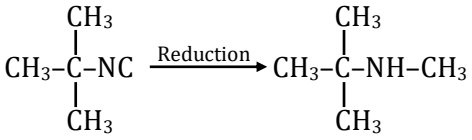
(1)  + KCN → No reaction

(2) 

(3) 

(4) 

28. Ans. (2)



(2°amine)

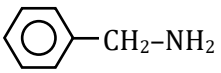
29. Ans. (2)

Nitrobenzene can't show F.C.R. so nitrobenzene can act as solvent in Friedel craft reaction.

Aniline form salt in product.

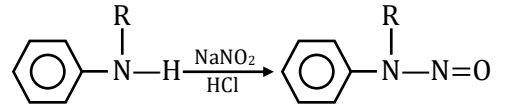
Toluene and Benzene show FCR so that can't be used as solvent in FCR.

30. Ans. (4)

For Diazo coupling reaction, we require diazonium salt to be coupled with highly activated aromatic compound like aniline phenol, β-Naphthol therefore,  does not show diazo coupling reaction due to +H effect of $-CH_2-NH_2$ group as it activate the ring less as compared to other given groups.

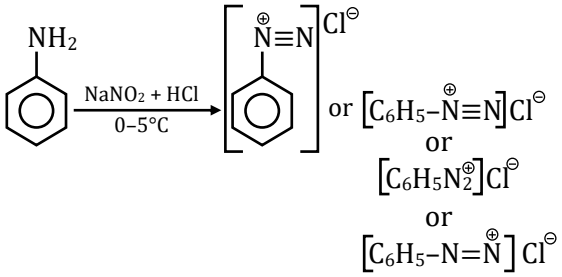
31. Ans. (3)

Secondary amines form N–nitroso derivative when treated with $NaNO_2$ and HCl

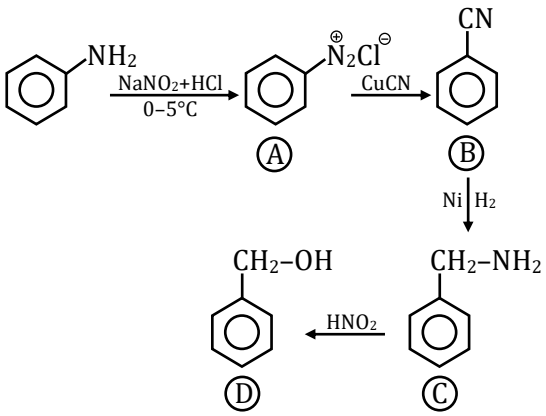


(2° amine) (N–nitroso derivative)

32. Ans. (4)

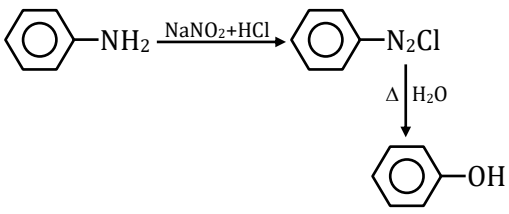


33. Ans. (1)



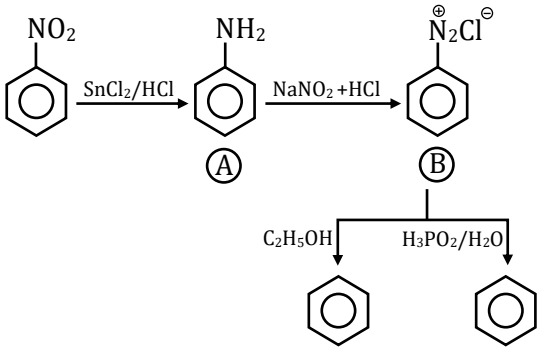
∴ Structure of product D is $C_6H_5-CH_2-OH$

34. Ans. (2)



∴ X is $-NH_2$, Y is $-OH$ → Both are O & P directing.

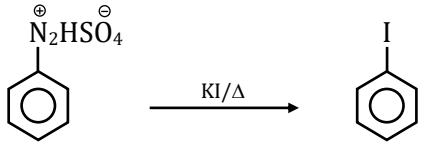
35. Ans. (3)



∴ From B, Benzene can be obtained by using ethanol as will as H_3PO_2.

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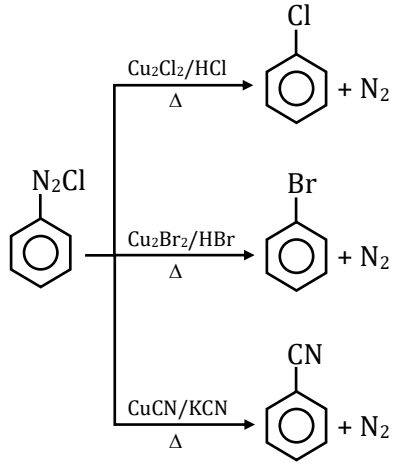
36. Ans. (3)



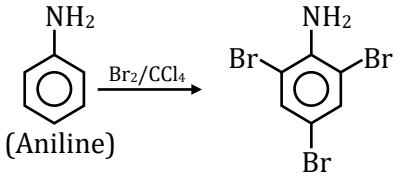
(Benzenediazonium acid sulphate) (Iodobenzene)

37. Ans. (1)

In Sandmeyer's reaction, –N=N–X group of diazonium salt is replaced by Halide group.



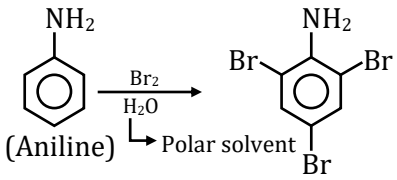
38. Ans. (2)



(Aniline) (2,4,6- Tribromoaniline)

Note:- Since, $-NH_2$ is a highly Activating group it is O & P directing and we get trisubstituted aniline.

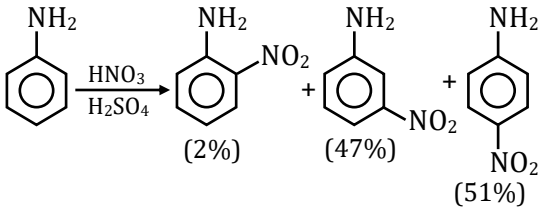
39. Ans. (3)



(Aniline) (2,4,6- Tribromoaniline)

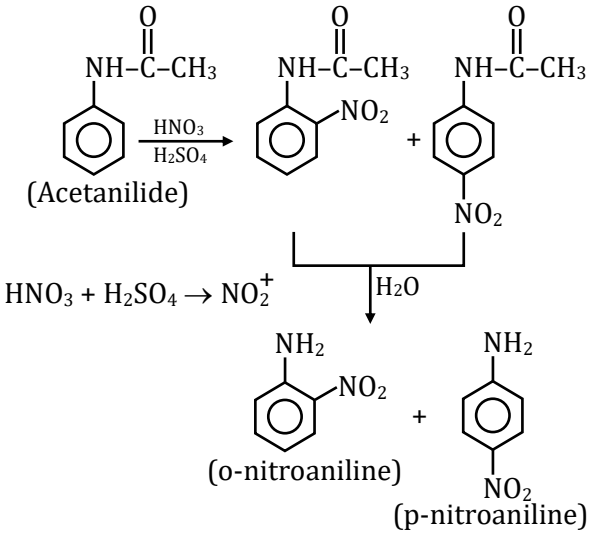
Note:- Since, $-NH_2$ is a highly Activating group it is O & P directing and we get trisubstituted aniline.

40. Ans. (4)

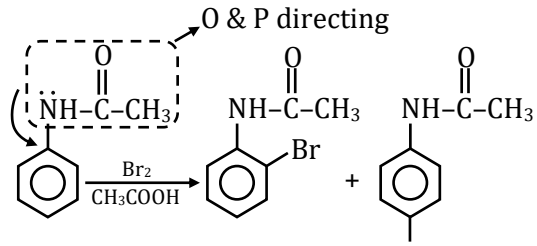


Meta substitution is also take place due to the formation of anilium ion.

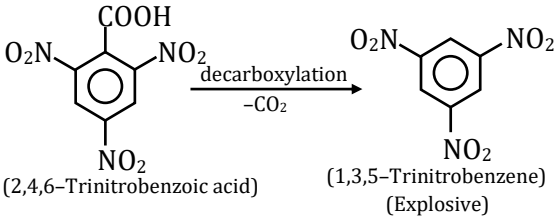
41. Ans. (3)



42. Ans. (3)



43. Ans. (1)

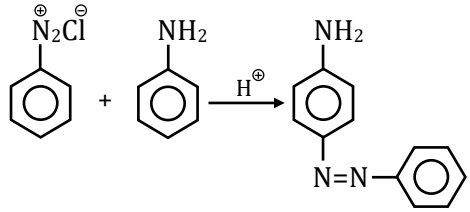


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Exercise-II (Previous Year Questions)

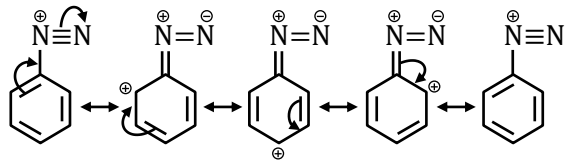
1. Ans. (4)



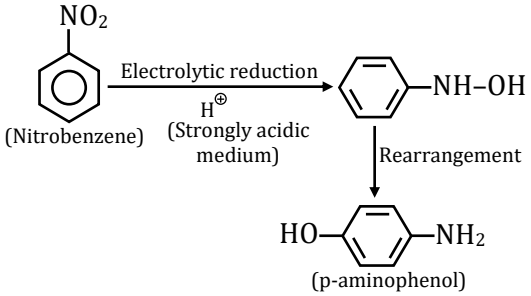
(Yellow colour dye)

2. Ans. (2)

Among all the given four options, $C_6H_5\ N_2^+X^-$ is the most stable diazonium salt because it shows resonance.



3. Ans. (4)

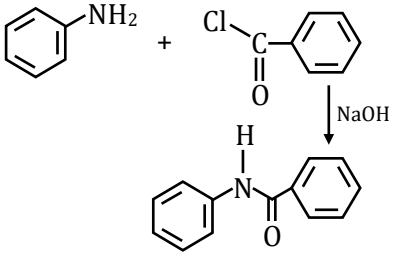


(Nitrobenzene)

(p-aminophenol)

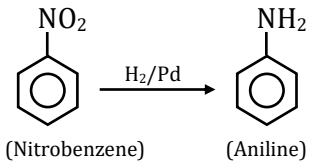
4. Ans. (2)

→ Benzoylation of amine is an example of Schotten-Baumann reaction.



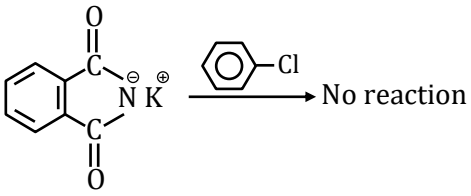
5. Ans. (2)

(1) Reduction of Nitrobenzene with H_2/Pd in ethanol



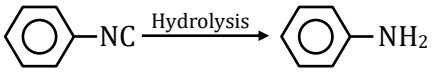
(Nitrobenzene) (Aniline)

(2) Potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution.

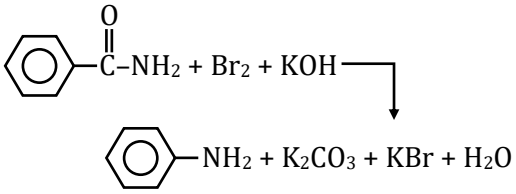


In Gabriel pthalimide reaction, Aniline cannot be formed.

(3) Hydrolysis of phenyl isocyanide with acidic solution.

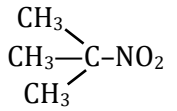


(4) Degradation of benzamide with bromine in alkaline solution.



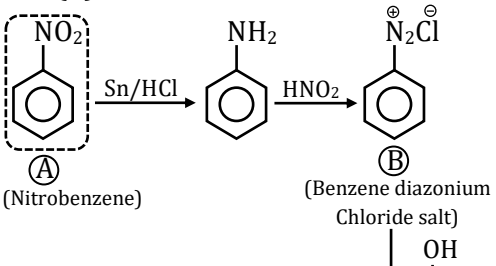
6. Ans. (1)

The nitro compounds which contains atleast one active H (α–H) can react with nitrous acid (HNO_2). Therefore,

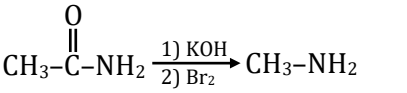


cannot react with nitrous acid.

7. Ans. (4)



8. Ans. (1)

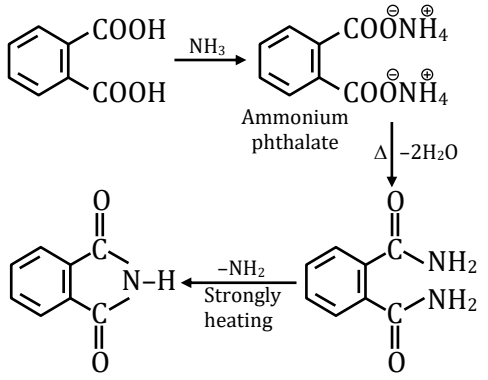


(Acetamide) (Methan amine)

∴ Hoffmann hypobromamide reaction is appropriate for above conversion.

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9. Ans. (2)



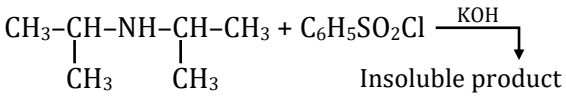
10. Ans. (1)

In Hinsberg method, mixture of amines is separated by using benzene sulphonyl chloride (Hinsberg reagent)

$C_6H_5SO_2Cl$ + 1° amine $\rightarrow$ Product $\xrightarrow{KOH}$ soluble (PPT.)

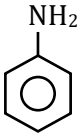
$C_6H_5SO_2Cl$ + 2° amine $\rightarrow$ Product $\xrightarrow{KOH}$ insoluble. (PPT.)

$\therefore$ Option 1 represents secondary amine

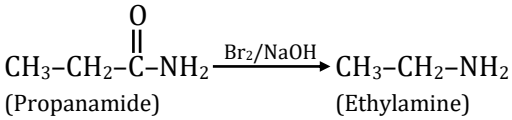


11. Ans. (2)

Carbylamine test is generally used to detect primary amines and at is also given by aniline.



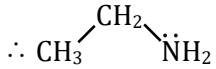
12. Ans. (1)



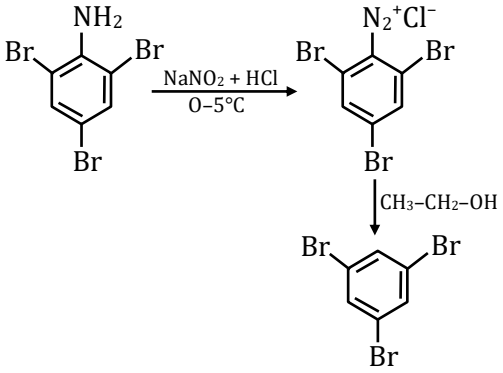
(Hoffmann bromamide degradation reaction)

13. Ans. (3)

When primary amine is treated with Hinsberg reagent, it gives soluble product with base.

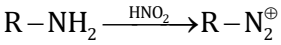
$\therefore$  will react with Hinsberg's reagent to give a solid which dissolves in alkali.

14. Ans. (2)

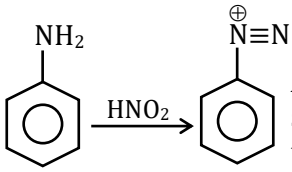


$\therefore$ R $\rightarrow$ $CH_3–CH_2–OH$

15. Ans. (2)

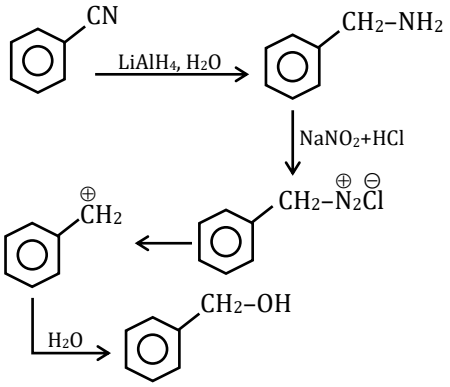


Alkyl diazonium ion (unstable)

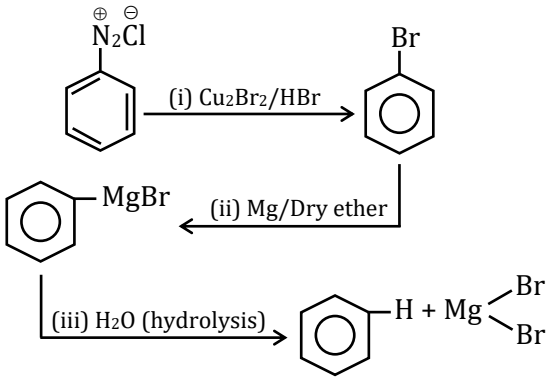


Aryl diazonium ion stable at low temperature (0–5°C)

16. Ans. (3)



17. Ans. (1)



18. Ans. (2)

(1) $CH_3-CN \xrightarrow[(ii)H_3O^{\oplus}]{(i)LiAlH_4} CH_3-CH_2-NH_2$

1° Amine

(2) $CH_3NC \xrightarrow[(ii)H_3O^{\oplus}]{(i)LiAlH_4} CH_3-NH-CH_3$

2° Amine

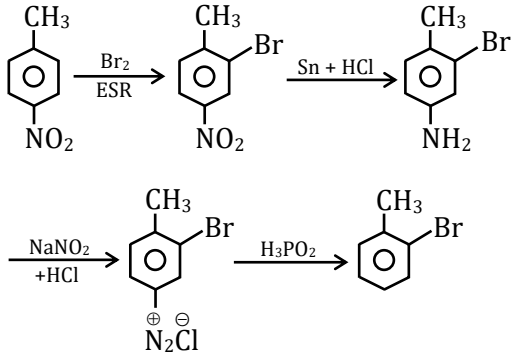
(3) $CH_3-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow[(ii)H_3O^{\oplus}]{(i)LiAlH_4} CH_3-CH_2-NH_2$

1° Amine

(4) $CH_3-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow{Br_2+OH^-} CH_3-NH_2$

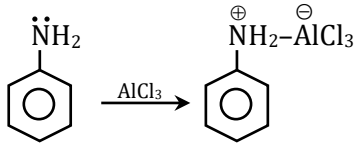
1° Amine

19. Ans. (3)



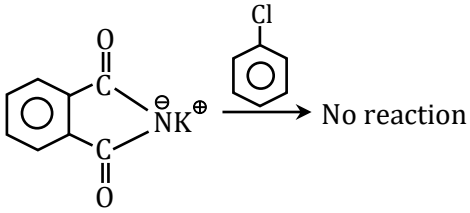
20. Ans. (1)

Statement-I :- Explanation : Aniline forms complex with Lewis acid ($AlCl_3$),

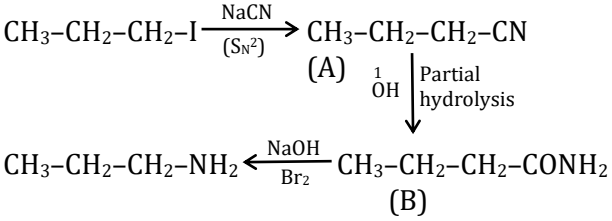


So statement-I is correct.

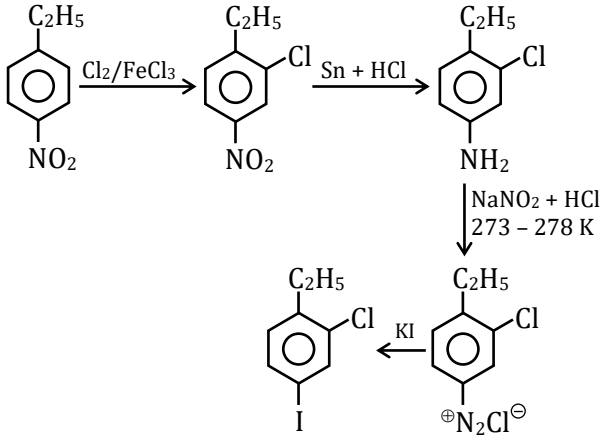
Statement-II :- Explanation : Aryl halide are not good substrate for nucleophilic substitution reaction so aniline does not give Gabriel synthesis.



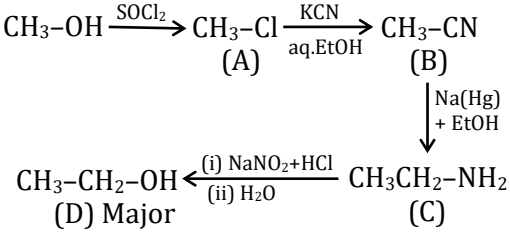
21. Ans. (1)



22. Ans. (3)



23. Ans. (3)



24. Ans. (1)

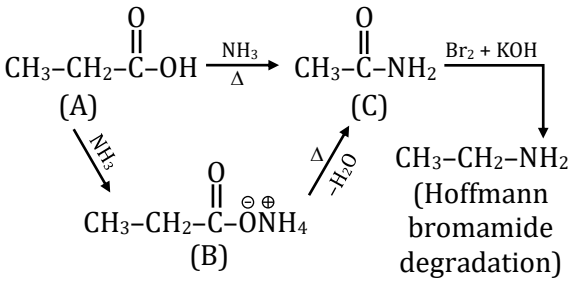
Aniline does not show FCR but gives positive carbyl amine test.

$Ph-NH_2 + CHCl_3 + 3KOH \xrightarrow{Heat}$

$Ph-NC + 3KCl + 3H_2O$

Exercise-III (Analytical Questions)

1. Ans. (4)



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2. Ans. (4)

$$C_2H_5Br \xrightarrow{AgCN} \underset{(A)}{C_2H_5NC} \xrightarrow{H_3O^+} HCOOH + \underset{(B)}{C_2H_5NH_2}$$

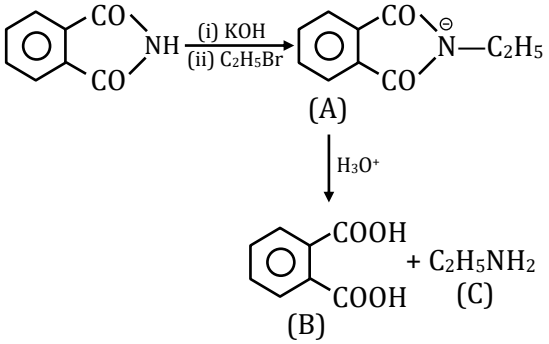
$$\underset{(B)}{C_2H_5NH_2} \xrightarrow[KOH]{CHCl_3} \underset{(A)}{C_2H_5NC} \xrightarrow{Reduction} \underset{(C)}{C_2H_5-NH-CH_3}$$

(A) ⟶ Carbylamino Ethane

(B) ⟶ Ethanamine

(C) ⟶ Ethyl methyl amine

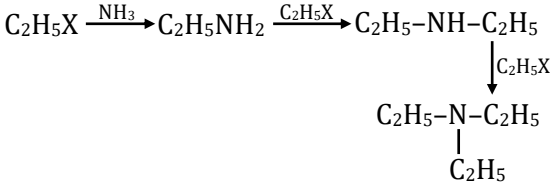
3. Ans. (2)



(B) ⟶ pthalic acid

(C) ⟶ Ethyl amine

4. Ans. (1)



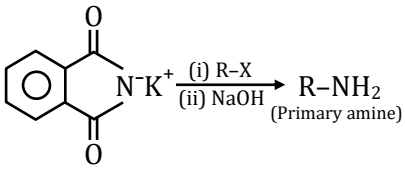
∴ In option 1, mixture of amine will be obtained.

→ $CH_3-CN \xrightarrow[EtOH]{Na(Hg)} CH_3-CH_2-NH_2$

(Primary amine)

→ $CH_3-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow[(ii)\ H_2O]{(i)\ LiAlH_4} CH_3-CH_2-NH_2$

(Primary amine)

→ 

∴ Option 1 is correct

5. Ans. (3)

→ Carbylamine reaction can be used to identify primary amino group in any given compound

→ Victor Meyer's is used to detect type of Alcohols.

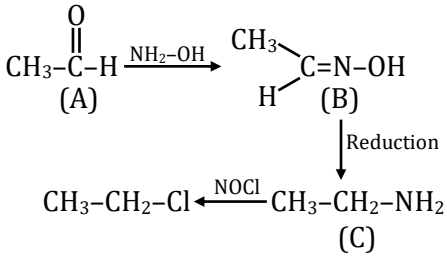
→ Iodoform test are the test given by aldehyde & ketones in which methyl keto $\left(CH_3-\overset{O}{\overset{\|}{C}}-\right)$ is present.

6. Ans. (1)

In amines, as the number of alkyl group on ammonium cation, the stability of ammonium action decreases because alkyl group provides steric hinderance to the water molecule from forming H–bonding in water for alkyl substituted ammonium cation will be

$$R\overset{\oplus}{N}H_3 > R_2\overset{\oplus}{N}H_2 > R_3\overset{\oplus}{N}H$$

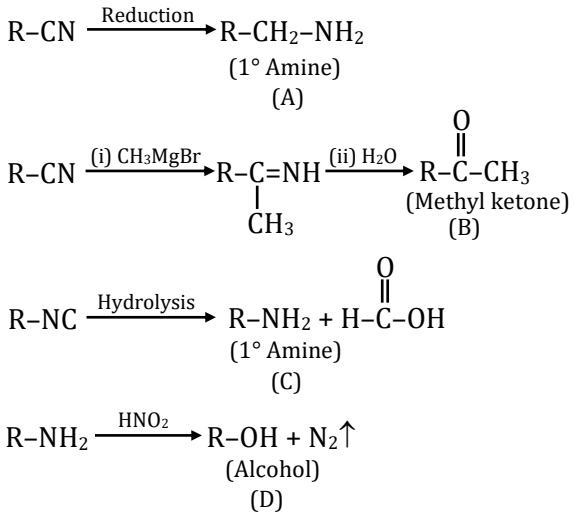
7. Ans. (4)



8. Ans. (4)

Basic nature of amine in aqueous medium can be explained by Inductive effect, solvation effect and steric hinderance.

9. Ans. (2)



10. Ans. (4)

(1) $C_2H_5NH_2 + CH_3-\overset{\overset{O}{\|}}{C}-Cl \rightarrow C_2H_5-NH-\overset{\overset{O}{\|}}{C}-CH_3$

(2) $C_2H_5NH_2 + CH_3-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-CH_3 \rightarrow CH_3-\overset{\overset{O}{\|}}{C}-NH-C_2H_5$

(3) $C_2H_5NH_2 + CH_3-\overset{\overset{O}{\|}}{C}-O-CH_3 \rightarrow CH_3-\overset{\overset{O}{\|}}{C}-NH-C_2H_5$

} Acylation

(4) $C_2H_5NH_2 + CH_3-\overset{\overset{O}{\|}}{C}-OH \rightarrow CH_3-COO^- + \overset{+}{C_2H_5}NH_3$ } No acylation

11. Ans. (3)

Tertiary amines are unable to give acylation reaction due to steric hinderance.

$\therefore$ In option 3, $C_2H_5-\underset{C_2H_5}{\overset{C_2H_5}{\underset{|}{\overset{|}{N}}}}$ represents tertiary amine, hence it will not give acylation reaction.

12. Ans. (3)

$(C_2H_5)_2NH + CHCl_3 + 3KOH \xrightarrow{X} C_2H_5-NC + 3KCl + 3H_2O$

It is incorrect reaction because $(C_2H_5)_2NH$ represents a secondary amine and carbylamine reaction is used to detect primary amines.

13. Ans. (4)

To distinguish primary, secondary and tertiary amine, following reaction can be used.

(a) Reaction with Nitrous acid

$R-NH_2 + HNO_2 \rightarrow R-OH + N_2\uparrow + H_2O$ (bubbles)

$R-NH_2 + HNO_2 \rightarrow R_2N-NO + H_2O$

Dialkyl nitrosoamine (bubbles)

$R_3N + HNO_2 \rightarrow R_3\overset{\oplus}{N}H\overset{-}{N}O_2$

Trialkyl ammonium Nitrate (Soluble in water)

(b) Reaction with Hinsberg reagant Following by NaOH

$C_6H_5SO_2Cl + 1°amine \rightarrow product \xrightarrow{NaOH}$ Soluble (ppt.)

$C_6H_5SO_2Cl + 2°amine \rightarrow product \xrightarrow{NaOH}$ Insoluble (ppt.)

$C_6H_5SO_2Cl + 3°amine \rightarrow$ No reaction

14. Ans. (1)

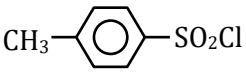
$C_6H_5SO_2Cl + 2°$ amine $\rightarrow$ product (ppt.) $\downarrow$ Alkali

Insoluble

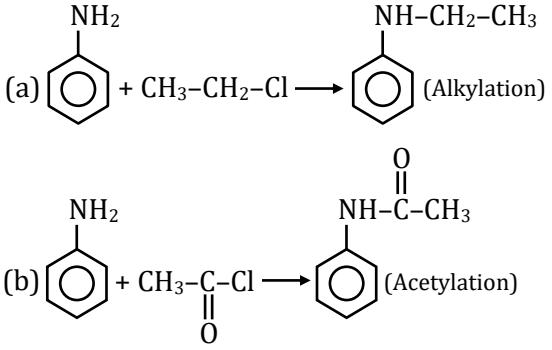
$\therefore$ $C_6H_5-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-N-C_2H_5$ represents 2° amine,

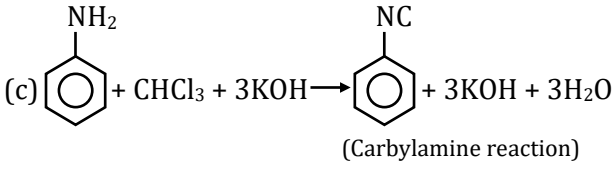
15. Ans. (3)

In Hinsberg test, Benzene sulphonyl chloride is replaced by p-toluene sulphonyl chloride.

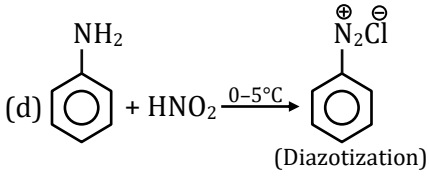


16. Ans. (1)





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(Diazotization)

(p) → Acetylation (q) → Alkylation

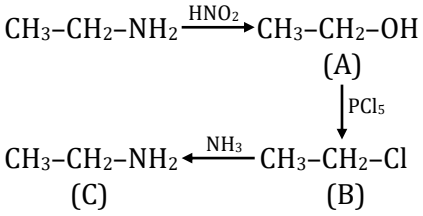
(r) → Diazotization (s) → Carbylamine reaction

(t) → Halogenation

∴ Correct answer is

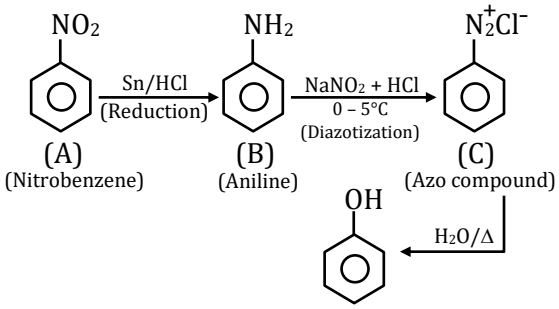
(a) → (q), (b) → (p), (c) → (s), (d) → (r)

17. Ans. (2)

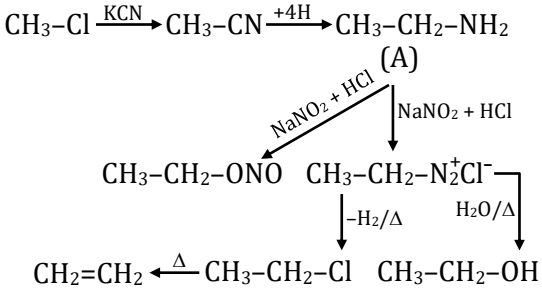


∴ End product (C) will be Ethylamine.

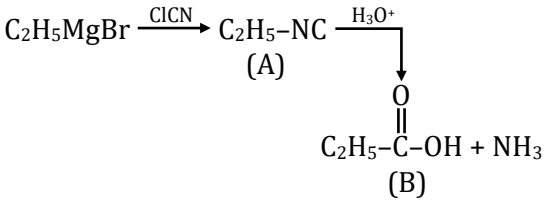
18. Ans. (2)

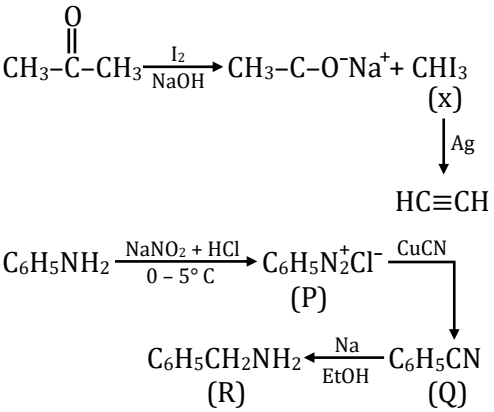


19. Ans. (4)



20. Ans. (1)





B → C_2H_5COOH

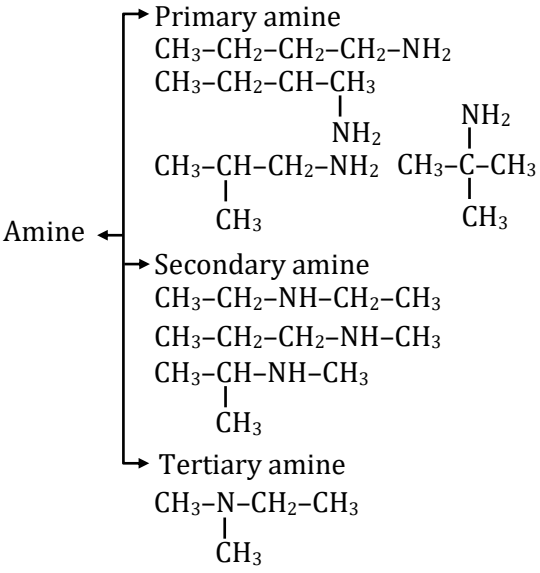
X → CHI_3

R → $C_6H_5CH_2NH_2$

21. Ans. (2)

Molecular Formula = $C_4H_{11}N$

D.U. = 0



∴ Total structural isomeric amines for $C_4H_{11}N$ = 8

22. Ans. (3)

Basic strength of amins in gaseous solution

$R_3–N > R_2NH > R–NH_2 > NH_3$

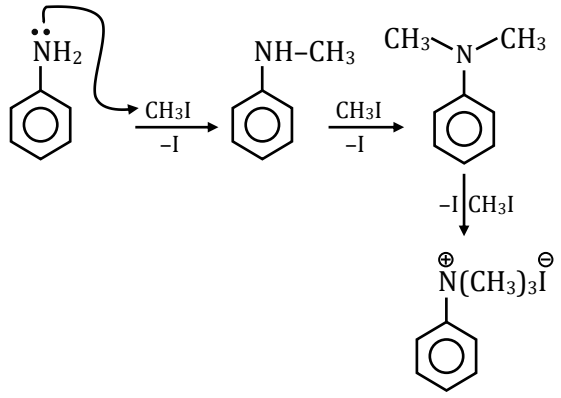
Basic strength of amines in an aq. solution

If R is $–CH_3$ $(CH_3)_2NH > CH_3–NH_2 > (CH_3)_3N > NH_3$

R is $–C_2H_5$ $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5–NH_2 > NH_3$

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23. Ans. (4)



24. Ans. (1)

Molecular Formula :- C_3H_9N

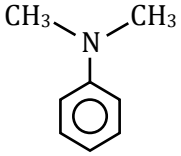
→ Primary amines when treated with HNO_2 gives alcohol and liberate N_2 gas.

→ Different primary amines with molecular formula C_3H_9N are 2.

$CH_3–CH_2–CH_2–NH_2$

$CH_3–\underset{\displaystyle CH_3}{\underset{|}{CH}}–NH_2$

25. Ans. (3)



(N, N Dimethyl aniline)
(i)



(Benzene amine)
(ii)

$CH_3–NH–CH_3$
(N–methyl methanamine)
(iii)

$CH_3–NH_2$
(Methanamine)
(iv)

∴ Order of basic strength :-
(iii) > (iv) > (i) > (ii)

∴ Given valves of pk_b :-
a = 9.38, b = 8.92, c = 3.38, d = 3.27

$$\text{Basic strength} \propto \frac{1}{pK_b}$$

∴ (i) → b, (ii) → a, (iii) → d, (iv) → c

26. Ans. (3)

Order of basic medium in aqueous medium can be defined by more than one type of effects.

$(CH_3)_2NH > CH_3–NH_2 > (CH_3)_3NH > NH_3$
2° 1° 3° 0°

More substituted ammonium ion gets stabilised due to I effect of alkyl group as it increases the availability of electrons to donate but on the other hand, more the number of Alkyl groups, more will be the steric hinderance to H–bonding in ammonium ion which decreases the solvation effect.

∴ a, c, d are correct.

27. Ans. (2)

Benzene diazonium fluoroborate ($C_6H_5\overset{\oplus}{N}_2B\overset{-}{F}_4$) is water insoluble because it is non–polar and it is also stable as diazonium ion is resonance stabilised.

28. Ans. (4)

(1) $ArN_2^{\oplus}Cl^- \xrightarrow[HCl]{Cu_2Cl_2} Ar–Cl + N_2$

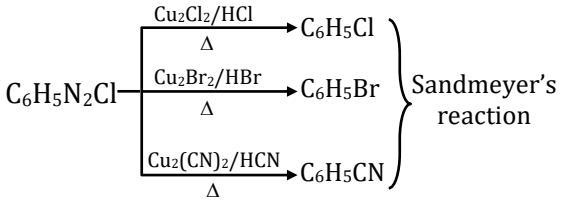
(2) $ArN_2^{\oplus}Cl^- \xrightarrow{Cu/HCl} Ar–Cl + N_2$

(3) $ArN_2^{\oplus}Cl^- \xrightarrow{KI} Ar–I + N_2$

(4) $ArN_2^{\oplus}Cl^- \xrightarrow{HBF_4} ArN_2^{\oplus}BF_4^-$

∴ reaction 4 do not displace N_2 from benzenediazonium salt.

29. Ans. (4)



Therefore, –I group can never be introduced in sandmeyer's reaction.

30. Ans. (4)

As we know, in diazocoupling reaction, Benzene diazonium salt reacts highly activated aromatic rings to form colour dyes.

∴ Aniline can give diazo coupling reaction but benzylamine cannot due to +M effect of $–NH_2$ group.

∴ Ethylamine can not give diazocoupling reaction but N–methyl aniline can give due to +M effect of $–NH_2$ group.

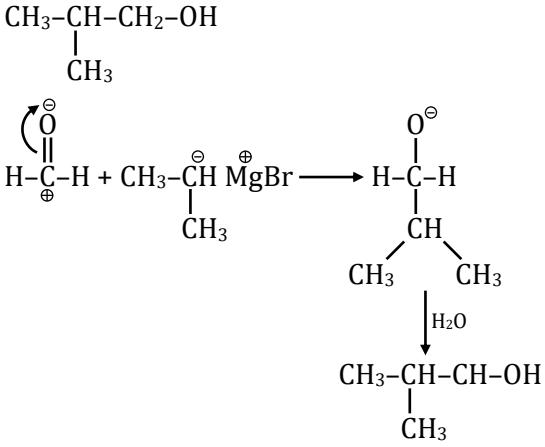
∴ Aniline and phenol both can give azo dye test but can be distinguish by the colour of dye.
Aniline when treated with benzene diazonium salt gives yellow dye whereas phenol gives orange dye.

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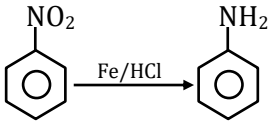
31. Ans. (3)

Product given

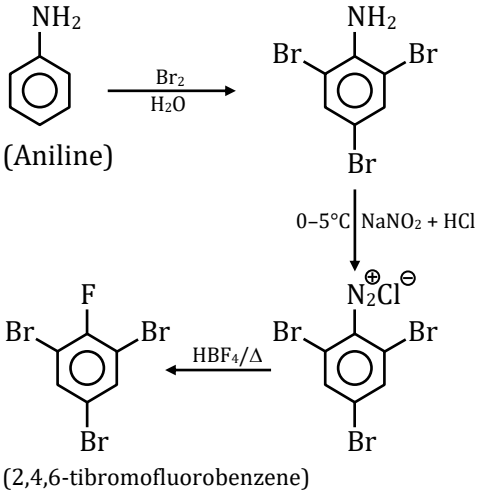


32. Ans. (2)

Given $\phi = C_6H_5$

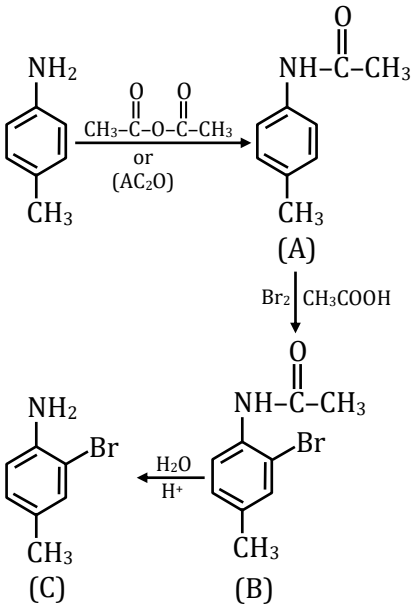


33. Ans. (2)



(2,4,6-tibromofluorobenzene)

34. Ans. (3)



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EXERCISE-I (Conceptual Questions) Build Up Your Understanding

BIOMOLECULES

1. Which is correct in following –
(1) Monosaccharides also known as sugars
(2) Polysaccharides are non sugars
(3) Maltose and Lactose are reducing sugar
(4) All of these

CBM015

2. Which of the following gives osazone different from the other three :-
(1) Glucose (2) Mannose
(3) Galactose (4) Fructose

CBM017

3. The change in the optical rotation of freshly prepared solution of glucose is known as :-
(1) Tautomerism
(2) Racemisation
(3) Specific rotation
(4) Mutarotation

CBM011

4. Structure of glycogen is similar to :-
(1) Amylose (2) Amylopectin
(3) Cellulose (4) Glucose

CBM016

5. Anomers of glucose (α-form & β-form) are differ in the stereochemistry at which carbon-
(1) C-1 (2) C-2
(3) C-3 (4) All of these

CBM018

6. Sucrose in presence of invertase on hydrolysis gave –
(1) Glucose (2) Fructose
(3) Ethyl alcohol (4) Both (1) and (2)

CBM019

7. Cellulose can not be tested by followings –
(1) Fehling's solution (2) Tollen's reagent
(3) Both of these (4) None of these

CBM022

8. Which of the following do not have hemiacetal group:-
(1) Fructose (2) Maltose
(3) Sucrose (4) Glucose

CBM024

9. Which of the following carbohydrates are branched polymer of glucose.
(1) Glycogen (2) Amylopectin
(3) Cellulose (4) Both (1) and (2)

CBM038

10. Reducing sugars are one which :
(1) Reduce fehling's solutions
(2) Not reduce tollen's reagent
(3) Have bonded aldehydic or ketonic groups
(4) All of these

CBM042

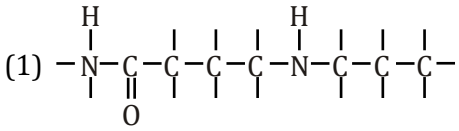
11. Which of the following are polysaccharides –
(a) Starch (b) Cellulose
(c) Dextrins (d) Glycogen
(1) a, b & c (2) a, b, d
(3) a & c (4) a, b, c, d

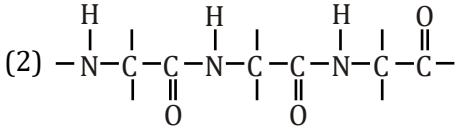
CBM021

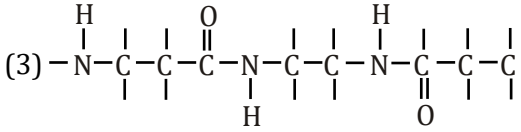
12. The helical structure of protein is stabilized by –
(1) Hydrogen bonds (2) Ether bonds
(3) Peptide bonds (4) Dipeptide bonds

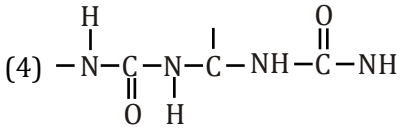
CBM004

13. Which of the following structures represents the peptide chain -

(1) 

(2) 

(3) 

(4) 

CBM005

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14. Which functional group participates in disulfide bond formation in proteins –
(1) Thioether (2) Thiol
(3) Thioester (4) Thiolactone
CBM006

15. Which is simplest amino acid –
(1) Alanine (2) Asparagine
(3) Glycine (4) Tyrosine
CBM008

16. Which of the following biomolecules is insoluble in water -
(1) α–Keratin (2) Haemoglobin
(3) Ribonuclease (4) Adenine
CBM009

17. Lysine; $H_2N-(CH_2)_4-\underset{\underset{NH_2}{|}}{CH}-COOH$ is :-
(1) α-Amino acid
(2) γ-amino acid
(3) Amino acid synthesised
(4) β-Amino acid
CBM013

18. In fibrous protein, polypeptide chains are held together by :-
(1) H-bond
(2) Disulphide linkage
(3) Electrostatic forces attraction
(4) Both (1) and (2)
CBM014

19. In amino acids, more number of amino than carboxyl groups makes it –
(1) Acidic (2) Basic
(3) Neutral (4) None of these
CBM025

20. Which amino acids are called non essential –
(1) Those which can be synthesized in the body.
(2) Those which have more amino groups as compared to carboxyl groups
(3) Those which have equal number of amino acid and carboxyl groups
(4) None of these
CBM026

21. Which of the following is not essential amino acid –
(1) Serine (2) Lysine
(3) Threonine (4) Tryptophan
CBM027

22. In acidic & alkaline solution amino acids exists as a –
(1) Positive ion & negative ion respectively
(2) Negative ion & positive ions respectively
(3) Neutral in both medium
(4) None of these
CBM028

23. In which of following shapes are found in tertiary structure of proteins –
(1) Fibrous (2) Globular
(3) Both of these (4) None of these
CBM029

24. The example of globular protein is
(1) Silk (2) Collagen
(3) Haemoglobin (4) All of these
CBM030

25. If a native protein is subjected to physical or chemical treatment which may disrupt its various forms without affecting its primary structure, are called –
(1) Inactive protein (2) Denatured protein
(3) Both of these (4) None of these
CBM031

26. The coagulation of egg protein while boiling of egg is called –
(1) Reversible protein denaturation
(2) Irreversible protein denaturation
(3) Renaturation
(4) None of these
CBM032

27. The correct statement in respect of protein haemoglobin is that it –
(1) Maintains blood sugar level
(2) Acts as an oxygen carrier in the blood
(3) Forms antibodies and offers resistance to diseases
(4) Acts as a catalyst for biological reactions
CBM002

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28. Which of the following is not an amino acid
(1) Histidine (2) Benzidine
(3) Alanine (4) Proline
CBM040

29. The nitrogenous base having two possible hydrogen bonding sites is –
(1) Thymine (2) Cytosine
(3) Guanine (4) None
CBM001

30. The presence or absence of hydroxy group on which carbon atom of sugar differentiates RNA and DNA?
(1) 3rd (2) 4th
(3) 1st (4) 2nd
CBM010

31. Which one of the following bases is not present in DNA ?
(1) Cytosine (2) Thymine
(3) Quinoline (4) Adenine
CBM012

32. RNA contains following pyrimidine bases –
(a) Thymine (b) Uracil
(c) Cytosine (d) Adenine
(1) b & c (2) a, b, c
(3) a, b, d (4) All of these
CBM036

33. Which of the following is correct about H–bonding in nucleotide –
(1) A–T G–C (2) A–G T–C
(3) G–T A–C (4) A–A T–T
CBM045

34. Mainly DNA is localized in –
(1) Cytoplasm (2) Nucleus
(3) Mitochondria (4) Chloroplasts
CBM035

35. DNA molecules can duplicate themselves-called –
(1) Replication (2) Translation
(3) Transcription (4) None of these
CBM037

36. Which of the following B group vitamins can be stored in our body.
(1) Vitamin B_1 (2) Vitamin B_2
(3) Vitamin B_6 (4) Vitamin B_{12}
CBM020

37. Vitamin K
(1) Is phylloquinone
(2) Soluble in oils and fats
(3) Deficiency lenthens the blood clotting
(4) All of these
CBM039

38. An enzyme molecule may contain protein and non-protein part. Non-protein part is known as:-
(1) Holoenzyme (2) Cofactor
(3) Inverted enzyme (4) None of these
CBM033

39. Vitamin C is called :
(1) Antisterility (2) Antiscurvy
(3) both of these (4) None of these
CBM041

40. Which vitamin is synthesize in human body from carotene
(1) Vitamin-A (2) Vitamin-C
(3) Vitamin-K (4) All of these
CBM044

41. Multiple deficiencies caused by lack of more than one vitamin are more common in human beings. This condition of vitamin deficiency is known as
(1) Avitaminosis (2) Xerophthalmia
(3) Convulsions (4) None of these
CBM043

42. The cell membranes are mainly composed of–
(1) Phospholipids
(2) Proteins
(3) Fats
(4) Carbohydrates
CBM007

43. On hydrolysis of proteins, the product is/are–
(1) Amino acids (2) Peptides
(3) Enzymes (4) Both (1) and (2)
CBM023

44. Cofactor which gets attached to the enzyme at the time of reaction are known as
(1) Coenzymes (2) Apoenzyme
(3) Prothetic group (4) None of these
CBM034

45. The hormone that helps in the conversion of glucose to glycogen is –
(1) Bile acids (2) Adrenaline
(3) Insulin (4) Cortisone
CBM003

EXERCISE-I (Conceptual Questions) ANSWER KEY

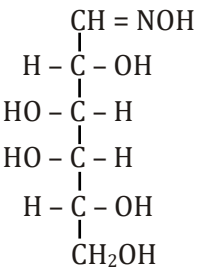
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**3**	**4**	**2**	**1**	**4**	**3**	**3**	**4**	**1**	**4**	**1**	**2**	**2**	**3**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**1**	**1**	**4**	**2**	**1**	**1**	**1**	**3**	**3**	**2**	**2**	**2**	**2**	**1**	**4**
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	**3**	**1**	**1**	**2**	**1**	**4**	**4**	**2**	**2**	**1**	**1**	**1**	**4**	**1**	**3**

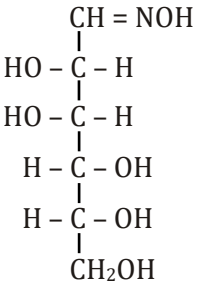
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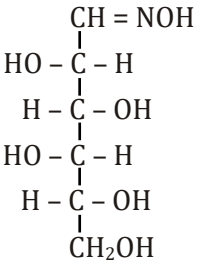
EXERCISE-II (Previous Year Questions) AIPMT/NEET

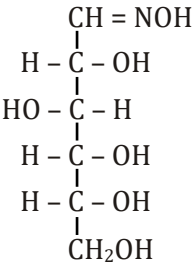
AIPMT 2014

1. D (+) glucose reacts with hydroxylamine and yields an oxime. The structure of the oxime would be :

(1) 

(2) 

(3) 

(4) 

CBM062

NEET-I 2016

2. In a protein molecule various amino acids are linked together by :

(1) α-glycosidic bond (2) β-glycosidic bond

(3) Peptide bond (4) Dative bond

CBM063

3. The **correct** statement regarding RNA and DNA, respectively is :

(1) The sugar component in RNA is arabinose and the sugar component in DNA is 2'-deoxyribose.

(2) The sugar component in RNA is ribose and the sugar component in DNA is 2'-deoxyribose.

(3) The sugar component in RNA is arabinose

(4) The sugar component in RNA is 2'-deoxyribose and the sugar component in DNA is arabinose.

CBM064

4. Which one given below is a non-reducing sugar ?

(1) Maltose (2) Lactose

(3) Glucose (4) Sucrose

CBM065

NEET-II 2016

5. The central dogma of molecular genetics states that the genetic information flows from :-

(1) DNA → RNA → Proteins

(2) DNA → RNA → Carbohydrates

(3) Amino acids → Proteins → DNA

(4) DNA → Carbohydrates → Proteins

CBM066

6. Which one of the following compounds shows the presence of intramolecular hydrogen bond?

(1) Cellulose

(2) Concentrated acetic acid

(3) H_2O_2

(4) HCN

CBM067

NEET(UG) 2018

7. The difference between amylose and amylopectin is

(1) Amylopectin have 1 → 4 α-linkage and 1→ 6 α-linkage

(2) Amylose have 1 → 4 α-linkage and 1→ 6 β-linkage

(3) Amylopectin have 1 → 4 α-linkage and 1→ 6 β-linkage

(4) Amylose is made up of glucose and galactose

CBM068

8. Which of the following compounds can form a zwitterion ?

(1) Aniline (2) Acetanilide

(3) Benzoic acid (4) Glycine

CBM127

NEET(UG) 2019

9. The non-essential amino acid among the following is:

(1) Valine (2) Leucine

(3) Alanine (4) Lysine

CBM069

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NEET(UG) 2019 (ODISHA)

10. Which structure(s) of proteins remains(s) intact during denaturation process ?
(1) Both secondary and tertiary structures
(2) Primary structure only
(3) Secondary structure only
(4) Tertiary structure only

CBM070

NEET(UG) 2020

11. Sucrose on hydrolysis gives :
(1) α-D-Fructose + β-D-Fructose
(2) β-D-Glucose + α-D-Fructose
(3) α-D-Glucose + β-D-Glucose
(4) α-D-Glucose + β-D-Fructose

CBM071

12. Which of the following is a basic amino acid:
(1) Lysine (2) Serine
(3) Alanie (4) Tyrosine

CBM072

NEET(UG) 2020 (COVID-19)

13. The reaction of concentrated sulphuric acid with carbohydrates ($C_{12}H_{22}O_{11}$) is an example of
(1) Dehydration (2) Oxidation
(3) Reduction (4) Sulphonation

CBM073

14. Deficiency of which vitamin causes osteomalacia?
(1) Vitamin A (2) Vitamin D
(3) Vitamin K (4) Vitamin E

CBM074

NEET(UG) 2021

15. The RBC deficiency is deficiency disease of:
(1) Vitamin B_{12} (2) Vitamin B_6
(3) Vitamin B_1 (4) Vitamin B_2

CBM075

NEET(UG) 2023

16. Given below are two statements:
Statement-I : A unit formed by the attachment of a base to l' position of sugar is known as nucleoside
Statement-II : When nucleoside is linked to phosphorous acid at 5'-position of sugar moiety, we get nucleotide.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement-I and Statement-II are false
(2) Statement-I is true but Statement-II is false
(3) Statement-I is false but Statement-II is true
(4) Both Statement-I and Statement-II are true

CBM114

17. Cheilosis occurs due to deficiency of_____.
(1) thiamine (2) nicotinamide
(3) pyridoxamine (4) riboflavin

CBM115

NEET(UG) 2024 (Re-examination)

18. Given below are two statements :
Statement-I : Glycogen is similar to amylose in its structure.
Statement-II : Glycogen is found in yeast and fungi also.
In the light of the above statements, choose the **correct** answer from the options given below :
(1) Statement-I is true but Statement-II is false
(2) Statement-I is false but Statement-II is true
(3) Both Statement-I and Statement-II are true
(4) Both Statement-I and Statement-II are false

CBM116

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	2	4	1	1	1	4	3	2	4	1	1	2	1
Question	16	17	18												
Answer	2	4	2												

EXERCISE-III (Analytical Questions) Master Your Understanding

1. Match the column.

Column-I (Reaction of Glucose)		Column-II (Product)	
(a)	Glucose $\xrightarrow{HI, \Delta}$	(p)	Osazone
(b)	Glucose $\xrightarrow{Br_2 \text{ water}}$	(q)	n-Hexane
(c)	Glucose $\xrightarrow{\text{Nitric acid}}$	(r)	Gluconic acid
(d)	Glucose $\xrightarrow[H^{\oplus}]{3Ph-NH-NH_2}$	(s)	Saccharic acid

(1) a – q, b – s, c – p, d - r
(2) a – q, b – p, c – r, d - s
(3) a – q, b – s, c – r, d - p
(4) a – q, b – r, c – s, d - p

CBM095

2. Which of the following reactions of glucose can be explained only by its cyclic structure.
(1) Glucose forms penta acetate
(2) Glucose reacts with hydroxylamine to form an oxime
(3) Penta acetate of glucose does not react with hydroxylamine
(4) Glucose is oxidised by nitric acid to gluconic acid

CBM085

3. When glucose was crystallized from a concentrated solution at 30°C & 98°C. It gave α & β form of glucose respectively what are the angle of rotation [α_D] for both forms –
(1) (+) 111° & (+)19.2°
(2) (+)19.2° & (–) 111°
(3) (+) 111° & (–)19.2°
(4) (–) 111° & (–)19.2°

CBM077

4. Which statement is incorrect in following –
(1) Sucrose is dextrorotatory
(2) After hydrolysis of sucrose, dextrorotatory glucose and laevorotatory fructose will obtained
(3) Laevorotation of fructose is more than dextrorotation of glucose
(4) Cane sugar gives nonequimolar mixture of D- (+)-glucose and D- (–)-fructose

CBM079

5. On hydrolysis of one mole of maltose two moles of D-glucose are obtained. These two glucose units are linked together through a α-glycoside linkage between –
(1) C-2 of one unit and C-4 of another unit
(2) C-1 of one unit and C-2 of another unit
(3) C-1 of one unit and C-4 of another unit
(4) C-2 of one unit and C-3 of another unit

CBM080

6. Which statement is/are incorrect –
(1) Natural starch has approximately 10-20% of amylose and 80-90% of amylopectin.
(2) Amylose is water soluble and gives blue colour with iodine
(3) Amylopectin is a branched chain polysaccharides insoluble in water and does not give blue colour with iodine
(4) Starch is not hydrolysed by enzyme amylase present in saliva

CBM081

7. Which is correct statement –
(1) Starch is polymer of α–glucose
(2) Amylose is a component of cellulose
(3) Proteins are composed of only one type of amino acid
(4) In cyclic structure of fructose, there are four carbons and one oxygen atom

CBM093

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8. Match the column.

Column-I (Carbohydrates)		Column-II (Source or name)	
(a)	Sucrose	(p)	Animal starch
(b)	Lactose	(q)	Cereals, roots, tubers
(c)	Glycogen	(r)	Milk sugar
(d)	Starch	(s)	Cane sugar

(1) a – r, b – s, c – q, d - p
(2) a – s, b – r, c – p, d - q
(3) a – s, b – p, c – r, d - q
(4) a – p, b – q, c – r, d - s

CBM094

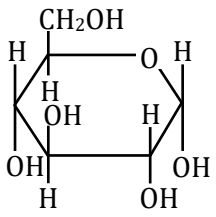
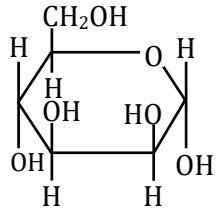
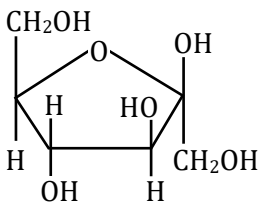
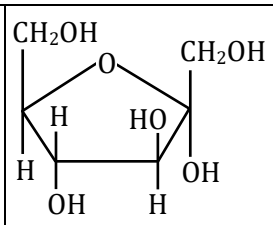
9. Match the column.

Column-I (Carbohydrates)		Column-II (Monosaccharide unit)	
(a)	Maltose	(p)	β-D-Glucose
(b)	Lactose	(q)	Two α-D-Glucose
(c)	Cellulose	(r)	β-D-Galactose & β-D-Glucose
(d)	Sucrose	(s)	α-D-Glucose & β-D-Fructose

(1) a – q, b – s, c – r, d - p
(2) a – q, b – r, c – p, d - s
(3) a – r, b – s, c – p, d - q
(4) a – r, b – p, c – q, d - s

CBM096

10. Match the column.

Column-I (Structure)		Column-II (Name)	
(a)		(p)	α-D-Mannose
(b)		(q)	β-D-Fructose
(c)		(r)	α-D-Glucose
(d)		(s)	α-D-Fructose

(1) a – p, b – r, c – q, d - s
(2) a – r, b – p, c – q, d - s
(3) a – p, b – r, c – s, d - q
(4) a – r, b – p, c – s, d - q

CBM097

11. Which of the following are non-reducing in nature-

(A) Cellulose (B) Sucrose
(C) Lactose (D) Starch

(1) A and B are correct
(2) A, B and C are correct
(3) A, B and D are correct
(4) A, C and D are correct

CBM098

12. In which of the following properties of an open chain structure of glucose could not be explained by Baeyer –

(1) Glucose contain aldehyde group, but does not give schiff's test
(2) Glucose does not react with sodium hydrogen sulphite and ammonia
(3) The pentaacetate of glucose does not react with hydroxy amine indicates absence of –CHO group
(4) All of these

CBM076

13. Which of the following statement is true for proteins –

(1) Proteins are high molecular mass complex biopolymers of amino acids present in all living cells
(2) Protein may contains phosphorous & iodine
(3) Proteins may contains iron, copper, zinc and manganese
(4) All of these

CBM082

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14. Which of following are essential amino acids–
(a) Aspartic acid (b) Leucine
(c) Valine (d) Glycine
(e) Alanine
(1) a, b & c (2) a, b, d & e
(3) b & c (4) a, c, d & e

CBM083

15. At a certain hydrogen ion concentration (pH), the dipolar ion exists as a neutral ion and does not migrate to either electrode. What is the name of pH point for amino acids –
(1) Isoelectric point (2) Isoprotonic point
(3) Isotonic point (4) None of these

CBM084

16. Protein found in biological system with definite configuration and biological activity is called –
(1) Amino acids (2) Native protein
(3) Conformer protein (4) Inactive protein

CBM086

17. Read given statements
(A) During denaturation, 2° & 3° structures of proteins are not destroyed.
(B) 1° structure of protein is destroyed during denaturation.
(C) Curdling of milk which is caused due to formation of lactic acid by bacteria present in milk is an example of denaturation.
Which statement(s) is/are incorrect in given options -
(1) A, C only (2) A, B only
(3) A, B, C (4) B, C only

CBM113

18. Which of following statements is incorrect –
(1) Proteins have a polyamide chain while nucleic acids contains a poly phosphate ester chain
(2) Nucleic acids are long chain polymers of nucleotides
(3) RNA is major source of genetic information which is copied into a DNA molecules
(4) Proteins are synthesized in a process involving translation of RNA.

CBM088

19. Complete hydrolysis of DNA or RNA yields following–
(1) Ribose in RNA & deoxyribose in DNA
(2) Hetrocyclic nitrogenous purines base
(3) Hetrocyclic nitrogenous pyrimidines
(4) All of these

CBM089

20. Dinucleotide is obtained by joining two nucleotides together by phosphodiester linkage. Between which carbon atoms of pentose sugar of nucleotides are these linkages presents.
(1) 5' and 3' (2) 1' and 5'
(3) 5' and 5' (4) 3' and 3'

CBM078

21. Nucleosides are –
(1) Base + sugar = Nucleoside
(2) N–glycosides of purine or pyrimidine bases with pentose sugar
(3) Both of these
(4) None of these

CBM090

22. RNA molecules in turn directs the synthesis of specific proteins which are characteristic of each kind of organism. This process is known as –
(1) Translation
(2) Transcription
(3) Replication
(4) None of these

CBM091

23. A single stand of DNA can act as a template on which a molecule of RNA is synthesized in a specific manner. This process is known as –
(1) Replication (2) Transcription
(3) Translation (4) None of these

CBM092

24. Which of following statement is not true for enzyme –
(1) Some enzymes can be non proteins also
(2) Prothetic groups which get attached to enzyme at the time of reaction are known as cofactor.
(3) Enzymes provides lower activation pathways there by increasing the rate of reaction
(4) None of these

CBM087

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25. Match the column I & II :-

Column-I (Vitamins)		Column-II (Sources)	
(a)	Vitamin-A	(p)	Exposure to sunlight
(b)	Vitamin-B12	(q)	Yeast, milk
(c)	Vitamin-B6	(r)	Meat
(d)	Vitamin-D	(s)	Fish liver oil

(1) a - s, b - r, c - q, d - p
(2) a - s, b - q, c - p, d - r
(3) a - q, b - s, c - r, d - p
(4) a - r, b - p, c - q, d - s

CBM106

26. Match the column I & II :-

Column-I (Vitamins)		Column-II (Disease)	
(a)	Vitamin-B1	(p)	Scurvy
(b)	Vitamin-B6	(q)	Blood clotting delay
(c)	Vitamin-C	(r)	Beri-Beri
(d)	Vitamin-K	(s)	Convulsions

(1) a - r, b - p, c - q, d - s
(2) a - s, b - q, c - p, d - r
(3) a - r, b - s, c - p, d - q
(4) a - q, b - r, c - s, d - p

CBM107

27. Match the column :-

Column-I		Column-II	
(a)	Fat soluble vitamin	(p)	Fingerprinting
(b)	Water soluble vitamin	(q)	Vitamin A and vitamin D
(c)	DNA	(r)	Uracil
(d)	RNA	(s)	Vitamin B and vitamin C

(1) a - s, b - s, c - r, d - p
(2) a - q, b - s, c - p, d - r
(3) a - q, b - r, c - p, d - s
(4) a - q, b - s, c - r, d - p

CBM108

28. Match the column I & II :-

Column-I		Column-II	
(a)	Heredity	(p)	Proteins
(b)	Chromosomes made up by	(q)	Transmission of inherent characters
(c)	Nucleic Acid	(r)	Increased fragility of RBCs
(d)	Vitamin-E deficiency	(s)	Long chain polymers of Nucleotides

(1) a - q, b - p, c - s, d - r
(2) a - q, b - p, c - r, d - s
(3) a - p, b - q, c - r, d - s
(4) a - p, b - q, c - s, d - r

CBM109

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**3**	**1**	**4**	**3**	**4**	**1**	**2**	**2**	**2**	**3**	**4**	**4**	**3**	**1**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28		
Answer	**2**	**2**	**3**	**4**	**1**	**3**	**1**	**2**	**2**	**1**	**3**	**2**	**1**		

Biomolecules

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (4)

Generally Mono saccharides → Sugars

Poly saccharides → Non-sugars

Reducing sugar → Mono saccharides

2. Ans. (3)

Glucose, Fructose, Mannose $\xrightarrow{\text{gives}}$ Same osazone when react with $Ph–NH–NH_2$ (Phenyl Hydrazine)

3. Ans. (4)

$\alpha-D-Glucose \rightleftharpoons Glucose \rightleftharpoons \beta-D-Glucose$

Mutarotation

4. Ans. (2)

Glycogen structure is similar to amylopectin which is highly branched.

5. Ans. (1)

Anomer of glucose means

C-1 H and –OH group change position

6. Ans. (4)

Sucrose $\xrightarrow{\text{Invertase}}$ Glucose + Fructose

7. Ans. (3)

Cellulose → Polysaccharide

Non-reducing in Nature

So No Reaction with Fehling solutions and Tollen reagent.

8. Ans. (3)

Hemiacetal Hemiketal



If Hemiacetal and Hemiketal structure is present then reducing in nature

Sucrose → Non reducing in nature

9. Ans. (4)

Branched polymer of glucose

Glycogen → Highly branched amylopectin

10. Ans. (1)

Reducing sugars→ Reduced fehling's solution

→ Reduced tollen's reagent

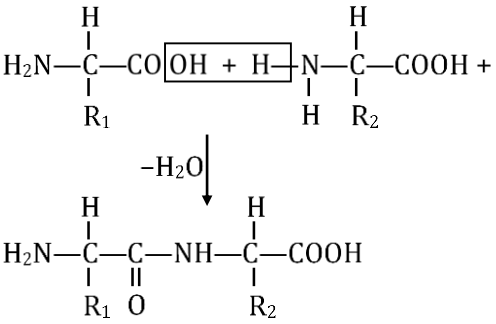
11. Ans. (4)

Polysaccharides → Starch Cellulose Dextrin Glycogen

12. Ans. (1)

Helical structure of protein is stabilized by Hydrogen bonds → α-Helix, → β-Helix

13. Ans. (2)



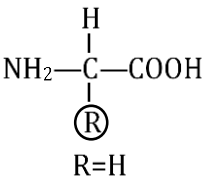
Peptide Bond

14. Ans. (2)

disulphide Bond formation in protein.

–S–S– Linkage [SH–functional group]

15. Ans. (3)



Simples amino acid → Glycine

16. Ans. (1)

Fibrous protein → Generally insoluble in water

Eg. Keratin (present in Hair, Wool, Silk)

17. Ans. (1)

Lysine → α-Amino acid

Relative position of Amino group with respect to carboxylic acid.

18. Ans. (4)

Fibrous protein → Hydrogen Bond→ Present

Disulphide Bond →Present

Electrostatic force attraction → Absent

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19. Ans. (2)

NH_2 group > Carboxylic acid group ⇒ Basic in nature

Eg. Histidine
Lysine
Arginine

20. Ans. (1)

Non-essential Amino acid → Synthesized in Body
Non-essential Amino acid → No need to take in diet

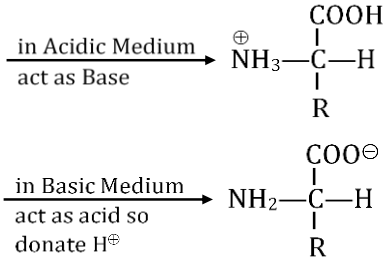
21. Ans. (1)

Non essential Amino acid → Serine

22. Ans. (1)

Amino acid → Acidic Medium → ⊕ Ve
Amino acid → Basic Medium → ⊖ Ve

Amino acid Have 2 group (1) NH_2
(2) –COOH



23. Ans. (3)

tertiary structure of proteins → Folding of secondary structure

Two major molecular shapes viz → Fibrous and globular

24. Ans. (3)

Globular protein → Haemoglobin Insulin Albumins

Fibrous protein → Keratin (present in Hair, Wool, Silk myosin (present in muscles)

25. Ans. (2)

Due to change in physical and chemical reaction $\xrightarrow{\text{Denaturation of protein}}$ disrupt it's various form (Except primary structure)

26. Ans. (2)

Egg $\xrightarrow{\text{boiling}}$ Coagulation of egg protein (Irreversible denaturation)

27. Ans. (2)

Haemoglobin → Oxygen carrier in the blood

28. Ans. (2)

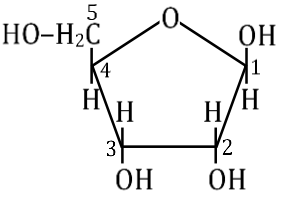
Benzidine is not a Amino acid

29. Ans. (1)

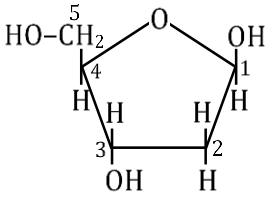
A = T
G ≡ C

30. Ans. (4)

RNA



DNA



OH group Absent at 2nd position

31. Ans. (3)

DNA → Adenine, Guanine
Thymine, Cytosine → Present
Quinoline → Absent

32. Ans. (1)

RNA → A , G (Purines)
RNA → U , C (Pyrimidine)

33. Ans. (1)

A = T
G ≡ C

34. Ans. (2)

Mainly DNA is localized in Nucleus

35. Ans. (1)

DNA $\xrightarrow[\text{Transcription}]{}$ RNA $\xrightarrow[\text{Translation}]{}$ Protein

36. Ans. (4)

Only Vitamin B_{12} ⟶ Store in Body
Water soluble vitamin B, vitamin C
Readily excreted in Urine and Cannot be stored (Except-Vit. B_{12})

37. Ans. (4)

Vitamin K → Fat soluble
→ Blood clotting
→ Chemical name is phylloquinone

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38. Ans. (2)

Enzyme → Protein part + Non protein part
(Holoenzyme) (Apoenzyme) (Cofactor coenzyme)

39. Ans. (2)

Vit. C → Anti scury

Vit E → Anti sterility

40. Ans. (1)

Carotene → Vit. A

41. Ans. (1)

Avitaminosis → Multiple deficiencies (more than one vitamin deficiency)

Xerophthalmia → Deficiency of Vit - A

Convulsions → Deficiency of Vit - B6

42. Ans. (1)

Cell Membranes $\xrightarrow{\text{Mainly composed}}$ Phospholipids

43. Ans. (4)

Protein $\xrightarrow{\text{Hydrolysis}}$ Peptides chain ⟶ Amino acid

44. Ans. (1)

Non protein part

Coenzyme — Made up of Organic molecule

Cofactor — Made up of Inorganic molecule

45. Ans. (3)

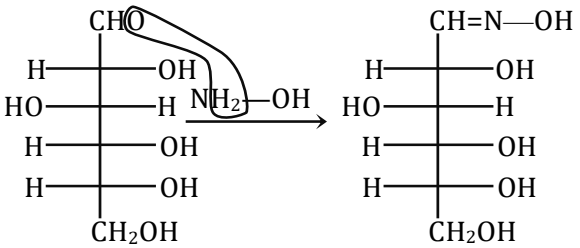
Glucose $\underset{\text{Harmon}}{\xrightarrow{\text{Insulin}}}$ Glycogen

Glycogen Harmon

⇒ Balance glucose level

Exercise-II (Previous Year Questions)

1. Ans. (4)



2. Ans. (3)

Amino acid $\xrightarrow{\text{Peptide Bond}}$ protein

3. Ans. (2)

RNA $\xrightarrow{\text{Sugar}}$ ribose

DNA ⟶ 2 – deoxyribose

4. Ans. (4)

Sucrose is non reducing sugar

5. Ans. (1)

Central dogma

DNA → RNA ⟶ Protein

6. Ans. (1)

Cellulose ⟶ Intramolecular H-Bonding possible

7. Ans. (1)

Amylose → 1 - 4 α - linkage

Amylopectin → 1 – 4 α linkage + 1 – 6 α - linkage Both

8. Ans. (4)

Generally Amino acid form zwitterion

∴ Glycine can form zwitterion

9. Ans. (3)

Non-essential amino acid → Alanine

10. Ans. (2)

Primary structure remain same during denaturation process

11. Ans. (4)

Sucrose = α – D Glucose + β – D Fructose

12. Ans. (1)

Basic Amino acid → lysine, arginine, histidine

13. Ans. (1)

$$C_{12}H_{22}O_{11} \xrightarrow[\text{Dehydrating agent}]{\text{Con. } H_2SO_4} 12C + 11H_2O$$

14. Ans. (2)

Osteomalacia → Vit. D. Deficiency

Vit. A → Xerophthalmia

E → Anti sterility hormone

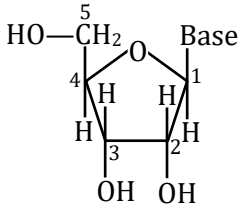
K → Blood dealy in clotting

15. Ans. (1)

Vit. B_{12} → Pericious Anaemia (RBC deficient in haemoglobin)

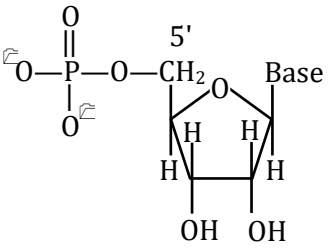
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16. Ans. (2)



Nucleoside

Base link with 1' position of sugar in nucleoside so statement I is correct



Nucleotide

→ When nucleoside is linked to **phosphoric acid** at 5' position of sugar moiety we get a Nucleotide

⇓

Statement II is Incorrect because not link with **phosphorous acid**.

17. Ans. (4)

Cheilosis (Fissuring at corners of mouth and lips) occurs due to deficiency of vitamin B_2 (Riboflavin)

18. Ans. (2)

Glycogen is similar to amylopectin in its structure.

Exercise-III (Analytical Questions)

1. Ans. (4)

Reaction		Product
Glucose	$\xrightarrow{HI\,\Delta}$	n-hexane
Glucose	$\xrightarrow{Br_2\ water}$	Gluconic acid
Glucose	$\xrightarrow{Nitric\ acid}$	Saccharic acid
Glucose	$\xrightarrow[H^+]{3Ph-NH-NH_2}$	Osozone

2. Ans. (3)

Option 1, 2, 4 explained by open chain.

Only option 3 is explained by cyclic st.

Penta acetate of glucose + Hydroxylamine

This reaction does not occur O.S. in this –CHO doesn't present in free state.

3. Ans. (1)

α-D Glucose	β-D-Glucose
Specific rotation	Specific rotation
$[\alpha]_D^{20}$ +111°	$[\alpha]_D^{20}$ +19.2°
30°C	98°C

4. Ans. (4)

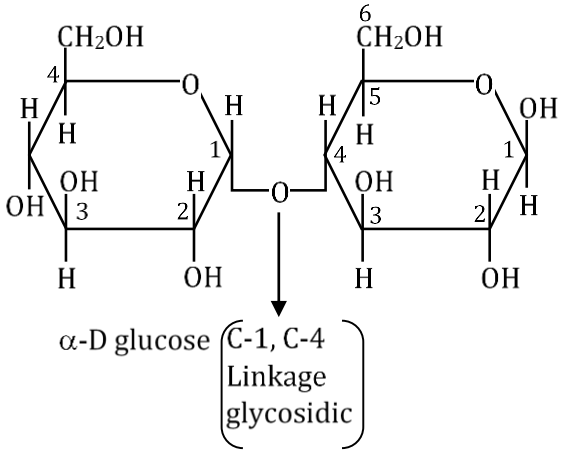
Cane sugar (Sucrose) $\xrightarrow{Hydrolysis}$ Glucose + Fructose

+52.50 −92.40

equimolar mixture

5. Ans. (3)

Maltose structure



α-D glucose (C-1, C-4 Linkage glycosidic)

6. Ans. (4)

Amylose	Amylopectin
(1) 10-20%	(1) 80-90%
(2) water soluble	(2) water insoluble
(3) Blue colour with Iodine	(3) No colour with Iodine

7. Ans. (1)

Starch is polymer of α-glucose

8. Ans. (2)

Carbohydrate	Source
Sucrose	Cane sugar
Lactose	Milk sugar
Glycogen	Animal starch
Starch	Cereal root, tuber

9. Ans. (2)

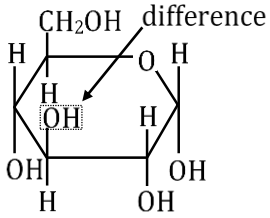
Carbohydrate		Monosaccharide unit
Maltose	⟶	Two α-D-Glucose
Lactose	⟶	β-D-Galactose & β-D-Glucose
Cellulose	⟶	β-D-Glucose
Sucrose	⟶	α-D-Glucose & β-D-Fructose

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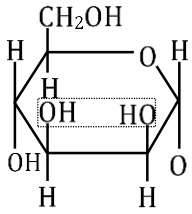
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10. Ans. (2)

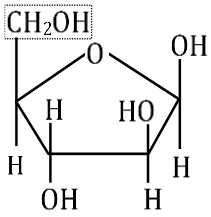
Structure Name



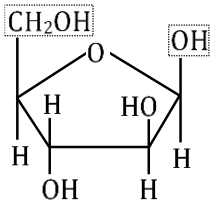
⟶ α-D-Glucose



⟶ α-D-Mannose



⟶ β-D-Fructose



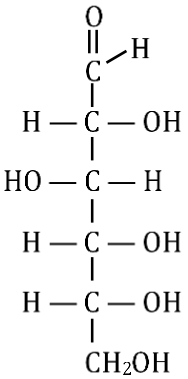
⟶ α-D-Fructose

11. Ans. (3)

Lactose have free aldehyde group
So, reducing in nature
Cellulose, sucrose & starch doesn't have free aldehyde & ketone
So, Non reducing in Nature.

12. Ans. (4)

Baeyer proposed the open chain structure of glucose



D-glucose

The open chain structure of glucose doesn't give reaction with sodium hydrogen sulphite & ammonia which absence of free –CHO group.

13. Ans. (4)

All are true for protein

14. Ans. (3)

Essential amino acid are those amino acid which are not produced by body & have to consumed externally.
Ex. Leucine & valine

15. Ans. (1)

Isoelectric point of amino acids at which amino acid has no net charge & it exist as neutral ion at certain pH.

16. Ans. (2)

Native protein

17. Ans. (2)

When protein is denatured, the different structure such as secondary & tertiary are denatured because they doesn't contain strong bonds. But primary structure of protein is not denatured because it consist of peptide bond which are strong in nature.
Casein protein present in the milk reacts with the lactic acid present in the curd which breaks the chemical bond of the milk & form new bond & breaking of chemical bond is denaturation.

18. Ans. (3)

DNA contains the major source of genetic information that is copied to another DNA molecule.

$$\text{DNA} \xrightarrow{\text{Transcription}} \text{RNA} \xrightarrow{\text{Translation}} \text{Protein}$$

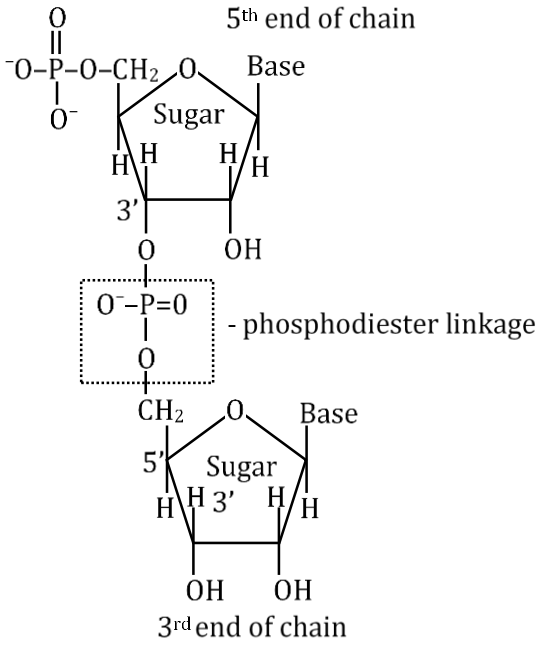
19. Ans. (4)

Nucleotides $\Rightarrow$ N_2 Base + Phosphate group + pentose sugar

RNA → A, G, U, C
DNA → A, G, T, C

20. Ans. (1)

2 Nucleoside ⟶



21. Ans. (3)

Nucleoside → Sugar + Base

Sugar is attach with Base with N-glycosidic linkage.

22. Ans. (1)

$DNA \xrightarrow{transcription} RNA \xrightarrow{Translation} Protein$

23. Ans. (2)

$DNA \xrightarrow{transcription} RNA \xrightarrow{Translation} Protein$

24. Ans. (2)

Option (1) Ribozyme non protein enzyme.

Ans is (2) option as the prosthetic that it attach to enzyme at the time of reaction is known as coenzyme.

Cofactor is combination of Coenzyme & Prosthetic group.

25. Ans. (1)

(Vitamins)		(Sources)
(a) Vitamin-A	⟶	(s) Fish liver oil
(b) Vitamin-B_{12}	⟶	(r) Meat
(c) Vitamin-B_6	⟶	(q) Yeast, milk
(d) Vitamin-D	⟶	(p) Exposure to sunlight

26. Ans. (3)

(Vitamins)		(Disease)
(a) Vitamin-B_1	⟶	(r) Beri-Beri
(b) Vitamin-B_6	⟶	(s) Convulsions
(c) Vitamin-C	⟶	(p) Scurvy
(d) Vitamin-K	⟶	(q) Blood clotting delay

27. Ans. (2)

(a) Fat soluble vitamin	⟶	(q) Vitamin A and vitamin D
(b) Water soluble Vitamin	⟶	(s) Vitamin B and vitamin C
(c) DNA	⟶	(p) Fingerprinting
(d) RNA	⟶	(r) Uracil

28. Ans. (1)

(a) Heredity	→	(q) Transmission of inherent characters
(b) Chromosomes made up by	→	(p) Proteins
(c) Nucleic Acid	→	(s) Long chain polymers of Nucleotides
(d) Vitamin-E deficiency	→	(r) Increased fragility of RBC

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EXERCISE - I (Conceptual Questions) Build Up Your Understanding

1. Ether and benzene can be separated by :-
(1) Filtration (2) Distillation
(3) Crystallization (4) Sublimation
CEC028

2. Phenol on reaction with dilute nitric acid, gives two products. Which method will be most effective for large scale separation
(1) Chromatographic separation
(2) Fractional crystallisation
(3) Steam distillation
(4) Sublimation
CEC086

3. Correct match :-

	Mixture		Purification technique
(A)	Chloroform and aniline	(P)	Distillation
(B)	Glycerol and Spentlye	(Q)	Chromatography
(C)	Aniline and Water	(R)	Steam distillation
(D)	Amino Acid	(S)	Distillation under reduced pressure

(1) A–P, B–Q, C–R, D–S (2) A–P, B–S, C–R, D–Q
(3) A–P, B–R, C–S, D–Q (4) A–R, B–S, C–P, D–Q
CEC087

4. The substance which can be commonly used as an adsorbent in column chromatography is
(1) Na_2O (2) NaCl (3) Al_2O_3 (4) Alum
CEC088

5. The purification method where solid substance change from vapour state without passing through the liquid state is called as
(1) Sublimation
(2) Crystallization
(3) Distillation
(4) Differential extraction
CEC089

6. Chloroform (B.P. 334 K) and aniline (B.P. 457 K) are easily separated by the technique of
(1) Crystallisation (2) Distillation
(3) Sublimation (4) Chromatography
CEC090

7. **Statement-I :** Silica gel and alumina are used as adsorbent for adsorption chromatography.
Statement-II : Paper chromatography is a type of partition chromatography.
In the light of above statements, select the correct option.
(1) Statement-I is true but statement-II is false
(2) Statement-I is false but statement-II is true
(3) Both statements-I and II are true
(4) Both statements-I and II are false
CEC091

8. In Lassaigne extract of an organic compound, both nitrogen and sulphur are present, which gives blood red colour with Fe^{3+}, due to formation of
(1) NaSCN
(2) $[Fe(CN)_5NOS]^{4-}$
(3) $[Fe(SCN)]^{2+}$
(4) $Fe_4[Fe(CN)_6]3\text{-}x\ H_2O$
CEC084

9. The formula of the purple/violet colour formed in Lassaigne's test for sulphur using sodium nitroprusside is
(1) $NaFe[Fe(CN)_6]$
(2) $Na[Cr(NH_3)_2(NCS)_4]$
(3) $Na_2[Fe(CN)_5(NO)]$
(4) $Na_4[Fe(CN)_5(NOS)]$
CEC085

10. In carius method of estimation of halogen, 0.15 g of an organic compound gave 0.12g of AgBr. Find out the percentage of bromine in the compound.
(1) 40% (2) 34.04%
(3) 43.04% (4) 53.04%
CEC092

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	**2**	**3**	**2**	**3**	**1**	**2**	**3**	**3**	**4**	**2**

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EXERCISE - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. In the Kjeldahl's method for estimation of nitrogen present in a soil sample, ammonia evolved from 0.75 g of sample neutralised 10 mL of 1 M H_2SO_4. The percentage of nitrogen in the soil is :

(1) 37.33 (2) 45.33 (3) 35.33 (4) 43.33

CEC035

AIPMT 2015

2. In Duma's method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is :-

(1) 18.20 (2) 16.76 (3) 15.76 (4) 17.36

CEC036

NEET(UG) 2017

3. The most suitable method of separation of 1 : 1 mixture of ortho and para-nitrophenols is :

(1) Chromatography (2) Crystallisation

(3) Steam distillation (4) Sublimation

CEC037

NEET(UG) 2020

4. Paper chromatography is an example of:

(1) Column chromatography

(2) Adsorption chromatography

(3) Partition chromatography

(4) Thin layer chromatography

CEC041

NEET(UG) 2020 (COVID-19)

5. A liquid compound (x) can be purified by steam distillation only if it is

(1) Steam volatile, immiscible with water

(2) Not steam volatile, miscible with water

(3) Steam volatile, miscible with water

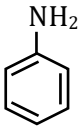
(4) Not steam volatile, immiscible with water

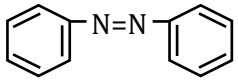
CEC044

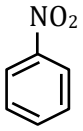
NEET(UG) 2022

6. The Kjeldahl's method for the estimation of nitrogen can be used to estimate the amount of nitrogen in which one of the following compounds ?

(1) 

(2) 

(3) 

(4) 

CEC093

NEET(UG) 2023

7. Match **List-I** with **List-II**

List-I (Mixtures/Sample)		**List-II (Technique used for purification)**	
(A)	Glycerol from spent lye	(I)	Steam distillation
(B)	Chloroform + Aniline	(II)	Fractional distillation
(C)	Fractions of crude oil	(III)	Distillation under reduced pressure
(D)	Aniline + Water	(IV)	Distillation

Choose the **correct** answer from the options given below :

(1) (A)-(III), (B)-(IV),(C)-(II),(D)-(I)

(2) (A)-(IV), (B)-(II),(C)-(I),(D)-(III)

(3) (A)-(I), (B)-(II),(C)-(III),(D)-(IV)

(4) (A)-(I), (B)-(III),(C)-(II),(D)-(IV)

CEC094

NEET(UG) 2024

8. On heating, some solid substances change from solid to vapour state without passing through liquid state. The technique used for the purification of such solid substances based on the above principle is known as :

(1) Crystallization (2) Sublimation

(3) Distillation (4) Chromatography

CEC095

NEET(UG) 2024 (Re-examination)

9. A steam volatile organic compound which is immiscible with water has a boiling point of 250°C. During steam distillation, a mixture of this organic compound and water will boil

(1) above 100°C but below 250°C

(2) above 250°C

(3) at 250°C

(4) close to but below 100°C

CEC096

10. On complete combustion, 0.3 g of an organic compound gave 0.2 g of CO_2 and 0.1 g of H_2O. The percentage composition of carbon and hydrogen in the compound, respectively is :

(1) 4.07 % and 15.02%

(2) 18.18% and 3.70%

(3) 15.02% and 4.07%

(4) 3.70% and 18.18%

CEC097

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	**1**	**2**	**3**	**3**	**1**	**2**	**1**	**2**	**4**	**2**

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EXERCISE – III (Analytical Questions) Master Your Understanding

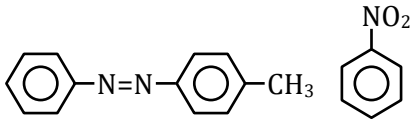
1. Match the column :-

Column-I Nature of compound/ impurity		**Column-II** Purification technique	
(a)	Liquids having sufficient difference in boiling point	(p)	Fractional distillation
(b)	Difference of boiling points of liquids is not much.	(q)	Distillation
(c)	Compounds which decompose at or below their boiling point	(r)	Steam distillation
(d)	Substances which are steam volatile and are immiscible with water	(s)	Distillation under reduced pressure

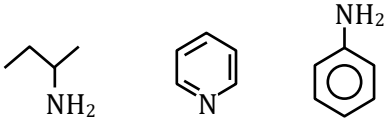
(1) a = q, b = p, c = r, d = s
(2) a = p, b = q, c = s, d = r
(3) a = q, b = p, c = s, d = r
(4) a = p, b = q, c = r, d = s

CEC080

2. For which of the following compounds Kjeldahl method is not used for estimation of nitrogen ?



(A) (B)



(C) (D) (E)

(1) A, B, D and E (2) A, B and D
(3) A and D (4) D and E

CEC079

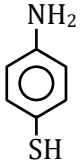
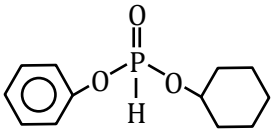
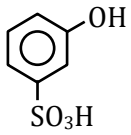
3. Match the column :-

Column-I Test product for element detection		**Column-II Colour**	
(a)	$[Fe(CN)_5NOS]^{4-}$	(p)	Prussian blue
(b)	$[Fe(SCN)]^{2+}$	(q)	Blood red
(c)	$(NH_4)_3PO_4 \cdot 12MoO_3$	(r)	Violet
(d)	$Fe_4[Fe(CN)_6]_3 \cdot xH_2O$	(s)	Yellow

(1) a = q, b = s, c = r, d = p
(2) a = p, b = q, c = r, d = s
(3) a = q, b = r, c = s, d = p
(4) a = r, b = q, c = s, d = p

CEC081

4. Match the column :-

Column-I (Compound)		**Column-II** (Observe test)	
(a)		(p)	Oxidised product gives yellow precipitate with HNO_3 followed by $(NH_4)_2MoO_4$
(b)		(q)	Sodium fusion extract gives blood red colour with Fe^{+3}
(c)		(r)	Sodium fusion extract gives black precipitate with $(CH_3COO)_2Pb$
(d)		(s)	Sodium fusion extract gives blue colour when treat with $FeSO_4$ followed by heating with conc. H_2SO_4

(1) a = q, b = s, c = p, d = r
(2) a = s, b = q, c = p, d = r
(3) a = r, b = p, c = q, d = s
(4) a = s, b = p, c = q, d = r

CEC082

5. Match the column :-

Column-I (Method of detection/estimation of element)		**Column-II** (Final form of element in compound)	
(a)	Kjeldahl's method	(p)	NH_3
(b)	Carius method	(q)	NaCN
(c)	Dumas method	(r)	N_2
(d)	Lassaigne's test	(s)	AgCl

(1) a = r, b = p, c = r, d = s
(2) a = q, b = p, c = r, d = s
(3) a = p, b = s, c = r, d = q
(4) a = r, b = s, c = p, d = q

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EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5
Answer	**3**	**2**	**4**	**2**	**3**

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Purification, Quantitative and Qualitative Analysis

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (2)

B.P. of Benzene = 353 K

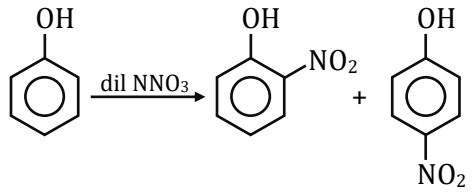
B.P. of Ether = 308 K

Difference of B.P. = 353 – 308 = 45 K

Since the difference in B.P. is greater than 40, these can be separated by simple distillation.

2. Ans. (3)

Steam distillation is used for separation and purification of organic compounds which are steam volatile and are insoluble in water.



3. Ans. (2)

Chloroform and aniline = Distillation

Glycerol and spentlye = Distillation under Reduced pressure

Aniline and water = steam distillation

Amino acid = chromatography

4. Ans. (3)

The most commonly used adsorbent in column chromatography is Al_2O_3 and silica gel because there have high surface area and porous in nature such that they absorb substance on to the surface by intermolecular forces.

5. Ans. (1)

The process of conversion of sublimable solid to vapour state directly by heating without passing through liquid state is called sublimation.

6. Ans. (2)

Mixture of chloroform and aniline can be separated by simple distillation as there two liquids have sufficient difference in boiling point.

7. Ans. (3)

Both statement I and II are true.

8. Ans. (3)

When both nitrogen and sulphur are present in an organic compound sodium thiocyanate is formed, which gives blood red colour with Fe^{+3}

$Na + C + N + S \rightarrow NaCNS$

$Fe^{+3} + SCN^- \rightarrow [Fe(SCN)]^{+2}$ (blood red)

9. Ans. (4)

Sodium fusion extract is treated with freshly prepared sodium nitroprusside, appearance of violet colour (purple) indicates presence of sulphur.

10. Ans. (2)

Mass of bromine = 80 u

Mass of Ag Br = 188 u

Weight of Ag Br = 0.12 g

Weight of organic compounds = 0.15 g

Percentage of Bromine

$$= \frac{\text{Molar mass of bromine} \times \text{weight of Ag Br}}{\text{Molar mass of Ag Br} \times \text{weight of O.C.}}$$

$$= \frac{80 \times 0.12}{188 \times 0.15} \times 100$$

Percentage of bromine = 34.04%

Exercise-II (Previous Year Questions)

1. Ans. (1)

W = 0.75g

10mol of 1M H_2SO_4

$\because$ Valancy Factor of $H_2SO_4 = 2$

$1M \longrightarrow 2 \times N$

$\Rightarrow$ milli eq. of $NH_3 = 2 \times 10$

$= 20$ millimoles

Moles of $NH_3 = \frac{20}{1000}$

Now % N $= \frac{14}{0.75} \times \frac{20}{1000} \times 100$

$= 37.33\%$

2. Ans. (2)

w = 0.25 g

$V_1 = 40$ ml, $T_1 = 300$k

725mm aq. Tension = 25 mm

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Act. pr. = 725 –25 = 700 mm

$\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}\Rightarrow V_2=\frac{700\times40}{300}\times\frac{273}{760}=33.526ml$

Now

$\therefore$ 22400ml $\longrightarrow$ 28g

1 ml $\longrightarrow \frac{28}{22400}$

33.526 ml $\longrightarrow \frac{28}{22400}\times33.526$

= 0.0419 weight of N_2

$\%\ N=\frac{0.0419}{0.25}\times100=16.76\%$

3. Ans. (3)

The most suitable method of separation of 1 : 1 mixture of ortho and para nitrophenols is steam distillation because ortho nitrophenol is steam volatile and para nitrophenol is steam non-volatile.

4. Ans. (3)

Paper chromatography is an example of partition chromatography.

5. Ans. (1)

A liquid compound (x) can be purified by steam distillation only if it is →

→ Steam volatile and immiscible with water.

6. Ans. (2)

This method is not applicable to compounds containing nitro (–NO_2), Nitroso (NO), Azo group (–N=N–), azoxy compounds $(-\overset{\uparrow}{\underset{O}{N}}=N-)$ And nitrogen present in the ring.

7. Ans. (1)

(A) Glyceral from spent lye → (iii) Distillation under reduced pressure

(B) Chloroform + aniline → (iv) distillation

(C) Fractions of crude oil → (ii) fractional distillation

(D) Aniline + water → (i) steam distillation

8. Ans. (2)

Sublimation

9. Ans. (4)

If one of the substances in the mixture is water and the other, a water insoluble substance, then the mixuture will boil close to but below 373K.

10. Ans. (2)

$C\% = \frac{12}{44}\times\frac{0.2}{0.3}\times100=18.18\%$

$H\% = \frac{2}{18}\times\frac{0.1}{0.3}\times100=3.70\%$

Exercise-III (Analytical Questions)

1. Ans. (3)

(a) Liquids having sufficient difference in boiling point → (q) Distillation

(b) Difference of boiling points of liquids is not much. → (p) Fractional distillation

(c) Compounds which decompose at or below their boiling point

→ (s) Distillation under reduced pressure

(d) Substances which are steam volatile and are immiscible with water

→ (r) Steam distillation

2. Ans. (2)

Kjeldahl method is not applicable to compounds containing nitrogen in nitro and azo groups and nitrogen present in the ring (eg. Pyridine) as nitrogen of these compounds does not change to ammonium sulphate under these conditions.

3. Ans. (4)

(a) $[Fe(CN)_5NOS]^{4-}$ → (r) Violet

(b) $[Fe(SCN)]^{2+}$ → (q) Blood red

(c) $(NH_4)_3PO_4\cdot12MoO_3$ → (s) yellow

(d) $Fe_4[Fe(CN)_6]_3\cdot xH_2O$ → (p) Prussian blue

4. Ans. (2)

(a) For N present in compound

→ (s) Na fusion extract gives blue colour

(b) For N & S present in compound

→ (q) Na fusion extract gives blood red colour

(c) For P present in compound

→ (p) Oxidised product gives yellow PPT with HNO_3 followed by $(NH_4)_2\ MoO_4$

(d) For s present in compound

→ (r) Na fusion extract gives black ppt with $(CH_3COO)_2$ Pb.

5. Ans. (3)

(a) Kjeldahl's method → (p) NH_3

(b) Carius method → (s) AgCl

(c) Dumas method → (r) N_2

(d) Lassaigne's Test→ (q) NaCN